Kube-OVN Document

v1.16.0

Kube-OVN Team

2025 Kube-OVN Team

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1. Kube-OVN



1.1 What is Kube-OVN?

Kube-OVN CNCF SDN

Kube-OVN Open Virtual Network OVN Open vSwitch Kubernetes OVN OVS Kubernetes SDN Kube-OVN Kubernetes Kubernetes

1.2 Why Kube-OVN?

Kubernetes OVN OVS

Kubernetes KubeVirt, Kube-OVN Kube-OVN SDN

SDN Kube-OVN

OVS/OVN SDN Kube-OVN IP / Underlay/Overlay VPC VPC EIP/FIP/SNAT VPC VPC L3 ECMP HA KubeVirt Non-Primary CNI BGP OVN IPSec QoS ACL ARM/x86

Kube-OVN

eBPF DPDK Kube-OVN OVN Kube-OVN Pod

Kube-OVN CPU

Kube-OVN

Kube-OVN Grafana Cilium eBPF NPM

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1.3 CNI

Kubernetes CNI

1.

2.

Kube-OVN CNI Kube-OVN Kube-OVN

1.3.1 eBPF

Cilium Calico eBPF

Kube-OVN Open vSwitch

1.3.2 CNI, Ingress, Service Mesh Observability All in One

Cilium

Kube-OVN CNI

1.3.3 OpenShift

ovn-kubernetes

OpenShift CNI Cluster Network Operator Kube-OVN RedHat Kubernetes

1.3.4 Kubernetes EKS/AKS/GKE

Kubernetes CNI

1.3.5 AI

Hostnetwork host-device

AI

1.4 OVN/ovn-kubernetes/Kube-OVN

1.4.1 OVN

OVN Open vSwitch OpenStack, Kubernetes CMS ovn-kubernetes Kube-OVN OVN Kubernetes

1.4.2 ovn-kubernetes

ovn-kubernetes OVN OVN Kubernetes CNI OpenShift OpenShift UDN, Multihoming, Hardware Acceleration

1.4.3 Kube-OVN

Kube-OVN IP Namespace OVN ovn-kubernetes annotation Pod join Underlay VPC KubeVirt

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1.5

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2.

2.1

Kube-OVN CNI Kubernetes

2.1.1

- Kubernetes >= 1.29
- Docker >= 1.12.6, Containerd >= 1.3.4
- : CentOS 7/8, Ubuntu 16.04/18.04/20.04
- Linux `geneve`, `openvswitch`, `ip_tables` `iptable_nat` Kube-OVN

1. 3.10.0-862 `netfilter` bug Kube-OVN CentOS bug Floating IPs broken after kernel upgrade to Centos/RHEL 7.5 - DNAT not working
2. Rocky Linux 8.6 4.18.0-372.9.1.el8.x86_64 TCP TCP connection failed in Rocky Linux 8.6 4.18.0-372.13.1.el8_6.x86_64
3. 4.4 `openvswitch` `openvswitch`
4. Geneve IPv6 `cat /proc/cmdline` bug Geneve tunnels don't work when ipv6 is disabled

2.1.2

- IPv6 `ipv6.disable=1` 0
- `kube-proxy` Kube-OVN Service ClusterIP `kube-apiserver`
- kubelet CNI , kubelet `--network-plugin=cni --cni-bin-dir=/opt/cni/bin --cni-conf-dir=/etc/cni/net.d`
- `/etc/cni/net.d/`

2.1.3

ovn-central	6641/tcp	ovn nb db server
ovn-central	6642/tcp	ovn sb db server
ovn-central	6643/tcp	ovn northd server
ovn-central	6644/tcp	ovn raft server
ovn-ic	6645/tcp	ovn ic nb db server
ovn-ic	6646/tcp	ovn ic sb db server
ovs-ovn	Geneve 6081/udp, STT 7471/tcp, Vxlan 4789/udp	
kube-ovn-controller	10660/tcp	
kube-ovn-daemon	10665/tcp	
kube-ovn-monitor	10661/tcp	

firewalld Packet Forwarding Masquerade

```
#    Packet Forwarding
firewall-cmd --add-forward --permanent
```


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```
#    IPv4 Masquerade
firewall-cmd --add-masquerade --permanent
#   Kube-OVN IPv6/        Masquerade
firewall-cmd --permanent --add-rich-rule 'rule family="ipv6" source address="fd00:10:16::/112" masquerade'

firewall-cmd --reload
```

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2.1.4

2.2

Kube-OVN Charts Kube-OVN Overlay

Underlay/Vlan Underlay

Kube-OVN

2.2.1

release

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/refs/tags/v1.16.16/dist/images/install.sh
```

master

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/images/install.sh
```

```
REGISTRY="docker.io/kubeovn"           #
VERSION="v1.16.16"                      #     /Tag
POD_CIDR="10.16.0.0/16"                #      CIDR     SVC/NODE/JOIN CIDR
SVC_CIDR="10.96.0.0/12"                #     apiserver   service-cluster-ip-range
JOIN_CIDR="100.64.0.0/16"              # Pod         CIDR     SVC/NODE/POD CIDR
LABEL="node-role.kubernetes.io/control-plane" #    OVN DB
IFACE=""                               #                        Kubernetes    Node IP
TUNNEL_TYPE="geneve"                   #           geneve, vxlan   stt stt       ovs
```

`IFACE=enp6s0f0,eth.*`

> root

`bash install.sh`

Kube-OVN

1. `[Step 4/6]` Pod
2. Kube-OVN

2.2.2 Helm Chart

Kube-OVN Helm Kube-OVN

IP

```
# kubectl get node -o wide
NAME                   STATUS     ROLES           AGE   VERSION   INTERNAL-IP   EXTERNAL-IP   OS-IMAGE             KERNEL-VERSION      CONTAINER-RUNTIME
kube-ovn-control-plane   NotReady   control-plane   20h   v1.26.0   172.18.0.3    <none>        Ubuntu 22.04.1 LTS   5.10.104-linuxkit   containerd://1.6.9
kube-ovn-worker          NotReady   <none>          20h   v1.26.0   172.18.0.2    <none>        Ubuntu 22.04.1 LTS   5.10.104-linuxkit   containerd://1.6.9
```


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label

```
# kubectl label node -lbeta.kubernetes.io/os=linux kubernetes.io/os=linux --overwrite
node/kube-ovn-control-plane not labeled
node/kube-ovn-worker not labeled

# kubectl label node -lnode-role.kubernetes.io/control-plane kube-ovn/role=master --overwrite
node/kube-ovn-control-plane labeled

#    label    dpdk         dpdk
# kubectl label node -lovn.kubernetes.io/ovs_dp_type!=userspace ovn.kubernetes.io/ovs_dp_type=kernel --overwrite
node/kube-ovn-control-plane labeled
node/kube-ovn-worker labeled
```

Helm Repo

```
# helm repo add kubeovn https://kubeovn.github.io/kube-ovn/
"kubeovn" has been added to your repositories

# helm repo list
NAME            URL
kubeovn         https://kubeovn.github.io/kube-ovn/

# helm repo update kubeovn
Hang tight while we grab the latest from your chart repositories...
...Successfully got an update from the "kubeovn" chart repository
Update Complete. ⎈Happy Helming!⎈

# helm search repo kubeovn
NAME                    CHART VERSION   APP VERSION     DESCRIPTION
kubeovn/kube-ovn        v1.16.16        v1.16.16         Helm chart for Kube-OVN
```

helm install Kube-OVN

Chart `values.yaml`

```
# helm install kube-ovn kubeovn/kube-ovn --wait -n kube-system --version v1.16.16
NAME: kube-ovn
LAST DEPLOYED: Thu Apr 24 08:30:13 2025
NAMESPACE: kube-system
STATUS: deployed
REVISION: 1
TEST SUITE: None
```

Helm `values.yaml`

```
helm upgrade -f values.yaml kube-ovn kubeovn/kube-ovn --wait -n kube-system --version v1.16.16
```

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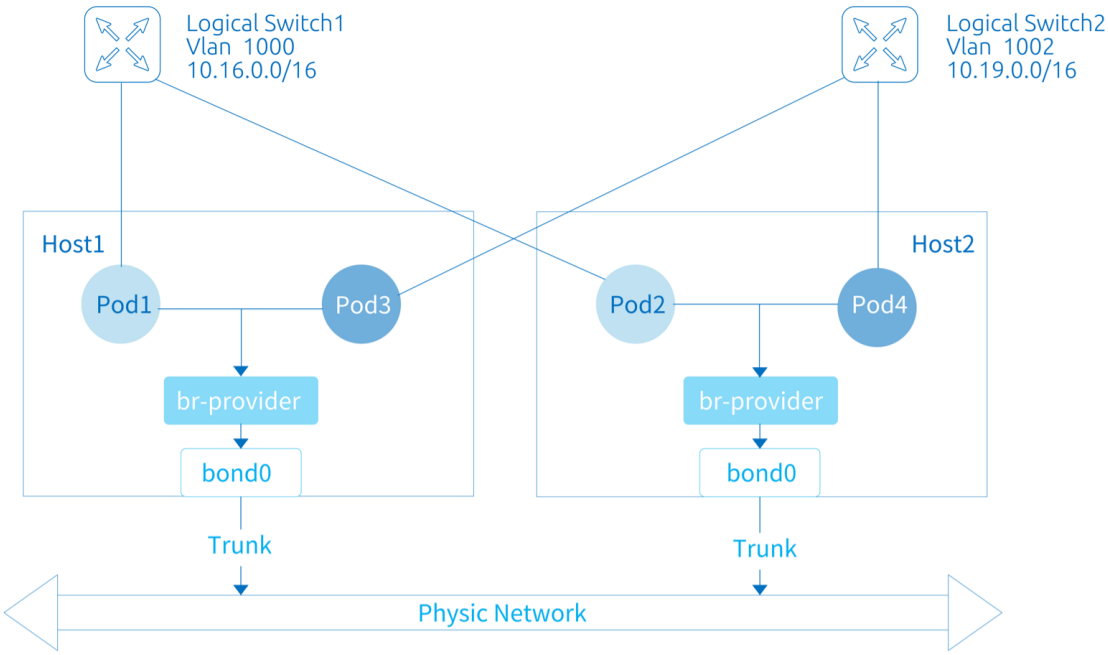
2.2.3

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2.3 Underlay

Kube-OVN Geneve Overlay

Kube-OVN Underlay



2.3.1

Overlay SNAT/EIP / L3 VPC Underlay

2.3.2 Macvlan

Kube-OVN Underlay Macvlan

1. Macvlan OVS Macvlan
2. Kube-OVN arp-proxy arp
3. Macvlan netfilter Service NetworkPolicy Kube-OVN OVS Service NetworkPolicy
4. Kube-OVN Underlay Macvlan IP QoS

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2.3.3

Underlay OVS OVS L2/L3 Vlan

1. OpenStack VM `PortSecurity`
2. VMware vSwitch `MAC Address Changes`, `Forged Transmits` `Promiscuous Mode Operation` `allow`
3. VMware NSX-T Underlay
4. Hyper-V `MAC Address Spoofing`
5. AWS GCE Mac Underlay Underlay VPC-CNI
6. Linux Bridge

Kube-OVN Underlay Underlay

Underlay

Kube-OVN Mac IP MTU OVS Bridge Underlay OVS Bridge `br-PROVIDER_NAME` `PROVIDER_NAME` `ProviderNetwork` provider

1. OVS Bridge NetworkManager DHCP
2. OVS Bridge OVS Kube-OVN OVS Underlay

2.3.4

Underlay Pod Underlay

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/install.sh
```

```
ENABLE_ARP_DETECT_IP_CONFLICT #            vlan    arp
NETWORK_TYPE                  #     vlan
VLAN_INTERFACE_NAME           #                  eth1
VLAN_ID                       #         VLAN Tag      0     VLAN
POD_CIDR                      #         CIDR     192.168.1.0/24
POD_GATEWAY                   #              192.168.1.1
EXCLUDE_IPS                   #                   IP       192.168.1.1..192.168.1.100
ENABLE_LB                     #    Underlay        Service       true
EXCHANGE_LINK_NAME            #        provider-network   OVS               false
LS_DNAT_MOD_DL_DST            # DNAT      MAC           Service         true
```

```
bash install.sh
```

2.3.5 CRD Underlay

Underlay Pod `ProviderNetwork` `Vlan` `Subnet`

ProviderNetwork

ProviderNetwork Underlay

ProviderNetwork :

```yaml
apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: net1
```


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```yaml
spec:
  defaultInterface: eth1
  customInterfaces:
    - interface: eth2
      nodes:
        - node1
  nodeSelector:
    matchLabels:
      kubernetes.io/arch: amd64
      network-type: underlay
    matchExpressions:
      - key: kubernetes.io/hostname
        operator: In
        values:
          - node1
          - node2
```

ProviderNetwork **12**

- `defaultInterface`: Bond Vlan ProviderNetwork excludeNodes br-net1 `br-NAME` OVS
- `customInterfaces`:
- `nodeSelector`: OVS `matchLabels` `matchExpressions`
- `excludeNodes`: `net1.provider-network.ovn.kubernetes.io/exclude=true` **`nodeSelector`** **`excludeNodes`** **`nodeSelector`**
- `autoCreateVlanSubinterfaces`: `false` `true` VLAN `<interface>.<vlanid>`

Key	Value	
net1.provider-network.ovn.kubernetes.io/ready	true	ProviderNetwork
net1.provider-network.ovn.kubernetes.io/interface	eth1	
net1.provider-network.ovn.kubernetes.io/mtu	1500	MTU

> IP IP OVS

VLAN

ProviderNetwork `autoCreateVlanSubinterfaces: true` VLAN VLAN

```yaml
apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: vlan-provider
spec:
  defaultInterface: eth0.341
  autoCreateVlanSubinterfaces: true
  customInterfaces:
    - interface: eth2.341
      nodes:
        - worker-node-1
        - worker-node-2
```

`<interface>.<vlanid>` VLAN VLAN ProviderNetwork

VLAN

Vlan Vlan Tag ProviderNetwork

VLAN

```yaml
apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
```


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```
  name: vlan1
spec:
  id: 0
  provider: net1
```

- `id`： VLAN ID/Tag Kube-OVN Vlan Vlan 0 vlan localnet
- `provider`： ProviderNetwork VLAN ProviderNetwork

Subnet

Vlan

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  protocol: IPv4
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  vlan: vlan1
  disableGatewayCheck: false
```

- `vlan` VLAN Subnet VLAN
- `disableGatewayCheck` Underlay `true`

2.3.6

IP

IP Pod IP Mac

2.3.7

Kube-OVN Underlay `spec.logicalGateway` `true`

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  protocol: IPv4
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  vlan: vlan1
  logicalGateway: true
```

Pod Kube-OVN Logical Router

2.3.8 Underlay Overlay

Underlay Overlay Overlay Pod NAT Underlay Pod IP Underlay Pod Overlay
Overlay Underlay Pod Pod IP Overlay Pod

Underlay Overlay `u2oInterconnection` `true` Kube-OVN Underlay IP Underlay `ovn-cluster`
Kube-OVN Underlay Overlay

> **Warning**
>
> Vlan Underlay Underlay u2o

IP

subnet IP Underlay Subnet `u2oInterconnectionIP`

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Underlay Subnet VPC

Underlay Subnet VPC Overlay Subnet VPC `u2oInterconnection` `true` `subnet.spec.vpc` VPC

2.3.9

IP Netplan Ubuntu Netplan renderer NetworkManager IP DHCP

```
network:
  renderer: NetworkManager
  ethernets:
    eth0:
      dhcp4: no
      addresses:
        - 172.16.143.129/24
  version: 2
```

IP netplan

```
netplan generate

nmcli connection reload netplan-eth0
nmcli device set eth0 managed yes
```

Kube-OVN IP OVS

NetworkManager CentOS

```
nmcli connection reload eth0
nmcli device set eth0 managed yes
nmcli -t -f GENERAL.STATE device show eth0 | grep -qw unmanaged || nmcli device reapply eth0
```

IP MAC

2.3.10

hairpin Pod

hairpin Pod Pod OVS MAC

hairpin Kube-OVN

Pod Pod

Pod 300 ARP OVS resubmit

```
2022-11-13T08:43:46.782Z|00222|ofproto_dpif_upcall(handler5)|WARN|Flow:
arp,in_port=331,vlan_tci=0x0000,dl_src=00:00:00:25:eb:39,dl_dst=ff:ff:ff:ff:ff:ff,arp_spa=10.213.131.240,arp_tpa=10.213.159.254,arp_op=1,arp_sha=00:00:00:25:e
b:39,arp_tha=ff:ff:ff:ff:ff:ff

bridge("br-int")
----------------
 0. No match.
     >>>> received packet on unknown port 331 <<<<
    drop

Final flow: unchanged
Megaflow: recirc_id=0,eth,arp,in_port=331,dl_src=00:00:00:25:eb:39
Datapath actions: drop
2022-11-13T08:44:34.077Z|00224|ofproto_dpif_xlate(handler5)|WARN|over 4096 resubmit actions on bridge br-int while processing
arp,in_port=13483,vlan_tci=0x0000,dl_src=00:00:00:59:ef:13,dl_dst=ff:ff:ff:ff:ff:ff,arp_spa=10.213.152.3,arp_tpa=10.213.159.254,arp_op=1,arp_sha=00:00:00:59:e
f:13,arp_tha=ff:ff:ff:ff:ff:ff
```

OVN NB `bcast_arp_req_flood` `false`

```
kubectl ko nbctl set NB_Global . options:bcast_arp_req_flood=false
```

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2.3.11

2.4 CNI

Kube-OVN CNI CNI Cilium Calico Flannel

2.4.1

CNI Kube-OVN CNI Pod CNI eth0

- CNI Kube-OVN VPC
-
- Pod
- Kube-OVN VPC NAT

2.4.2

1. **CNI** Kubernetes CNI Cilium Calico Flannel
2. **Multus CNI**
3. **Kube-OVN**

1. CNI
2. Multus CNI
3. Kube-OVN

2.4.3

Helm Chart v2

```
# values.yaml
cni:
  nonPrimaryCNI: true
```

Helm

```
helm install kube-ovn ./charts/kube-ovn-v2 \
  --namespace kube-system \
  --set cni.nonPrimaryCNI=true
```

Helm Chart v1

```
# values.yaml
cni_conf:
  NON_PRIMARY_CNI: true
```

Helm

```
helm install kube-ovn ./charts/kube-ovn \
  --namespace kube-system \
  --set cni_conf.NON_PRIMARY_CNI=true
```

`kube-ovn-controller` Deployment `kube-ovn-cni` DaemonSet Pod

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```
# kube-ovn-controller Deployment
containers:
- name: kube-ovn-controller
  args:
  - --non-primary-cni-mode=true
```

```
# kube-ovn-cni DaemonSet
containers:
- name: kube-ovn-cni
  args:
  - --non-primary-cni-mode=true
```

2.4.4

NADs

NAD

```
apiVersion: k8s.cni.cncf.io/v1
kind: NetworkAttachmentDefinition
metadata:
  name: kube-ovn-vpc-network
  namespace: default
spec:
  config: |
    {
      "cniVersion": "0.3.1",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "kube-ovn-vpc-network.default.ovn"
    }
```

VPC

VPC

```
apiVersion: kubeovn.io/v1
kind: Vpc
metadata:
  name: vpc-secondary
spec:
  namespaces:
  - default
---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet-secondary
spec:
  vpc: vpc-secondary
  cidr: "10.100.0.0/16"
  gateway: "10.100.0.1"
  provider: kube-ovn-vpc-network.default.ovn
```

Pod

Pod

```
apiVersion: v1
kind: Pod
metadata:
  name: multi-network-pod
  annotations:
    k8s.v1.cni.cncf.io/networks: default/kube-ovn-vpc-network
spec:
  containers:
  - name: app
    image: nginx
```

2.4.5

eth0

- CNI Cilium Calico
-

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- Kubernetes

net1 net2...

- Kube-OVN
-
- VPC
- Kube-OVN QoS NAT

2.4.6

1.

```
#    Pod
apiVersion: v1
kind: Pod
metadata:
  name: frontend
  annotations:
    k8s.v1.cni.cncf.io/networks: |
      [
        {"name": "default/public-network"},
        {"name": "default/internal-network"}
      ]
```

2.

```
#    A Pod
apiVersion: v1
kind: Pod
metadata:
  name: tenant-a-app
  annotations:
    k8s.v1.cni.cncf.io/networks: default/tenant-a-network

---
#    B Pod
apiVersion: v1
kind: Pod
metadata:
  name: tenant-b-app
  annotations:
    k8s.v1.cni.cncf.io/networks: default/tenant-b-network
```

3. VPC NAT

```
apiVersion: kubeovn.io/v1
kind: VpcNatGateway
metadata:
  name: vpc-nat-gw
spec:
  vpc: vpc-secondary
  subnet: subnet-secondary
  lanIp: "10.100.0.254"
```

2.4.7

Pod

```
metadata:
  annotations:
    k8s.v1.cni.cncf.io/networks: |
      [
```


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```
        {"name": "default/kube-ovn-network-1", "interface": "net1"},
        {"name": "default/kube-ovn-network-2", "interface": "net2"}
      ]
```

IP

IP

```
#    IP
metadata:
  annotations:
    k8s.v1.cni.cncf.io/networks: |
      [{
        "name": "default/kube-ovn-network",
        "ips": ["10.100.0.100/16"]
      }]
```

QoS

QoS

```
metadata:
  annotations:
    ovn.kubernetes.io/ingress_rate: "1000"
    ovn.kubernetes.io/egress_rate: "1000"
```

2.4.8

1. **Pod** - NAD - kube-ovn-cni - Multus
2. - VPC - Pod -
3. **IP** - CIDR - IP - IPAM

```
#    Pod
kubectl get pod <pod-name> -o yaml | grep -A 10 "networks-status"

#    Pod
kubectl exec <pod-name> -- ip addr show

#
kubectl exec <pod-name> -- ip route show

#    kube-ovn-cni
kubectl logs -n kube-system daemonset/kube-ovn-cni
```

2.4.9

1. Kubernetes
2.
3. **DNS** Pod DNS

2.4.10

1. IP
2.
3.
4.
5.

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2.4.11

- Multus CNI
-
- Kube-OVN

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2.4.12

2.5 Talos

Talos Linux Kubernetes Linux

2.5.1 Helm Chart Kube-OVN

Talos Linux Kube-OVN

```
helm install kube-ovn kubeovn/kube-ovn --wait \
    -n kube-system \
    --version v1.16.16 \
    --set OVN_DIR=/var/lib/ovn \
    --set OPENVSWITCH_DIR=/var/lib/openvswitch \
    --set DISABLE_MODULES_MANAGEMENT=true \
    --set cni_conf.MOUNT_LOCAL_BIN_DIR=false
```

Underlay Helm Chart

```
helm install kubeovn kubeovn/kube-ovn --wait \
    -n kube-system \
    --version v1.16.16 \
    --set OVN_DIR=/var/lib/ovn \
    --set OPENVSWITCH_DIR=/var/lib/openvswitch \
    --set DISABLE_MODULES_MANAGEMENT=true \
    --set cni_conf.MOUNT_LOCAL_BIN_DIR=false \
    --set networking.NETWORK_TYPE=vlan \
    --set networking.vlan.VLAN_INTERFACE_NAME=enp0s5f1 \
    --set networking.vlan.VLAN_ID=0 \    # VLAN_ID=0      VLAN tag    untagged      chart      100
    --set networking.NET_STACK=ipv4 \
    --set-json networking.EXCLUDE_IPS='"172.99.99.11..172.99.99.99"' \
    --set-json ipv4.POD_CIDR='"172.99.99.0/24"' \
    --set-json ipv4.POD_GATEWAY='"172.99.99.1"'
```

Note

VLAN Bond Bridge Underlay Underlay Talos `ignore=true`

```
machine:
  network:
    interfaces:
      - interface: enp0s5f1
        ignore: true
```

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2.5.2

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2.6

Kube-OVN pre-v0.0.1
Kube-OVN

2.6.1

- Linkerd
- Elasticsearch
- EMQX
- KubeSphere

2.6.2 OpenVswitch/OVN

Kube-OVN OpenVswitch OVN

- OVN
- OpenVswitch

ovn-architecture

2.6.3 Kube-OVN

Kube-OVN

Kube-OVN kubectl bash
Kube-OVN

2.6.4

Kube-OVN

Kube-OVN Kube-OVN E2E

2.6.5

Kube-OVN OpenTelemetry DeepFlow

2.6.6

Kube-OVN 7*24 Github Issue
Github Issue AI AI

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2.6.7

2.7

Kube-OVN Kube-OVN OVS issue Kube-OVN

2.7.1 Kubernetes

Script Uninstall **Helm Uninstall**

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/cleanup.sh
bash cleanup.sh

helm uninstall kube-ovn -n kube-system
```

2.7.2

ovsdb openvswitch

```
rm -rf /var/run/openvswitch
rm -rf /var/run/ovn
rm -rf /etc/origin/openvswitch/
rm -rf /etc/origin/ovn/
rm -rf /etc/cni/net.d/01-kube-ovn.conflist
#       v1.10         00-kube-ovn.conflist
rm -rf /etc/cni/net.d/00-kube-ovn.conflist
rm -rf /var/log/openvswitch
rm -rf /var/log/ovn
rm -fr /var/log/kube-ovn
```

2.7.3

iptable/ipset

```
reboot
```

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2.7.4

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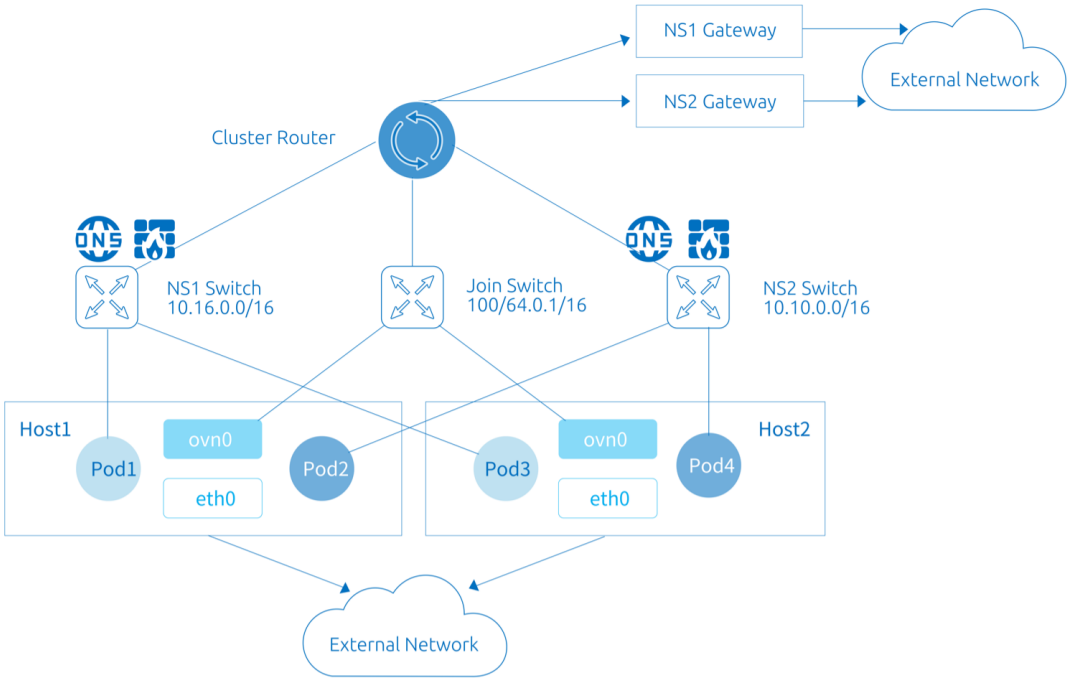
3.

3.1

Kube-OVN Kube-OVN IP Namespace Namespace Pod IP CIDR NAT

CNI Kube-OVN

VPC IP VPC peering VPN IP



Overlay Underlay

3.1.1

Kube-OVN Namespace IP CIDR

Overlay NAT Flannel

Underlay arping

spec default true `ovn-default`

```
# kubectl get subnet ovn-default -o yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  creationTimestamp: "2019-08-06T09:33:43Z"
  generation: 1
  name: ovn-default
  resourceVersion: "1571334"
```


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```
  selfLink: /apis/kubeovn.io/v1/subnets/ovn-default
  uid: 7e2451f8-fb44-4f7f-b3e0-cfd27f6fd5d6
spec:
  cidrBlock: 10.16.0.0/16
  default: true
  excludeIps:
  - 10.16.0.1
  gateway: 10.16.0.1
  gatewayType: distributed
  natOutgoing: true
  private: false
  protocol: IPv4
```

3.1.2 Join

Kubernetes Node Pod Overlay Kube-OVN `join` Node ovn0 `join` Pod

Pods Nodes ovn0 Node Pod ovn0 ovn0

join CIDR Join

Join

> Pod `hostport` `externalTrafficPolicy: Local` NodePort Service
>
> `join` CIDR

```
# kubectl get subnet join -o yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  creationTimestamp: "2019-08-06T09:33:43Z"
  generation: 1
  name: join
  resourceVersion: "1571333"
  selfLink: /apis/kubeovn.io/v1/subnets/join
  uid: 9c744810-c678-4d50-8a7d-b8ec12ef91b8
spec:
  cidrBlock: 100.64.0.0/16
  default: false
  excludeIps:
  - 100.64.0.1
  gateway: 100.64.0.1
  gatewayNode: ""
  gatewayType: ""
  natOutgoing: false
  private: false
  protocol: IPv4
```

Node ovn0

```
# ifconfig ovn0
ovn0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1420
        inet 100.64.0.4  netmask 255.255.0.0  broadcast 100.64.255.255
        inet6 fe80::800:ff:fe40:5  prefixlen 64  scopeid 0x20<link>
        ether 0a:00:00:40:00:05  txqueuelen 1000  (Ethernet)
        RX packets 18  bytes 1428 (1.3 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 19  bytes 1810 (1.7 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

3.1.3

Namespace

```
cat <<EOF | kubectl create -f -
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  protocol: IPv4
  cidrBlock: 10.66.0.0/16
  excludeIps:
```


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```
  - 10.66.0.1..10.66.0.10
  - 10.66.0.101..10.66.0.151
  gateway: 10.66.0.1
  gatewayType: distributed
  natOutgoing: true
  routeTable: ""
  namespaces:
  - ns1
  - ns2
EOF
```

- `cidrBlock`: CIDR VPC Subnet CIDR
- `excludeIps`: IP Underlay
- `gateway` Overlay Kube-OVN Underlay
- `namespaces`: Namespace Namespace Pod
- `routeTable`:

```
# kubectl create ns ns1
namespace/ns1 created

# kubectl run nginx --image=docker.io/library/nginx:alpine -n ns1
deployment.apps/nginx created

# kubectl get pod -n ns1 -o wide
NAME                     READY   STATUS    RESTARTS   AGE   IP           NODE    NOMINATED NODE   READINESS GATES
nginx-74d5899f46-n8wtg   1/1     Running   0          10s   10.66.0.11   node1   <none>           <none>
```

Workload

Pod Namespace IP Namespace Workload Pod Annotatior `ovn.kubernetes.io/logical_switch`

```
apiVersion: v1
kind: Pod
metadata:
  name: another-subnet
  annotations:
    ovn.kubernetes.io/logical_switch: subnet1
spec:
  containers:
  - name: another-subnet
    image: docker.io/library/nginx:alpine
```

Workload Deployment StatefulSet `ovn.kubernetes.io/logical_switch` Annotation `spec.template.metadata.annotations`

3.1.4 Overlay

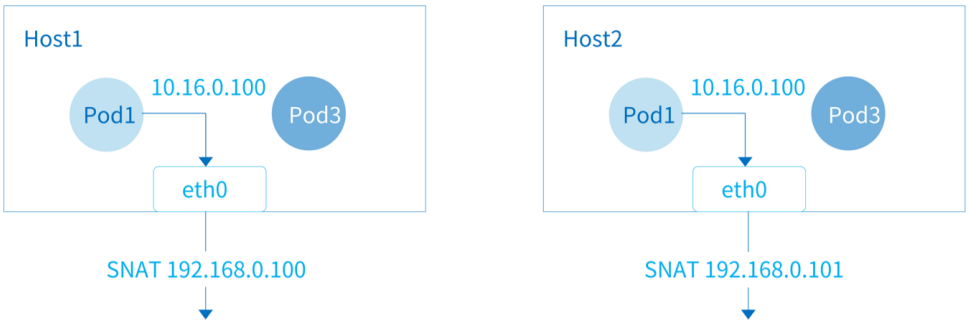
> VPC Overlay Underlay VPC

Overlay Pod Kube-OVN

`natOutgoing` Pod snat

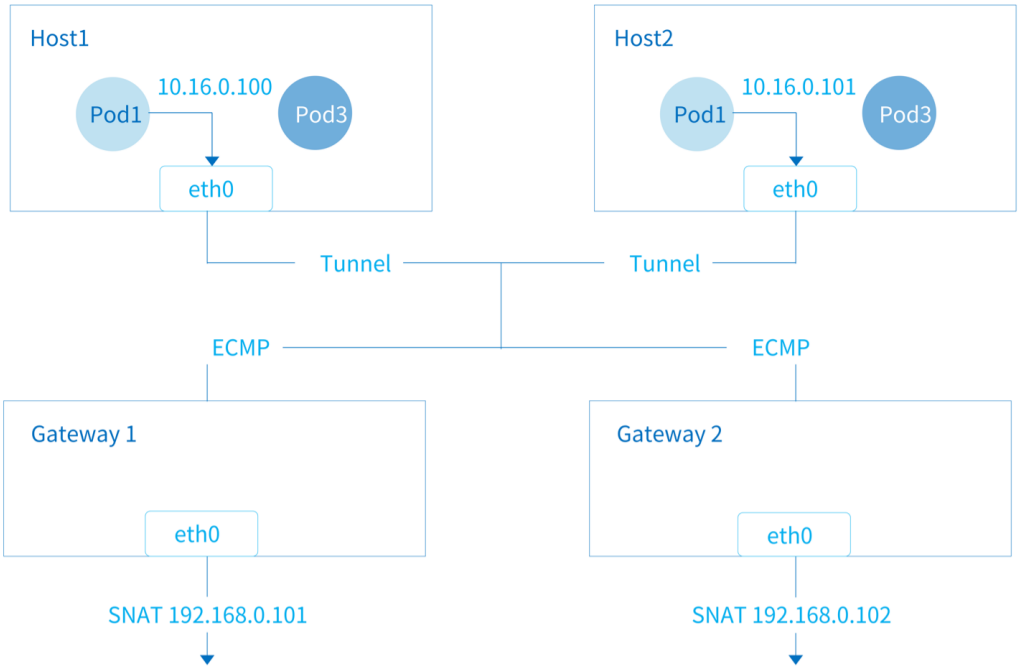
Node Node Pod `ovn0` `natOutgoing` `true` Pod IP

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`gatewayType` `distributed`

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: distributed
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
  - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: distributed
  natOutgoing: true
```



IP Pod `ovn0` `natOutgoing` `true` Pod IP

`gatewayType` `centralized` `gatewayNode` Kubernetes NodeName `gatewayNode`

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: centralized
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
  - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: centralized
  gatewayNode: "node1,node2"
  natOutgoing: true
```

- `gatewayNode` `kube-ovn-worker:172.18.0.2, kube-ovn-control-plane:172.18.0.3`
- ECMP ECMP
- Kube-OVN v1.12.0 subnet crd spec `enableEcmp` ECMP ECMP `kube-ovn-controller` Deployment `enable-ecmp` v1.12.0

ECMP kube-ovn-controller ping 5s 5s-10s Node Ready

`gatewayNodeSelectors`

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gatewayNodeSelectors

- `gatewayNode` `gatewayNode` `gatewayNodeSelectors`
- OR
-

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: centralized-selector
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
  - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: centralized
  gatewayNodeSelectors:
    - matchLabels:
        role: gateway
    - matchExpressions:
        - key: node-type
          operator: In
          values: ["gateway", "egress"]
  natOutgoing: true
```

3.1.5 ACL

Warning

Kube-OVN NetworkPolicy Network Policy API Subnet ACL Security Group OVN ACL NetworkPolicy NetworkPolicy API

ACL Kube-OVN Subnet ACL

Subnet ACL OVN ACL ovn-nb ACL Table `match` ovn-sb Logical Flow Table

IP `10.10.0.2` Pod ACL

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: acl
spec:
  allowEWTraffic: false
  acls:
    - action: drop
      direction: to-lport
      match: ip4.dst == 10.10.0.2 && ip
      priority: 1002
    - action: allow-related
      direction: from-lport
      match: ip4.src == 10.10.0.2 && ip
      priority: 1002
  cidrBlock: 10.10.0.0/24
```

ACL `allowEWTraffic: true`

3.1.6

ACL ACL

Kube-OVN Pod

CRD `private` true `allowSubnets` `allowSubnets`

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```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: private
spec:
  protocol: IPv4
  default: false
  namespaces:
  - ns1
  - ns2
  cidrBlock: 10.69.0.0/16
  private: true
  allowSubnets:
  - 10.16.0.0/16
  - 10.18.0.0/16
```

3.1.7 Underlay

> Underlay

- `vlan`: Underlay Subnet Vlan CR Underlay
- `logicalGateway`: Underlay OVN Underlay Overlay `false`

3.1.8

`kube-ovn-cni` Pod ICMP ARP Underlay ICMP

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: disable-gw-check
spec:
  disableGatewayCheck: true
```

3.1.9 Multicast-Snoop

subnet Pod OVN Pod subnet multicast snoop OVN `South Database` `Multicast_Group`

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: sample1
spec:
  enableMulticastSnoop: true
```

3.1.10 Subnet MTU

Subnet Pod MTU Subnet Pod

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: sample1
spec:
  mtu: 1300
```

3.1.11

- IP
- VPC NAT
- QoS
-
- DHCP
-
-

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- IP

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3.1.12

3.2

Kube-OVN Pod Namespace IP Mac Kube-OVN

- Pod IP/Mac
- IP/MAC
- Workload IP Pool
- StatefulSet
- KubeVirt VM
- Multus

3.2.1 Pod IP Mac

Pod annotation Pod IP/Mac, `kube-ovn-controller`

```
apiVersion: v1
kind: Pod
metadata:
  name: static-ip
  annotations:
    ovn.kubernetes.io/ip_address: 10.16.0.15   #           10.16.0.15,fd00:10:16::15
    ovn.kubernetes.io/mac_address: 00:00:00:53:6B:B6
spec:
  containers:
  - name: static-ip
    image: docker.io/library/nginx:alpine
```

annotation Pod IP/Mac

1. IP/Mac IP/Mac
2. IP CIDR
3. IP Mac

3.2.2 IP/MAC

Pod VM Multus IP/MAC Multus `k8s.v1.cni.cncf.io/networks` `interface` annotation

- `<nadName>.<nadNamespace>.kubernetes.io/ip_address.<interfaceName>` IP
- `<nadName>.<nadNamespace>.kubernetes.io/mac_address.<interfaceName>` MAC

`<nadName>` `<nadNamespace>` NetworkAttachmentDefinition `<interfaceName>` Multus `interface` annotation annotation `ovn.kubernetes.io/ip_address` / `ovn.kubernetes.io/mac_address`

NAD IP/MAC

```
apiVersion: v1
kind: Pod
metadata:
  name: multi-if-static
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: '[{"name": "attachnet", "namespace": "default", "interface": "net1"}, {"name": "attachnet", "namespace": "default",
"interface": "net2"}]'
    attachnet.default.kubernetes.io/ip_address.net1: 172.17.0.100
    attachnet.default.kubernetes.io/mac_address.net1: 00:00:00:53:6B:B1
    attachnet.default.kubernetes.io/ip_address.net2: 172.17.0.101
    attachnet.default.kubernetes.io/mac_address.net2: 00:00:00:53:6B:B2
spec:
  containers:
  - name: multi-if-static
    image: docker.io/library/nginx:alpine
```


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3.2.3 Workload IP Pool

Kube-OVN annotation `ovn.kubernetes.io/ip_pool` Workload Deployment/StatefulSet/DaemonSet/Job/CronJob IP `kube-ovn-controller` `ovn.kubernetes.io/ip_pool` IP

IP Pool Annotation `template` `annotation` Kubernetes Workload Workload

Deployment IP

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ippool
  labels:
    app: ippool
spec:
  replicas: 2
  selector:
    matchLabels:
      app: ippool
  template:
    metadata:
      labels:
        app: ippool
      annotations:
        ovn.kubernetes.io/ip_pool: 10.16.0.15,10.16.0.16,10.16.0.17 #
10.16.0.15,fd00:10:16::000E;10.16.0.16,fd00:10:16::000F;10.16.0.17,fd00:10:16::0010
    spec:
      containers:
      - name: ippool
        image: docker.io/library/nginx:alpine
```

Workload IP

1. `ovn.kubernetes.io/ip_pool` IP CIDR
2. `ovn.kubernetes.io/ip_pool` IP IP
3. `ovn.kubernetes.io/ip_pool` IP replicas Pod Workload `ovn.kubernetes.io/ip_pool` IP

3.2.4 StatefulSet

StatefulSet IP Workload `ovn.kubernetes.io/ip_pool` Pod IP

StatefulSet Kube-OVN

1. Pod `ovn.kubernetes.io/ip_pool` IP StatefulSet web web-0 `ovn.kubernetes.io/ip_pool` IP web-1 IP
2. StatefulSet Pod OVN logical_switch_port Pod interface Pod IP/Mac StatefulSet Volume
3. 2 `ovn.kubernetes.io/ip_pool` StatefulSet Pod IP/Mac StatefulSet

StatefulSet

```yaml
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: web
spec:
  serviceName: "nginx"
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: docker.io/library/nginx:alpine
        ports:
        - containerPort: 80
          name: web
```


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StatefulSet Pod Pod IP

StatefulSet Pod IP

StatefulSet IP Pod Name Statefulset `ovn.kubernetes.io/ip_pool` Annotation Pod IP

StatefulSet Pod IP StatefulSet scale 0 Annotation StatefulSet

3.2.5 KubeVirt VM

KubeVirt VM `kube-ovn-controller` StatefulSet Pod IP VM
VM IP

3.2.6 Multus

Multus Pod Kube-OVN annotation Kube-OVN CNI Kube-OVN IPAM CNI

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3.2.7

3.3 IP

IP IPPool Subnet IPAM IP Namespace Workload

3.3.1

```yaml
apiVersion: kubeovn.io/v1
kind: IPPool
metadata:
  name: pool-1
spec:
  subnet: ovn-default
  ips:
  - "10.16.0.201"
  - "10.16.0.210/30"
  - "10.16.0.220..10.16.0.230"
  namespaces:
  - ns-1
  enableAddressSet: true
```

Workload

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ippool
  labels:
    app: ippool
spec:
  replicas: 2
  selector:
    matchLabels:
      app: ippool
  template:
    metadata:
      labels:
        app: ippool
      annotations:
        ovn.kubernetes.io/ip_pool: pool-1
    spec:
      containers:
      - name: ippool
        image: docker.io/library/nginx:alpine
```

subnet
ips IP .. IPv6
namespaces Pod IP IP
enableAddressSet AddressSet false true ACL AddressSet

3.3.2

1. Workload IP Pool IP IP
2. IP `.spec.ips` IP IP `.spec.ips` CIDR
3. IP IP
4. IP `.spec.ips`
5. IP IP IP IP IP IP
6. IP IP
7. IP Namespace

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8. IP `.spec.enableAddressSet` `false` `true` IP OVN NB AddressSet IP IP AddressSet NetworkPolicy VPC AddressSet

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3.3.3

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3.4

Pod Annotations

```
apiVersion: v1
kind: Pod
metadata:
  name: custom-routes
  annotations:
    ovn.kubernetes.io/routes: |
      [{
        "dst": "192.168.0.101/24",
        "gw": "10.16.0.254"
      }, {
        "gw": "10.16.0.254"
      }]
spec:
  containers:
  - name: nginx
    image: docker.io/library/nginx:alpine
```

`dst`

Deployment DaemonSet StatefulSet Annotation `.spec.template.metadata.annotations`

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: custom-routes
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
      annotations:
        ovn.kubernetes.io/routes: |
          [{
            "dst": "192.168.0.101/24",
            "gw": "10.16.0.254"
          }, {
            "gw": "10.16.0.254"
          }]
    spec:
      containers:
      - name: nginx
        image: docker.io/library/nginx:alpine
```

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3.4.1

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3.5 EIP SNAT

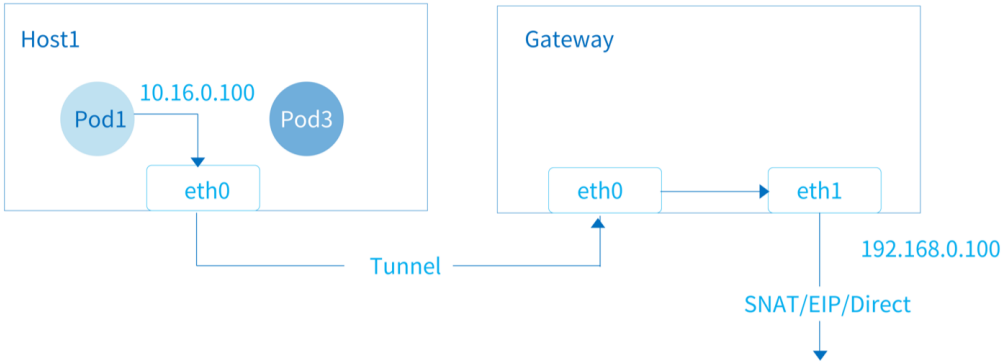
> VPC VPC NAT

- VPC Iptables NAT
- VPC OVN NAT

> VPC ,

- VPC OVN NAT pod annotation
- VPC OVN NAT provider-network vlan subnet CRD ovn-eip ovn-dnat ovn-fip ovn-snat CRD EIP NAT VPC OVN NAT
- VPC Iptables NAT VPC Iptables NAT

Kube-OVN OVN L3 Gateway Pod SNAT EIP SNAT Pod IP EIP Pod IP EIP Pod Pod EIP



3.5.1

- OVN L3 Gateway OVS Overlay Underlay
- NAT Underlay
- EIP SNAT

3.5.2

`kube-system` ConfigMap `ovn-external-gw-config`

```yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-external-gw-config
  namespace: kube-system
data:
  type: "centralized"
  enable-external-gw: "true"
  external-gw-nodes: "kube-ovn-worker"
  external-gw-nic: "eth1"
  external-gw-addr: "172.56.0.1/16"
```


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```
  nic-ip: "172.56.0.254/16"
  nic-mac: "16:52:f3:13:6a:25"
```

- `enable-external-gw`: SNAT EIP
- `type`: `centralized` `distributed` `centralized` `distributed`
- `external-gw-nodes`: `centralized`
- `external-gw-nic`:
- `external-gw-addr`: IP
- `nic-ip`,`nic-mac`: IP Mac IP Mac

3.5.3 OVN OVS

OVN-NB `ovn-external` `ovn-cluster-ovn-external` chassis

```
# kubectl ko nbctl show
switch 3de4cea7-1a71-43f3-8b62-435a57ef16a6 (external)
    port localnet.external
        type: localnet
        addresses: ["unknown"]
    port external-ovn-cluster
        type: router
        router-port: ovn-cluster-external
router e1eb83ad-34be-4ed5-9a02-fcc8b1d357c4 (ovn-cluster)
    port ovn-cluster-external
        mac: "ac:1f:6b:2d:33:f1"
        networks: ["172.56.0.100/16"]
        gateway chassis: [a5682814-2e2c-46dd-9c1c-6803ef0dab66]
```

OVS `br-external`

```
# kubectl ko vsctl ${gateway node name} show
e7d81150-7743-4d6e-9e6f-5c688232e130
    Bridge br-external
        Port br-external
            Interface br-external
                type: internal
        Port eth1
            Interface eth1
        Port patch-localnet.external-to-br-int
            Interface patch-localnet.external-to-br-int
                type: patch
                options: {peer=patch-br-int-to-localnet.external}
```

3.5.4 Pod EIP SNAT

Pod `ovn.kubernetes.io/snat` `ovn.kubernetes.io/eip` annotation SNAT EIP

```
apiVersion: v1
kind: Pod
metadata:
  name: pod-snat
  annotations:
    ovn.kubernetes.io/snat: 172.56.0.200
spec:
  containers:
  - name: pod-snat
    image: docker.io/library/nginx:alpine
---
apiVersion: v1
kind: Pod
metadata:
  name: pod-eip
  annotations:
    ovn.kubernetes.io/eip: 172.56.0.233
spec:
  containers:
  - name: pod-eip
    image: docker.io/library/nginx:alpine
```

kubectl Pod EIP SNAT `ovn.kubernetes.io/routed` annotation

```
kubectl annotate pod pod-eip ovn.kubernetes.io/eip=172.56.0.221 --overwrite
kubectl annotate pod pod-eip ovn.kubernetes.io/routed-
```

EIP SNAT `ovn.kubernetes.io/routed` annotation

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3.5.5

`kube-ovn-controller` SNAT EIP

- `--external-gateway-config-ns` : Configmap `ovn-external-gw-config` Namespace `kube-system`
- `--external-gateway-net` : `external`
- `--external-gateway-vlanid` : Vlan Tag 0 Vlan

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3.5.6

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3.6 QoS

Kube-OVN Pod QoS

- QoS
- `linux-netem` QoS

> Pod QoS Namespace Subnet QoS

3.6.1 QoS

QoS Pod annotation Pod `Mbit/s`

```
apiVersion: v1
kind: Pod
metadata:
  name: qos
  namespace: ls1
  annotations:
    ovn.kubernetes.io/ingress_rate: "3"
    ovn.kubernetes.io/egress_rate: "1"
spec:
  containers:
  - name: qos
    image: docker.io/library/nginx:alpine
```

annotation QoS

```
kubectl annotate --overwrite  pod nginx-74d5899f46-d7qkn ovn.kubernetes.io/ingress_rate=3
```

QoS

```
kind: DaemonSet
apiVersion: apps/v1
metadata:
  name: perf
  namespace: ls1
  labels:
    app: perf
spec:
  selector:
    matchLabels:
      app: perf
  template:
    metadata:
      labels:
        app: perf
    spec:
      containers:
      - name: nginx
        image: docker.io/kubeovn/perf
```

Pod iperf3 server

```
# kubectl exec -it perf-4n4gt -n ls1 sh
# iperf3 -s
-----------------------------------------------------------
Server listening on 5201
-----------------------------------------------------------
```

Pod Pod

```
# kubectl exec -it perf-d4mqc -n ls1 sh
# iperf3 -c 10.66.0.12
Connecting to host 10.66.0.12, port 5201
[  4] local 10.66.0.14 port 51544 connected to 10.66.0.12 port 5201
[ ID] Interval           Transfer     Bandwidth       Retr  Cwnd
[  4]   0.00-1.00   sec  86.4 MBytes   725 Mbits/sec    3    350 KBytes
[  4]   1.00-2.00   sec  89.9 MBytes   754 Mbits/sec  118    473 KBytes
[  4]   2.00-3.00   sec   101 MBytes   848 Mbits/sec  184    586 KBytes
[  4]   3.00-4.00   sec   104 MBytes   875 Mbits/sec  217    671 KBytes
[  4]   4.00-5.00   sec   111 MBytes   935 Mbits/sec  175    772 KBytes
```


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```
[  4]   5.00-6.00   sec   100 MBytes   840 Mbits/sec  658    598 KBytes
[  4]   6.00-7.00   sec   106 MBytes   890 Mbits/sec  742    668 KBytes
[  4]   7.00-8.00   sec   102 MBytes   857 Mbits/sec  764    724 KBytes
[  4]   8.00-9.00   sec  97.4 MBytes   817 Mbits/sec  1175    764 KBytes
[  4]   9.00-10.00  sec   111 MBytes   934 Mbits/sec  1083    838 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth       Retr
[  4]   0.00-10.00  sec  1010 MBytes   848 Mbits/sec  5119             sender
[  4]   0.00-10.00  sec  1008 MBytes   846 Mbits/sec                  receiver

iperf Done.
```

Pod QoS

```
kubectl annotate --overwrite  pod perf-4n4gt -n ls1 ovn.kubernetes.io/ingress_rate=30
```

Pod Pod

```
# iperf3 -c 10.66.0.12
Connecting to host 10.66.0.12, port 5201
[  4] local 10.66.0.14 port 52372 connected to 10.66.0.12 port 5201
[ ID] Interval           Transfer     Bandwidth       Retr  Cwnd
[  4]   0.00-1.00   sec  3.66 MBytes  30.7 Mbits/sec    2   76.1 KBytes
[  4]   1.00-2.00   sec  3.43 MBytes  28.8 Mbits/sec    0    104 KBytes
[  4]   2.00-3.00   sec  3.50 MBytes  29.4 Mbits/sec    0    126 KBytes
[  4]   3.00-4.00   sec  3.50 MBytes  29.3 Mbits/sec    0    144 KBytes
[  4]   4.00-5.00   sec  3.43 MBytes  28.8 Mbits/sec    0    160 KBytes
[  4]   5.00-6.00   sec  3.43 MBytes  28.8 Mbits/sec    0    175 KBytes
[  4]   6.00-7.00   sec  3.50 MBytes  29.3 Mbits/sec    0    212 KBytes
[  4]   7.00-8.00   sec  3.68 MBytes  30.9 Mbits/sec    0    294 KBytes
[  4]   8.00-9.00   sec  3.74 MBytes  31.4 Mbits/sec    0    398 KBytes
[  4]   9.00-10.00  sec  3.80 MBytes  31.9 Mbits/sec    0    526 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth       Retr
[  4]   0.00-10.00  sec  35.7 MBytes  29.9 Mbits/sec    2             sender
[  4]   0.00-10.00  sec  34.5 MBytes  29.0 Mbits/sec                  receiver

iperf Done.
```

3.6.2 linux-netem QoS

> RHEL netem yum install -y kernel-modules-extra && modprobe sch_netem

Pod annotation `linux-netem` QoS netem QoS Pod Ingress

- `ovn.kubernetes.io/latency` Pod ms
- `ovn.kubernetes.io/jitter` Pod ms
- `ovn.kubernetes.io/limit` `qdisc` 1000
- `ovn.kubernetes.io/loss` float 20 20%

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3.6.3

3.7

Kube-OVN IPv4 IPv6

Kubernetes Kubernetes

3.7.1

CIDR `cidr=<IPv4 CIDR>,<IPv6 CIDR>` CIDR IPv4 IPv6

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: ovn-test
spec:
  cidrBlock: 10.16.0.0/16,fd00:10:16::/64
  excludeIps:
  - 10.16.0.1
  - fd00:10:16::1
  gateway: 10.16.0.1,fd00:10:16::1
```

`install.sh` `/112` ULA `/64` /Service CIDR

```
POD_CIDR="10.16.0.0/16,fd00:10:16::/112"
JOIN_CIDR="100.64.0.0/16,fd00:100:64::/112"
```

3.7.2 Pod

Pod IPv4 IPv6 Pod annotation :

```
apiVersion: v1
kind: Pod
metadata:
  annotations:
    ovn.kubernetes.io/allocated: "true"
    ovn.kubernetes.io/cidr: 10.16.0.0/16,fd00:10:16::/64
    ovn.kubernetes.io/gateway: 10.16.0.1,fd00:10:16::1
    ovn.kubernetes.io/ip_address: 10.16.0.9,fd00:10:16::9
    ovn.kubernetes.io/logical_switch: ovn-default
    ovn.kubernetes.io/mac_address: 00:00:00:14:88:09
    ovn.kubernetes.io/routed: "true"
...
podIP: 10.16.0.9
  podIPs:
  - ip: 10.16.0.9
  - ip: fd00:10:16::9
```

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3.7.3

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3.8 Webhook

Webhook Kube-OVN CRD Webhook IP Subnet CIDR

Webhook Subnet Pod Kube-OVN Webhook Pod

3.8.1 Cert-Manager

Webhook cert-manager Webhook cert-manager

cert-manager:

```
kubectl apply -f https://github.com/cert-manager/cert-manager/releases/download/v1.8.0/cert-manager.yaml
```

cert-manager cert-manager

3.8.2 Webhook

Webhook yaml :

```
# kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/yamls/webhook.yaml
deployment.apps/kube-ovn-webhook created
service/kube-ovn-webhook created
validatingwebhookconfiguration.admissionregistration.k8s.io/kube-ovn-webhook created
certificate.cert-manager.io/kube-ovn-webhook-serving-cert created
issuer.cert-manager.io/kube-ovn-webhook-selfsigned-issuer created
```

3.8.3 Webhook

Pod Pod IP `10.16.0.15`

```
# kubectl get pod -o wide
NAME                  READY   STATUS    RESTARTS   AGE     IP           NODE              NOMINATED NODE   READINESS GATES
static-7584848b74-fw9dm   1/1     Running   0          2d13h   10.16.0.15   kube-ovn-worker   <none>
```

yaml IP Pod

```
apiVersion: v1
kind: Pod
metadata:
  annotations:
    ovn.kubernetes.io/ip_address: 10.16.0.15
    ovn.kubernetes.io/mac_address: 00:00:00:53:6B:B6
  labels:
    app: static
  managedFields:
  name: staticip-pod
  namespace: default
spec:
  containers:
  - image: docker.io/library/nginx:alpine
    imagePullPolicy: IfNotPresent
    name: qatest
```

yaml Pod IP

```
# kubectl apply -f pod-static.yaml
Error from server (annotation static-ip 10.16.0.15 is conflict with ip crd static-7584848b74-fw9dm.default, ip 10.16.0.15): error when creating "pod-static.yaml": admission webhook "pod-ip-validating.kube-ovn.io" denied the request: annotation static-ip 10.16.0.15 is conflict with ip crd static-7584848b74-fw9dm.default, ip 10.16.0.15
```

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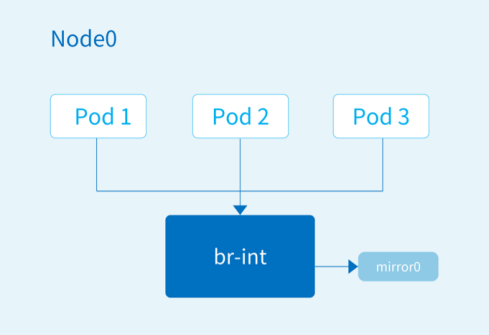
GitHub 

3.8.4

3.9

NPM

CPU 5%~10% CPU



3.9.1

`kube-ovn-cni` DaemonSet

- `--enable-mirror=true`
- `--mirror-iface=mirror0`: br-int Kube-OVN `mirror0`

tcpdump `mirror0`

```
tcpdump -ni mirror0
```

3.9.2 Pod

Pod Pod `ovn.kubernetes.io/mirror` annotation Pod

```
apiVersion: v1
kind: Pod
metadata:
  name: mirror-pod
  namespace: ls1
  annotations:
    ovn.kubernetes.io/mirror: "true"
spec:
  containers:
  - name: mirror-pod
    image: docker.io/library/nginx:alpine
```


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3.9.3

1. Pod to Pod in the same Nodes

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	12.7 us	289 Mbits/sec	12.6 us	(1.8%)	77.9 Mbits/sec
128	15.5 us	517 Mbits/sec	12.7 us	(0%)	155 Mbits/sec
512	12.2 us	1.64 Gbits/sec	12.4 us	(0%)	624 Mbits/sec
1k	13 us	2.96 Gbits/sec	11.4 us	(0.53%)	1.22 Gbits/sec
4k	18 us	7.67 Gbits/sec	25.7 us	(0.41%)	1.50 Gbits/sec

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	11.9 us	324 Mbits/sec	12.2 us	(0.22%)	102 Mbits/sec
128	10.5 us	582 Mbits/sec	9.5 us	(0.21%)	198 Mbits/sec
512	11.6 us	1.84 Gbits/sec	9.32 us	(0.091%)	827 Mbits/sec
1k	10.5 us	3.44 Gbits/sec	10 us	(1.2%)	1.52 Gbits/sec
4k	16.7 us	8.52 Gbits/sec	18.2 us	(1.3%)	2.42 Gbits/sec

2. Pod to Pod in the different Nodes

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	258 us	143 Mbits/sec	237 us	(61%)	28.5 Mbits/sec
128	240 us	252 Mbits/sec	231 us	(64%)	54.9 Mbits/sec
512	236 us	763 Mbits/sec	256 us	(68%)	194 Mbits/sec
1k	242 us	969 Mbits/sec	225 us	(62%)	449 Mbits/sec
4k	352 us	1.12 Gbits/sec	382 us	(0.71%)	21.4 Mbits/sec

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	278 us	140 Mbits/sec	227 us	(24%)	59.6 Mbits/sec
128	249 us	265 Mbits/sec	265 us	(23%)	114 Mbits/sec
512	233 us	914 Mbits/sec	235 us	(21%)	468 Mbits/sec
1k	238 us	1.14 Gbits/sec	240 us	(15%)	891 Mbits/sec
4k	370 us	1.25 Gbits/sec	361 us	(0.43%)	7.54 Mbits/sec

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3. Node to Node

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	205 us	162 Mbits/sec	183 us	(11%)	74.2 Mbits/sec
128	222 us	280 Mbits/sec	206 us	(6.3%)	155 Mbits/sec
512	220 us	1.04 Gbits/sec	177 us	(20%)	503 Mbits/sec
1k	213 us	2.06 Gbits/sec	201 us	(8.6%)	1.14 Gbits/sec
4k	280 us	5.01 Gbits/sec	315 us	(37%)	1.20 Gbits/sec

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	204 us	157 Mbits/sec	204 us	(8.8%)	81.9 Mbits/sec
128	213 us	262 Mbits/sec	225 us	(19%)	136 Mbits/sec
512	220 us	1.02 Gbits/sec	227 us	(21%)	486 Mbits/sec
1k	217 us	1.79 Gbits/sec	218 us	(29%)	845 Mbits/sec
4k	275 us	5.27 Gbits/sec	336 us	(34%)	1.21 Gbits/sec

4. Pod to the Node where the Pod is located

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	12.2 us	295 Mbits/sec	12.7 us	(0.27%)	74.1 Mbits/sec
128	14.1 us	549 Mbits/sec	10.6 us	(0.41%)	153 Mbits/sec
512	13.5 us	1.83 Gbits/sec	12.7 us	(0.23%)	586 Mbits/sec
1k	12 us	2.69 Gbits/sec	13 us	(1%)	1.16 Gbits/sec
4k	18.9 us	4.51 Gbits/sec	21.8 us	(0.42%)	1.81 Gbits/sec

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	10.4 us	335 Mbits/sec	12.2 us	(0.75%)	95.4 Mbits/sec
128	12.1 us	561 Mbits/sec	11.3 us	(0.25%)	194 Mbits/sec
512	11.6 us	1.87 Gbits/sec	10.7 us	(0.66%)	745 Mbits/sec
1k	12.7 us	3.12 Gbits/sec	10.9 us	(1.2%)	1.46 Gbits/sec
4k	16.5 us	8.23 Gbits/sec	17.9 us	(1.5%)	2.51 Gbits/sec

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5. Pod to the Node where the Pod is not located

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	234 us	153 Mbits/sec	232 us	(63%)	29.4 Mbits/sec
128	237 us	261 Mbits/sec	238 us	(49%)	76.1 Mbits/sec
512	231 us	701 Mbits/sec	238 us	(57%)	279 Mbits/sec
1k	256 us	1.05 Gbits/sec	228 us	(56%)	524 Mbits/sec
4k	330 us	1.08 Gbits/sec	359 us	(1.5%)	35.7 Mbits/sec

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	283 us	141 Mbits/sec	230 us	(26%)	55.8 Mbits/sec
128	234 us	255 Mbits/sec	234 us	(25%)	113 Mbits/sec
512	246 us	760 Mbits/sec	234 us	(22%)	458 Mbits/sec
1k	268 us	1.23 Gbits/sec	242 us	(20%)	879 Mbits/sec
4k	326 us	1.20 Gbits/sec	369 us	(0.5%)	7.87 Mbits/sec

6. Pod to the cluster ip service

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	237 us	133 Mbits/sec	213 us	(65%)	25.5 Mbits/sec
128	232 us	271 Mbits/sec	222 us	(62%)	54.8 Mbits/sec
512	266 us	800 Mbits/sec	234 us	(60%)	232 Mbits/sec
1k	248 us	986 Mbits/sec	239 us	(50%)	511 Mbits/sec
4k	314 us	1.03 Gbits/sec	367 us	(0.6%)	13.2 Mbits/sec

TCP-Conn-Number	QPS	Avg-Resp-Time	Stdev-Resp-Time	Max-Resp-Time
10	14305.17	0.87ms	1.48ms	24.46ms
100	29082.07	3.87ms	4.35ms	102.85ms

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Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	241 us	145 Mbits/sec	225 us	(19%)	60.2 Mbits/sec
128	245 us	261 Mbits/sec	212 us	(15%)	123 Mbits/sec
512	252 us	821 Mbits/sec	219 us	(14%)	499 Mbits/sec
1k	253 us	1.08 Gbits/sec	242 us	(16%)	852 Mbits/sec
4k	320 us	1.32 Gbits/sec	360 us	(0.47%)	6.70 Mbits/sec

TCP-Conn-Number	QPS	Avg-Resp-Time	Stdev-Resp-Time	Max-Resp-Time
10	13634.07	0.96ms	1.72ms	30.07ms
100	30215.23	3.59ms	3.20ms	77.56ms

7. Host to the Node port service where the Pod is not located on the target Node

TCP-Conn-Number	QPS	Avg-Resp-Time	Stdev-Resp-Time	Max-Resp-Time
10	14802.73	0.88ms	1.66ms	31.49ms
100	29809.58	3.78ms	4.12ms	105.34ms

TCP-Conn-Number	QPS	Avg-Resp-Time	Stdev-Resp-Time	Max-Resp-Time
10	14273.33	0.90ms	1.60ms	37.16ms
100	30757.81	3.62ms	3.41ms	59.78ms

8. Host to the Node port service where the Pod is located on the target Node

TCP-Conn-Number	QPS	Avg-Resp-Time	Stdev-Resp-Time	Max-Resp-Time
10	15402.39	802.50us	1.42ms	30.91ms
100	29424.66	4.05ms	4.31ms	90.60ms

TCP-Conn-Number	QPS	Avg-Resp-Time	Stdev-Resp-Time	Max-Resp-Time
10	14649.21	0.91ms	1.72ms	43.92ms
100	32143.61	3.66ms	3.76ms	67.02ms

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3.9.4

3.10 LoadBalancer Service

Kube-OVN VPC VPC VPC

VPC VPC VPC `LoadBalancer` Service LoadBalancerIP VPC Service

1. `multus-cni` `macvlan cni`
2. LoadBalancer Service VPC `vpc-nat-gw` macvlan
3. `VPC` VPC LoadBalancer VPC VPC

3.10.1 VPC LoadBalancer Service

kube-system namespace deployment `kube-ovn-controller` `args` `--enable-lb-svc=true` false

```
containers:
- args:
  - /kube-ovn/start-controller.sh
  - --default-cidr=10.16.0.0/16
  - --default-gateway=10.16.0.1
  - --default-gateway-check=true
  - --enable-lb-svc=true                #       true
```

NetworkAttachmentDefinition CRD

yaml `net-attach-def`

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: lb-svc-attachment
  namespace: kube-system
spec:
  # master                    eth0 / ens192 / bond0
  config: '{
      "cniVersion": "0.3.0",
      "type": "macvlan",
      "master": "eth0",
      "mode": "bridge"
    }'
```

`eth0` `master`

Subnet

Subnet LoadBalancer Service LoadBalancerIP Underlay Subnet

yaml

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: attach-subnet
spec:
  protocol: IPv4
  provider: lb-svc-attachment.kube-system    # provider            net-attach-def     Name.Namespace
  cidrBlock: 172.18.0.0/16
  gateway: 172.18.0.1
  excludeIps:
  - 172.18.0.0..172.18.0.10
```

Subnet `provider` `ovn` `.ovn` Kube-OVN `logical switch`

`provider` `ovn` `.ovn` Kube-OVN IPAM IP

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LoadBalancer Service

yaml LoadBalancer Service

```yaml
apiVersion: v1
kind: Service
metadata:
  annotations:
    lb-svc-attachment.kube-system.kubernetes.io/logical_switch: attach-subnet   #
    ovn.kubernetes.io/attachmentprovider: lb-svc-attachment.kube-system          #
  labels:
    app: dynamic
  name: test-service
  namespace: default
spec:
  loadBalancerIP: 172.18.0.18                                                    #
  ports:
    - name: test
      protocol: TCP
      port: 80
      targetPort: 80
  selector:
    app: dynamic
  sessionAffinity: None
  type: LoadBalancer
```

yaml annotation `ovn.kubernetes.io/attachmentprovider` `net-attach-def` Name.Namespace annotation Pod `net-attach-def`

annotation annotation key net-attach-def `Name.Namespace.kubernetes.io/logical_switch` LoadBalancerIP LoadBalancerIP

LoadBalancerIP `spec.loadBalancerIP`

yaml Service Service Namespace Pod

```
# kubectl get pod
NAME                                    READY   STATUS    RESTARTS   AGE
lb-svc-test-service-6869d98dd8-cjvll    1/1     Running   0          107m
# kubectl get svc
NAME             TYPE           CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
test-service     LoadBalancer   10.109.201.193   172.18.0.18   80:30056/TCP   107m
```

`service.spec.loadBalancerIP` service external-ip

Pod yaml

```yaml
# kubectl get pod -o yaml lb-svc-test-service-6869d98dd8-cjvll
apiVersion: v1
kind: Pod
metadata:
  annotations:
    k8s.v1.cni.cncf.io/network-status: |-
      [{
          "name": "kube-ovn",
          "ips": [
              "10.16.0.2"
          ],
          "default": true,
          "dns": {}
      },{
          "name": "default/test-service",
          "interface": "net1",
          "mac": "ba:85:f7:02:9f:42",
          "dns": {}
      }]
    k8s.v1.cni.cncf.io/networks: default/test-service
    k8s.v1.cni.cncf.io/networks-status: |-
      [{
          "name": "kube-ovn",
          "ips": [
              "10.16.0.2"
          ],
          "default": true,
          "dns": {}
      },{
          "name": "default/test-service",
          "interface": "net1",
          "mac": "ba:85:f7:02:9f:42",
          "dns": {}
      }]
    ovn.kubernetes.io/allocated: "true"
    ovn.kubernetes.io/cidr: 10.16.0.0/16
    ovn.kubernetes.io/gateway: 10.16.0.1
```


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```
    ovn.kubernetes.io/ip_address: 10.16.0.2
    ovn.kubernetes.io/logical_router: ovn-cluster
    ovn.kubernetes.io/logical_switch: ovn-default
    ovn.kubernetes.io/mac_address: 00:00:00:45:F4:29
    ovn.kubernetes.io/pod_nic_type: veth-pair
    ovn.kubernetes.io/routed: "true"
    test-service.default.kubernetes.io/allocated: "true"
    test-service.default.kubernetes.io/cidr: 172.18.0.0/16
    test-service.default.kubernetes.io/gateway: 172.18.0.1
    test-service.default.kubernetes.io/ip_address: 172.18.0.18
    test-service.default.kubernetes.io/logical_switch: attach-subnet
    test-service.default.kubernetes.io/mac_address: 00:00:00:AF:AA:BF
    test-service.default.kubernetes.io/pod_nic_type: veth-pair
```

Service

```
# kubectl get svc -o yaml test-service
apiVersion: v1
kind: Service
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"v1","kind":"Service","metadata":{"annotations":{"test-service.default.kubernetes.io/logical_switch":"attach-subnet"},"labels":
{"app":"dynamic"},"name":"test-service","namespace":"default"},"spec":{"ports":[{"name":"test","port":80,"protocol":"TCP","targetPort":80}],"selector":
{"app":"dynamic"},"sessionAffinity":"None","type":"LoadBalancer"}}
    ovn.kubernetes.io/vpc: ovn-cluster
    test-service.default.kubernetes.io/logical_switch: attach-subnet
  creationTimestamp: "2022-06-15T09:01:58Z"
  labels:
    app: dynamic
  name: test-service
  namespace: default
  resourceVersion: "38485"
  uid: 161edee1-7f6e-40f5-9e09-5a52c44267d0
spec:
  allocateLoadBalancerNodePorts: true
  clusterIP: 10.109.201.193
  clusterIPs:
  - 10.109.201.193
  externalTrafficPolicy: Cluster
  internalTrafficPolicy: Cluster
  ipFamilies:
  - IPv4
  ipFamilyPolicy: SingleStack
  ports:
  - name: test
    nodePort: 30056
    port: 80
    protocol: TCP
    targetPort: 80
  selector:
    app: dynamic
  sessionAffinity: None
  type: LoadBalancer
status:
  loadBalancer:
    ingress:
    - ip: 172.18.0.18
```

3.10.2 LoadBalancerIP

yaml, Pod Service Endpoints

```
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: dynamic
  name: dynamic
  namespace: default
spec:
  replicas: 2
  selector:
    matchLabels:
      app: dynamic
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
    type: RollingUpdate
  template:
    metadata:
      creationTimestamp: null
      labels:
        app: dynamic
    spec:
      containers:
      - image: docker.io/library/nginx:alpine
```

```
        imagePullPolicy: IfNotPresent
        name: nginx
      dnsPolicy: ClusterFirst
      restartPolicy: Always
```

Service `LoadBalancerIP:Port`

```
# curl 172.18.0.18:80
<html>
<head>
        <title>Hello World!</title>
        <link href='//fonts.googleapis.com/css?family=Open+Sans:400,700' rel='stylesheet' type='text/css'>
        <style>
        body {
                background-color: white;
                text-align: center;
                padding: 50px;
                font-family: "Open Sans","Helvetica Neue",Helvetica,Arial,sans-serif;
        }
        #logo {
                margin-bottom: 40px;
        }
        </style>
</head>
<body>
                <h1>Hello World!</h1>
                                <h3>Links found</h3>
        <h3>I am on  dynamic-7d8d7874f5-hsgc4</h3>
        <h3>Cookie                  =</h3>
                                    <b>KUBERNETES</b> listening in 443 available at tcp://10.96.0.1:443<br />
                                        <h3>my name is hanhouchao!</h3>
                        <h3> RequestURI='/'</h3>
</body>
</html>
```

Service Pod

```
# ip a
4: net1@if62: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether ba:85:f7:02:9f:42 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 172.18.0.18/16 scope global net1
       valid_lft forever preferred_lft forever
    inet6 fe80::b885:f7ff:fe02:9f42/64 scope link
       valid_lft forever preferred_lft forever
36: eth0@if37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue state UP group default
    link/ether 00:00:00:45:f4:29 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.16.0.2/16 brd 10.16.255.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fe45:f429/64 scope link
       valid_lft forever preferred_lft forever

# ip rule
0: from all lookup local
32764: from all iif eth0 lookup 100
32765: from all iif net1 lookup 100
32766: from all lookup main
32767: from all lookup default

# ip route show table 100
default via 172.18.0.1 dev net1
10.109.201.193 via 10.16.0.1 dev eth0
172.18.0.0/16 dev net1 scope link

# iptables -t nat -L -n -v
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 DNAT       tcp  --  *      *       0.0.0.0/0            172.18.0.18          tcp dpt:80 to:10.109.201.193:80

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 MASQUERADE  all  --  *      *       0.0.0.0/0            10.109.201.193
```

lb service Pod nodeSelector

`ovn-vpc-nat-config` ConfigMap `nodeSelector` LoadBalancer service Pod

```
apiVersion: v1
data:
  image: docker.io/kubeovn/vpc-nat-gateway:v1.16.16
  nodeSelector: |
    kubernetes.io/hostname: kube-ovn-control-plane
```


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```
    kubernetes.io/os: linux
kind: ConfigMap
metadata:
  name: ovn-vpc-nat-config
  namespace: kube-system
```

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3.10.3

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3.11

Kube-OVN Prometheus

kube-prometheus CRD Prometheus kube-prometheus CRD Kube-OVN Kube-OVN

3.11.1 Prometheus Monitor

Kube-OVN Prometheus Monitor CRD

```
#
kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/pinger-monitor.yaml
# kube-ovn-controller
kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/controller-monitor.yaml
# kube-ovn-cni
kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/cni-monitor.yaml
# ovn
kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/ovn-monitor.yaml
```

Prometheus 15s yaml `interval`

3.11.2 Grafana

Kube-OVN Grafana Dashboard

Dashboard

```
#
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/pinger-grafana.json
# kube-ovn-controller
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/controller-grafana.json
# kube-ovn-cni
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/cni-grafana.json
# ovn
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/ovn-grafana.json
# ovs
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/ovs-grafana.json
```

Grafana Prometheus Dashboard

`kube-ovn-controller`

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`kube-ovn-pinger`



`kube-ovn-cni`



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3.11.3

3.12

3.12.1 NetworkPolicy

NetworkPolicy Kubernetes Pod Pod Kube-OVN OVN ACL Kubernetes NetworkPolicy NetworkPolicy

Kube-OVN

Kube-OVN OVN Open Virtual Network NetworkPolicy OVN

PORT GROUP

NetworkPolicy Kube-OVN Port Group `podSelector` Pod Port Group `< >.< > - .`

ADDRESS SET

Address Set NetworkPolicy IP NetworkPolicy Kube-OVN `podSelector` `namespaceSelector` `ipBlock` IP Address Set

NetworkPolicy Ingress Egress Allow Except Address Set

- `< >.< >.ingress.allow` IP
- `< >.< >.ingress.except` IP
- `< >.< >.egress.allow` IP
- `< >.< >.egress.except` IP

ACL

ACL OVN Kube-OVN NetworkPolicy OVN ACL Port Group ACL

Kube-OVN NetworkPolicy ACL

- Ingress Allow 2001
- Egress Allow 2001
- Default Deny 2000

Kube-OVN Kubernetes NetworkPolicy OVN ACL

Kube-OVN

- **NetworkPolicy** Kubernetes
- **Network Policy API** AdminNetworkPolicy BaselineAdminNetworkPolicy
- **Subnet ACL**
- **Security Group**

OVN ACL NetworkPolicy Network Policy API

NAMED PORT

NetworkPolicy Named Port

```
ports:
- protocol: TCP
  port: http
```


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Kube-OVN Named Port **Named Port**

Pod Named Port NetworkPolicy Pod A `http` 8080 Pod B `http` 8081 `port: http` NetworkPolicy

Named Port Pod

IPBLOCK EXCEPT

NetworkPolicy `ipBlock` `except` IP

```
egress:
- to:
  - ipBlock:
      cidr: 10.0.0.0/8
      except:
      - 10.0.1.0/24
      - 10.0.2.0/24
```

OVN `except` `except` ACL OVN

`except` IP

- CIDR
- `podSelector` `namespaceSelector` IP

NetworkPolicy

Kube-OVN NetworkPolicy

> **Warning**
>
> NetworkPolicy CPU
>
> `ovn.kubernetes.io/acl_log_meter_rate` annotation ACL

NetworkPolicy annotation `ovn.kubernetes.io/enable_log`

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: default-deny-ingress
  namespace: default
  annotations:
    ovn.kubernetes.io/enable_log: "true"
spec:
  podSelector: {}
  policyTypes:
  - Ingress
```

Drop

Pod `/var/log/ovn/ovn-controller.log`

```
# tail -f /var/log/ovn/ovn-controller.log
2022-07-20T05:55:03.229Z|00394|acl_log(ovn_pinctrl0)|INFO|name="np/default-deny-ingress.default/IPv4/0", verdict=drop, severity=warning, direction=to-lport:
udp,vlan_tci=0x0000,dl_src=00:00:00:21:b7:d1,dl_dst=00:00:00:8d:0b:86,nw_src=10.16.0.10,nw_dst=10.16.0.7,nw_tos=0,nw_ecn=0,nw_ttl=63,tp_src=54343,tp_dst=53
```

IP IP

Kube-OVN v1.13.0 `ovn.kubernetes.io/log_acl_actions` annotation Allow

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```yaml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-from-client
  namespace: default
  annotations:
    ovn.kubernetes.io/enable_log: "true"
    ovn.kubernetes.io/log_acl_actions: "allow"
spec:
  podSelector:
    matchLabels:
      app: web
  policyTypes:
  - Ingress
  ingress:
  - from:
    - podSelector:
        matchLabels:
          app: client
```

`ovn.kubernetes.io/log_acl_actions`

- `drop`
- `allow`
- `allow,drop`

```
# tail -f /var/log/ovn/ovn-controller.log
2024-08-14T09:27:49.590Z|00004|acl_log(ovn_pinctrl0)|INFO|name="np/allow-from-client.default/ingress/IPv4/0", verdict=allow, severity=info, direction=to-
lport:
icmp,vlan_tci=0x0000,dl_src=96:7b:b0:2f:a0:1a,dl_dst=a6:e5:1b:c2:1b:f8,nw_src=10.16.0.7,nw_dst=10.16.0.12,nw_tos=0,nw_ecn=0,nw_ttl=64,nw_frag=no,icmp_type=8,icm
```

`ovn.kubernetes.io/acl_log_meter_rate` annotation

```yaml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-from-client
  namespace: default
  annotations:
    ovn.kubernetes.io/enable_log: "true"
    ovn.kubernetes.io/log_acl_actions: "allow"
    ovn.kubernetes.io/acl_log_meter_rate: "100"
spec:
  podSelector:
    matchLabels:
      app: web
  policyTypes:
  - Ingress
  ingress:
  - from:
    - podSelector:
        matchLabels:
          app: client
```

`ovn.kubernetes.io/acl_log_meter_rate` / `100` 100

annotation `ovn.kubernetes.io/enable_log` `false`

```
kubectl annotate networkpolicy -n default allow-from-client ovn.kubernetes.io/enable_log=false --overwrite
```

Kube-OVN

- **standard** NetworkPolicy IP
- **lax** TCP/UDP/SCTP ICMP L4 IP DHCP UDP

NetworkPolicy annotation

```yaml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: example-policy
  namespace: default
  annotations:
    ovn.kubernetes.io/network_policy_enforcement: "lax"
spec:
  podSelector: {}
  policyTypes:
  - Ingress
```

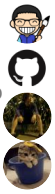
Kube-OVN `--np-enforcement`

- NetworkPolicy

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3.12.2 NetworkPolicy

Kube-OVN NetworkPolicy `podSelector` Pod OVN

Pod Provider

- `ovn.kubernetes.io/network_policy_for`

Pod

```
metadata:
  annotations:
    ovn.kubernetes.io/network_policy_for: "ovn,default/net-a,default/net-b"
```

- `ovn` OVN Provider
- `<namespace>/<net-attach-def>`

- `ovn`
- `default/net-a`
- `ovn,default/net-a`

Provider

- OVN Provider
-
- Provider
-

`<namespace>/<net-attach-def>` Kube-OVN Provider

- `<nad-name>.<nad-namespace>.ovn`

Pod

- OVN Provider `ovn`
- `default/net-a`
- `default/net-b`

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- `network_policy_for`
- `ovn net-a net-b`
- `network_policy_for: default/net-a`
- `net-a`
- `network_policy_for: ovn,default/net-b`
- `ovn net-b`

- Provider/ Kubernetes NetworkPolicy
-

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3.12.3

Kubernetes NetworkPolicy L3 L4 AdminNetworkPolicy (ANP) Pod DNSNameResolver CoreDNS

NetworkPolicy OVN AddressSet IP IP OVN AddressSet DNS

1. kube-ovn-controller AdminNetworkPolicy DNSNameResolver CR
2. CoreDNS DNSNameResolver CR IP DNSNameResolver status
3. kube-ovn-controller DNSNameResolver CR status AddressSet

IP Deny Allow Allow Deny

ANP BANP CRD

AdminNetworkPolicy CRD

```
kubectl apply -f https://raw.githubusercontent.com/kubernetes-sigs/network-policy-api/refs/heads/main/config/crd/experimental/
policy.networking.k8s.io_adminnetworkpolicies.yaml
kubectl apply -f https://raw.githubusercontent.com/kubernetes-sigs/network-policy-api/refs/heads/main/config/crd/experimental/
policy.networking.k8s.io_baselineadminnetworkpolicies.yaml
```

DNSNAMERESOLVER

DNSNameResolver

```
kubectl apply -f https://raw.githubusercontent.com/kubeovn/dnsnameresolver/refs/heads/main/manifest/crd.yaml
kubectl apply -f https://raw.githubusercontent.com/kubeovn/dnsnameresolver/refs/heads/main/manifest/rbac.yaml
kubectl apply -f https://raw.githubusercontent.com/kubeovn/dnsnameresolver/refs/heads/main/manifest/cm.yaml
```

COREDNS

DNSNameResolver CoreDNS tag `dev` kubeovn/dnsnameresolver releases

```
kubectl set image deployment/coredns coredns=kubeovn/dnsnameresolver:<release-tag> -n kube-system
```

CoreDNS

```
kubectl get pod -n kube-system -l k8s-app=kube-dns
```

ANP

kube-ovn-controller

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: kube-ovn-controller
spec:
  template:
    spec:
      containers:
      - name: kube-ovn-controller
        args:
        - --enable-anp=true
        - --enable-dns-name-resolver=true
        # ...
```


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```yaml
apiVersion: policy.networking.k8s.io/v1alpha1
kind: AdminNetworkPolicy
metadata:
  name: deny-external-domains
spec:
  priority: 55
  subject:
    namespaces:
      matchLabels:
        kubernetes.io/metadata.name: kube-system
  egress:
  - action: Deny
    name: deny-baidu-google
    to:
    - domainNames:
      - '*.baidu.com.'
      - '*.google.com.'
```

`priority`	
`subject`	Pod
`egress`	
`action`	`Allow` `Deny` `Pass`
`domainNames`	`.`

kube-ovn-pinger

```
#
kubectl exec -it -n kube-system kube-ovn-pinger-xxxxx -- ping baidu.com
```

DNS ACL

DNSNameResolver

```
# kubectl get dnsnameresolver
NAME                                DNS NAME        RESOLVED IPS
anp-deny-external-domains-88dc32ab  *.google.com.
anp-deny-external-domains-fb3029ce  *.baidu.com.    220.181.7.203
```

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3.12.4

Pod Pod Kube-OVN IPAM MAC IP

- Pod IP
- Pod MAC ARP
-

OVN Port Security Pod Kube-OVN OVN

- MAC IP
- OVN MAC IP
- IPAM
- OVN

OVN

Pod `ovn.kubernetes.io/port_security` annotation

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: secure-pod
  annotations:
    ovn.kubernetes.io/port_security: "true"
spec:
  containers:
  - name: nginx
    image: docker.io/library/nginx:alpine
```

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4. KubeVirt

4.1 VM IP

IP VM VM IP

CNI

- IP VM VM IP
- IP Node VM IP
- IP VM IP

KubeVirt `masquerade` iptables VM VM IP `masquerade` `bridge`

- Pod IP VM IP Pod IP
- `masquerade` iptables `bridge`
- `masquerade`
- `masquerade` conntrack

Kube-OVN KubeVirt `bridge` `managedTap` IP VM IP VM annotation VM IP
Kube-OVN Multus

4.1.1 IP VM

VM IP IP VM Kube-OVN IPAM VM VM IP

`bridge` VM IP

1. VM

```yaml
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: testvm
spec:
  runStrategy: Always
  template:
    metadata:
      labels:
        kubevirt.io/size: small
        kubevirt.io/domain: testvm
      annotations:
        kubevirt.io/allow-pod-bridge-network-live-migration: "true"
    spec:
      domain:
        devices:
          disks:
            - name: containerdisk
              disk:
                bus: virtio
            - name: cloudinitdisk
              disk:
                bus: virtio
          interfaces:
          - name: default
            bridge: {}
        resources:
          requests:
            memory: 64M
      networks:
      - name: default
        pod: {}
      volumes:
        - name: containerdisk
          containerDisk:
            image: quay.io/kubevirt/cirros-container-disk-demo
        - name: cloudinitdisk
```


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```
          cloudInitNoCloud:
            userDataBase64: SGkuXG4=
```

1. VM

```
kubectl get vmi testvm
```

1. VM

```
virtctl restart testvm
```

1. VM

```
virtctl migrate testvm
```

bridge VM IP

4.1.2 IP/Mac

VM IP/Mac VM annotation VM IP KubeVirt VM IP/MAC `<nadName>.<nadNamespace>.kubernetes.io/ip_address.<interfaceName>` `<nadName>.<nadNamespace>.kubernetes.io/mac_address.<interfaceName>` - IP/MAC

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: testvm
spec:
  runStrategy: Always
  template:
    metadata:
      labels:
        kubevirt.io/size: small
        kubevirt.io/domain: testvm
      annotations:
        ovn.kubernetes.io/ip_address: 10.16.0.15 #(1)
        ovn.kubernetes.io/mac_address: 00:00:00:53:6B:B6 #(2)
        kubevirt.io/allow-pod-bridge-network-live-migration: "true"
    spec:
      domain:
        devices:
          disks:
            - name: containerdisk
              disk:
                bus: virtio
            - name: cloudinitdisk
              disk:
                bus: virtio
          interfaces:
          - name: default
            bridge: {}
        resources:
          requests:
            memory: 64M
      networks:
      - name: default
        pod: {}
      volumes:
        - name: containerdisk
          containerDisk:
            image: quay.io/kubevirt/cirros-container-disk-demo
        - name: cloudinitdisk
          cloudInitNoCloud:
            userDataBase64: SGkuXG4=
```

1. 🙋 IP
2. 🙋 Mac

> ⚠ **Warning**
>
> KubeVirt VM API `spec.template.spec.domain.devices.interfaces.macAddress` Mac Mac Kube-OVN annotation Mac Mac

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4.1.3 VM IP

Kube-OVN VM IP IP VM

VM IP

1. VM Annotation IP
2. `virtctl restart <vm name>` VM IP

4.1.4

VM IP VM `ovn.kubernetes.io/logical_switch`

```yaml
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: testvm
spec:
  runStrategy: Always
  template:
    metadata:
      labels:
        kubevirt.io/size: small
        kubevirt.io/domain: testvm
      annotations:
        ovn.kubernetes.io/logical_switch: subnet1 #(1)
        kubevirt.io/allow-pod-bridge-network-live-migration: "true"
    spec:
      domain:
        devices:
          disks:
            - name: containerdisk
              disk:
                bus: virtio
            - name: cloudinitdisk
              disk:
                bus: virtio
          interfaces:
          - name: default
            bridge: {}
        resources:
          requests:
            memory: 64M
      networks:
      - name: default
        pod: {}
      volumes:
        - name: containerdisk
          containerDisk:
            image: quay.io/kubevirt/cirros-container-disk-demo
        - name: cloudinitdisk
          cloudInitNoCloud:
            userDataBase64: SGkuXG4=
```

1. 🙋 VM

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4.1.5

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4.2

KubeVirt Clone Label Annotation Kube-OVN Annotation IP MAC

4.2.1 Annotation

`VirtualMachineClone` Kube-OVN Annotation

```yaml
kind: VirtualMachineClone
apiVersion: "clone.kubevirt.io/v1beta1"
metadata:
  name: testclone
spec:
  source:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-source
  target:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-target
  template:
    annotationFilters:
      - "ovn.kubernetes.io/*"
```

4.2.2

> **Note**
>
> `patches` KubeVirt 1.6

IP

```yaml
kind: VirtualMachineClone
apiVersion: "clone.kubevirt.io/v1beta1"
metadata:
  name: testclone
spec:
  source:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-source
  target:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-target
  patches:
  - '{"op": "replace", "path": "/spec/template/metadata/annotations/ovn.kubernetes.io~1ip_address", "value": "10.16.0.15"}'
```

```yaml
kind: VirtualMachineClone
apiVersion: "clone.kubevirt.io/v1beta1"
metadata:
  name: testclone
spec:
  source:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-source
  target:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-target
  patches:
  - '{"op": "remove", "path": "/spec/template/metadata/annotations/ovn.kubernetes.io~1ip_address"}'
```

Annotation KubeVirt Clone API

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4.2.3

4.3

KubeVirt Bridge DHCP virt-launcher KubeVirt IPv4 DHCP Bridge KubeVirt VM RA IPv6 Kube-OVN DHCP RA KubeVirt DHCP/RA

KubeVirt 1.4.0 Network Binding Plugin Bridge `managedTap` KubeVirt DHCP `managedTap` Kube-OVN DHCP/RA VM

4.3.1 DHCP

Kube-OVN Subnet DHCP IPv6 RA YAML

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: dual-stack-subnet
spec:
  cidrBlock: "10.244.0.0/16,fd00:10:244::/64"
  enableDHCP: true
  enableIPv6RA: true
```

4.3.2 `managedTap`

KubeVirt `managedTap` Network Binding Plugin:

```
# kubectl patch kubevirts -n kubevirt kubevirt --type=json -p=\
'[{"op": "add", "path": "/spec/configuration/network",   "value": {
    "binding": {
        "managedtap": {
            "domainAttachmentType": "managedTap"
        }
    }
}}]'
```

4.3.3 `managedTap`

```yaml
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: dual-stack-vm
  namespace: default
spec:
  running: false
  template:
    spec:
      domain:
        devices:
          interfaces:
            - name: default
              binding:
                name: managedtap
      networks:
      - name: default
        pod: {}
```

VM DHCP IPv6 RA IPv4/IPv6

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4.3.4

4.4

KubeVirt

- KubeVirt Bridge
- KubeVirt
- IP
-

Kube-OVN 0.5 TCP

4.4.1

VM Spec `kubevirt.io/allow-pod-bridge-network-live-migration: "true"` annotation Kube-OVN

1. VM

```
kubectl apply -f - <<EOF
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: testvm
spec:
  runStrategy: Always
  template:
    metadata:
      labels:
        kubevirt.io/size: small
        kubevirt.io/domain: testvm
      annotations:
        kubevirt.io/allow-pod-bridge-network-live-migration: "true"
    spec:
      domain:
        devices:
          disks:
            - name: containerdisk
              disk:
                bus: virtio
            - name: cloudinitdisk
              disk:
                bus: virtio
          interfaces:
          - name: default
            bridge: {}
        resources:
          requests:
            memory: 64M
      networks:
      - name: default
        pod: {}
      volumes:
        - name: containerdisk
          containerDisk:
            image: quay.io/kubevirt/cirros-container-disk-demo
        - name: cloudinitdisk
          cloudInitNoCloud:
            userDataBase64: SGkuXG4=
EOF
```

1. SSH

```
# password: gocubsgo
virtctl ssh cirros@testvm
ping 8.8.8.8
```

1.

```
virtctl migrate testvm
```

VM SSH ping

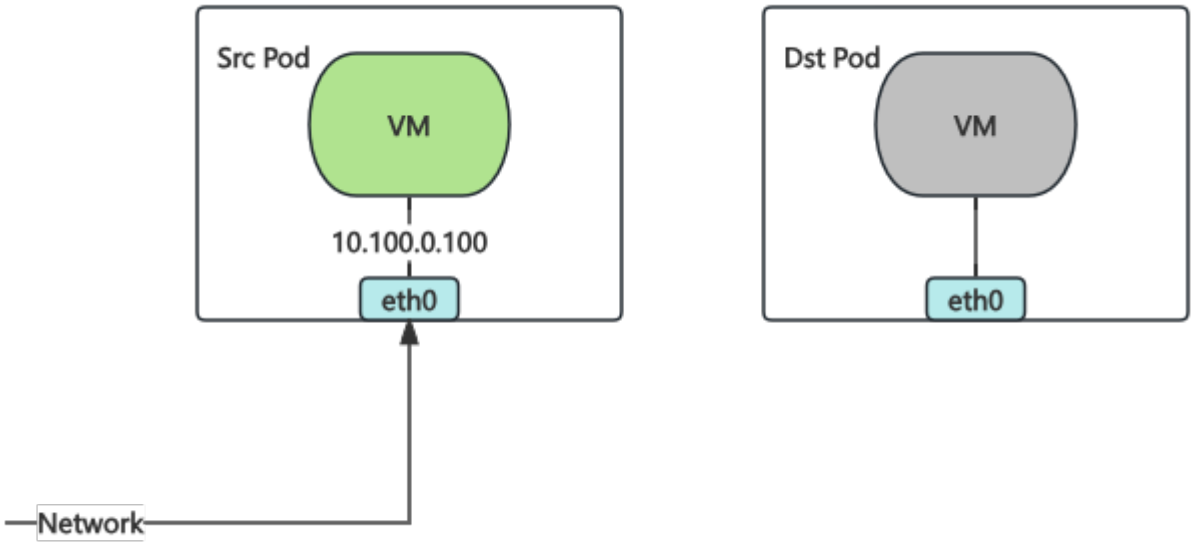
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4.4.2

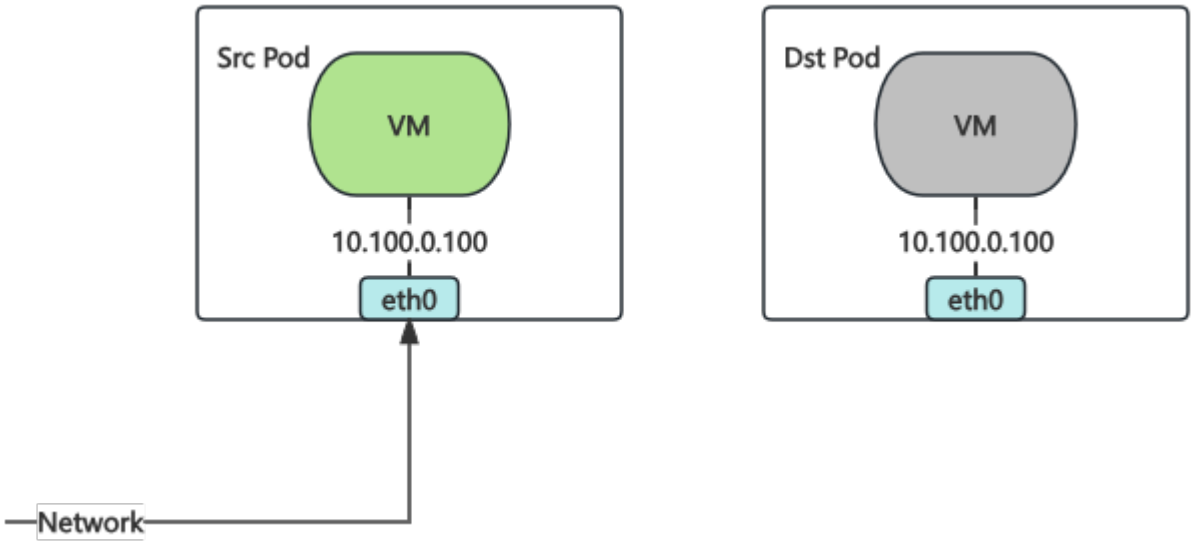
Kube-OVN Live migration - Reducing downtime with multchassis port bindings

IP

1. KubeVirt Pod

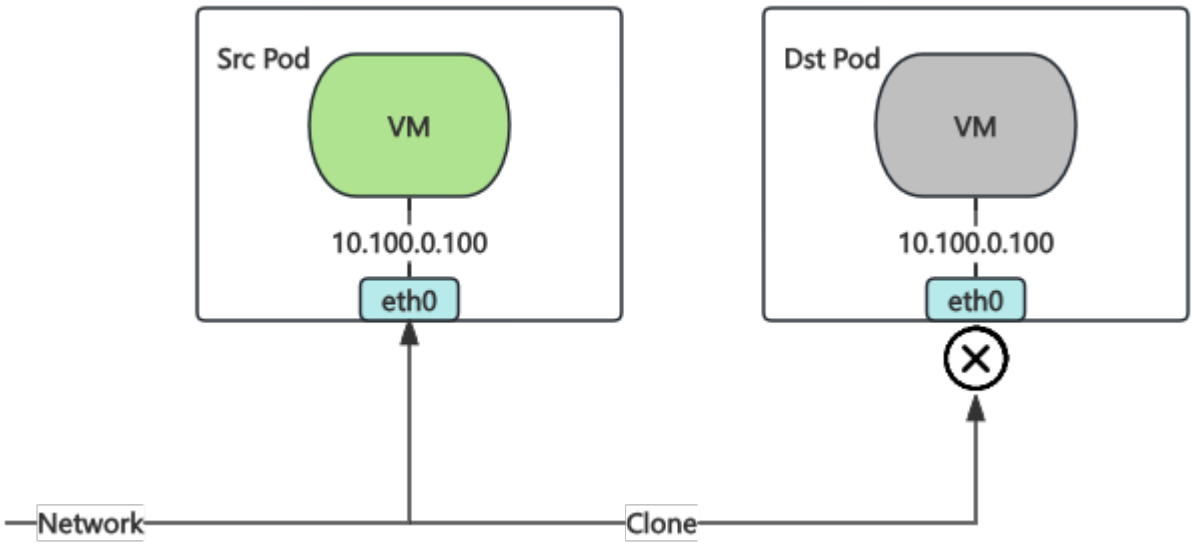


1. Kube-OVN Pod Pod Pod

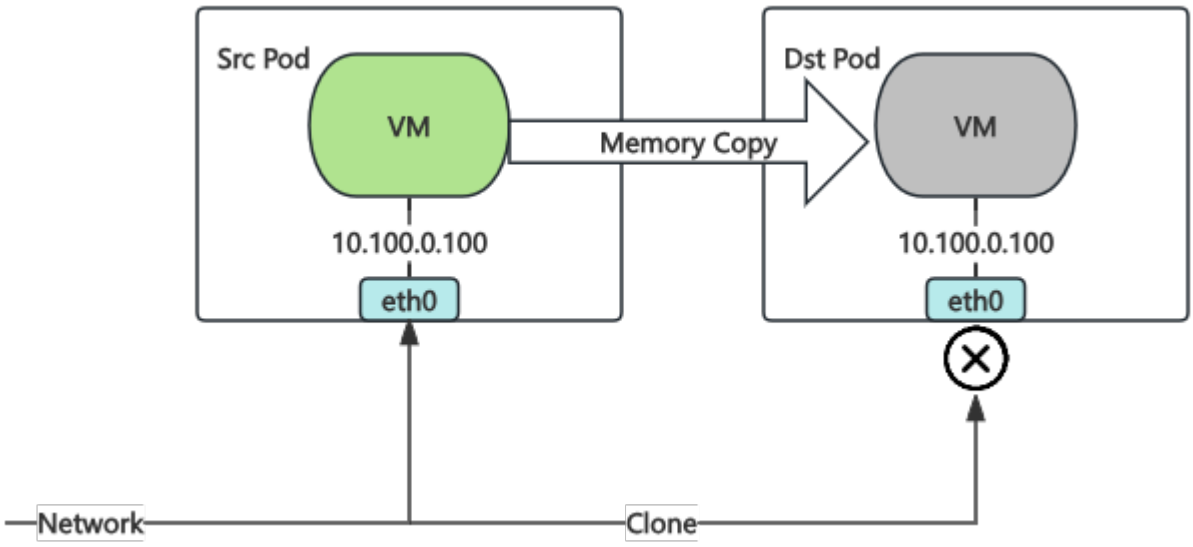


1. Kube-OVN Pod Pod Pod Pod

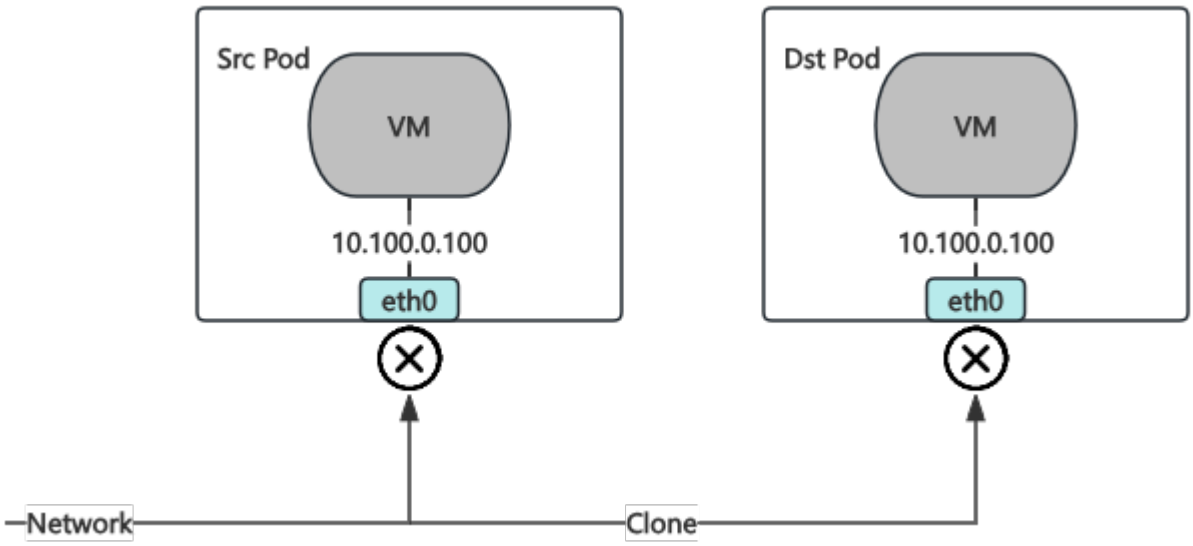
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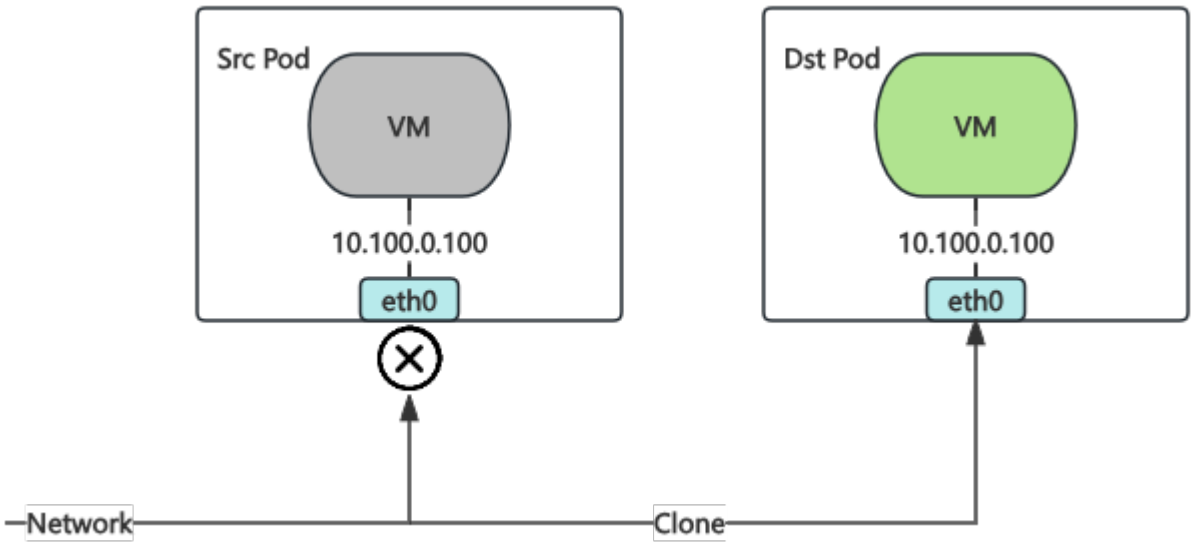
1. KubeVirt VM



1. KubeVirt Pod Pod

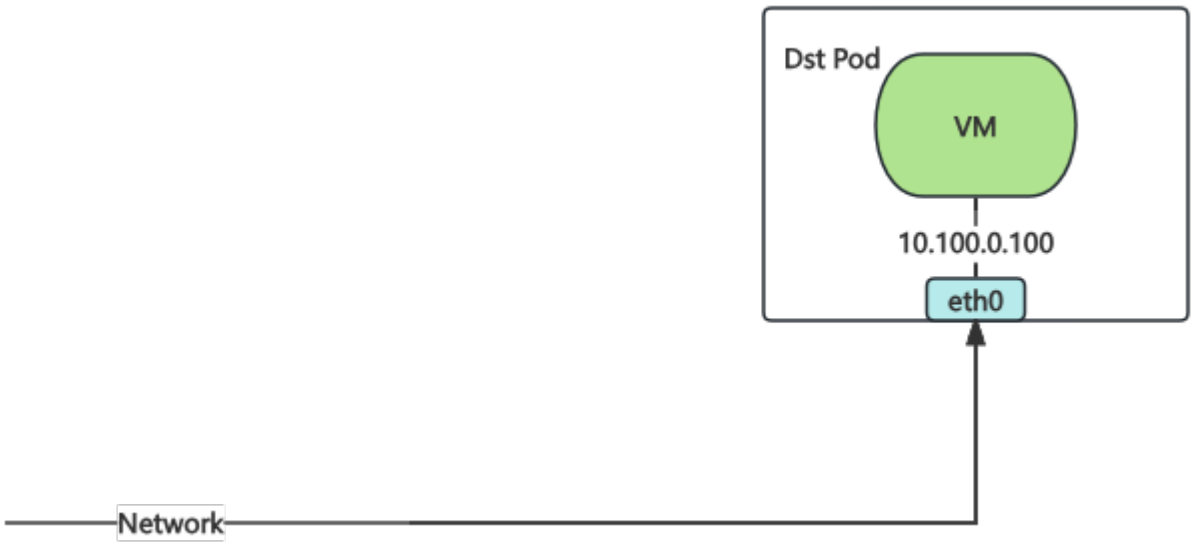


1. KubeVirt Pod libvirt RARP Pod Pod



1. KubeVirt Pod Kube-OVN Watch Migration CR

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5 6 libvirt RARP 0.5 TCP

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4.4.3

4.5

Kube-OVN Multus Dynamic Networks Controller KubeVirt v1.4.0 VM

4.5.1

Thick Multus

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/refs/heads/master/deployments/multus-daemonset-thick.yml
```

Multus Dynamic Networks Controller

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-dynamic-networks-controller/refs/heads/main/manifests/dynamic-networks-controller.yaml
```

4.5.2

NetworkAttachmentDefinition

`provider` `ovn`

```yaml
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: attachnet
  namespace: default
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "attachnet.default.ovn"
    }'
```

- `spec.config.type` : `kube-ovn` CNI Kube-OVN
- `server_socket` : Kube-OVN socket `/run/openvswitch/kube-ovn-daemon.sock`
- `provider` : NetworkAttachmentDefinition `<name>.<namespace>.ovn` , Kube-OVN Subnet ovn

Kube-OVN Subnet

Kube-OVN `provider` NetworkAttachmentDefinition `<name>.<namespace>.ovn` `ovn` Kube-OVN Subnet

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: attachnet
spec:
  protocol: IPv4
  provider: attachnet.default.ovn
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
  - 172.17.0.0..172.17.0.10
```

4.5.3 VM

yaml VM

```yaml
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: vm-fedora
spec:
  runStrategy: Always
  template:
    spec:
      domain:
```


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```
        devices:
          disks:
          - disk:
              bus: virtio
            name: containerdisk
          interfaces:
          - masquerade: {}
            name: defaultnetwork
          rng: {}
        resources:
          requests:
            memory: 1024M
      networks:
      - name: defaultnetwork
        pod: {}
      terminationGracePeriodSeconds: 0
      volumes:
      - containerDisk:
          image: quay.io/kubevirt/fedora-with-test-tooling-container-disk:devel
        name: containerdisk
```

VM Spec

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: vm-fedora
template:
  spec:
    domain:
      devices:
        interfaces:
        - name: defaultnetwork
          masquerade: {}
          # new interface
        - name: dyniface1
          bridge: {}
    networks:
    - name: defaultnetwork
      pod: {}
      # new network
    - name: dyniface1
      multus:
        networkName: attachnet
```

`interface` `state` `absent`

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: vm-fedora
template:
  spec:
    domain:
      devices:
        interfaces:
          - name: defaultnetwork
            masquerade: {}
          # set the interface state to absent
          - name: dyniface1
            state: absent
            bridge: {}
    networks:
      - name: defaultnetwork
        pod: {}
      - name: dyniface1
        multus:
          networkName: attachnet
```

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GitHub 

4.5.4

4.6 DHCP

`managedTap` `SR-IOV` `DPDK` KubeVirt DHCP Kube-OVN OVN DHCP Pod DHCP KubeVirt DHCP IP Kube-OVN DHCP DHCPv6, IPv6RA, DNS TFTP DHCP DHCP

> **Warning**
>
> `bridge` KubeVirt DHCP Kube-OVN DHCP Kube-OVN DHCP Kube-OVN DHCP `managedTap` `bridge` `managedTap` managedTap

4.6.1 DHCP

DHCP

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: sn-dual
spec:
  cidrBlock: "10.0.0.0/24,240e::a00/120"
  default: false
  disableGatewayCheck: true
  disableInterConnection: false
  excludeIps:
    - 10.0.0.1
    - 240e::a01
  gateway: 10.0.0.1,240e::a01
  gatewayNode: ''
  gatewayType: distributed
  natOutgoing: false
  private: false
  protocol: Dual
  provider: ovn
  vpc: vpc-test
  enableDHCP: true
  dhcpV4Options: "lease_time=3600,router=10.0.0.1,server_id=169.254.0.254,server_mac=00:00:00:2E:2F:B8"
  dhcpV6Options: "server_id=00:00:00:2E:2F:C5"
  enableIPv6RA: true
  ipv6RAConfigs: "address_mode=dhcpv6_stateful,max_interval=30,min_interval=5,send_periodic=true"
```

- `enableDHCP`: DHCP
- `dhcpV4Options`, `dhcpV6Options`: ovn-nb DHCP DHCP Options `"lease_time=3600, router=$ipv4_gateway, server_id=169.254.0.254, server_mac=$random_mac"` `server_id=$random_mac`
- `enableIPv6RA`: DHCPv6
- `ipv6RAConfigs` ovn-nb Logical_Router_Port Logical Router Port `address_mode=dhcpv6_stateful, max_interval=30, min_interval=5, send_periodic=true`

4.6.2 Pod DHCP

DHCP Kube-OVN Pod Pod DHCP Pod DHCP DHCP `enableDHCP`

```yaml
#     provider   "ovn"
ovn.kubernetes.io/dhcp-v4-options: "lease_time=3600,router=10.0.0.1,dns_server=8.8.8.8"
ovn.kubernetes.io/dhcp-v6-options: "server_id=00:00:00:00:00:01"

#      provider   "net1.ns1.ovn"
net1.ns1.ovn.kubernetes.io/dhcp-v4-options: "lease_time=7200"
```

key `<provider>.kubernetes.io/dhcp-v4-options` `<provider>.kubernetes.io/dhcp-v6-options` `<provider>` provider `ovn` Multus provider `<net-attach-def-name>.<namespace>.ovn`

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Pod DHCPv4

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: dhcp-pod
  annotations:
    ovn.kubernetes.io/dhcp-v4-options: "lease_time=3600,router=10.0.0.1,dns_server=8.8.8.8"
    ovn.kubernetes.io/dhcp-v6-options: "server_id=00:00:00:00:00:01"
spec:
  containers:
    - name: test
      image: docker.io/library/nginx:alpine
```

Multus provider DHCP

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: multi-nic-dhcp-pod
  annotations:
    k8s.v1.cni.cncf.io/networks: '[{"name": "net1", "namespace": "ns1"}]'
    ovn.kubernetes.io/dhcp-v4-options: "lease_time=3600,router=10.0.0.1"
    net1.ns1.ovn.kubernetes.io/dhcp-v4-options: "lease_time=7200,router=10.0.1.1"
spec:
  containers:
    - name: test
      image: docker.io/library/nginx:alpine
```

DHCP

1. **Pod** `<provider>.kubernetes.io/dhcp-v4-options` `<provider>.kubernetes.io/dhcp-v6-options` Pod DHCP
2. Subnet CRD `dhcpV4Options` `dhcpV6Options` DHCP

Pod DHCP Pod DHCP

> **Note**
>
> Pod DHCP DHCP Pod

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4.6.3

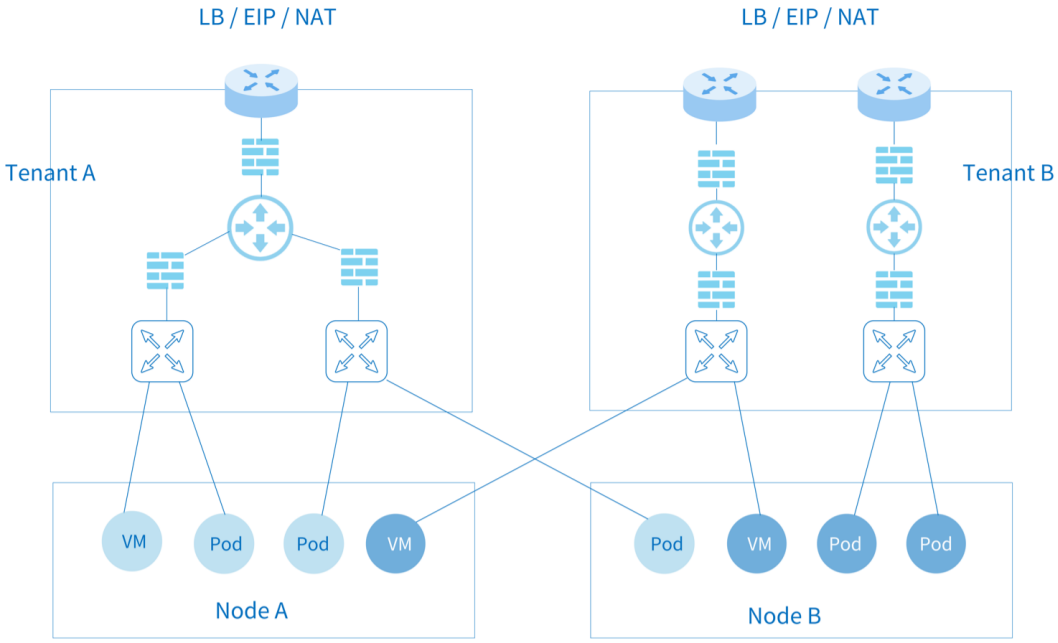
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5. VPC

5.1 VPC

Kube-OVN VPC VPC Subnet EIP

> VPC Kubernetes Pod NodePort DNS Kubernetes
> Kube-OVN VPC VPC Subnet Kubernetes VPC EIP NAT VPC ACL
> VPC VPC VPC Underlay VPC Underlay



5.1.1

Kube-OVN VPC OVN IP VPC OVN Datapath ID
Datapath ID IP VPC IP OVN Architecture Design Decisions

5.1.2 VPC

VPC

```yaml
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: test-vpc-1
spec:
  namespaces:
  - ns1
---
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: test-vpc-2
spec:
  namespaces:
  - ns2
```

- `namespaces` Namespace VPC

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VPC CIDR:

```
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net1
spec:
  vpc: test-vpc-1
  cidrBlock: 10.0.1.0/24
  protocol: IPv4
  namespaces:
    - ns1
---
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net2
spec:
  vpc: test-vpc-2
  cidrBlock: 10.0.1.0/24
  protocol: IPv4
  namespaces:
    - ns2
```

Namespace Pod:

```
apiVersion: v1
kind: Pod
metadata:
  namespace: ns1
  name: vpc1-pod
spec:
  containers:
    - name: vpc1-pod
      image: docker.io/library/nginx:alpine
---
apiVersion: v1
kind: Pod
metadata:
  namespace: ns2
  name: vpc2-pod
spec:
  containers:
    - name: vpc2-pod
      image: docker.io/library/nginx:alpine
```

Pod CIDR VPC Pod

5.1.3 VPC

VPC VPC VPC IP SNAT DNAT

VPC Multus-CNI multus-cni

Note

VPC VPC

VPC VPC NAT OVN Egress Gateway VPC NAT Kube-OVN VPC NAT Pod VPC Macvlan Pod iptables

OVN OVN NAT OVN BFD OVN OVN

Egress Gateway VPC NAT

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: ovn-vpc-external-network
spec:
  protocol: IPv4
  provider: ovn-vpc-external-network.kube-system
  cidrBlock: 192.168.0.0/24
  gateway: 192.168.0.1  # IP address of the physical gateway
  excludeIps:
```


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```
  - 192.168.0.1..192.168.0.10
---
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-vpc-external-network
  namespace: kube-system
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "macvlan",
      "master": "eth1",
      "mode": "bridge",
      "ipam": {
        "type": "kube-ovn",
        "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
        "provider": "ovn-vpc-external-network.kube-system"
      }
    }'

          `ipam`        VPC     net1     Kube-OVN    IP
---
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-vpc-external-network
  namespace: kube-system
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "macvlan",
      "master": "eth1",
      "mode": "bridge"
    }'
```

- Subnet Macvlan VPC IP
- VPC Macvlan `NetworkAttachmentDefinition` `master`
- `name`

Macvlan L2/L3 Vlan

1. OpenStack VM `PortSecurity`
2. VMware vSwitch `MAC Address Changes`, `Forged Transmits` `Promiscuous Mode Operation` `allow`
3. Hyper-V `MAC Address Spoofing`
4. AWS GCE Mac Macvlan
5. Macvlan Macvlan VpcNATGateway Pod Pod
6. Trunk Macvlan

VPC

VPC `kube-system` `ovn-vpc-nat-gw-config` `ovn-vpc-nat-config` ConfigMap `image` Pod `nodeSelector` `nodeSelector` `LoadBalancer` Service lb-svc VpcNATGateway Pod `VpcNatGateway.spec.selector`

```
---
kind: ConfigMap
apiVersion: v1
metadata:
  name: ovn-vpc-nat-config
  namespace: kube-system
data:
  image: 'docker.io/kubeovn/vpc-nat-gateway:v1.16.16'
  nodeSelector: |
    kubernetes.io/hostname: kube-ovn-control-plane
---
kind: ConfigMap
apiVersion: v1
metadata:
  name: ovn-vpc-nat-gw-config
  namespace: kube-system
data:
  enable-vpc-nat-gw: 'true'
```

- `image`: Pod
- `enable-vpc-nat-gw` VPC

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VPC

```
kind: VpcNatGateway
apiVersion: kubeovn.io/v1
metadata:
  name: gw1
spec:
  vpc: test-vpc-1
  subnet: net1
  lanIp: 10.0.1.254
  selector:
    - "kubernetes.io/hostname: kube-ovn-worker"
    - "kubernetes.io/os: linux"
  externalSubnets:
    - ovn-vpc-external-network
  noDefaultEIP: false
```

- `vpc` VpcNatGateway VPC
- `subnet` VPC Subnet VPC Pod `lanIp`
- `lanIp` `subnet` IP VPC Pod IP VPC VpcNatGateway `nextHopIP` `lanIp`
- `selector` VpcNatGateway Pod Kubernetes NodeSelector
- `externalSubnets` VPC `ovn-vpc-external-network`
- `noDefaultEIP` VPC EIP `false` v1.15 BGP `true` Underlay

- `tolerations` VPC
- `affinity` VPC Pod

VPC-NAT-GW

1. nat gw pod net1 arp ping eip arp ping

EIP

EIP IP VPC DNAT SNAT IP

EIP

```
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eip-random
spec:
  natGwDp: gw1
```

EIP

```
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eip-static
spec:
  natGwDp: gw1
  v4ip: 192.168.0.100
```

EIP

```
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eip-random
spec:
  natGwDp: gw1
  externalSubnet: ovn-vpc-external-network
```

- `externalSubnet` EIP `ovn-vpc-external-network` IptablesEIP VpcNatGateway `externalSubnets` EIP

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DNAT

DNAT EIP VPC IP

```
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eipd01
spec:
  natGwDp: gw1

---
kind: IptablesDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: dnat01
spec:
  eip: eipd01
  externalPort: '8888'
  internalIp: 10.0.1.10
  internalPort: '80'
  protocol: tcp
```

SNAT

SNAT VPC Pod EIP SNAT

```
---
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eips01
spec:
  natGwDp: gw1
---
kind: IptablesSnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: snat01
spec:
  eip: eips01
  internalCIDR: 10.0.1.0/24
```

IP

IP VPC IP EIP EIP VPC IP VPC IP SNAT EIP

```
---
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eipf01
spec:
  natGwDp: gw1

---
kind: IptablesFIPRule
apiVersion: kubeovn.io/v1
metadata:
  name: fip01
spec:
  eip: eipf01
  internalIp: 10.0.1.5
```

5.1.4

VPC Kube-OVN

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: test-vpc-1
spec:
  staticRoutes:
    - cidr: 0.0.0.0/0
      nextHopIP: 10.0.1.254
      policy: policyDst
```


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```
    - cidr: 172.31.0.0/24
      nextHopIP: 10.0.1.253
      policy: policySrc
      routeTable: "rtb1"
```

- `policy`： `policyDst` `policySrc`
- CIDR
- `routeTable`：

OVN Logical Router Policy

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: test-vpc-1
spec:
  policyRoutes:
    - action: drop
      match: ip4.src==10.0.1.0/24 && ip4.dst==10.0.1.250
      priority: 11
    - action: reroute
      match: ip4.src==10.0.1.0/24
      nextHopIP: 10.0.1.252
      priority: 10
```

5.1.5

Kubernetes Service Kubernetes Service IP VPC VPC Kubernetes Service

Kube-OVN `SwitchLBRule`

`SwitchLBRule`

```
apiVersion: kubeovn.io/v1
kind: SwitchLBRule
metadata:
  name:  cjh-slr-nginx
spec:
  vip: 1.1.1.1
  sessionAffinity: ClientIP
  namespace: default
  selector:
    - app: nginx
  ports:
  - name: dns
    port: 8888
    targetPort: 80
    protocol: TCP
```

- `vip`
- `namespace` Pod Namespace
- `sessionAffinity` Service `sessionAffinity`
- `selector` Service `selector`
- `ports` Service `port`

```
# kubectl get slr
NAME               VIP         PORT(S)                 SERVICE                           AGE
vpc-dns-test-cjh2  10.96.0.3   53/UDP,53/TCP,9153/TCP  kube-system/slr-vpc-dns-test-cjh2  88m
```

5.1.6 vpc-dns

VPC VPC VPC Pod coredns VPC CoreDNS Service Kube-OVN vpc-dns

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```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-nad
  namespace: default
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "ovn-nad.default.ovn"
    }'
```

vpc-dns ConfigMap

kube-system configmap vpc-dns vpc-dns

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: vpc-dns-config
  namespace: kube-system
data:
  coredns-vip: 10.96.0.3
  enable-vpc-dns: "true"
  nad-name: ovn-nad
  nad-provider: ovn-nad.default.ovn
```

- `enable-vpc-dns` `true` `false` `true`
- `coredns-image` dns coredns
- `coredns-vip` coredns lb vip
- `nad-name` `network-attachment-definitions`
- `nad-provider` provider
- `k8s-service-host` coredns k8s apiserver ip
- `k8s-service-port` coredns k8s apiserver port

vpc-dns

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  labels:
    kubernetes.io/bootstrapping: rbac-defaults
  name: system:vpc-dns
rules:
  - apiGroups:
    - ""
    resources:
    - endpoints
    - services
    - pods
    - namespaces
    verbs:
    - list
    - watch
  - apiGroups:
    - discovery.k8s.io
    resources:
    - endpointslices
    verbs:
    - list
    - watch
---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  annotations:
    rbac.authorization.kubernetes.io/autoupdate: "true"
  labels:
    kubernetes.io/bootstrapping: rbac-defaults
  name: vpc-dns
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: ClusterRole
  name: system:vpc-dns
subjects:
```


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```
- kind: ServiceAccount
  name: vpc-dns
  namespace: kube-system
---
apiVersion: v1
kind: ServiceAccount
metadata:
  name: vpc-dns
  namespace: kube-system
---
apiVersion: v1
kind: ConfigMap
metadata:
  name: vpc-dns-corefile
  namespace: kube-system
data:
  Corefile: |
    .:53 {
        errors
        health {
          lameduck 5s
        }
        ready
        kubernetes cluster.local in-addr.arpa ip6.arpa {
          pods insecure
          fallthrough in-addr.arpa ip6.arpa
        }
        prometheus :9153
        forward . /etc/resolv.conf {
          prefer_udp
        }
        cache 30
        loop
        reload
        loadbalance
    }
```

vpc-dns

```
kind: VpcDns
apiVersion: kubeovn.io/v1
metadata:
  name: test-cjh1
spec:
  vpc: cjh-vpc-1
  subnet: cjh-subnet-1
```

- `vpc` dns vpc
- `subnet` dns

```
[root@hci-dev-mst-1 kubeovn]# kubectl get vpc-dns
NAME        ACTIVE   VPC         SUBNET
test-cjh1   false    cjh-vpc-1   cjh-subnet-1
test-cjh2   true     cjh-vpc-1   cjh-subnet-2
```

- `ACTIVE`：`true` dns `false`

- VPC DNS
- VPC vpc-dns VPC subnet vpc-dns `true` `false`
- `true` vpc-dns `false` vpc-dns

5.1.7

VPC VPC

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: test-vpc-1
spec:
  namespaces:
```


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```
  - ns1
  defaultSubnet: test
```

- `defaultSubnet` VPC

Namespace `ovn.kubernetes.io/logical_switch` `ovn.kubernetes.io/logical_switch` Pod

VPC Pod livenessProbe readinessProbe

VPC Pod kubelet VPC Pod Kube-OVN TProxy kubelet VPC Pod

DaemonSet `kube-ovn-cni` `--enable-tproxy=true`

```
spec:
  template:
    spec:
      containers:
      - args:
        - --enable-tproxy=true
```

1. VPC Pod IP
2. `tcpSocket` `httpGet`

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5.1.8

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5.2 VPC Egress Gateway

> **Note**
>
> VPC VPC
>
> VPC VPC NAT OVN Egress Gateway VPC NAT Kube-OVN VPC NAT Pod VPC Macvlan Pod iptables
>
> OVN OVN NAT OVN BFD OVN OVN
>
> Egress Gateway VPC NAT

VPC Egress Gateway VPC VPC Pod

- ECMP Active-Active
- BFD <1s
- IPv6
- Namespace Pod
- Node Egress Gateway

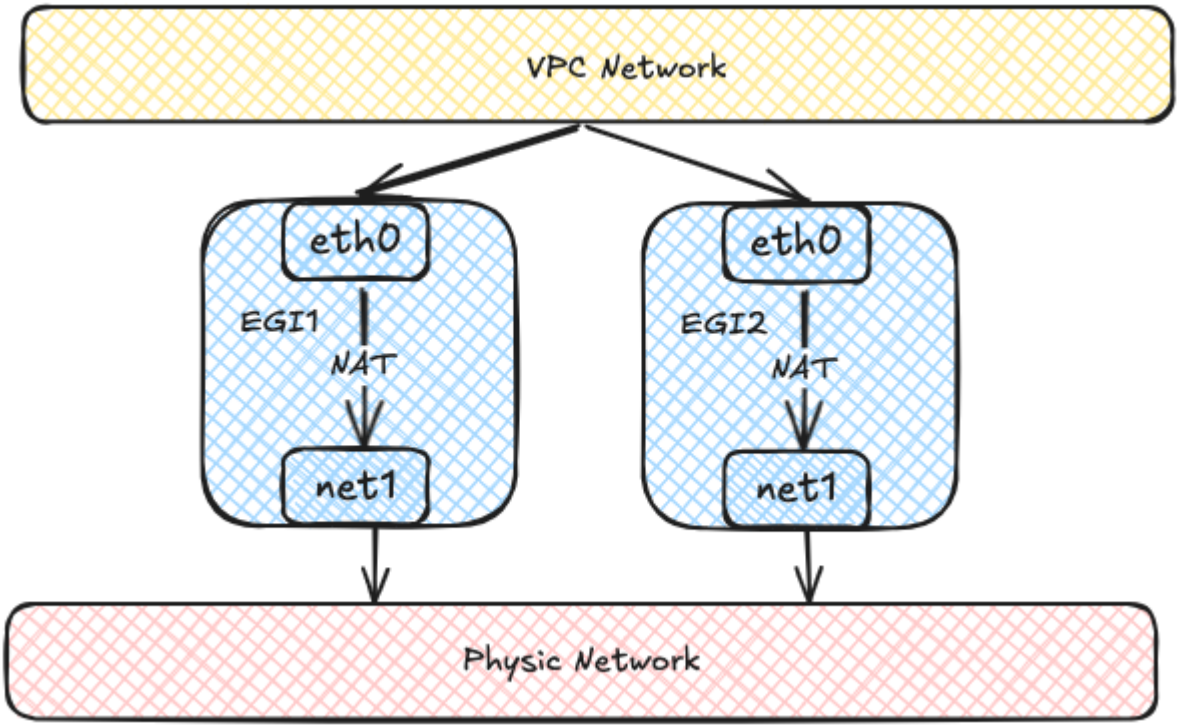
VPC Egress Gateway BGP EVPN/VXLAN Egress Gateway BGP/EVPN

VPC Egress Gateway

- Macvlan Underlay
- Gateway Egress IP
- SNAT EIP DNAT
-

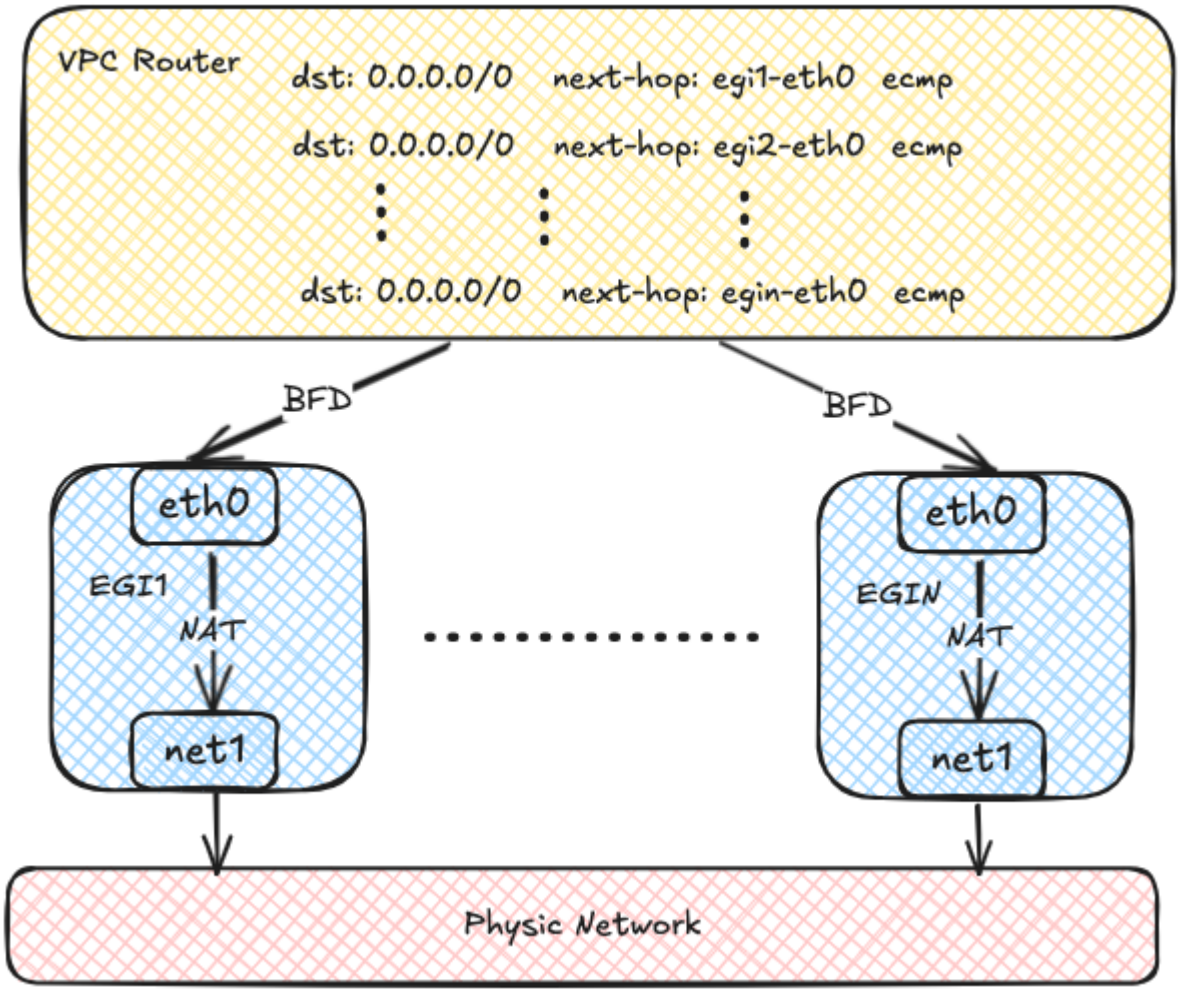
5.2.1

Egress Gateway Pod Pod VPC Macvlan Egress Gateway NAT

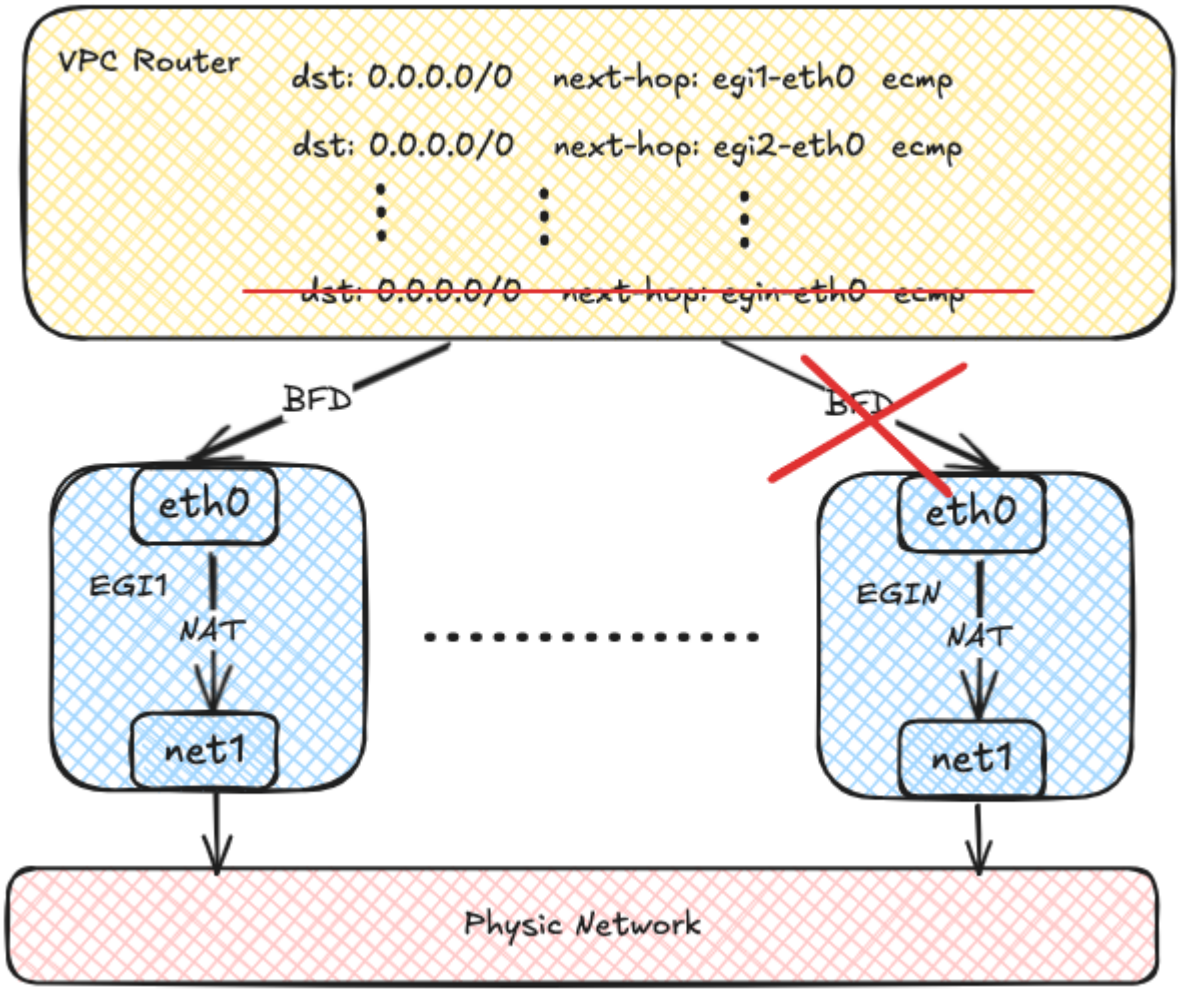


Egress Gateway OVN VPC Pod OVN Egress Gateway Egress Gateway

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OVN BFD Egress Gateway Egress Gateway OVN



5.2.2

VPC Egress Gateway VPC NAT Gateway Multus-CNI

> VPC Egress Gateway ConfigMap

5.2.3

NetworkAttachmentDefinition

VPC Egress Gateway VPC NetworkAttachmentDefinition `macvlan` Kube-OVN IPAM

```
apiVersion: k8s.cni.cncf.io/v1
kind: NetworkAttachmentDefinition
metadata:
  name: eth1
  namespace: default
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "macvlan",
      "master": "eth1",
      "mode": "bridge",
      "ipam": {
        "type": "kube-ovn",
        "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
        "provider": "eth1.default"
      }
    }'
---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: macvlan1
```


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```
spec:
  protocol: IPv4
  provider: eth1.default
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
    - 172.17.0.0..172.17.0.10
```

CNI NetworkAttachmentDefinition VPC Egress Gateway

VPC Egress Gateway

VPC Egress Gateway

```
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway1
  namespace: default
spec:
  vpc: ovn-cluster
  replicas: 1
  externalSubnet: macvlan1
  policies:
    - snat: true
      subnets:
        - ovn-default
```

default VPC `ovn-cluster` gateway1 VPC Egress Gateway `ovn-cluster` `ovn-default` 10.16.0.0/16 Pod `macvlan1` SNAT

VPC Egress Gateway

```
$ kubectl get veg gateway1
NAME       VPC           REPLICAS   BFD ENABLED   EXTERNAL SUBNET   PHASE       READY   AGE
gateway1   ovn-cluster   1          false         macvlan1          Completed   true    13s
```

```
kubectl get veg gateway1 -o wide
NAME       VPC           REPLICAS   BFD ENABLED   EXTERNAL SUBNET   PHASE       READY   INTERNAL IPS     EXTERNAL IPS      WORKING NODES         AGE
gateway1   ovn-cluster   1          false         macvlan1          Completed   true    ["10.16.0.12"]   ["172.17.0.11"]   ["kube-ovn-worker"]   82s
```

```
$ kubectl get deployment -l ovn.kubernetes.io/vpc-egress-gateway=gateway1
NAME       READY   UP-TO-DATE   AVAILABLE   AGE
gateway1   1/1     1            1           4m40s

$ kubectl get pod -l ovn.kubernetes.io/vpc-egress-gateway=gateway1 -o wide
NAME                        READY   STATUS    RESTARTS   AGE     IP           NODE              NOMINATED NODE   READINESS GATES
gateway1-b9f8b4448-76lhm   1/1     Running   0          4m48s   10.16.0.12   kube-ovn-worker   <none>           <none>
```

Pod IP iptables

```
$ kubectl exec gateway1-b9f8b4448-76lhm -c gateway -- ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: net1@if13: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 62:d8:71:90:7b:86 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 172.17.0.11/16 brd 172.17.255.255 scope global net1
       valid_lft forever preferred_lft forever
    inet6 fe80::60d8:71ff:fe90:7b86/64 scope link
       valid_lft forever preferred_lft forever
17: eth0@if18: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue state UP group default
    link/ether 36:7c:6b:c7:82:6b brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.16.0.12/16 brd 10.16.255.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::347c:6bff:fec7:826b/64 scope link
       valid_lft forever preferred_lft forever

$ kubectl exec gateway1-b9f8b4448-76lhm -c gateway -- ip route show
default via 172.17.0.1 dev net1
10.16.0.0/16 dev eth0 proto kernel scope link src 10.16.0.12
```


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```
172.17.0.0/16 dev net1 proto kernel scope link src 172.17.0.11

$ kubectl exec gateway1-b9f8b4448-76lhm -c gateway -- iptables -t nat -S
-P PREROUTING ACCEPT
-P INPUT ACCEPT
-P OUTPUT ACCEPT
-P POSTROUTING ACCEPT
-A POSTROUTING -s 10.16.0.0/16 -j MASQUERADE --random-fully
```

Gateway Pod

```
$ kubectl exec -ti gateway1-b9f8b4448-76lhm -c gateway -- bash
nobody@gateway1-b9f8b4448-76lhm:/kube-ovn$ tcpdump -i any -nnve icmp and host 172.17.0.1
tcpdump: data link type LINUX_SLL2
tcpdump: listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
06:50:58.936528 eth0  In  ifindex 17 92:26:b8:9e:f2:1c ethertype IPv4 (0x0800), length 104: (tos 0x0, ttl 63, id 30481, offset 0, flags [DF], proto ICMP (1),
length 84)
    10.16.0.9 > 172.17.0.1: ICMP echo request, id 37989, seq 0, length 64
06:50:58.936574 net1  Out ifindex 2 62:d8:71:90:7b:86 ethertype IPv4 (0x0800), length 104: (tos 0x0, ttl 62, id 30481, offset 0, flags [DF], proto ICMP (1),
length 84)
    172.17.0.11 > 172.17.0.1: ICMP echo request, id 39449, seq 0, length 64
06:50:58.936613 net1  In  ifindex 2 02:42:39:79:7f:08 ethertype IPv4 (0x0800), length 104: (tos 0x0, ttl 64, id 26701, offset 0, flags [none], proto ICMP
(1), length 84)
    172.17.0.1 > 172.17.0.11: ICMP echo reply, id 39449, seq 0, length 64
06:50:58.936621 eth0  Out ifindex 17 36:7c:6b:c7:82:6b ethertype IPv4 (0x0800), length 104: (tos 0x0, ttl 63, id 26701, offset 0, flags [none], proto ICMP
(1), length 84)
    172.17.0.1 > 10.16.0.9: ICMP echo reply, id 37989, seq 0, length 64
```

OVN Logical Router VPC

```
$ kubectl ko nbctl lr-policy-list ovn-cluster
Routing Policies
     31000                            ip4.dst == 10.16.0.0/16           allow
     31000                           ip4.dst == 100.64.0.0/16           allow
     30000                              ip4.dst == 172.18.0.2         reroute               100.64.0.3
     30000                              ip4.dst == 172.18.0.3         reroute               100.64.0.2
     30000                              ip4.dst == 172.18.0.4         reroute               100.64.0.4
     29100                            ip4.src == 10.16.0.0/16         reroute               10.16.0.12
     29000 ip4.src == $ovn.default.kube.ovn.control.plane_ip4         reroute               100.64.0.2
     29000       ip4.src == $ovn.default.kube.ovn.worker2_ip4         reroute               100.64.0.4
     29000        ip4.src == $ovn.default.kube.ovn.worker_ip4         reroute               100.64.0.3
```

`.spec.replicas`

```
$ kubectl scale veg gateway1 --replicas=2
vpcegressgateway.kubeovn.io/gateway1 scaled

$ kubectl get veg gateway1
NAME       VPC           REPLICAS   BFD ENABLED   EXTERNAL SUBNET   PHASE       READY   AGE
gateway1   ovn-cluster   2          false         macvlan           Completed   true    39m

$ kubectl get pod -l ovn.kubernetes.io/vpc-egress-gateway=gateway1 -o wide
NAME                        READY   STATUS    RESTARTS   AGE   IP           NODE               NOMINATED NODE   READINESS GATES
gateway1-b9f8b4448-76lhm    1/1     Running   0          40m   10.16.0.12   kube-ovn-worker    <none>           <none>
gateway1-b9f8b4448-zd4dl    1/1     Running   0          64s   10.16.0.13   kube-ovn-worker2   <none>           <none>

$ kubectl ko nbctl lr-policy-list ovn-cluster
Routing Policies
     31000                            ip4.dst == 10.16.0.0/16           allow
     31000                           ip4.dst == 100.64.0.0/16           allow
     30000                              ip4.dst == 172.18.0.2         reroute               100.64.0.3
     30000                              ip4.dst == 172.18.0.3         reroute               100.64.0.2
     30000                              ip4.dst == 172.18.0.4         reroute               100.64.0.4
     29100                            ip4.src == 10.16.0.0/16         reroute               10.16.0.12, 10.16.0.13
     29000 ip4.src == $ovn.default.kube.ovn.control.plane_ip4         reroute               100.64.0.2
     29000       ip4.src == $ovn.default.kube.ovn.worker2_ip4         reroute               100.64.0.4
     29000        ip4.src == $ovn.default.kube.ovn.worker_ip4         reroute               100.64.0.3
```

Egress Gateway IP

`externalIPs` `nodeSelector` Egress Gateway Pods Egress IP

```yaml
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway1
  namespace: default
spec:
  vpc: ovn-cluster
  replicas: 2
  externalSubnet: macvlan1
  policies:
    - snat: true
      subnets:
        - ovn-default
```


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```
  externalIPs:
    - 172.17.0.10
    - 172.17.0.11
  nodeSelector:
    - matchLabels:
        kubernetes.io/hostname: kube-ovn-worker
```

IP `internalIPPool` `externalIPPool` `IPPool` IPPool `internalSubnet` `externalSubnet` `internalIPs` `externalIPs`

```
spec:
  internalIPPool: veg-internal
  externalIPPool: veg-external
```

BFD

BFD VPC BFD LRP VPC BFD Port

```
apiVersion: kubeovn.io/v1
kind: Vpc
metadata:
  name: vpc1
spec:
  bfdPort:
    enabled: true
    ip: 10.255.255.255
---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  vpc: vpc1
  protocol: IPv4
  cidrBlock: 192.168.0.0/24
```

BFD Port OVN LR BFD LRP

```
$ kubectl ko nbctl show vpc1
router 0c1d1e8f-4c86-4d96-88b2-c4171c7ff824 (vpc1)
    port bfd@vpc1
        mac: "8e:51:4b:16:3c:90"
        networks: ["10.255.255.255"]
    port vpc1-subnet1
        mac: "de:c9:5c:38:7a:61"
        networks: ["192.168.0.1/24"]
```

VPC Egress Gateway `.spec.bfd.enabled` `true`

```
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway2
  namespace: default
spec:
  vpc: vpc1
  replicas: 2
  internalSubnet: subnet1
  externalSubnet: macvlan
  bfd:
    enabled: true
  policies:
    - snat: true
      ipBlocks:
        - 192.168.0.0/24
```

VPC Egress Gateway

```
$ kubectl get veg gateway2 -o wide
NAME       VPC    REPLICAS   BFD ENABLED   EXTERNAL SUBNET   PHASE       READY   INTERNAL IPS                    EXTERNAL IPS                    WORKING
NODES                            AGE
gateway2   vpc1   2          true          macvlan           Completed   true    ["192.168.0.2","192.168.0.3"]   ["172.17.0.13","172.17.0.14"]   ["kube-ovn-
worker","kube-ovn-worker2"]   58s

$ kubectl get pod -l ovn.kubernetes.io/vpc-egress-gateway=gateway2 -o wide
NAME                        READY   STATUS    RESTARTS   AGE     IP            NODE              NOMINATED NODE   READINESS GATES
gateway2-fcc6b8b87-8lgvx    1/1     Running   0          2m18s   192.168.0.3   kube-ovn-worker2  <none>           <none>
gateway2-fcc6b8b87-wmww6    1/1     Running   0          2m18s   192.168.0.2   kube-ovn-worker   <none>           <none>

$ kubectl ko nbctl lr-route-list vpc1
IPv4 Routes
Route Table <main>:
```


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```
          192.168.0.0/24              192.168.0.2 src-ip ecmp ecmp-symmetric-reply bfd
          192.168.0.0/24              192.168.0.3 src-ip ecmp ecmp-symmetric-reply bfd

$ kubectl ko nbctl list bfd
_uuid               : 223ede10-9169-4c7d-9524-a546e24bfab5
detect_mult         : 3
dst_ip              : "192.168.0.2"
external_ids        : {af="4", vendor=kube-ovn, vpc-egress-gateway="default/gateway2"}
logical_port        : "bfd@vpc1"
min_rx              : 1000
min_tx              : 1000
options             : {}
status              : up

_uuid               : b050c75e-2462-470b-b89c-7bd38889b758
detect_mult         : 3
dst_ip              : "192.168.0.3"
external_ids        : {af="4", vendor=kube-ovn, vpc-egress-gateway="default/gateway2"}
logical_port        : "bfd@vpc1"
min_rx              : 1000
min_tx              : 1000
options             : {}
status              : up
```

Pod BFD

```
$ kubectl exec gateway2-fcc6b8b87-8lgvx -c bfdd -- bfdd-control status
There are 1 sessions:
Session 1
 id=1 local=192.168.0.3 (p) remote=10.255.255.255 state=Up

$ kubectl exec gateway2-fcc6b8b87-wmww6 -c bfdd -- bfdd-control status
There are 1 sessions:
Session 1
 id=1 local=192.168.0.2 (p) remote=10.255.255.255 state=Up
```

VPC BFD PORT

`enabled`	`boolean`	`false`	BFD Port	`true`
`ip`	`string`	-	BFD Port IP IPv6	`169.255.255.2`
`nodeSelector`	`object`	-	BFD Port BFD Port OVN HA Chassis Group Active/ Backup Active nodeSelector Kube-OVN `kubectl ko nbctl list ha_chassis_group` OVN HA Chassis Group	-
`nodeSelector.matchLabels`	`dict/map`	-		-
`nodeSelector.matchExpressions`	`object array`	-		-

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VPC EGRESS GATEWAY

Spec

`vpc`	`string`	VPC ovn-cluster	VPC	`vpc1`
`replicas`	`integer/int32`	`1`		`2`
`prefix`	`string`	-	Deployment	`veg-`
`image`	`string`	-	Deployment	`docker.io/kubeovn/kube-`
`internalSubnet`	`string`	VPC	VPC	`subnet1`
`externalSubnet`	`string`	-		`ext1`
`internalIPs`	`string array`	-	VPC IP IPv6 IP `<replicas> + 1` Pod	`10.16.0.101` / `fd00::11`
`externalIPs`	`string array`	-	IP IPv6 IP `<replicas> + 1` Pod	`10.16.0.101` / `fd00::11`
`internalIPPool`	`string`	-	`IPPool` IPPool `internalSubnet` `internalIPs`	`veg-internal`
`externalIPPool`	`string`	-	`IPPool` IPPool `externalSubnet` `externalIPs`	`veg-external`
`bfd`	`object`	-	BFD	-
`policies`	`object array`	-	Egress `selectors`	-
`selectors`	`object array`	-	Namespace Selector Pod Selector Egress Pod SNAT/ MASQUERADE `policies`	-
`nodeSelector`	`object array`	-	Deployment/Pod	-
`trafficPolicy`	`string`	`Cluster`	`Cluster` / `Local` **BFD** `Local` Egress VPC	`Local`

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			Egress Gateway VPC Egress Gateway Egress	
`bandwidth`	`object`	-		-
`bandwidth.ingress`	`integer` `string`	-	VPC	`1024` / `1Gi`
`bandwidth.egress`	`integer` `string`	-	VPC	`1024` / `1Gi`

Mbps `M` `Mi` `G` `Gi` quantity bit/s Mbps `100M` `100Mi` `1G` `1Gi` 100 105 1000 1074 Mbps

9223372036854 Mbps `ingress` `egress`

```
spec:
  bandwidth:
    ingress: 1024
    egress: 1Gi
```



0 9223372036854 Mbps Kube-OVN `vpc-egress-gateways.kubeovn.io` CRD `x-kubernetes-int-or-string: true`

Helm Chart `crds/` CRD `kube-ovn-v2` Chart `KUBE_OVN_VERSION` Kube-OVN CRD bundle

```
kubectl apply -f "https://raw.githubusercontent.com/kubeovn/kube-ovn/${KUBE_OVN_VERSION}/charts/kube-ovn-v2/crds/kube-ovn-crd.yaml"
```

quantity Mbps

BFD

`enabled`	`boolean`	`false`	BFD	`true`
`minRX`	`integer/int32`	`1000`	BFD minRX ms	`500`
`minTX`	`integer/int32`	`1000`	BFD minTX ms	`500`
`multiplier`	`integer/int32`	`3`	BFD multiplier	`1`

Egress

`snat`	`boolean`	`false`	SNAT/ MASQUERADE	`true`
`ipBlocks`	`string array`	-	Gateway IP IPv6	`192.168.0.1` / `192.168.0.0/24`
`subnets`	`string array`	-	Gateway VPC IPv6	`subnet1`

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Selectors

namespaceSelector	object	-	Namespace Namespace	-
namespaceSelector.matchLabels	dict/map	-		-
namespaceSelector.matchExpressions	object array	-		-
podSelector	object	-	Pod Pod	-
podSelector.matchLabels	dict/map	-		-
podSelector.matchExpressions	object array	-		-

matchLabels	dict/map	-	-
matchExpressions	object array	-	-
matchFields	object array	-	-

Status

ready	boolean	Gateway	true
phase	string	Gateway	Pending / Processing / Completed
internalIPs	string array	VPC IP	-
externalIPs	string array	IP	-
workload	object		-
workload.apiVersion	string	API	apps/v1
workload.kind	string		Deployment
workload.name	string		gateway1
workload.nodes	string array		-
conditions	object array	-	-

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5.2.4

5.3 Egress Gateway BGP EVPN

BGP EVPN

- L3VPN L2VPN
- FRR Pod
- BGP BFD

5.3.1

VPC Egress Gateway BGP EVPN/VXLAN Egress Gateway Pod FRR Free Range Routing BGP EVPN Gateway init BgpConf EvpnConf FRR

BGP Egress Gateway BGP EVPN init Pod Linux VRF bridge VXLAN L2VPN EVPN VXLAN Egress Gateway Pod Kubernetes Geneve

> BGP kube-ovn-speaker Pod/ /Service

5.3.2

BGP/EVPN VPC Egress Gateway Multus-CNI

BGP EVPN BGP

5.3.3

BgpConf

BgpConf BGP

```yaml
apiVersion: kubeovn.io/v1
kind: BgpConf
metadata:
  name: bgp-conf-6502
spec:
  localASN: 65002
  peerASN: 65001
  neighbours:
    - 10.0.1.1
  holdTime: 90s
  keepaliveTime: 30s
  connectTime: 10s
  ebgpMultiHop: true
```

AS 65002 AS 65001 BGP 10.0.1.1 EBGP Multi-Hop

EvpnConf

EvpnConf EVPN EvpnConf BgpConf

```yaml
apiVersion: kubeovn.io/v1
kind: EvpnConf
metadata:
  name: evpn-conf-1016
spec:
  vni: 1016
  routeTargets:
    - "65000:1016"
```


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VNI 1016 EVPN Route Target `65000:1016`

BGP/EVPN VPC Egress Gateway

VPC Egress Gateway NetworkAttachmentDefinition

VpcEgressGateway `.spec` `bgpConf` `evpnConf`

```yaml
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway1
  namespace: default
spec:
  vpc: ovn-cluster
  replicas: 1
  externalIPs:
    - 10.0.1.13
  internalIPs:
    - 10.16.0.13
  externalSubnet: macvlan1
  bgpConf: bgp-conf-6502
  evpnConf: evpn-conf-1016
  policies:
    - snat: false
      subnets:
        - ovn-default
    - snat: false
      ipBlocks:
        - 10.17.0.0/16
```

BGP EVPN VPC Egress Gateway Gateway Pod FRR BGP EVPN VXLAN BGP/EVPN `snat` `false`

BGP EVPN

BGP EVPN/VXLAN `bgpConf` `evpnConf` FRR BGP IPv4 Unicast

```yaml
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway-bgp-only
  namespace: default
spec:
  vpc: ovn-cluster
  replicas: 1
  externalSubnet: macvlan1
  bgpConf: bgp-conf-6502
  policies:
    - snat: true
      subnets:
        - ovn-default
```


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5.3.4

BgpConf

`localASN`	`uint32`	-	AS	`65002`
`peerASN`	`uint32`	-	AS	`65001`
`routerId`	`string`	Pod IP	BGP Router ID	`10.0.1.13`
`neighbours`	`string array`	-	BGP IP	`10.0.1.1`
`password`	`string`	-	BGP	`secret`
`holdTime`	`string (duration)`	-	BGP Hold	`90s`
`keepaliveTime`	`string (duration)`	-	BGP Keepalive	`30s`
`connectTime`	`string (duration)`	-	BGP	`10s`
`ebgpMultiHop`	`boolean`	`false`	EBGP Multi-Hop	`true`

EvpnConf

`vni`	`uint32`	-	VXLAN Network Identifier	`1016`
`routeTargets`	`string array`	-	Route Target	`65000:1016`

VpcEgressGateway

VpcEgressGateway `.spec` BGP/EVPN VpcEgressGateway Egress Gateway

`bgpConf`	`string`	-	BgpConf Egress Gateway BGP	`bgp-conf-6502`
`evpnConf`	`string`	-	EvpnConf `bgpConf` Egress Gateway EVPN/VXLAN	`evpn-conf-1016`

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5.3.5

5.4 VPC QoS

Kube-OVN QoSPolicy CRD VPC

5.4.1

QoSPolicy `priority` QoS

EIP QoS	1	EIP
NATGW QoS	2	NATGW IP
NATGW net1 QoS	3	

- `shared=false` QoS `shared=true`
- EIP QoS NATGW QoS

5.4.2 EIP QoS

EIP 1Mbps 1 `shared=false` QoSPolicy EIP QoSPolicy QoS

QoSPolicy

```yaml
apiVersion: kubeovn.io/v1
kind: QoSPolicy
metadata:
  name: qos-eip-example
spec:
  shared: false
  bindingType: EIP
  bandwidthLimitRules:
  - name: eip-ingress
    rateMax: "1" # Mbps
    burstMax: "1" # Mbps
    priority: 1
    direction: ingress
  - name: eip-egress
    rateMax: "1" # Mbps
    burstMax: "1" # Mbps
    priority: 1
    direction: egress
```

IptablesEIP

```yaml
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eip-1
spec:
  natGwDp: gw1
  qosPolicy: qos-eip-example
```

`.spec.qosPolicy`

5.4.3 QoS EIP

`label` qos eip

```
# kubectl get eip  -l ovn.kubernetes.io/qos=qos-eip-example
NAME    IP            MAC               NAT   NATGWDP   READY
eip-1   172.18.11.24   00:00:00:34:41:0B   fip   gw1       true
```


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5.4.4 VPC NATGW net1 QoS

VPC NATGW net1 10Mbps 3 `shared=true` QoSPolicy QoSPolicy

QoSPolicy

```yaml
apiVersion: kubeovn.io/v1
kind: QoSPolicy
metadata:
  name: qos-natgw-example
spec:
  shared: true
  bindingType: NATGW
  bandwidthLimitRules:
  - name: net1-ingress
    interface: net1
    rateMax: "10" # Mbps
    burstMax: "10" # Mbps
    priority: 3
    direction: ingress
  - name: net1-egress
    interface: net1
    rateMax: "10" # Mbps
    burstMax: "10" # Mbps
    priority: 3
    direction: egress
```

VpcNatGateway

```yaml
kind: VpcNatGateway
apiVersion: kubeovn.io/v1
metadata:
  name: gw1
spec:
  vpc: test-vpc-1
  subnet: net1
  lanIp: 10.0.1.254
  qosPolicy: qos-natgw-example
  selector:
    - "kubernetes.io/hostname: kube-ovn-worker"
    - "kubernetes.io/os: linux"
```

`.spec.qosPolicy`

5.4.5 net1 QoS

net1 5Mbps 2 `shared=true` QoSPolicy QoSPolicy

QoSPolicy

```yaml
apiVersion: kubeovn.io/v1
kind: QoSPolicy
metadata:
  name: qos-natgw-example
spec:
  shared: true
  bindingType: NATGW
  bandwidthLimitRules:
  - name: net1-extip-ingress
    interface: net1
    rateMax: "5" # Mbps
    burstMax: "5" # Mbps
    priority: 2
    direction: ingress
    matchType: ip
    matchValue: src 172.18.11.22/32
  - name: net1-extip-egress
    interface: net1
    rateMax: "5" # Mbps
    burstMax: "5" # Mbps
    priority: 2
    direction: egress
    matchType: ip
    matchValue: dst 172.18.11.23/32
```

VpcNatGateway

```yaml
kind: VpcNatGateway
apiVersion: kubeovn.io/v1
metadata:
  name: gw1
spec:
```

```
  vpc: test-vpc-1
  subnet: net1
  lanIp: 10.0.1.254
  qosPolicy: qos-natgw-example
  selector:
    - "kubernetes.io/hostname: kube-ovn-worker"
    - "kubernetes.io/os: linux"
```

5.4.6 QoS NATGW

`label` qos eip

```
# kubectl get vpc-nat-gw  -l ovn.kubernetes.io/qos=qos-natgw-example
NAME   VPC          SUBNET   LANIP
gw1    test-vpc-1   net1     10.0.1.254
```

5.4.7 qos

```
# kubectl get qos -A
NAME               SHARED   BINDINGTYPE
qos-eip-example    false    EIP
qos-natgw-example  true     NATGW
```

5.4.8

- QoS QoS QoS EIP NATGW `spec.qosPolicy`

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5.4.9

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5.5 VPC

Kubernetes Service VPC Service

1. Service IP VPC
2. IP

Kube-OVN 1.11 `SwitchLBRule` CRD VPC

`SwitchLBRule` VPC

5.5.1 `Selector`

`selector` `label` `pod`

`SwitchLBRule`

```yaml
apiVersion: kubeovn.io/v1
kind: SwitchLBRule
metadata:
  name:  cjh-slr-nginx
spec:
  vip: 1.1.1.1
  sessionAffinity: ClientIP
  namespace: default
  selector:
    - app: nginx
  ports:
  - name: dns
    port: 8888
    targetPort: 80
    protocol: TCP
```

- `selector`, `sessionAffinity` `port` Kubernetes Service
- `vip` IP
- `namespace` `selector` Pod

Kube-OVN `SwitchLBRule` Pod Pod VPC L2 LB

5.5.2 `Endpoints`

`endpoints` `selector` `kubevirt` `vm`

`SwitchLBRule`

```yaml
apiVersion: kubeovn.io/v1
kind: SwitchLBRule
metadata:
  name:  cjh-slr-nginx
spec:
  vip: 1.1.1.1
  sessionAffinity: ClientIP
  namespace: default
  endpoints:
    - 192.168.0.101
    - 192.168.0.102
    - 192.168.0.103
  ports:
  - name: dns
    port: 8888
    targetPort: 80
    protocol: TCP
```

- `sessionAffinity` `port` Kubernetes Service
- `vip` IP
- `namespace`
- `endpoints` IP

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`selector` `endpoints`, `selector`

5.5.3

`OVN` `IPv4`

[Health Checks](https://www.ovn.org/support/dist-docs/ovn-nb.5.html)

`ovn` `SwitchLBRule` `SwitchLBRule` `VPC` `subnet` `vip` `ip_port_mappings` `load_balancer_health_check`

- `vip` `subnet` `subnet` `SwitchLBRule`
- `Selector`

```
root@server:~# kubectl get po -o wide -n vulpecula
NAME                     READY   STATUS    RESTARTS   AGE     IP          NODE     NOMINATED NODE   READINESS GATES
nginx-78d9578975-f4qn4   1/1     Running   3          4d16h   10.16.0.4   worker   <none>           <none>
nginx-78d9578975-t8tm5   1/1     Running   3          4d16h   10.16.0.6   worker   <none>           <none>

#    slr
root@server:~# cat << END > slr.yaml
apiVersion: kubeovn.io/v1
kind: SwitchLBRule
metadata:
  name:  nginx
  namespace:  vulpecula
spec:
  vip: 1.1.1.1
  sessionAffinity: ClientIP
  namespace: default
  selector:
    - app: nginx
  ports:
  - name: dns
    port: 8888
    targetPort: 80
    protocol: TCP
END
root@server:~# kubectl apply -f slr.yaml
root@server:~# kubectl get slr
NAME              VIP       PORT(S)    SERVICE                       AGE
vulpecula-nginx   1.1.1.1   8888/TCP   default/slr-vulpecula-nginx   3d21h
```

`subnet` `vip`

```
#       vip

root@server:~# kubectl get vip
NAME          NS    V4IP        MAC                V6IP    PMAC   SUBNET        READY   TYPE
vulpecula-subnet    10.16.0.2   00:00:00:39:95:C1   <nil>          vulpecula-subnet   true
```

`Load_Balancer_Health_Check` `Service_Monitor`

```
root@server:~# kubectl ko nbctl list Load_Balancer
_uuid               : 3cbb6d43-44aa-4028-962f-30d2dba9f0b8
external_ids        : {}
health_check        : [5bee3f12-6b54-411c-9cc8-c9def8f67356]
ip_port_mappings    : {"10.16.0.4"="nginx-78d9578975-f4qn4.default:10.16.0.2", "10.16.0.6"="nginx-78d9578975-t8tm5.default:10.16.0.2"}
name                : cluster-tcp-session-loadbalancer
options             : {affinity_timeout="10800"}
protocol            : tcp
selection_fields    : [ip_src]
vips                : {"1.1.1.1:8888"="10.16.0.4:80,10.16.0.6:80"}

root@server:~# kubectl ko nbctl list Load_Balancer_Health_Check
_uuid               : 5bee3f12-6b54-411c-9cc8-c9def8f67356
external_ids        : {switch_lb_subnet=vulpecula-subnet}
options             : {failure_count="3", interval="5", success_count="3", timeout="20"}
vip                 : "1.1.1.1:8888"

root@server:~# kubectl ko sbctl list Service_Monitor
_uuid               : 1bddc541-cc49-44ea-9935-a4208f627a91
external_ids        : {}
ip                  : "10.16.0.4"
logical_port        : nginx-78d9578975-f4qn4.default
options             : {failure_count="3", interval="5", success_count="3", timeout="20"}
port                : 80
protocol            : tcp
```

```
src_ip              : "10.16.0.2"
src_mac             : "c6:d4:b8:08:54:e7"
status              : online

_uuid               : 84dd24c5-e1b4-4e97-9daa-13687ed59785
external_ids        : {}
ip                  : "10.16.0.6"
logical_port        : nginx-78d9578975-t8tm5.default
options             : {failure_count="3", interval="5", success_count="3", timeout="20"}
port                : 80
protocol            : tcp
src_ip              : "10.16.0.2"
src_mac             : "c6:d4:b8:08:54:e7"
status              : online
```

`vip`

```
root@server:~# kubectl exec -it -n vulpecula nginx-78d9578975-t8tm5 -- curl 1.1.1.1:8888
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

`pod`

```
kubectl delete po nginx-78d9578975-f4qn4
kubectl get po -o wide -n vulpecula
NAME                     READY   STATUS    RESTARTS   AGE     IP          NODE     NOMINATED NODE   READINESS GATES
nginx-78d9578975-lxmvh   1/1     Running   0          31s     10.16.0.8   worker   <none>           <none>
nginx-78d9578975-t8tm5   1/1     Running   3          4d16h   10.16.0.6   worker   <none>           <none>
```

`Load_Balancer_Health_Check` `Service_Monitor`

```
root@server:~# kubectl ko nbctl list Load_Balancer
_uuid               : 3cbb6d43-44aa-4028-962f-30d2dba9f0b8
external_ids        : {}
health_check        : [5bee3f12-6b54-411c-9cc8-c9def8f67356]
ip_port_mappings    : {"10.16.0.4"="nginx-78d9578975-f4qn4.default:10.16.0.2", "10.16.0.6"="nginx-78d9578975-t8tm5.default:10.16.0.2",
"10.16.0.8"="nginx-78d9578975-lxmvh.default:10.16.0.2"}
name                : cluster-tcp-session-loadbalancer
options             : {affinity_timeout="10800"}
protocol            : tcp
selection_fields    : [ip_src]
vips                : {"1.1.1.1:8888"="10.16.0.6:80,10.16.0.8:80"}

root@server:~# kubectl ko nbctl list Load_Balancer_Health_Check
_uuid               : 5bee3f12-6b54-411c-9cc8-c9def8f67356
external_ids        : {switch_lb_subnet=vulpecula-subnet}
options             : {failure_count="3", interval="5", success_count="3", timeout="20"}
vip                 : "1.1.1.1:8888"

root@server:~# kubectl ko sbctl list Service_Monitor
_uuid               : 84dd24c5-e1b4-4e97-9daa-13687ed59785
external_ids        : {}
ip                  : "10.16.0.6"
logical_port        : nginx-78d9578975-t8tm5.default
options             : {failure_count="3", interval="5", success_count="3", timeout="20"}
port                : 80
protocol            : tcp
src_ip              : "10.16.0.2"
src_mac             : "c6:d4:b8:08:54:e7"
status              : online

_uuid               : 5917b7b7-a999-49f2-a42d-da81f1eeb28f
external_ids        : {}
ip                  : "10.16.0.8"
logical_port        : nginx-78d9578975-lxmvh.default
options             : {failure_count="3", interval="5", success_count="3", timeout="20"}
port                : 80
protocol            : tcp
src_ip              : "10.16.0.2"
src_mac             : "c6:d4:b8:08:54:e7"
status              : online
```

`SwitchLBRule` `Load_Balancer_Health_Check` `Service_Monitor` `vip`

```
root@server:~# kubectl delete -f slr.yaml
switchlbrule.kubeovn.io "vulpecula-nginx" deleted
root@server:~# kubectl get vip
No resources found
```

```
root@server:~# kubectl ko sbctl list Service_Monitor
root@server:~#
root@server:~# kubectl ko nbctl list Load_Balancer_Health_Check
root@server:~#
```

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5.5.4

5.6 VPC DNS

VPC VPC VPC VPC coredns VPC Kubernetes `vpc-dns` CRD

CRD coredns Pod VPC VPC VPC VPC

> **Note**
>
> DNS Pod VM Webhook `/etc/resolv.conf`

5.6.1 vpc-dns

```yaml
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  labels:
    kubernetes.io/bootstrapping: rbac-defaults
  name: system:vpc-dns
rules:
  - apiGroups:
    - ""
    resources:
    - endpoints
    - services
    - pods
    - namespaces
    verbs:
    - list
    - watch
  - apiGroups:
    - discovery.k8s.io
    resources:
    - endpointslices
    verbs:
    - list
    - watch
---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  annotations:
    rbac.authorization.kubernetes.io/autoupdate: "true"
  labels:
    kubernetes.io/bootstrapping: rbac-defaults
  name: vpc-dns
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: ClusterRole
  name: system:vpc-dns
subjects:
- kind: ServiceAccount
  name: vpc-dns
  namespace: kube-system
---
apiVersion: v1
kind: ServiceAccount
metadata:
  name: vpc-dns
  namespace: kube-system
---
apiVersion: v1
kind: ConfigMap
metadata:
  name: vpc-dns-corefile
  namespace: kube-system
data:
  Corefile: |
    .:53 {
        errors
        health {
          lameduck 5s
        }
        ready
        kubernetes cluster.local in-addr.arpa ip6.arpa {
          pods insecure
          fallthrough in-addr.arpa ip6.arpa
        }
        prometheus :9153
        forward . /etc/resolv.conf {
          prefer_udp
        }
        cache 30
```


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```
        loop
        reload
        loadbalance
    }
```

nat-gw-pod

5.6.2

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-nad
  namespace: default
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "ovn-nad.default.ovn"
    }'
```

5.6.3 vpc-dns Configmap

kube-system configmap vpc-dns vpc-dns

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: vpc-dns-config
  namespace: kube-system
data:
  coredns-vip: 10.96.0.3
  enable-vpc-dns: "true"
  nad-name: ovn-nad
  nad-provider: ovn-nad.default.ovn
```

- `enable-vpc-dns` `true`
- `coredns-image` dns coredns
- `coredns-vip` coredns lb vip
- `nad-name` `network-attachment-definitions`
- `nad-provider` provider
- `k8s-service-host` coredns k8s apiserver ip apiserver
- `k8s-service-port` coredns k8s apiserver port apiserver

5.6.4 vpc-dns

vpc-dns yaml

```
kind: VpcDns
apiVersion: kubeovn.io/v1
metadata:
  name: test-cjh1
spec:
  vpc: cjh-vpc-1
  subnet: cjh-subnet-1
  replicas: 2
```

- `vpc` dns vpc
- `subnet` dns
- `replicas` : vpc dns deployment replicas

```
# kubectl get vpc-dns
NAME        ACTIVE   VPC         SUBNET
test-cjh1   false    cjh-vpc-1   cjh-subnet-1
test-cjh2   true     cjh-vpc-1   cjh-subnet-2
```


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`ACTIVE : true` dns `false`

VPC DNS

- VPC vpc-dns VPC subnet vpc-dns `true` `false`
- `true` vpc-dns `false` vpc-dns

5.6.5

vpc-dns Pod label `app=vpc-dns` vpc-dns Pod

```
# kubectl -n kube-system get pods -l app=vpc-dns
NAME                                READY   STATUS    RESTARTS   AGE
vpc-dns-test-cjh1-7b878d96b4-g5979   1/1     Running   0          28s
vpc-dns-test-cjh1-7b878d96b4-ltmf9   1/1     Running   0          28s
```

slr

```
# kubectl -n kube-system get slr
NAME               VIP         PORT(S)                 SERVICE                             AGE
vpc-dns-test-cjh1   10.96.0.3   53/UDP,53/TCP,9153/TCP   kube-system/slr-vpc-dns-test-cjh1   113s
```

VPC Pod dns :

```
nslookup kubernetes.default.svc.cluster.local 10.96.0.3
```

VPC switch lb rule VPC Pod

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5.6.6

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5.7 SecurityGroup

Kube-OVN Pod

> **Warning**
>
> Kube-OVN NetworkPolicy Network Policy API Subnet ACL Security Group OVN ACL NetworkPolicy NetworkPolicy API

5.7.1

```yaml
apiVersion: kubeovn.io/v1
kind: SecurityGroup
metadata:
  name: sg-example
spec:
  allowSameGroupTraffic: true
  egressRules:
  - ipVersion: ipv4
    policy: allow
    priority: 1
    protocol: all
    remoteAddress: 10.16.0.13 # 10.16.0.0/16
    remoteType: address
  ingressRules:
  - ipVersion: ipv4
    policy: deny
    priority: 1
    protocol: icmp
    remoteAddress: 10.16.0.14
    remoteType: address
```

Kube-OVN

Pod `ovn.kubernetes.io/security_groups` annotation

```yaml
    ovn.kubernetes.io/security_groups: sg-example
```

Port Security

5.7.2

`tier` ACL ACL

- **`tier`** `0` `1` `0` tier `0` tier `0` `policy: pass` ACL tier `1`
- **`policy: pass`** `allow` `deny` tier `1` `pass`

tier-0 tier-1

```yaml
apiVersion: kubeovn.io/v1
kind: SecurityGroup
metadata:
  name: sg-tier0
spec:
  tier: 0
  allowSameGroupTraffic: true
  ingressRules:
  - ipVersion: ipv4
    policy: pass
    priority: 1
    protocol: tcp
    remoteAddress: 10.16.0.0/16
```


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```
    remoteType: address
  - ipVersion: ipv4
    policy: deny
    priority: 2
    protocol: all
    remoteAddress: 0.0.0.0/0
    remoteType: address
---
apiVersion: kubeovn.io/v1
kind: SecurityGroup
metadata:
  name: sg-tier1
spec:
  tier: 1
  allowSameGroupTraffic: true
  ingressRules:
  - ipVersion: ipv4
    policy: allow
    priority: 1
    protocol: tcp
    remoteAddress: 10.16.0.0/16
    remoteType: address
    portRangeMin: 80
    portRangeMax: 443
```

`sg-tier0` `10.16.0.0/16` TCP tier 1 `sg-tier1` 80-443 TCP

Pod annotation

```
apiVersion: v1
kind: Pod
metadata:
  labels:
    app: web
  annotations:
    ovn.kubernetes.io/security_groups: 'sg-tier0,sg-tier1'
  name: multi-tier-pod
  namespace: default
spec:
  nodeName: kube-ovn-worker
  containers:
  - image: docker.io/library/nginx:alpine
    imagePullPolicy: IfNotPresent
    name: nginx
```

5.7.3

`localAddress`

- **`localAddress`** IP CIDR
- **`sourcePortRangeMin`** / **`sourcePortRangeMax`** 1-65535 TCP UDP

```
apiVersion: kubeovn.io/v1
kind: SecurityGroup
metadata:
  name: sg-local-filter
spec:
  allowSameGroupTraffic: true
  ingressRules:
  - ipVersion: ipv4
    policy: allow
    priority: 1
    protocol: tcp
    remoteAddress: 10.16.0.0/16
    remoteType: address
    portRangeMin: 8080
    portRangeMax: 8080
    localAddress: 10.16.0.100
    sourcePortRangeMin: 1024
    sourcePortRangeMax: 65535
```

`10.16.0.0/16` `10.16.0.100` 8080 TCP 1024-65535

5.7.4

- ACL OVN ACL ACL

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- `priority` 1-16384 ACL ACL `ACL = 18484 -` ACL

- `tier` `0` `1` `policy: pass` tier `0` tier `1`
- Kube-OVN CNI Pod Pod Pod ContainerCreating Running

5.7.5

YAML Pod annotation

```yaml
apiVersion: v1
kind: Pod
metadata:
  labels:
    app: static
  annotations:
    ovn.kubernetes.io/security_groups: 'sg-example'
  name: sg-test-pod
  namespace: default
spec:
  nodeName: kube-ovn-worker
  containers:
  - image: docker.io/library/nginx:alpine
    imagePullPolicy: IfNotPresent
    name: qatest
```

```
# kubectl get pod -o wide
NAME                   READY   STATUS              RESTARTS   AGE     IP           NODE                  NOMINATED NODE   READINESS GATES
sg-test-pod            0/1     ContainerCreating   0          5h32m   <none>       kube-ovn-worker       <none>           <none>
test-99fff7f86-52h9r   1/1     Running             0          5h41m   10.16.0.14   kube-ovn-control-plane   <none>           <none>
test-99fff7f86-qcgjw   1/1     Running             0          5h43m   10.16.0.13   kube-ovn-worker       <none>           <none>
```

kubectl describe pod Pod

```
# kubectl describe pod sg-test-pod
Name:         sg-test-pod
Namespace:    default
Priority:     0
Node:         kube-ovn-worker/172.18.0.2
Start Time:   Tue, 28 Feb 2023 10:29:36 +0800
Labels:       app=static
Annotations:  ovn.kubernetes.io/allocated: true
              ovn.kubernetes.io/cidr: 10.16.0.0/16
              ovn.kubernetes.io/gateway: 10.16.0.1
              ovn.kubernetes.io/ip_address: 10.16.0.15
              ovn.kubernetes.io/logical_router: ovn-cluster
              ovn.kubernetes.io/logical_switch: ovn-default
              ovn.kubernetes.io/mac_address: 00:00:00:FA:17:97
              ovn.kubernetes.io/pod_nic_type: veth-pair
              ovn.kubernetes.io/port_security: true
              ovn.kubernetes.io/routed: true
              ovn.kubernetes.io/security_groups: sg-allow-reject
Status:       Pending
IP:
IPs:          <none>
.
.
.
Events:
  Type     Reason                  Age                   From     Message
  ----     ------                  ----                  ----     -------
  Warning  FailedCreatePodSandBox  5m3s (x70 over 4h59m)  kubelet  (combined from similar events): Failed to create pod sandbox: rpc error: code = Unknown
desc = failed to setup network for sandbox "40636e0c7f1ade5500fa958486163d74f2e2300051a71522a9afd7ba0538afb6": plugin type="kube-ovn" failed (add): RPC
failed; request ip return 500 configure nic failed 10.16.0.15 network not ready after 200 ping 10.16.0.1
```

```yaml
apiVersion: kubeovn.io/v1
kind: SecurityGroup
metadata:
  name: sg-gw-both
spec:
  allowSameGroupTraffic: true
  egressRules:
  - ipVersion: ipv4
    policy: allow
    priority: 2
    protocol: all
    remoteAddress: 10.16.0.13
    remoteType: address
```


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```
  - ipVersion: ipv4
    policy: allow
    priority: 1
    protocol: all
    remoteAddress: 10.16.0.1
    remoteType: address
  ingressRules:
  - ipVersion: ipv4
    policy: deny
    priority: 2
    protocol: icmp
    remoteAddress: 10.16.0.14
    remoteType: address
  - ipVersion: ipv4
    policy: allow
    priority: 1
    protocol: icmp
    remoteAddress: 10.16.0.1
    remoteType: address
```

yaml Pod Pod

```
apiVersion: v1
kind: Pod
metadata:
  labels:
    app: static
  annotations:
    ovn.kubernetes.io/security_groups: 'sg-gw-both'
  name: sg-gw-both
  namespace: default
spec:
  nodeName: kube-ovn-worker
  containers:
  - image: docker.io/library/nginx:alpine
    imagePullPolicy: IfNotPresent
    name: qatest
```

Pod

```
# kubectl get pod -o wide
NAME                READY   STATUS              RESTARTS   AGE     IP          NODE                NOMINATED NODE   READINESS GATES
sg-test-pod         0/1     ContainerCreating   0          5h41m   <none>      kube-ovn-worker     <none>           <none>
sg-gw-both          1/1     Running             0          5h37m   10.16.0.19  kube-ovn-worker     <none>           <none>
```

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5.7.6

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5.8 OVN EIP FIP SNAT DNAT

Note

VPC VPC

VPC VPC NAT OVN Egress Gateway VPC NAT Kube-OVN VPC NAT Pod VPC Macvlan Pod iptables

OVN OVN NAT OVN BFD OVN OVN

Egress Gateway VPC NAT

VPC OVN NAT `provider-network vlan (external) subnet` VPC EIP/SNAT

5.8.1

- `kube-ovn-controller` `kube-ovn-cni` `ovn-external-gw-config` `VPC spec enableExternal`
- `CRD` `provider-network` `vlan` `subnet` `VPC spec extraExternalSubnets` `ovn-eip` `ovn-dnat` `ovn-fip` `ovn-snat` CRD

```
graph LR

pod-->subnet-->vpc-->lrp--bind-->gw-chassis-->snat-->lsp-->external-subnet
lrp-.-peer-.-lsp
```

Pod SNAT Pod Fip

```
graph LR


pod-->subnet-->vpc-->lrp--bind-->local-chassis-->snat-->lsp-->external-subnet


lrp-.-peer-.-lsp
```

Pod FIP (dnat_and_snat)

CRD iptables nat gw

- ovn eip: ip underlay provider network vlan subnet
- ovn fip dnat snat VPC ip vip
- ovn snat VPC ip snat
- ovn dnat router lb , ip + VPC endpoints

5.8.2 1.

OpenStack Neutron ovn provider network VPC EIP/SNAT

vlan vlan 0 vlan id

```
#
# 1. kube-ovn-controller
          - --external-gateway-vlanid=204
          - --external-gateway-switch=external204

# 2. kube-ovn-cni         :
          - --external-gateway-switch=external204
```


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```
###                vlan id                    underlay
```

- provider network vlan subnet
- VPC enable_eip_snat vlan subnet ip ipam
- VPC enable_eip_snat , pod annotation fip snat
- VPC enable_eip_snat vlan subnet VPC eip snat

1.1 underlay

```
#    provider-network  vlan  subnet
# cat 01-provider-network.yaml

apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: external204
spec:
  defaultInterface: vlan

# cat 02-vlan.yaml

apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
  name: vlan204
spec:
  id: 204
  provider: external204

# cat 03-vlan-subnet.yaml

apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: external204
spec:
  protocol: IPv4
  cidrBlock: 10.5.204.0/24
  gateway: 10.5.204.254
  vlan: vlan204
  excludeIps:
  - 10.5.204.1..10.5.204.100
```

1.2 VPC eip_snat

```
#      VPC     underlay    provider subnet
# cat 00-centralized-external-gw-no-ip.yaml

apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-external-gw-config
  namespace: kube-system
data:
  enable-external-gw: "true"
  external-gw-nodes: "pc-node-1,pc-node-2,pc-node-3"
  type: "centralized"
  external-gw-nic: "vlan" #      ovs
  external-gw-addr: "10.5.204.254/24" # underlay       ip
```

logical router port (lrp) ip mac underlay lrp ovn eip

ip lrp ovn-eip lrp ovn eip

1.3 VPC eip snat fip

node

```
#           external-gw-nodes
kubectl label nodes pc-node-1 pc-node-2 pc-node-3 ovn.kubernetes.io/external-gw=true
```

```
# cat 00-ns.yml

apiVersion: v1
```

```
kind: Namespace
metadata:
  name: vpc1

# cat 01-vpc-ecmp-enable-external-bfd.yml

kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc1
spec:
  namespaces:
  - vpc1
  enableExternal: true
  staticRoutes:
    - cidr: 0.0.0.0/0
      nextHopIP: 10.5.204.254
      policy: policyDst

# VPC    enableExternal       lrp

# cat 02-subnet.yml

apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: vpc1-subnet1
spec:
  cidrBlock: 192.168.0.0/24
  default: false
  disableGatewayCheck: false
  disableInterConnection: true
  enableEcmp: true
  gatewayNode: ""
  gatewayType: distributed
  #gatewayType: centralized
  natOutgoing: false
  private: false
  protocol: IPv4
  provider: ovn
  vpc: vpc1
  namespaces:
  - vpc1

#                   subnet
```

```
# kubectl ko nbctl show vpc1

router 87ad06fd-71d5-4ff8-a1f0-54fa3bba1a7f (vpc1)
    port vpc1-vpc1-subnet1
        mac: "00:00:00:ED:8E:C7"
        networks: ["192.168.0.1/24"]
    port vpc1-external204
        mac: "00:00:00:EF:05:C7"
        networks: ["10.5.204.105/24"]
        gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
    nat 21d853b0-f7b4-40bd-9a53-31d2e2745739
        external ip: "10.5.204.115"
        logical ip: "192.168.0.0/24"
        type: "snat"
```

```
# kubectl ko nbctl lr-route-list vpc1

IPv4 Routes
Route Table <main>:
                0.0.0.0/0               10.5.204.254 dst-ip
#    VPC CRD
```

> enableExternal VPC CRD

1.4

1.4.1 UNDERLAY

eip snat fip 1 eip snat fip

```
#    provider-network  vlan  subnet
# cat 01-extra-provider-network.yaml
apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: extra
spec:
  defaultInterface: vlan
# cat 02-extra-vlan.yaml
```

```
apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
  name: vlan0
spec:
  id: 0
  provider: extra
# cat 03-extra-vlan-subnet.yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: extra
spec:
  protocol: IPv4
  cidrBlock: 10.10.204.0/24
  gateway: 10.10.204.254
  vlan: vlan0
  excludeIps:
  - 10.10.204.1..10.10.204.100
```

1.4.2 VPC

```
apiVersion: kubeovn.io/v1
kind: Vpc
metadata:
  name: vpc1
spec:
  namespaces:
  - vpc1
  enableExternal: true  #    enableExternal   VPC               external   ls
  extraExternalSubnets: #    extraExternalSubnets
  - extra
```

```
# kubectl ko nbctl show vpc1
router 87ad06fd-71d5-4ff8-a1f0-54fa3bba1a7f (vpc1)
    port vpc1-vpc1-subnet1
        mac: "00:00:00:ED:8E:C7"
        networks: ["192.168.0.1/24"]
    port vpc1-external204
        mac: "00:00:00:EF:05:C7"
        networks: ["10.5.204.105/24"]
        gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
    port vpc1-extra
        mac: "00:00:00:EF:6A:C7"
        networks: ["10.10.204.105/24"]
        gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
```

5.8.3 2. ovn-eip

iptables-eip ovn-eip type

- nat: ovn dnat fip, snat nat
- lrp: underlay lrp ip dnat snat
- lsp: ovn bfd ecmp ovs internal port ecmp

```
---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-static
spec:
  externalSubnet: external204
  type: nat

#       eip           fip
```

externalSubnet external204

externalSubnet extra

2.1 ovn-fip pod fip

```
# kubectl get po -o wide -n vpc1 vpc-1-busybox01
NAME              READY   STATUS    RESTARTS   AGE     IP            NODE
vpc-1-busybox01   1/1     Running   0          3d15h   192.168.0.2   pc-node-2

# kubectl get ip vpc-1-busybox01.vpc1
```


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```
NAME                   V4IP          V6IP   MAC                NODE        SUBNET
vpc-1-busybox01.vpc1   192.168.0.2          00:00:00:0A:DD:27  pc-node-2   vpc1-subnet1

---

kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-static
spec:
  externalSubnet: external204
  type: nat

---
kind: OvnFip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-static
spec:
  ovnEip: eip-static
  ipName: vpc-1-busybox01.vpc1  #       ip crd
  type: "centralized"           # centralized    distributed

--
#          VPC       ip

kind: OvnFip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-static
spec:
  ovnEip: eip-static
  vpc: vpc1
  v4Ip: 192.168.0.2
  type: "centralized"           # centralized    distributed
```

```
# kubectl get ofip
NAME          VPC    V4EIP          V4IP          READY   IPTYPE   IPNAME
eip-for-vip   vpc1   10.5.204.106   192.168.0.3   true    vip      test-fip-vip
eip-static    vpc1   10.5.204.101   192.168.0.2   true             vpc-1-busybox01.vpc1
# kubectl get ofip eip-static
NAME         VPC    V4EIP          V4IP          READY   IPTYPE   IPNAME
eip-static   vpc1   10.5.204.101   192.168.0.2   true             vpc-1-busybox01.vpc1

[root@pc-node-1 03-cust-vpc]# ping 10.5.204.101
PING 10.5.204.101 (10.5.204.101) 56(84) bytes of data.
64 bytes from 10.5.204.101: icmp_seq=2 ttl=62 time=1.21 ms
64 bytes from 10.5.204.101: icmp_seq=3 ttl=62 time=0.624 ms
64 bytes from 10.5.204.101: icmp_seq=4 ttl=62 time=0.368 ms
^C
--- 10.5.204.101 ping statistics ---
4 packets transmitted, 3 received, 25% packet loss, time 3049ms
rtt min/avg/max/mdev = 0.368/0.734/1.210/0.352 ms
[root@pc-node-1 03-cust-vpc]#

#       node ping    VPC    pod     ip
```

```
#     ip
# kubectl ko nbctl show vpc1
router 87ad06fd-71d5-4ff8-a1f0-54fa3bba1a7f (vpc1)
    port vpc1-vpc1-subnet1
        mac: "00:00:00:ED:8E:C7"
        networks: ["192.168.0.1/24"]
    port vpc1-external204
        mac: "00:00:00:EF:05:C7"
        networks: ["10.5.204.105/24"]
        gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
    nat 813523e7-c68c-408f-bd8c-cba30cb2e4f4
        external ip: "10.5.204.101"
        logical ip: "192.168.0.2"
        type: "dnat_and_snat"
```

2.2 ovn-fip vip fip

vip kubevirt vip keepalived kube-vip

fip VPC vip vip

```
#     vip eip    eip     vip
# cat vip.yaml

apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: test-fip-vip
spec:
  subnet: vpc1-subnet1
```

```
# cat 04-fip.yaml

---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-for-vip
spec:
  externalSubnet: external204
  type: nat

---
kind: OvnFip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-for-vip
spec:
  ovnEip: eip-for-vip
  ipType: vip         #       fip     pod ip              vip
  ipName: test-fip-vip

---
#          VPC       ip

kind: OvnFip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-for-vip
spec:
  ovnEip: eip-for-vip
  ipType: vip         #       fip     pod ip              vip
  vpc: vpc1
  v4Ip: 192.168.0.3
```

```
# kubectl get ofip
NAME          VPC    V4EIP          V4IP          READY   IPTYPE   IPNAME
eip-for-vip   vpc1   10.5.204.106   192.168.0.3   true    vip      test-fip-vip


[root@pc-node-1 fip-vip]# ping  10.5.204.106
PING 10.5.204.106 (10.5.204.106) 56(84) bytes of data.
64 bytes from 10.5.204.106: icmp_seq=1 ttl=62 time=0.694 ms
64 bytes from 10.5.204.106: icmp_seq=2 ttl=62 time=0.436 ms

#   node    ping


# pod     ip

[root@pc-node-1 fip-vip]# kubectl -n vpc1 exec -it vpc-1-busybox03 -- bash
[root@vpc-1-busybox03 /]#
[root@vpc-1-busybox03 /]#
[root@vpc-1-busybox03 /]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
1568: eth0@if1569: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 00:00:00:56:40:e5 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 192.168.0.5/24 brd 192.168.0.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet 192.168.0.3/24 scope global secondary eth0  #      vip
       valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fe56:40e5/64 scope link
       valid_lft forever preferred_lft forever

[root@vpc-1-busybox03 /]# tcpdump -i eth0 host  192.168.0.3 -netvv
tcpdump: listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
00:00:00:ed:8e:c7 > 00:00:00:56:40:e5, ethertype IPv4 (0x0800), length 98: (tos 0x0, ttl 62, id 44830, offset 0, flags [DF], proto ICMP (1), length 84)
    10.5.32.51 > 192.168.0.3: ICMP echo request, id 177, seq 1, length 64
00:00:00:56:40:e5 > 00:00:00:ed:8e:c7, ethertype IPv4 (0x0800), length 98: (tos 0x0, ttl 64, id 43962, offset 0, flags [none], proto ICMP (1), length 84)
    192.168.0.3 > 10.5.32.51: ICMP echo reply, id 177, seq 1, length 64

# pod        fip     icmp
```

5.8.4 3. ovn-snat

3.1 ovn-snat subnet cidr

iptables-snat

```
# cat 03-subnet-snat.yaml

---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
```

```
  name: snat-for-subnet-in-vpc
spec:
  externalSubnet: external204
  type: nat

---
kind: OvnSnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: snat-for-subnet-in-vpc
spec:
  ovnEip: snat-for-subnet-in-vpc
  vpcSubnet: vpc1-subnet1 # eip

---
#          VPC       subnet cidr

kind: OvnSnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: snat-for-subnet-in-vpc
spec:
  ovnEip: snat-for-subnet-in-vpc
  vpc: vpc1
  v4IpCidr: 192.168.0.0/24 #        cidr      ip
```

externalSubnet extra

3.2 ovn-snat pod ip

iptables-snat

```
# cat 03-pod-snat.yaml

---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: snat-for-pod-vpc-ip
spec:
  externalSubnet: external204
  type: nat

---
kind: OvnSnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: snat01
spec:
  ovnEip: snat-for-pod-vpc-ip
  ipName: vpc-1-busybox02.vpc1 # eip      pod ip

---
#          VPC       ip

kind: OvnSnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: snat-for-subnet-in-vpc
spec:
  ovnEip: snat-for-subnet-in-vpc
  vpc: vpc1
  v4IpCidr: 192.168.0.4
```

externalSubnet extra

snat

```
# kubectl ko nbctl show vpc1
router 87ad06fd-71d5-4ff8-a1f0-54fa3bba1a7f (vpc1)
    port vpc1-vpc1-subnet1
        mac: "00:00:00:ED:8E:C7"
        networks: ["192.168.0.1/24"]
    port vpc1-external204
        mac: "00:00:00:EF:05:C7"
        networks: ["10.5.204.105/24"]
        gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
    nat 21d853b0-f7b4-40bd-9a53-31d2e2745739
        external ip: "10.5.204.115"
        logical ip: "192.168.0.0/24"
        type: "snat"
    nat da77a11f-c523-439c-b1d1-72c664196a0f
        external ip: "10.5.204.116"
        logical ip: "192.168.0.4"
        type: "snat"
```

```
[root@pc-node-1 03-cust-vpc]# kubectl get po -A -o wide  | grep busy
vpc1            vpc-1-busybox01                              1/1     Running   0               3d15h   192.168.0.2   pc-node-2   <none>           <none>
vpc1            vpc-1-busybox02                              1/1     Running   0               17h     192.168.0.4   pc-node-1   <none>           <none>
vpc1            vpc-1-busybox03                              1/1     Running   0               17h     192.168.0.5   pc-node-1   <none>           <none>
vpc1            vpc-1-busybox04                              1/1     Running   0               17h     192.168.0.6   pc-node-3   <none>           <none>
vpc1            vpc-1-busybox05                              1/1     Running   0               17h     192.168.0.7   pc-node-1   <none>           <none>

# kubectl exec -it -n vpc1            vpc-1-busybox04   bash
kubectl exec [POD] [COMMAND] is DEPRECATED and will be removed in a future version. Use kubectl exec [POD] -- [COMMAND] instead.
[root@vpc-1-busybox04 /]#
[root@vpc-1-busybox04 /]#
[root@vpc-1-busybox04 /]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
17095: eth0@if17096: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 00:00:00:76:94:55 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 192.168.0.6/24 brd 192.168.0.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fe76:9455/64 scope link
       valid_lft forever preferred_lft forever
[root@vpc-1-busybox04 /]# ping 223.5.5.5
PING 223.5.5.5 (223.5.5.5) 56(84) bytes of data.
64 bytes from 223.5.5.5: icmp_seq=1 ttl=114 time=22.2 ms
64 bytes from 223.5.5.5: icmp_seq=2 ttl=114 time=21.8 ms

[root@pc-node-1 03-cust-vpc]# kubectl exec -it -n vpc1            vpc-1-busybox02   bash
kubectl exec [POD] [COMMAND] is DEPRECATED and will be removed in a future version. Use kubectl exec [POD] -- [COMMAND] instead.
[root@vpc-1-busybox02 /]#
[root@vpc-1-busybox02 /]#
[root@vpc-1-busybox02 /]#
[root@vpc-1-busybox02 /]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
1566: eth0@if1567: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 00:00:00:0b:e9:d0 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 192.168.0.4/24 brd 192.168.0.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fe0b:e9d0/64 scope link
       valid_lft forever preferred_lft forever
[root@vpc-1-busybox02 /]# ping 223.5.5.5
PING 223.5.5.5 (223.5.5.5) 56(84) bytes of data.
64 bytes from 223.5.5.5: icmp_seq=2 ttl=114 time=22.7 ms
64 bytes from 223.5.5.5: icmp_seq=3 ttl=114 time=22.6 ms
64 bytes from 223.5.5.5: icmp_seq=4 ttl=114 time=22.1 ms
^C
--- 223.5.5.5 ping statistics ---
4 packets transmitted, 3 received, 25% packet loss, time 3064ms
rtt min/avg/max/mdev = 22.126/22.518/22.741/0.278 ms

#        pod           snat
```

5.8.5 4. ovn-dnat

4.1 ovn-dnat pod dnat

```yaml
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
spec:
  externalSubnet: underlay

---
kind: OvnDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
spec:
  ovnEip: eip-dnat
  ipName: vpc-1-busybox01.vpc1 #       pod ip crd
  protocol: tcp
  internalPort: "22"
  externalPort: "22"


---
#         VPC       ip

kind: OvnDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
```

```
spec:
  ovnEip: eip-dnat
  protocol: tcp
  internalPort: "22"
  externalPort: "22"
  vpc: vpc1
  v4Ip: 192.168.0.3
```

externalSubnet extra

OvnDnatRule IptablesDnatRule

```
# kubectl get oeip eip-dnat
NAME       V4IP        V6IP   MAC                TYPE   READY
eip-dnat   10.5.49.4          00:00:00:4D:CE:49  dnat   true

# kubectl get odnat
NAME                EIP                PROTOCOL   V4EIP       V4IP          INTERNALPORT   EXTERNALPORT   IPNAME                          READY
eip-dnat            eip-dnat           tcp        10.5.49.4   192.168.0.3   22             22             vpc-1-busybox01.vpc1            true
```

4.2 ovn-dnat vip dnat

```
kind: OvnDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
spec:
  ipType: vip  #       dnat     pod ip              vip
  ovnEip: eip-dnat
  ipName: test-dnat-vip
  protocol: tcp
  internalPort: "22"
  externalPort: "22"

---
#          VPC       ip

kind: OvnDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
spec:
  ipType: vip  #       dnat     pod ip              vip
  ovnEip: eip-dnat
  ipName: test-dnat-vip
  protocol: tcp
  internalPort: "22"
  externalPort: "22"
  vpc: vpc1
  v4Ip: 192.168.0.4
```

OvnDnatRule IptablesDnatRule

```
# kubectl get vip test-dnat-vip
NAME            V4IP          PV4IP   MAC                PMAC   V6IP   PV6IP   SUBNET         READY
test-dnat-vip   192.168.0.4           00:00:00:D0:C0:B5                        vpc1-subnet1   true

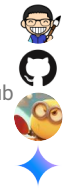
# kubectl get oeip eip-dnat
NAME       V4IP        V6IP   MAC                TYPE   READY
eip-dnat   10.5.49.4          00:00:00:4D:CE:49  dnat   true

# kubectl get odnat eip-dnat
NAME       EIP        PROTOCOL   V4EIP       V4IP          INTERNALPORT   EXTERNALPORT   IPNAME          READY
eip-dnat   eip-dnat   tcp        10.5.49.4   192.168.0.4   22             22             test-dnat-vip   true
```

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5.8.6

5.9 OVN SNAT ECMP BFD L3 HA

VPC OVN SNAT ECMP Gateway Node ovnext0

- bfd
- hash

```
graph LR

pod-->vpc-subnet-->vpc-->snat-->ecmp-->external-subnet-->gw-node1-ovnext0--> node1-external-switch
external-subnet-->gw-node2-ovnext0--> node2-external-switch
external-subnet-->gw-node3-ovnext0--> node3-external-switch
```

ovn-eip-fip-snat.md install.sh provider-network vlan subnet

lsp ovn-eip vpc enable_bfd bfd ecmp

5.9.1 1.

1.1 underlay

1.2 vpc eip_snat

1.3 VPC eip snat fip

ovn-eip-fip-snat.md VPC ecmp bfd

VPC 2 ovn-eip

```
# cat gw-node-eip.yaml
---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: pc-node-1
spec:
  externalSubnet: external204
  type: lsp

---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: pc-node-2
spec:
  externalSubnet: external204
  type: lsp

---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: pc-node-3
spec:
  externalSubnet: external204
  type: lsp
```

vpc ecmp vpc bfd enable bfd lrp ovn eip bfd

5.9.2 2. vpc ecmp bfd L3 HA

```
# cat 01-vpc-ecmp-enable-external-bfd.yml
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc1
spec:
  namespaces:
  - vpc1
  enableExternal: true
  enableBfd: true # bfd              bfd ecmp
  #enableBfd: false
```


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```
# cat 02-subnet.yml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: vpc1-subnet1
spec:
  cidrBlock: 192.168.0.0/24
  default: false
  disableGatewayCheck: false
  disableInterConnection: true
  enableEcmp: true  #      ecmp
  gatewayNode: ""
  gatewayType: distributed
  #gatewayType: centralized
  natOutgoing: false
  private: false
  protocol: IPv4
  provider: ovn
  vpc: vpc1
  namespaces:
  - vpc1
```

:

1. vpc ecmp ecmp bfd vpc enableBfd subnet enableEcmp ecmp bfd
2.
3. VPC VPC VPC snat
4. vpc subnet enableEcmp gatewayType
5. EnableExternal vpc
6. EnableExternal EnableBfd

```
#         ovn
#    vpc
# k get vpc
NAME          ENABLEEXTERNAL   ENABLEBFD   STANDBY   SUBNETS                                NAMESPACES
ovn-cluster   true                         true      ["external204","join","ovn-default"]
vpc1          true             true        true      ["vpc1-subnet1"]                       ["vpc1"]

#    vpc     ENABLEBFD
#     vpc

# 1.     bfd
# k ko nbctl list bfd
_uuid               : be7df545-2c4c-4751-878f-b3507987f050
detect_mult         : 3
dst_ip              : "10.5.204.121"
external_ids        : {}
logical_port        : vpc1-external204
min_rx              : 100
min_tx              : 100
options             : {}
status              : up

_uuid               : 684c4489-5b59-4693-8d8c-3beab93f8093
detect_mult         : 3
dst_ip              : "10.5.204.109"
external_ids        : {}
logical_port        : vpc1-external204
min_rx              : 100
min_tx              : 100
options             : {}
status              : up

_uuid               : f0f62077-2ae9-4e79-b4f8-a446ec6e784c
detect_mult         : 3
dst_ip              : "10.5.204.108"
external_ids        : {}
logical_port        : vpc1-external204
min_rx              : 100
min_tx              : 100
options             : {}
status              : up

###      status        up

# 2.       bfd
# k ko nbctl lr-route-list vpc1
IPv4 Routes
Route Table <main>:
           192.168.0.0/24             10.5.204.108 src-ip ecmp ecmp-symmetric-reply bfd
           192.168.0.0/24             10.5.204.109 src-ip ecmp ecmp-symmetric-reply bfd
           192.168.0.0/24             10.5.204.121 src-ip ecmp ecmp-symmetric-reply bfd

# 3.
```

```
# k ko nbctl find Logical_Router_Static_Route  policy=src-ip options=ecmp_symmetric_reply="true"
_uuid               : 3aacb384-d5ee-4b14-aebf-59e8c11717ba
bfd                 : 684c4489-5b59-4693-8d8c-3beab93f8093
external_ids        : {}
ip_prefix           : "192.168.0.0/24"
nexthop             : "10.5.204.109"
options             : {ecmp_symmetric_reply="true"}
output_port         : []
policy              : src-ip
route_table         : ""

_uuid               : 18bcc585-bc05-430b-925b-ef673c8e1aef
bfd                 : f0f62077-2ae9-4e79-b4f8-a446ec6e784c
external_ids        : {}
ip_prefix           : "192.168.0.0/24"
nexthop             : "10.5.204.108"
options             : {ecmp_symmetric_reply="true"}
output_port         : []
policy              : src-ip
route_table         : ""

_uuid               : 7d0a4e6b-cde0-4110-8176-fbaf19738498
bfd                 : be7df545-2c4c-4751-878f-b3507987f050
external_ids        : {}
ip_prefix           : "192.168.0.0/24"
nexthop             : "10.5.204.121"
options             : {ecmp_symmetric_reply="true"}
output_port         : []
policy              : src-ip
route_table         : ""
```

```
#
[root@pc-node-1 ~]# ip netns exec ovnext ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
1541: ovnext0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue state UNKNOWN group default qlen 1000
    link/ether 00:00:00:ab:bd:87 brd ff:ff:ff:ff:ff:ff
    inet 10.5.204.108/24 brd 10.5.204.255 scope global ovnext0
       valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:feab:bd87/64 scope link
       valid_lft forever preferred_lft forever
[root@pc-node-1 ~]#
[root@pc-node-1 ~]# ip netns exec ovnext route -n
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0         10.5.204.254    0.0.0.0         UG    0      0        0 ovnext0
10.5.204.0      0.0.0.0         255.255.255.0   U     0      0        0 ovnext0

##           internal port unerlay    pod     ns                        ns
[root@pc-node-1 ~]# ip netns exec ovnext bfdd-control status
There are 1 sessions:
Session 1
 id=1 local=10.5.204.108 (p) remote=10.5.204.122 state=Up

##      lrp bfd          lrp ecmp
[root@pc-node-1 ~]# ip netns exec ovnext ping -c1 223.5.5.5
PING 223.5.5.5 (223.5.5.5) 56(84) bytes of data.
64 bytes from 223.5.5.5: icmp_seq=1 ttl=115 time=21.6 ms

#
```

ovnext ns

```
[root@pc-node-3 ~]#  ip netns exec ovnext bash
[root@pc-node-3 ~]# tcpdump -i ovnext0 host 223.5.5.5 -netvv
dropped privs to tcpdump
tcpdump: listening on ovnext0, link-type EN10MB (Ethernet), capture size 262144 bytes
00:00:00:2d:f8:ce > 00:00:00:fd:b2:a4, ethertype IPv4 (0x0800), length 98: (tos 0x0, ttl 63, id 57978, offset 0, flags [DF], proto ICMP (1), length 84)
    10.5.204.102 > 223.5.5.5: ICMP echo request, id 22, seq 71, length 64
00:00:00:fd:b2:a4 > dc:ef:80:5a:44:1a, ethertype IPv4 (0x0800), length 98: (tos 0x0, ttl 62, id 57978, offset 0, flags [DF], proto ICMP (1), length 84)
    10.5.204.102 > 223.5.5.5: ICMP echo request, id 22, seq 71, length 64
^C
2 packets captured
2 packets received by filter
0 packets dropped by kernel
[root@pc-node-3 ~]#

#        down            pod
#         3
```

5.9.3 3. bfd

vpc enable_eip_snat

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```
# cat 01-vpc-ecmp-enable-external-bfd.yml
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc2
spec:
  namespaces:
  - vpc2
  enableExternal: true
  #enableBfd: true
  enableBfd: false

##   bfd

# k ko nbctl lr-route-list vpc2
IPv4 Routes
Route Table <main>:
               0.0.0.0/0             10.5.204.254 dst-ip

#
#    nbctl list bfd       lrp     bfd
#    ovnext ns      bfd
#         vpc subnet     ping    (  )
#           (  )
```

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5.9.4

5.10 VPC

VPC VPC VPC NAT

5.10.1

1. VPC
2. VPC CIDR
3. VPC VPC

5.10.2

VPC VPC Subnet Subnet CIDR

```yaml
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc-1
spec: {}
---
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net1
spec:
  vpc: vpc-1
  cidrBlock: 10.0.0.0/16
---
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc-2
spec: {}
---
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net2
spec:
  vpc: vpc-2
  cidrBlock: 172.31.0.0/16
```

VPC `vpcPeerings`

```yaml
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc-1
spec:
  vpcPeerings:
    - remoteVpc: vpc-2
      localConnectIP: 169.254.0.1/30
  staticRoutes:
    - cidr: 172.31.0.0/16
      nextHopIP: 169.254.0.2
      policy: policyDst
---
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc-2
spec:
  vpcPeerings:
    - remoteVpc: vpc-1
      localConnectIP: 169.254.0.2/30
  staticRoutes:
    - cidr: 10.0.0.0/16
      nextHopIP: 169.254.0.1
      policy: policyDst
```

- `remoteVpc`: VPC
- `localConnectIP`: IP CIDR IP CIDR
- `cidr` Subnet CIDR
- `nextHopIP` VPC `localConnectIP`

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Subnet Pod

```
apiVersion: v1
kind: Pod
metadata:
  annotations:
    ovn.kubernetes.io/logical_switch: net1
  name: vpc-1-pod
spec:
  containers:
    - name: vpc-1-pod
      image: docker.io/library/nginx:alpine
---
apiVersion: v1
kind: Pod
metadata:
  annotations:
    ovn.kubernetes.io/logical_switch: net2
  name: vpc-2-pod
spec:
  containers:
    - name: vpc-2-pod
      image: docker.io/library/nginx:alpine
```

```
# kubectl exec -it vpc-1-pod -- ping $(kubectl get pod vpc-2-pod -o jsonpath='{.status.podIP}')
PING 172.31.0.2 (172.31.0.2): 56 data bytes
64 bytes from 172.31.0.2: seq=0 ttl=62 time=0.655 ms
64 bytes from 172.31.0.2: seq=1 ttl=62 time=0.086 ms
64 bytes from 172.31.0.2: seq=2 ttl=62 time=0.098 ms
^C
--- 172.31.0.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.086/0.279/0.655 ms
# kubectl exec -it vpc-2-pod -- ping $(kubectl get pod vpc-1-pod -o jsonpath='{.status.podIP}')
PING 10.0.0.2 (10.0.0.2): 56 data bytes
64 bytes from 10.0.0.2: seq=0 ttl=62 time=0.594 ms
64 bytes from 10.0.0.2: seq=1 ttl=62 time=0.093 ms
64 bytes from 10.0.0.2: seq=2 ttl=62 time=0.088 ms
^C
--- 10.0.0.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.088/0.258/0.594 ms
```

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5.10.3

6.

6.1 kubectl

Kube-OVN kubectl OVN OVN OVS tcpdump

6.1.1

Kube-OVN kubectl

`kubectl-ko`

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/kubectl-ko
```

`$PATH`

```
mv kubectl-ko /usr/local/bin/kubectl-ko
```

```
chmod +x /usr/local/bin/kubectl-ko
```

```
# kubectl plugin list
The following compatible plugins are available:

/usr/local/bin/kubectl-ko
```

6.1.2

`kubectl ko`

```
# kubectl ko
kubectl ko {subcommand} [option...]
Available Subcommands:
  [nb|sb] [status|kick|backup|dbstatus|restore]     ovn-db operations show cluster status, kick stale server, backup database, get db consistency status or
restore ovn nb db when met 'inconsistent data' error
  nbctl [ovn-nbctl options ...]    invoke ovn-nbctl
  sbctl [ovn-sbctl options ...]    invoke ovn-sbctl
  vsctl {nodeName} [ovs-vsctl options ...]   invoke ovs-vsctl on the specified node
  ofctl {nodeName} [ovs-ofctl options ...]   invoke ovs-ofctl on the specified node
  dpctl {nodeName} [ovs-dpctl options ...]   invoke ovs-dpctl on the specified node
  appctl {nodeName} [ovs-appctl options ...]   invoke ovs-appctl on the specified node
  tcpdump {namespace/podname} [tcpdump options ...]     capture pod traffic
  {trace|ovn-trace} ...    trace ovn microflow of specific packet
    {trace|ovn-trace} {namespace/podname} {target ip address} [target mac address] {icmp|tcp|udp} [target tcp/udp port]    trace ICMP/TCP/UDP
    {trace|ovn-trace} {namespace/podname} {target ip address} [target mac address] arp {request|reply}                     trace ARP request/reply
    {trace|ovn-trace} {node//nodename} {target ip address} [target mac address] {icmp|tcp|udp} [target tcp/udp port]       trace ICMP/TCP/UDP
    {trace|ovn-trace} {node//nodename} {target ip address} [target mac address] arp {request|reply}                         trace ARP request/reply
  diagnose {all|node|subnet|IPPorts} [nodename|subnetName|{proto1}-{IP1}-{Port1},{proto2}-{IP2}-{Port2}]    diagnose connectivity of all nodes or a specific
node or specify subnet's ds pod or IPPorts like 'tcp-172.18.0.2-53,udp-172.18.0.3-53'
  env-check    check the environment configuration
  reload    restart all kube-ovn components
  log {kube-ovn|ovn|ovs|linux|all}    save log to ./kubectl-ko-log/
  perf [image] performance test default image is docker.io/kubeovn/test:v1.13.0
  icnbctl [ovn-nbctl options ...]    invoke ovn-ic-nbctl
  icsbctl [ovn-sbctl options ...]    invoke ovn-ic-sbctl
```

[nb | sb] [status | kick | backup | dbstatus | restore]

OVN

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OVN leader `ovs-appctl cluster/status` :

```
# kubectl ko nb status
306b
Name: OVN_Northbound
Cluster ID: 9a87 (9a872522-3e7d-47ca-83a3-d74333e1a7ca)
Server ID: 306b (306b256b-b5e1-4eb0-be91-4ca96adf6bad)
Address: tcp:[172.18.0.2]:6643
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 280309 ms ago, reason: timeout
Last Election won: 280309 ms ago
Election timer: 5000
Log: [139, 139]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-8723 ->8723 <-85d6 ->85d6
Disconnections: 0
Servers:
    85d6 (85d6 at tcp:[172.18.0.4]:6643) next_index=139 match_index=138 last msg 763 ms ago
    8723 (8723 at tcp:[172.18.0.3]:6643) next_index=139 match_index=138 last msg 763 ms ago
    306b (306b at tcp:[172.18.0.2]:6643) (self) next_index=2 match_index=138
status: ok
```

`Server` `match_index` `last msg` Server

OVN `172.18.0.3` :

```
# kubectl ko nb kick 8723
started removal
```

```
# kubectl ko nb status
306b
Name: OVN_Northbound
Cluster ID: 9a87 (9a872522-3e7d-47ca-83a3-d74333e1a7ca)
Server ID: 306b (306b256b-b5e1-4eb0-be91-4ca96adf6bad)
Address: tcp:[172.18.0.2]:6643
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 324356 ms ago, reason: timeout
Last Election won: 324356 ms ago
Election timer: 5000
Log: [140, 140]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-85d6 ->85d6
Disconnections: 2
Servers:
    85d6 (85d6 at tcp:[172.18.0.4]:6643) next_index=140 match_index=139 last msg 848 ms ago
    306b (306b at tcp:[172.18.0.2]:6643) (self) next_index=2 match_index=139
status: ok
```

OVN

```
# kubectl ko nb backup
tar: Removing leading `/' from member names
backup ovn-nb db to /root/ovnnb_db.060223191654183154.backup
```

```
# kubectl ko nb dbstatus
status: ok
```

`inconsistent data`

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`inconsistent data`

```
# kubectl ko nb restore
deployment.apps/ovn-central scaled
ovn-central original replicas is 3
first nodeIP is 172.18.0.5
ovs-ovn pod on node 172.18.0.5 is ovs-ovn-8jxv9
ovs-ovn pod on node 172.18.0.3 is ovs-ovn-sjzb6
ovs-ovn pod on node 172.18.0.4 is ovs-ovn-t87zk
backup nb db file
restore nb db file, operate in pod ovs-ovn-8jxv9
deployment.apps/ovn-central scaled
finish restore nb db file and ovn-central replicas
recreate ovs-ovn pods
pod "ovs-ovn-8jxv9" deleted
pod "ovs-ovn-sjzb6" deleted
pod "ovs-ovn-t87zk" deleted
```

[nbctl | sbctl] [options ...]

OVN leader `ovn-nbctl` `ovn-sbctl` OVN ovn-nbctl(8) ovn-sbctl(8)

```
# kubectl ko nbctl show
switch c7cd17e8-ceee-4a91-9bb3-e5a313fe1ece (snat)
    port snat-ovn-cluster
        type: router
        router-port: ovn-cluster-snat
switch 20e0c6d0-023a-4756-aec5-200e0c60f95d (join)
    port node-liumengxin-ovn3-192.168.137.178
        addresses: ["00:00:00:64:FF:A8 100.64.0.4"]
    port node-liumengxin-ovn1-192.168.137.176
        addresses: ["00:00:00:AF:98:62 100.64.0.2"]
    port node-liumengxin-ovn2-192.168.137.177
        addresses: ["00:00:00:D9:58:B8 100.64.0.3"]
    port join-ovn-cluster
        type: router
        router-port: ovn-cluster-join
switch 0191705c-f827-427b-9de3-3c3b7d971ba5 (central)
    port central-ovn-cluster
        type: router
        router-port: ovn-cluster-central
switch 2a45ff05-388d-4f85-9daf-e6fccd5833dc (ovn-default)
    port alertmanager-main-0.monitoring
        addresses: ["00:00:00:6C:DF:A3 10.16.0.19"]
    port kube-state-metrics-5d6885d89-4nf8h.monitoring
        addresses: ["00:00:00:6F:02:1C 10.16.0.15"]
    port fake-kubelet-67c55dfd89-pv86k.kube-system
        addresses: ["00:00:00:5C:12:E8 10.16.19.177"]
    port ovn-default-ovn-cluster
        type: router
        router-port: ovn-cluster-ovn-default
router 212f73dd-d63d-4d72-864b-a537e9afbee1 (ovn-cluster)
    port ovn-cluster-snat
        mac: "00:00:00:7A:82:8F"
        networks: ["172.22.0.1/16"]
    port ovn-cluster-join
        mac: "00:00:00:F8:18:5A"
        networks: ["100.64.0.1/16"]
    port ovn-cluster-central
        mac: "00:00:00:4D:8C:F5"
        networks: ["192.101.0.1/16"]
    port ovn-cluster-ovn-default
        mac: "00:00:00:A3:F8:18"
        networks: ["10.16.0.1/16"]
```

vsctl {nodeName} [options ...]

`nodeName` `ovs-ovn` `ovs-vsctl` `vswitchd` OVS ovs-vsctl(8)

```
# kubectl ko vsctl kube-ovn-01 show
0d4c4675-c9cc-440a-8c1a-878e17f81b88
    Bridge br-int
        fail_mode: secure
        datapath_type: system
        Port a2c1a8a8b83a_h
            Interface a2c1a8a8b83a_h
        Port "4fa5c4cbb1a5_h"
            Interface "4fa5c4cbb1a5_h"
        Port ovn-eef07d-0
            Interface ovn-eef07d-0
                type: stt
                options: {csum="true", key=flow, remote_ip="192.168.137.178"}
        Port ovn0
```


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```
            Interface ovn0
                type: internal
        Port mirror0
            Interface mirror0
                type: internal
        Port ovn-efa253-0
            Interface ovn-efa253-0
                type: stt
                options: {csum="true", key=flow, remote_ip="192.168.137.177"}
        Port br-int
            Interface br-int
                type: internal
    ovs_version: "2.17.2"
```

ofctl {nodeName} [options ...]

`nodeName` `ovs-ovn` `ovs-ofctl` OpenFlow OVS ovs-ofctl(8)

```
# kubectl ko ofctl kube-ovn-01 dump-flows br-int
NXST_FLOW reply (xid=0x4): flags=[more]
 cookie=0xcf3429e6, duration=671791.432s, table=0, n_packets=0, n_bytes=0, idle_age=65534, hard_age=65534, priority=100,in_port=2 actions=load:0x4-
>NXM_NX_REG13[],load:0x9->NXM_NX_REG11[],load:0xb->NXM_NX_REG12[],load:0x4->OXM_OF_METADATA[],load:0x1->NXM_NX_REG14[],resubmit(,8)
 cookie=0xc91413c6, duration=671791.431s, table=0, n_packets=907489, n_bytes=99978275, idle_age=0, hard_age=65534, priority=100,in_port=7 actions=load:0x1-
>NXM_NX_REG13[],load:0x9->NXM_NX_REG11[],load:0xb->NXM_NX_REG12[],load:0x4->OXM_OF_METADATA[],load:0x4->NXM_NX_REG14[],resubmit(,8)
 cookie=0xf180459, duration=671791.431s, table=0, n_packets=17348582, n_bytes=2667811214, idle_age=0, hard_age=65534, priority=100,in_port=6317
actions=load:0xa->NXM_NX_REG13[],load:0x9->NXM_NX_REG11[],load:0xb->NXM_NX_REG12[],load:0x4->OXM_OF_METADATA[],load:0x9->NXM_NX_REG14[],resubmit(,8)
 cookie=0x7806dd90, duration=671791.431s, table=0, n_packets=3235428, n_bytes=833821312, idle_age=0, hard_age=65534, priority=100,in_port=1 actions=load:0xd-
>NXM_NX_REG13[],load:0x9->NXM_NX_REG11[],load:0xb->NXM_NX_REG12[],load:0x4->OXM_OF_METADATA[],load:0x3->NXM_NX_REG14[],resubmit(,8)
...
```

dpctl {nodeName} [options ...]

`nodeName` `ovs-ovn` `ovs-dpctl` OVS datapath OVS ovs-dpctl(8)

```
# kubectl ko dpctl kube-ovn-01 show
system@ovs-system:
  lookups: hit:350805055 missed:21983648 lost:73
  flows: 105
  masks: hit:1970748791 total:22 hit/pkt:5.29
  port 0: ovs-system (internal)
  port 1: ovn0 (internal)
  port 2: mirror0 (internal)
  port 3: br-int (internal)
  port 4: stt_sys_7471 (stt: packet_type=ptap)
  port 5: eeb4d9e51b5d_h
  port 6: a2c1a8a8b83a_h
  port 7: 4fa5c4cbb1a5_h
```

appctl {nodeName} [options ...]

`nodeName` `ovs-ovn` `ovs-appctl` daemon OVS ovs-appctl(8)

```
# kubectl ko appctl kube-ovn-01 vlog/list
                console    syslog    file
                -------    ------    ------
backtrace         OFF        ERR       INFO
bfd               OFF        ERR       INFO
bond              OFF        ERR       INFO
bridge            OFF        ERR       INFO
bundle            OFF        ERR       INFO
bundles           OFF        ERR       INFO
...
```

tcpdump {namespace/podname} [tcpdump options ...]

`namespace/podname` `kube-ovn-cni` `tcpdump` veth

```
# kubectl ko tcpdump default/ds1-l6n7p icmp
+ kubectl exec -it kube-ovn-cni-wlg4s -n kube-ovn -- tcpdump -nn -i d7176fe7b4e0_h icmp
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on d7176fe7b4e0_h, link-type EN10MB (Ethernet), capture size 262144 bytes
06:52:36.619688 IP 100.64.0.3 > 10.16.0.4: ICMP echo request, id 2, seq 1, length 64
06:52:36.619746 IP 10.16.0.4 > 100.64.0.3: ICMP echo reply, id 2, seq 1, length 64
06:52:37.619588 IP 100.64.0.3 > 10.16.0.4: ICMP echo request, id 2, seq 2, length 64
06:52:37.619630 IP 10.16.0.4 > 100.64.0.3: ICMP echo reply, id 2, seq 2, length 64
06:52:38.619933 IP 100.64.0.3 > 10.16.0.4: ICMP echo request, id 2, seq 3, length 64
06:52:38.619973 IP 10.16.0.4 > 100.64.0.3: ICMP echo reply, id 2, seq 3, length 64
```


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trace [arguments ...]

Pod OVN Openflow

```
kubectl ko trace {namespace/podname} {target ip address} [target mac address] {icmp|tcp|udp} [target tcp/udp port]
kubectl ko trace {namespace/podname} {target ip address} [target mac address] arp {request|reply}
kubectl ko trace {node//nodename} {target ip address} [target mac address] {icmp|tcp|udp} [target tcp/udp port]
kubectl ko trace {node//nodename} {target ip address} [target mac address] arp {request|reply}
```

```
# kubectl ko trace default/ds1-l6n7p 8.8.8.8 icmp
+ kubectl exec ovn-central-5bc494cb5-np9hm -n kube-ovn -- ovn-trace --ct=new ovn-default 'inport == "ds1-l6n7p.default" && ip.ttl == 64 && icmp && eth.src ==
0a:00:00:10:00:05 && ip4.src == 10.16.0.4 && eth.dst == 00:00:00:B8:CA:43 && ip4.dst == 8.8.8.8'
#
icmp,reg14=0xf,vlan_tci=0x0000,dl_src=0a:00:00:10:00:05,dl_dst=00:00:00:b8:ca:43,nw_src=10.16.0.4,nw_dst=8.8.8.8,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=0,icmp_
code=0

ingress(dp="ovn-default", inport="ds1-l6n7p.default")
-----------------------------------------------------
 0. ls_in_port_sec_l2 (ovn-northd.c:4143): inport == "ds1-l6n7p.default" && eth.src == {0a:00:00:10:00:05}, priority 50, uuid 39453393
    next;
 1. ls_in_port_sec_ip (ovn-northd.c:2898): inport == "ds1-l6n7p.default" && eth.src == 0a:00:00:10:00:05 && ip4.src == {10.16.0.4}, priority 90, uuid 81bcd485
    next;
 3. ls_in_pre_acl (ovn-northd.c:3269): ip, priority 100, uuid 7b4f4971
    reg0[0] = 1;
    next;
 5. ls_in_pre_stateful (ovn-northd.c:3396): reg0[0] == 1, priority 100, uuid 36cdd577
    ct_next;

ct_next(ct_state=new|trk)
-------------------------
 6. ls_in_acl (ovn-northd.c:3759): ip && (!ct.est || (ct.est && ct_label.blocked == 1)), priority 1, uuid 7608af5b
    reg0[1] = 1;
    next;
10. ls_in_stateful (ovn-northd.c:3995): reg0[1] == 1, priority 100, uuid 2aba1b90
    ct_commit(ct_label=0/0x1);
    next;
16. ls_in_l2_lkup (ovn-northd.c:4470): eth.dst == 00:00:00:b8:ca:43, priority 50, uuid 5c9c3c9f
    outport = "ovn-default-ovn-cluster";
    output;

...
```

trace Underlay Mac

```
kubectl ko trace default/virt-handler-7lvml 8.8.8.8 82:7c:9f:83:8c:01 icmp
```

diagnose {all|node|subnet|IPPorts} [nodename|subnetName|{proto1}-{IP1}-{Port1},{proto2}-{IP2}-{Port2}]

`kube-ovn-pinger`

```
# kubectl ko diagnose all
switch c7cd17e8-ceee-4a91-9bb3-e5a313fe1ece (snat)
    port snat-ovn-cluster
        type: router
        router-port: ovn-cluster-snat
switch 20e0c6d0-023a-4756-aec5-200e0c60f95d (join)
    port node-liumengxin-ovn3-192.168.137.178
        addresses: ["00:00:00:64:FF:A8 100.64.0.4"]
    port node-liumengxin-ovn1-192.168.137.176
        addresses: ["00:00:00:AF:98:62 100.64.0.2"]
    port join-ovn-cluster
        type: router
        router-port: ovn-cluster-join
switch 0191705c-f827-427b-9de3-3c3b7d971ba5 (central)
    port central-ovn-cluster
        type: router
        router-port: ovn-cluster-central
switch 2a45ff05-388d-4f85-9daf-e6fccd5833dc (ovn-default)
    port ovn-default-ovn-cluster
        type: router
        router-port: ovn-cluster-ovn-default
    port prometheus-k8s-1.monitoring
        addresses: ["00:00:00:AA:37:DF 10.16.0.23"]
router 212f73dd-d63d-4d72-864b-a537e9afbee1 (ovn-cluster)
    port ovn-cluster-snat
        mac: "00:00:00:7A:82:8F"
        networks: ["172.22.0.1/16"]
    port ovn-cluster-join
        mac: "00:00:00:F8:18:5A"
        networks: ["100.64.0.1/16"]
```


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```
    port ovn-cluster-central
        mac: "00:00:00:4D:8C:F5"
        networks: ["192.101.0.1/16"]
    port ovn-cluster-ovn-default
        mac: "00:00:00:A3:F8:18"
        networks: ["10.16.0.1/16"]
Routing Policies
     31000                           ip4.dst == 10.16.0.0/16           allow
     31000                          ip4.dst == 100.64.0.0/16           allow
     30000                        ip4.dst == 192.168.137.177         reroute               100.64.0.3
     30000                        ip4.dst == 192.168.137.178         reroute               100.64.0.4
     29000                ip4.src == $ovn.default.fake.6_ip4         reroute              100.64.0.22
     29000                ip4.src == $ovn.default.fake.7_ip4         reroute              100.64.0.21
     29000                ip4.src == $ovn.default.fake.8_ip4         reroute              100.64.0.23
     29000 ip4.src == $ovn.default.liumengxin.ovn3.192.168.137.178_ip4         reroute               100.64.0.4
     20000 ip4.src == $ovn.default.liumengxin.ovn1.192.168.137.176_ip4 && ip4.dst != $ovn.cluster.overlay.subnets.IPv4         reroute
100.64.0.2
     20000 ip4.src == $ovn.default.liumengxin.ovn2.192.168.137.177_ip4 && ip4.dst != $ovn.cluster.overlay.subnets.IPv4         reroute
100.64.0.3
     20000 ip4.src == $ovn.default.liumengxin.ovn3.192.168.137.178_ip4 && ip4.dst != $ovn.cluster.overlay.subnets.IPv4         reroute
100.64.0.4
IPv4 Routes
Route Table <main>:
               0.0.0.0/0                100.64.0.1 dst-ip
UUID                                 LB                  PROTO      VIP                    IPs
e9bcfd9d-793e-4431-9073-6dec96b75d71    cluster-tcp-load    tcp        10.100.209.132:10660    192.168.137.176:10660
                                                            tcp        10.101.239.192:6641     192.168.137.177:6641
                                                            tcp        10.101.240.101:3000     10.16.0.7:3000
                                                            tcp        10.103.184.186:6642     192.168.137.177:6642
35d2b7a5-e3a7-485a-a4b7-b4970eb0e63b    cluster-tcp-sess    tcp        10.100.158.128:8080     10.16.0.10:8080,10.16.0.5:8080,10.16.63.30:8080
                                                            tcp        10.107.26.215:8080      10.16.0.19:8080,10.16.0.20:8080,10.16.0.21:8080
                                                            tcp        10.107.26.215:9093      10.16.0.19:9093,10.16.0.20:9093,10.16.0.21:9093
                                                            tcp        10.98.187.99:8080       10.16.0.22:8080,10.16.0.23:8080
                                                            tcp        10.98.187.99:9090       10.16.0.22:9090,10.16.0.23:9090
f43303e4-89aa-4d3e-a3dc-278a552fe27b    cluster-udp-load    udp        10.96.0.10:53           10.16.0.4:53,10.16.0.9:53
_uuid               : 06776304-5a96-43ed-90c4-c4854c251699
addresses           : []
external_ids        : {vendor=kube-ovn}
name                : node_liumengxin_ovn2_192.168.137.177_underlay_v6

_uuid               : 62690625-87d5-491c-8675-9fd83b1f433c
addresses           : []
external_ids        : {vendor=kube-ovn}
name                : node_liumengxin_ovn1_192.168.137.176_underlay_v6

_uuid               : b03a9bae-94d5-4562-b34c-b5f6198e180b
addresses           : ["10.16.0.0/16", "100.64.0.0/16", "172.22.0.0/16", "192.101.0.0/16"]
external_ids        : {vendor=kube-ovn}
name                : ovn.cluster.overlay.subnets.IPv4

_uuid               : e1056f3a-24cc-4666-8a91-75ee6c3c2426
addresses           : []
external_ids        : {vendor=kube-ovn}
name                : ovn.cluster.overlay.subnets.IPv6

_uuid               : 3e5d5fff-e670-47b2-a2f5-a39f4698a8c5
addresses           : []
external_ids        : {vendor=kube-ovn}
name                : node_liumengxin_ovn3_192.168.137.178_underlay_v6
_uuid               : 2d85dbdc-d0db-4abe-b19e-cc806d32b492
action              : drop
direction           : from-lport
external_ids        : {}
label               : 0
log                 : false
match               : "inport==@ovn.sg.kubeovn_deny_all && ip"
meter               : []
name                : []
options             : {}
priority            : 2003
severity            : []

_uuid               : de790cc8-f155-405f-bb32-5a51f30c545f
action              : drop
direction           : to-lport
external_ids        : {}
label               : 0
log                 : false
match               : "outport==@ovn.sg.kubeovn_deny_all && ip"
meter               : []
name                : []
options             : {}
priority            : 2003
severity            : []
Chassis "e15ed4d4-1780-4d50-b09e-ea8372ed48b8"
    hostname: liumengxin-ovn1-192.168.137.176
    Encap stt
        ip: "192.168.137.176"
        options: {csum="true"}
    Port_Binding node-liumengxin-ovn1-192.168.137.176
    Port_Binding perf-6vxkn.default
    Port_Binding kube-state-metrics-5d6885d89-4nf8h.monitoring
    Port_Binding alertmanager-main-0.monitoring
    Port_Binding kube-ovn-pinger-6ftdf.kube-system
```

```
    Port_Binding fake-kubelet-67c55dfd89-pv86k.kube-system
    Port_Binding prometheus-k8s-0.monitoring
Chassis "eef07da1-f8ad-4775-b14d-bd6a3b4eb0d5"
    hostname: liumengxin-ovn3-192.168.137.178
    Encap stt
        ip: "192.168.137.178"
        options: {csum="true"}
    Port_Binding kube-ovn-pinger-7twb4.kube-system
    Port_Binding prometheus-adapter-86df476d87-rl88g.monitoring
    Port_Binding prometheus-k8s-1.monitoring
    Port_Binding node-liumengxin-ovn3-192.168.137.178
    Port_Binding perf-ff475.default
    Port_Binding alertmanager-main-1.monitoring
    Port_Binding blackbox-exporter-676d976865-tvsjd.monitoring
Chassis "efa253c9-494d-4719-83ae-b48ab0f11c03"
    hostname: liumengxin-ovn2-192.168.137.177
    Encap stt
        ip: "192.168.137.177"
        options: {csum="true"}
    Port_Binding grafana-6c4c6b8fb7-pzd2c.monitoring
    Port_Binding node-liumengxin-ovn2-192.168.137.177
    Port_Binding alertmanager-main-2.monitoring
    Port_Binding coredns-6789c94dd8-9jqsz.kube-system
    Port_Binding coredns-6789c94dd8-25d4r.kube-system
    Port_Binding prometheus-operator-7bbc99fc8b-wgjm4.monitoring
    Port_Binding prometheus-adapter-86df476d87-gdxmc.monitoring
    Port_Binding perf-fjnws.default
    Port_Binding kube-ovn-pinger-vh2xg.kube-system
ds kube-proxy ready
kube-proxy ready
deployment ovn-central ready
deployment kube-ovn-controller ready
ds kube-ovn-cni ready
ds ovs-ovn ready
deployment coredns ready
ovn-nb leader check ok
ovn-sb leader check ok
ovn-northd leader check ok
### kube-ovn-controller recent log

### start to diagnose node liumengxin-ovn1-192.168.137.176
#### ovn-controller log:
2022-06-03T00:56:44.897Z|16722|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:06:44.912Z|16723|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:16:44.925Z|16724|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:26:44.936Z|16725|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:36:44.959Z|16726|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:46:44.974Z|16727|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:56:44.988Z|16728|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T02:06:45.001Z|16729|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T02:16:45.025Z|16730|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T02:26:45.040Z|16731|inc_proc_eng|INFO|User triggered force recompute.

#### ovs-vswitchd log:
2022-06-02T23:03:00.137Z|00079|dpif(handler1)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,dl_src=00:00:00:f8:07:c8,dl_dst=00:00:00:fa:1e:50,nw_src=10.16.0.5,nw_dst=10.16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0
icmp_csum:f9d1
 with metadata skb_priority(0),tunnel(tun_id=0x160017000004,src=192.168.137.177,dst=192.168.137.176,ttl=64,tp_src=38881,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.0.5,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0
2022-06-02T23:23:31.840Z|00080|dpif(handler1)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,dl_src=00:00:00:f8:07:c8,dl_dst=00:00:00:fa:1e:50,nw_src=10.16.0.5,nw_dst=10.16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0
icmp_csum:15b2
 with metadata skb_priority(0),tunnel(tun_id=0x160017000004,src=192.168.137.177,dst=192.168.137.176,ttl=64,tp_src=38881,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.0.5,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0
2022-06-03T00:09:15.659Z|00081|dpif(handler1)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,dl_src=00:00:00:dc:e3:63,dl_dst=00:00:00:fa:1e:50,nw_src=10.16.63.30,nw_dst=10.16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0
icmp_csum:e5a5
 with metadata skb_priority(0),tunnel(tun_id=0x150017000004,src=192.168.137.178,dst=192.168.137.176,ttl=64,tp_src=9239,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.63.30,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0
2022-06-03T00:30:13.409Z|00064|dpif(handler2)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,dl_src=00:00:00:f8:07:c8,dl_dst=00:00:00:fa:1e:50,nw_src=10.16.0.5,nw_dst=10.16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0
icmp_csum:6b4a
 with metadata skb_priority(0),tunnel(tun_id=0x160017000004,src=192.168.137.177,dst=192.168.137.176,ttl=64,tp_src=38881,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.0.5,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0
2022-06-03T02:02:33.832Z|00082|dpif(handler1)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,dl_src=00:00:00:f8:07:c8,dl_dst=00:00:00:fa:1e:50,nw_src=10.16.0.5,nw_dst=10.16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0
icmp_csum:a819
 with metadata skb_priority(0),tunnel(tun_id=0x160017000004,src=192.168.137.177,dst=192.168.137.176,ttl=64,tp_src=38881,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.0.5,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0

#### ovs-vsctl show results:
0d4c4675-c9cc-440a-8c1a-878e17f81b88
    Bridge br-int
        fail_mode: secure
        datapath_type: system
        Port a2c1a8a8b83a_h
            Interface a2c1a8a8b83a_h
        Port "4fa5c4cbb1a5_h"
            Interface "4fa5c4cbb1a5_h"
        Port ovn-eef07d-0
            Interface ovn-eef07d-0
                type: stt
                options: {csum="true", key=flow, remote_ip="192.168.137.178"}
        Port ovn0
```

```
            Interface ovn0
                type: internal
        Port "04d03360e9a0_h"
            Interface "04d03360e9a0_h"
        Port eeb4d9e51b5d_h
            Interface eeb4d9e51b5d_h
        Port mirror0
            Interface mirror0
                type: internal
        Port "8e5d887ccd80_h"
            Interface "8e5d887ccd80_h"
        Port ovn-efa253-0
            Interface ovn-efa253-0
                type: stt
                options: {csum="true", key=flow, remote_ip="192.168.137.177"}
        Port "17512d5be1f1_h"
            Interface "17512d5be1f1_h"
        Port br-int
            Interface br-int
                type: internal
    ovs_version: "2.17.2"

#### pinger diagnose results:
I0603 10:35:04.349404   17619 pinger.go:19]
--------------------------------------------------------------------------------
Kube-OVN:
  Version:       v1.16.16
  Build:         2022-04-24_08:02:50
  Commit:        git-73f9d15
  Go Version:    go1.17.8
  Arch:          amd64
--------------------------------------------------------------------------------
I0603 10:35:04.376797   17619 config.go:166] pinger config is &{KubeConfigFile: KubeClient:0xc000493380 Port:8080 DaemonSetNamespace:kube-system
DaemonSetName:kube-ovn-pinger Interval:5 Mode:job ExitCode:0 InternalDNS:kubernetes.default ExternalDNS: NodeName:liumengxin-ovn1-192.168.137.176
HostIP:192.168.137.176 PodName:kube-ovn-pinger-6ftdf PodIP:10.16.0.10 PodProtocols:[IPv4] ExternalAddress: NetworkMode:kube-ovn PollTimeout:2 PollInterval:15
SystemRunDir:/var/run/openvswitch DatabaseVswitchName:Open_vSwitch DatabaseVswitchSocketRemote:unix:/var/run/openvswitch/db.sock DatabaseVswitchFileDataPath:/
etc/openvswitch/conf.db DatabaseVswitchFileLogPath:/var/log/openvswitch/ovsdb-server.log DatabaseVswitchFilePidPath:/var/run/openvswitch/ovsdb-server.pid
DatabaseVswitchFileSystemIDPath:/etc/openvswitch/system-id.conf ServiceVswitchdFileLogPath:/var/log/openvswitch/ovs-vswitchd.log ServiceVswitchdFilePidPath:/
var/run/openvswitch/ovs-vswitchd.pid ServiceOvnControllerFileLogPath:/var/log/ovn/ovn-controller.log ServiceOvnControllerFilePidPath:/var/run/ovn/ovn-
controller.pid}
I0603 10:35:04.449166   17619 exporter.go:75] liumengxin-ovn1-192.168.137.176: exporter connect successfully
I0603 10:35:04.554011   17619 ovn.go:21] ovs-vswitchd and ovsdb are up
I0603 10:35:04.651293   17619 ovn.go:33] ovn_controller is up
I0603 10:35:04.651342   17619 ovn.go:39] start to check port binding
I0603 10:35:04.749613   17619 ovn.go:135] chassis id is 1d7f3d6c-eec5-4b3c-adca-2969d9cdfd80
I0603 10:35:04.763487   17619 ovn.go:49] port in sb is [node-liumengxin-ovn1-192.168.137.176 perf-6vxkn.default kube-state-metrics-5d6885d89-4nf8h.monitoring
alertmanager-main-0.monitoring kube-ovn-pinger-6ftdf.kube-system fake-kubelet-67c55dfd89-pv86k.kube-system prometheus-k8s-0.monitoring]
I0603 10:35:04.763583   17619 ovn.go:61] ovs and ovn-sb binding check passed
I0603 10:35:05.049309   17619 ping.go:259] start to check apiserver connectivity
I0603 10:35:05.053666   17619 ping.go:268] connect to apiserver success in 4.27ms
I0603 10:35:05.053786   17619 ping.go:129] start to check pod connectivity
I0603 10:35:05.249590   17619 ping.go:159] ping pod: kube-ovn-pinger-6ftdf 10.16.0.10, count: 3, loss count 0, average rtt 16.30ms
I0603 10:35:05.354135   17619 ping.go:159] ping pod: kube-ovn-pinger-7twb4 10.16.63.30, count: 3, loss count 0, average rtt 1.81ms
I0603 10:35:05.458460   17619 ping.go:159] ping pod: kube-ovn-pinger-vh2xg 10.16.0.5, count: 3, loss count 0, average rtt 1.92ms
I0603 10:35:05.458523   17619 ping.go:83] start to check node connectivity
```

diagnose subnet subnet daemonset `kube-ovn-pinger` daemonset pod daemonset

diagnose IPPorts `kube-ovn-pinger` pod IP Port

reload

Kube-OVN

```
# kubectl ko reload
pod "ovn-central-8684dd94bd-vzgcr" deleted
Waiting for deployment "ovn-central" rollout to finish: 0 of 1 updated replicas are available...
deployment "ovn-central" successfully rolled out
pod "ovs-ovn-bsnvz" deleted
pod "ovs-ovn-m9b98" deleted
pod "kube-ovn-controller-8459db5ff4-64c62" deleted
Waiting for deployment "kube-ovn-controller" rollout to finish: 0 of 1 updated replicas are available...
deployment "kube-ovn-controller" successfully rolled out
pod "kube-ovn-cni-2klnh" deleted
pod "kube-ovn-cni-t2jz4" deleted
Waiting for daemon set "kube-ovn-cni" rollout to finish: 0 of 2 updated pods are available...
Waiting for daemon set "kube-ovn-cni" rollout to finish: 1 of 2 updated pods are available...
daemon set "kube-ovn-cni" successfully rolled out
pod "kube-ovn-pinger-ln72z" deleted
pod "kube-ovn-pinger-w8lrk" deleted
Waiting for daemon set "kube-ovn-pinger" rollout to finish: 0 of 2 updated pods are available...
Waiting for daemon set "kube-ovn-pinger" rollout to finish: 1 of 2 updated pods are available...
daemon set "kube-ovn-pinger" successfully rolled out
pod "kube-ovn-monitor-7fb67d5488-7q6zb" deleted
Waiting for deployment "kube-ovn-monitor" rollout to finish: 0 of 1 updated replicas are available...
deployment "kube-ovn-monitor" successfully rolled out
```


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log

kube-ovn Kube-OVN OVN Open vSwitch log linux debug

```
# kubectl ko log all
Collecting kube-ovn logging files
Collecting ovn logging files
Collecting openvswitch logging files
Collecting linux dmesg files
Collecting linux iptables-legacy files
Collecting linux iptables-nft files
Collecting linux route files
Collecting linux link files
Collecting linux neigh files
Collecting linux memory files
Collecting linux top files
Collecting linux sysctl files
Collecting linux netstat files
Collecting linux addr files
Collecting linux ipset files
Collecting linux tcp files
Collected files have been saved in the directory /root/kubectl-ko-log
```

```
# tree kubectl-ko-log/
kubectl-ko-log/
|-- kube-ovn-control-plane
|   |-- kube-ovn
|   |   |-- kube-ovn-cni.log
|   |   |-- kube-ovn-monitor.log
|   |   `-- kube-ovn-pinger.log
|   |-- linux
|   |   |-- addr.log
|   |   |-- dmesg.log
|   |   |-- ipset.log
|   |   |-- iptables-legacy.log
|   |   |-- iptables-nft.log
|   |   |-- link.log
|   |   |-- memory.log
|   |   |-- neigh.log
|   |   |-- netstat.log
|   |   |-- route.log
|   |   |-- sysctl.log
|   |   |-- tcp.log
|   |   `-- top.log
|   |-- openvswitch
|   |   |-- ovs-vswitchd.log
|   |   `-- ovsdb-server.log
|   `-- ovn
|       |-- ovn-controller.log
|       |-- ovn-northd.log
|       |-- ovsdb-server-nb.log
|       `-- ovsdb-server-sb.log
```

perf [image]

Kube-OVN

1.
2. Hostnetwork
3.
4. OVN-NB, OVN-SB, OVN-Northd leader

image Pod `docker.io/kubeovn/test:v1.13.0`

```
# kubectl ko perf
============================== Prepareing Performance Test Resources ==============================
pod/test-client created
pod/test-host-client created
pod/test-server created
pod/test-host-server created
service/test-server created
pod/test-client condition met
pod/test-host-client condition met
pod/test-host-server condition met
pod/test-server condition met
===================================================================================================
============================ Start Pod Network Unicast Performance Test ============================
Size            TCP Latency     TCP Bandwidth   UDP Latency     UDP Lost Rate   UDP Bandwidth
64              82.8 us         97.7 Mbits/sec  67.6 us         (0%)            8.42 Mbits/sec
128             85.4 us         167 Mbits/sec   67.2 us         (0%)            17.2 Mbits/sec
```


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```
512             85.8 us         440 Mbits/sec   68.7 us         (0%)            68.4 Mbits/sec
1k              85.1 us         567 Mbits/sec   68.7 us         (0%)            134 Mbits/sec
4k              138 us          826 Mbits/sec   78.1 us         (1.4%)          503 Mbits/sec
================================================================================================
============================== Start Host Network Performance Test ==============================
Size            TCP Latency     TCP Bandwidth   UDP Latency     UDP Lost Rate   UDP Bandwidth
64              49.7 us         120 Mbits/sec   37.9 us         (0%)            18.6 Mbits/sec
128             49.7 us         200 Mbits/sec   38.1 us         (0%)            35.5 Mbits/sec
512             51.9 us         588 Mbits/sec   38.9 us         (0%)            142 Mbits/sec
1k              51.7 us         944 Mbits/sec   37.2 us         (0%)            279 Mbits/sec
4k              74.9 us         1.66 Gbits/sec  39.9 us         (0%)            1.20 Gbits/sec
================================================================================================
============================ Start Service Network Performance Test ============================
Size            TCP Latency     TCP Bandwidth   UDP Latency     UDP Lost Rate   UDP Bandwidth
64              111 us          96.3 Mbits/sec  88.4 us         (0%)            7.59 Mbits/sec
128             83.7 us         150 Mbits/sec   69.2 us         (0%)            16.9 Mbits/sec
512             87.4 us         374 Mbits/sec   75.8 us         (0%)            60.9 Mbits/sec
1k              88.2 us         521 Mbits/sec   73.1 us         (0%)            123 Mbits/sec
4k              148 us          813 Mbits/sec   77.6 us         (0.0044%)       451 Mbits/sec
================================================================================================
========================== Start Pod Multicast Network Performance Test ==========================
Size            UDP Latency     UDP Lost Rate   UDP Bandwidth
64              0.014 ms        (0.17%)         5.80 Mbits/sec
128             0.012 ms        (0%)            11.4 Mbits/sec
512             0.016 ms        (0%)            46.1 Mbits/sec
1k              0.023 ms        (0.073%)        89.8 Mbits/sec
4k              0.035 ms        (1.3%)          126 Mbits/sec
================================================================================================
========================== Start Host Multicast Network Performance ==========================
Size            UDP Latency     UDP Lost Rate   UDP Bandwidth
64              0.007 ms        (0%)            9.95 Mbits/sec
128             0.005 ms        (0%)            21.8 Mbits/sec
512             0.008 ms        (0%)            86.8 Mbits/sec
1k              0.013 ms        (0.045%)        168 Mbits/sec
4k              0.010 ms        (0.31%)         242 Mbits/sec
================================================================================================
================================== Start Leader Recover Time Test ==================================
Delete ovn central nb pod
pod "ovn-central-5cb9c67d75-tlz9w" deleted
Waiting for ovn central nb pod running
============================== OVN nb Recovery takes 3.305236803 s ==============================
Delete ovn central sb pod
pod "ovn-central-5cb9c67d75-szx4c" deleted
Waiting for ovn central sb pod running
============================== OVN sb Recovery takes 3.462698535 s ==============================
Delete ovn central northd pod
pod "ovn-central-5cb9c67d75-zqmqv" deleted
Waiting for ovn central northd pod running
============================ OVN northd Recovery takes 2.691291403 s ============================
================================================================================================
================================== Remove Performance Test Resource ==================================
rm -f unicast-test-client.log
rm -f unicast-test-host-client.log
rm -f unicast-test-client.log
kubectl ko nbctl lb-del test-server
rm -f multicast-test-server.log
kubectl exec ovs-ovn-gxdrf -n kube-system -- ip maddr del 01:00:5e:00:00:64 dev eth0
kubectl exec ovs-ovn-h57bf -n kube-system -- ip maddr del 01:00:5e:00:00:64 dev eth0
rm -f multicast-test-host-server.log
pod "test-client" deleted
pod "test-host-client" deleted
pod "test-host-server" deleted
pod "test-server" deleted
service "test-server" deleted
================================================================================================
```

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6.1.3

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6.2

Kubernetes `ovs-ovn` `ovn-controller` `ovn-central` Kubernetes

`ovn-central` ovn-central

6.2.1

```
 # kubectl drain kube-ovn-worker --ignore-daemonsets --force
 node/kube-ovn-worker cordoned
 WARNING: ignoring DaemonSet-managed Pods: kube-system/kube-ovn-cni-zt74b, kube-system/kube-ovn-pinger-5rxfs, kube-system/kube-proxy-jpmnm, kube-system/ovs-
ovn-v2kll
 evicting pod kube-system/coredns-64897985d-qsgpt
 evicting pod local-path-storage/local-path-provisioner-5ddd94ff66-llss6
 evicting pod kube-system/kube-ovn-controller-8459db5ff4-94lxb
 pod/kube-ovn-controller-8459db5ff4-94lxb evicted
 pod/coredns-64897985d-qsgpt evicted
 pod/local-path-provisioner-5ddd94ff66-llss6 evicted
 node/kube-ovn-worker drained
```

6.2.2 kubelet docker

`ovs-ovn` `ovn-central`

```
systemctl stop kubelet
systemctl stop docker
```

Containerd

> containerd CRI Pod/ `ovs-ovn`

containerd `ovs-ovn`

1. kubelet

   ```
   systemctl stop kubelet
   ```

2. ovs-ovn Pod

   ```
   crictl rm -f $(crictl ps | grep openvswitch | awk '{print $1}')
   ```

3. containerd

   ```
   systemctl stop containerd
   ```

6.2.3 Node

```
rm -rf /var/run/openvswitch
rm -rf /var/run/ovn
rm -rf /etc/origin/openvswitch/
rm -rf /etc/origin/ovn/
rm -rf /etc/cni/net.d/00-kube-ovn.conflist
rm -rf /etc/cni/net.d/01-kube-ovn.conflist
rm -rf /var/log/openvswitch
rm -rf /var/log/ovn
```

6.2.4 kubectl

```
kubectl delete no kube-ovn-01
```


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6.2.5 ovn-sb

`kube-ovn-worker`

```
# kubectl ko sbctl show
Chassis "b0564934-5a0d-4804-a4c0-476c93596a17"
  hostname: kube-ovn-worker
  Encap geneve
      ip: "172.18.0.2"
      options: {csum="true"}
  Port_Binding kube-ovn-pinger-5rxfs.kube-system
Chassis "6a29de7e-d731-4eaf-bacd-2f239ee52b28"
  hostname: kube-ovn-control-plane
  Encap geneve
      ip: "172.18.0.3"
      options: {csum="true"}
  Port_Binding coredns-64897985d-nbfln.kube-system
  Port_Binding node-kube-ovn-control-plane
  Port_Binding local-path-provisioner-5ddd94ff66-h4tn9.local-path-storage
  Port_Binding kube-ovn-pinger-hf2p6.kube-system
  Port_Binding coredns-64897985d-fhwlw.kube-system
```

6.2.6 chassis

uuid Chassis id

```
# kubectl ko sbctl chassis-del b0564934-5a0d-4804-a4c0-476c93596a17
# kubectl ko sbctl show
Chassis "6a29de7e-d731-4eaf-bacd-2f239ee52b28"
  hostname: kube-ovn-control-plane
  Encap geneve
      ip: "172.18.0.3"
      options: {csum="true"}
  Port_Binding coredns-64897985d-nbfln.kube-system
  Port_Binding node-kube-ovn-control-plane
  Port_Binding local-path-provisioner-5ddd94ff66-h4tn9.local-path-storage
  Port_Binding kube-ovn-pinger-hf2p6.kube-system
  Port_Binding coredns-64897985d-fhwlw.kube-system
```

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6.2.7

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6.3 ovn-central

`ovn-central` `ovn-nb` `ovn-sb` etcd Raft `ovn-central`

6.3.1 ovn-central

`kube-ovn-control-plane2` `ovn-central`

```
# kubectl -n kube-system get pod -o wide | grep central
ovn-central-6bf58cbc97-2cdhg                     1/1     Running   0             21m   172.18.0.3   kube-ovn-control-plane    <none>           <none>
ovn-central-6bf58cbc97-crmfp                     1/1     Running   0             21m   172.18.0.5   kube-ovn-control-plane2   <none>           <none>
ovn-central-6bf58cbc97-lxmpl                     1/1     Running   0             21m   172.18.0.4   kube-ovn-control-plane3   <none>           <none>
```

ovn-nb

ID

```
# kubectl ko nb status
1b9a
Name: OVN_Northbound
Cluster ID: 32ca (32ca07fb-739b-4257-b510-12fa18e7cce8)
Server ID: 1b9a (1b9a5d76-e69b-410c-8085-39943d0cd38c)
Address: tcp:[172.18.0.3]:6643
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 2135194 ms ago, reason: timeout
Last Election won: 2135188 ms ago
Election timer: 5000
Log: [135, 135]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-d64b ->d64b <-4984 ->4984
Disconnections: 0
Servers:
    4984 (4984 at tcp:[172.18.0.4]:6643) next_index=135 match_index=134 last msg 1084 ms ago
    1b9a (1b9a at tcp:[172.18.0.3]:6643) (self) next_index=2 match_index=134
    d64b (d64b at tcp:[172.18.0.5]:6643) next_index=135 match_index=134 last msg 1084 ms ago
status: ok
```

`kube-ovn-control-plane2` IP `172.18.0.5` ID `d64b` ovn-nb

```
# kubectl ko nb kick d64b
started removal
```

```
# kubectl ko nb status
1b9a
Name: OVN_Northbound
Cluster ID: 32ca (32ca07fb-739b-4257-b510-12fa18e7cce8)
Server ID: 1b9a (1b9a5d76-e69b-410c-8085-39943d0cd38c)
Address: tcp:[172.18.0.3]:6643
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 2297649 ms ago, reason: timeout
Last Election won: 2297643 ms ago
Election timer: 5000
Log: [136, 136]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-4984 ->4984
Disconnections: 2
Servers:
    4984 (4984 at tcp:[172.18.0.4]:6643) next_index=136 match_index=135 last msg 1270 ms ago
    1b9a (1b9a at tcp:[172.18.0.3]:6643) (self) next_index=2 match_index=135
status: ok
```


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ovn-sb

ovn-sb ID

```
kubectl ko sb status
3722
Name: OVN_Southbound
Cluster ID: d4bd (d4bd37a4-0400-499f-b4df-b4fd389780f0)
Server ID: 3722 (3722d5ae-2ced-4820-a6b2-8b744d11fb3e)
Address: tcp:[172.18.0.3]:6644
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 2395317 ms ago, reason: timeout
Last Election won: 2395316 ms ago
Election timer: 5000
Log: [130, 130]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-e9f7 ->e9f7 <-6e84 ->6e84
Disconnections: 0
Servers:
    e9f7 (e9f7 at tcp:[172.18.0.5]:6644) next_index=130 match_index=129 last msg 1006 ms ago
    6e84 (6e84 at tcp:[172.18.0.4]:6644) next_index=130 match_index=129 last msg 1004 ms ago
    3722 (3722 at tcp:[172.18.0.3]:6644) (self) next_index=2 match_index=129
status: ok
```

`kube-ovn-control-plane2` IP `172.18.0.5` ID `e9f7` ovn-sb

```
# kubectl ko sb kick e9f7
started removal
```

```
# kubectl ko sb status
3722
Name: OVN_Southbound
Cluster ID: d4bd (d4bd37a4-0400-499f-b4df-b4fd389780f0)
Server ID: 3722 (3722d5ae-2ced-4820-a6b2-8b744d11fb3e)
Address: tcp:[172.18.0.3]:6644
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 2481636 ms ago, reason: timeout
Last Election won: 2481635 ms ago
Election timer: 5000
Log: [131, 131]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-6e84 ->6e84
Disconnections: 2
Servers:
    6e84 (6e84 at tcp:[172.18.0.4]:6644) next_index=131 match_index=130 last msg 642 ms ago
    3722 (3722 at tcp:[172.18.0.3]:6644) (self) next_index=2 match_index=130
status: ok
```

ovn-central

ovn-central `NODE_IPS`

```
kubectl label node kube-ovn-control-plane2 kube-ovn/role-
kubectl scale deployment -n kube-system ovn-central --replicas=2
kubectl set env deployment/ovn-central -n kube-system NODE_IPS="172.18.0.3,172.18.0.4"
kubectl rollout status deployment/ovn-central -n kube-system
```

ovn-central

`ovs-ovn`

```
# kubectl set env daemonset/ovs-ovn -n kube-system OVN_DB_IPS="172.18.0.3,172.18.0.4"
daemonset.apps/ovs-ovn env updated
# kubectl delete pod -n kube-system -lapp=ovs
pod "ovs-ovn-4f6jc" deleted
```

```
pod "ovs-ovn-csn2w" deleted
pod "ovs-ovn-mpbmb" deleted
```

`kube-ovn-controller`

```
# kubectl set env deployment/kube-ovn-controller -n kube-system OVN_DB_IPS="172.18.0.3,172.18.0.4"
deployment.apps/kube-ovn-controller env updated

# kubectl rollout status deployment/kube-ovn-controller -n kube-system
Waiting for deployment "kube-ovn-controller" rollout to finish: 1 of 3 updated replicas are available...
Waiting for deployment "kube-ovn-controller" rollout to finish: 2 of 3 updated replicas are available...
deployment "kube-ovn-controller" successfully rolled out
```

`kube-ovn-control-plane2`

```
rm -rf /etc/origin/ovn
```

Kubernetes

6.3.2 ovn-central

Kubernetes `ovn-central`

`/etc/origin/ovn` `ovnnb_db.db` `ovnsb_db.db`

```
rm -rf /etc/origin/ovn
```

ovn-central

`ovn-central`

```
# kubectl ko nb status
1b9a
Name: OVN_Northbound
Cluster ID: 32ca (32ca07fb-739b-4257-b510-12fa18e7cce8)
Server ID: 1b9a (1b9a5d76-e69b-410c-8085-39943d0cd38c)
Address: tcp:[172.18.0.3]:6643
Status: cluster member
Role: leader
Term: 44
Leader: self
Vote: self

Last Election started 1855739 ms ago, reason: timeout
Last Election won: 1855729 ms ago
Election timer: 5000
Log: [147, 147]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: ->4984 <-4984
Disconnections: 0
Servers:
    4984 (4984 at tcp:[172.18.0.4]:6643) next_index=147 match_index=146 last msg 367 ms ago
    1b9a (1b9a at tcp:[172.18.0.3]:6643) (self) next_index=140 match_index=146
status: ok

# kubectl ko sb status
3722
Name: OVN_Southbound
Cluster ID: d4bd (d4bd37a4-0400-499f-b4df-b4fd389780f0)
Server ID: 3722 (3722d5ae-2ced-4820-a6b2-8b744d11fb3e)
Address: tcp:[172.18.0.3]:6644
Status: cluster member
Role: leader
Term: 33
Leader: self
Vote: self

Last Election started 1868589 ms ago, reason: timeout
Last Election won: 1868579 ms ago
Election timer: 5000
Log: [142, 142]
Entries not yet committed: 0
```

```
Entries not yet applied: 0
Connections: ->6e84 <-6e84
Disconnections: 0
Servers:
    6e84 (6e84 at tcp:[172.18.0.4]:6644) next_index=142 match_index=141 last msg 728 ms ago
    3722 (3722 at tcp:[172.18.0.3]:6644) (self) next_index=134 match_index=141
status: ok
```

ovn-central `NODE_IPS`

```
kubectl label node kube-ovn-control-plane2 kube-ovn/role=master
kubectl scale deployment -n kube-system ovn-central --replicas=3
kubectl set env deployment/ovn-central -n kube-system NODE_IPS="172.18.0.3,172.18.0.4,172.18.0.5"
kubectl rollout status deployment/ovn-central -n kube-system
```

ovn-central

`ovs-ovn`

```
# kubectl set env daemonset/ovs-ovn -n kube-system OVN_DB_IPS="172.18.0.3,172.18.0.4,172.18.0.5"
daemonset.apps/ovs-ovn env updated
# kubectl delete pod -n kube-system -lapp=ovs
pod "ovs-ovn-4f6jc" deleted
pod "ovs-ovn-csn2w" deleted
pod "ovs-ovn-mpbmb" deleted
```

`kube-ovn-controller`

```
# kubectl set env deployment/kube-ovn-controller -n kube-system OVN_DB_IPS="172.18.0.3,172.18.0.4,172.18.0.5"
deployment.apps/kube-ovn-controller env updated

# kubectl rollout status deployment/kube-ovn-controller -n kube-system
Waiting for deployment "kube-ovn-controller" rollout to finish: 1 of 3 updated replicas are available...
Waiting for deployment "kube-ovn-controller" rollout to finish: 2 of 3 updated replicas are available...
deployment "kube-ovn-controller" successfully rolled out
```

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6.4 OVN

6.4.1

kubectl backup

```
# kubectl ko nb backup
tar: Removing leading `/' from member names
backup ovn-nb db to /root/ovnnb_db.060223191654183154.backup

# kubectl ko sb backup
tar: Removing leading `/' from member names
backup ovn-sb db to /root/ovnsb_db.060223191654183154.backup
```

6.4.2

`/var/log/ovn/ovn-northd.log`

```
 * ovn-northd is not running
ovsdb-server: ovsdb error: error reading record 2739 from OVN_Northbound log: record 2739 advances commit index to 6308 but last log index is 6307
 * Starting ovsdb-nb
```

`OVN_Northbound` `OVN_Southbound` `OVN_Northbound` ovn-nb

```
# kubectl ko nb status
9182
Name: OVN_Northbound
Cluster ID: e75f (e75fa340-49ed-45ab-990e-26cb865ebc85)
Server ID: 9182 (9182e8dd-b5b0-4dd8-8518-598cc1e374f3)
Address: tcp:[10.0.128.61]:6643
Status: cluster member
Role: leader
Term: 1454
Leader: self
Vote: self

Last Election started 1732603 ms ago, reason: timeout
Last Election won: 1732587 ms ago
Election timer: 1000
Log: [7332, 12512]
Entries not yet committed: 1
Entries not yet applied: 1
Connections: ->f080 <-f080 <-e631 ->e631
Disconnections: 1
Servers:
    f080 (f080 at tcp:[10.0.129.139]:6643) next_index=12512 match_index=12510 last msg 63 ms ago
    9182 (9182 at tcp:[10.0.128.61]:6643) (self) next_index=10394 match_index=12510
    e631 (e631 at tcp:[10.0.131.173]:6643) next_index=12512 match_index=0
```

```
kubectl ko nb kick e631
```

```
mv /etc/origin/ovn/ovnnb_db.db /tmp
```

`ovn-central` Pod

```
kubectl delete pod -n kube-system ovn-central-xxxx
```


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6.4.3

leader

ovn-central

`ovn-central` `ovn-central`

```
kubectl scale deployment -n kube-system ovn-central --replicas=0
```

`ovsdb-tool cluster-to-standalone`

`ovn-central` `NODE_IPS` `/etc/origin/ovn`

`docker`

```
docker run -it -v /etc/origin/ovn:/etc/ovn kubeovn/kube-ovn:v1.16.16 bash
cd /etc/ovn/
ovsdb-tool cluster-to-standalone ovnnb_db_standalone.db ovnnb_db.db
ovsdb-tool cluster-to-standalone ovnsb_db_standalone.db ovnsb_db.db
```

containerd docker `ctr`

```
ctr -n k8s.io image pull docker.io/kubeovn/kube-ovn:v1.16.16
ctr -n k8s.io run --rm -t \
  --mount type=bind,src=/etc/origin/ovn,dst=/etc/ovn,options=rbind:rw \
  docker.io/kubeovn/kube-ovn:v1.16.16 ovn-recover bash
cd /etc/ovn/
ovsdb-tool cluster-to-standalone ovnnb_db_standalone.db ovnnb_db.db
ovsdb-tool cluster-to-standalone ovnsb_db_standalone.db ovnsb_db.db
exit
```

`ovn-central` Pod `ovsdb-tool cluster-to-standalone` `kubectl cp`

ovn-central

```
mv /etc/origin/ovn/ovnnb_db.db /tmp
mv /etc/origin/ovn/ovnsb_db.db /tmp
```

`ovnnb_db.db` `ovnsb_db.db` `ovn-central` `NODE_IPS` `/etc/origin/ovn/`

```
mv /etc/origin/ovn/ovnnb_db_standalone.db /etc/origin/ovn/ovnnb_db.db
mv /etc/origin/ovn/ovnsb_db_standalone.db /etc/origin/ovn/ovnsb_db.db
```

`ovn-central`

```
kubectl scale deployment -n kube-system ovn-central --replicas=3
kubectl rollout status deployment/ovn-central -n kube-system
```

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6.4.4

6.5 CIDR

CIDR

> CIDR Pod CIDR Join CIDR Join CIDR

6.5.1

`kubectl edit` `cidrBlock` `gateway` `excludeIps`

```
kubectl edit subnet test-subnet
```

6.5.2 Namespace Pod

`test` Namespace

```
for pod in $(kubectl get pod --no-headers -n test --field-selector spec.restartPolicy=Always -o custom-columns=NAME:.metadata.name,HOST:spec.hostNetwork |
awk '{if ($2!="true") print $1}'); do
  kubectl delete pod "$pod" -n test --ignore-not-found
done
```

host Pod

```
for ns in $(kubectl get ns --no-headers -o custom-columns=NAME:.metadata.name); do
  for pod in $(kubectl get pod --no-headers -n "$ns" --field-selector spec.restartPolicy=Always -o custom-columns=NAME:.metadata.name,HOST:spec.hostNetwork |
awk '{if ($2!="true") print $1}'); do
    kubectl delete pod "$pod" -n "$ns" --ignore-not-found
  done
done
```

6.5.3

CIDR `kube-ovn-controller` Deployment

```
args:
- --default-cidr=10.17.0.0/16
- --default-gateway=10.17.0.1
- --default-exclude-ips=10.17.0.1
```

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6.5.4

6.6 Join CIDR

Join CIDR

> Join CIDR Pod

6.6.1 Join

```
kubectl patch subnet join --type='json' -p '[{"op": "replace", "path": "/metadata/finalizers", "value": []}]'
kubectl delete subnet join
```

6.6.2

```
kubectl annotate node ovn.kubernetes.io/allocated=false --all --overwrite
```

6.6.3 Join

`kube-ovn-controller` Join

```
kubectl edit deployment -n kube-system kube-ovn-controller
```

```
args:
- --node-switch-cidr=100.51.0.0/16
```

`kube-ovn-controller` `join`

```
kubectl delete pod -n kube-system -lapp=kube-ovn-controller
```

Join

```
# kubectl get subnet
NAME          PROVIDER   VPC           PROTOCOL   CIDR            PRIVATE   NAT     DEFAULT   GATEWAYTYPE   V4USED   V4AVAILABLE   V6USED   V6AVAILABLE
EXCLUDEIPS
join          ovn        ovn-cluster   IPv4       100.51.0.0/16   false     false   false     distributed   2        65531         0        0
["100.51.0.1"]
ovn-default   ovn        ovn-cluster   IPv4       10.17.0.0/16    false     true    true      distributed   5        65528         0        0
["10.17.0.1"]
```

6.6.4 ovn0

`ovn0` `kube-ovn-cni`

```
kubectl delete pod -n kube-system -l app=kube-ovn-cni
```

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6.6.5

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6.7

`kube-ovn.yaml`

```
vi kube-ovn.yaml
# ...
        - name: kube-ovn-controller
          image: "docker.io/kubeovn/kube-ovn:v1.16.16"
          imagePullPolicy: IfNotPresent
          args:
          - /kube-ovn/start-controller.sh
          - --v=3
# ...
#
```

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6.7.1

6.8

Kube-OVN

6.8.1

1.
   - Pod Pod
   - Pod
   - Pod Service
   - Pod
   - Pod
   - `kube-ovn-pinger`
2.
   - Pod
   - kube-ovn-cni CNI
   - ovs-ovn OVS
   - kube-ovn-controller
   - ovn-central OVN
   - dmesg
   - `netstat -s`
3. CPU, IO

   `kubectl ko logs`

6.8.2 Pod IP

Pod Running kubectl describe Pod `duplicate IPv4 address <ip> found on logical switch port <port>`

1. Pod IP Pod IP IP IP IP Pod
2. kube-ovn-controller IP
3. kube-ovn-controller IP
4. `kubectl ko nbctl show` OVN IP
5. OVN Kubernetes IP `kubectl ko nbctl lsp-del <port>`

6.8.3 Pod ping gateway failed

Pod Running kubectl describe Pod `network <ip> with gateway <gw ip> is not ready for interface eth0 after 30 checks`

1. `kubectl ko sbctl show` Pod
2. ovn-central ovs-ovn ovn-central ovs-ovn

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3. Pod
4. Underlay Underlay

6.8.4 Pod

VPC Pod

1. `kubectl ko trace` OVN ACL
2. ACL ACL
3. ACL Subnet stats
4. Subnet Spec
5. `kube-ovn-controller`

6.8.5 Pod IP CIDR

Pod IP CIDR

1. `/etc/cni/net.d/` Kube-OVN CNI
2.
3. kubelet Pod

6.8.6 Debug Pod

kubectl debug Pod ContainerCreating Pod Event network not ready no address allocated

debug Pod yaml yaml Annotation

```
ovn.kubernetes.io/ip_address
ovn.kubernetes.io/mac_address
ovn.kubernetes.io/allocated
ovn.kubernetes.io/routed
```

debug Pod yaml debug Pod

6.8.7 ARM

ARM Offload

`netstat`

```
# netstat -us
IcmpMsg:
    InType0: 22
    InType3: 24
    InType8: 117852
    OutType0: 117852
```


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```
    OutType3: 29
    OutType8: 22
Udp:
    3040636 packets received
    0 packets to unknown port received.
    4 packet receive errors
    602 packets sent
    0 receive buffer errors
    0 send buffer errors
    InCsumErrors: 4
UdpLite:
IpExt:
    InBcastPkts: 10244
    InOctets: 4446320361
    OutOctets: 1496815600
    InBcastOctets: 3095950
    InNoECTPkts: 7683903
```

`InCsumErrors`

`tx offload` TCP

```
ethtool -K eth0 tx off
```

CentOS 7 `4.19.90-25.16.v2101`

6.8.8 Pod Service

Pod Service `dmesg`

```
netlink: Unknown conntrack attr (type=6, max=5)
openvswitch: netlink: Flow actions may not be safe on all matching packets.
```

OVS NAT

1. OVS
2. Overlay `kube-ovn-controller` `--enable-lb=false` OVN LB kube-proxy Service

6.8.9 ovn-central

v1.11.x 1 Pod OVN NB SB Kube-OVN ovsdb-server/compact

v1.13 `ENABLE_COMPACT` `false` `dist/images/install.sh` `start-db.sh` ovn-central `true`

```
- name: ENABLE_COMPACT
  value: "false"
```

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6.8.10

7.

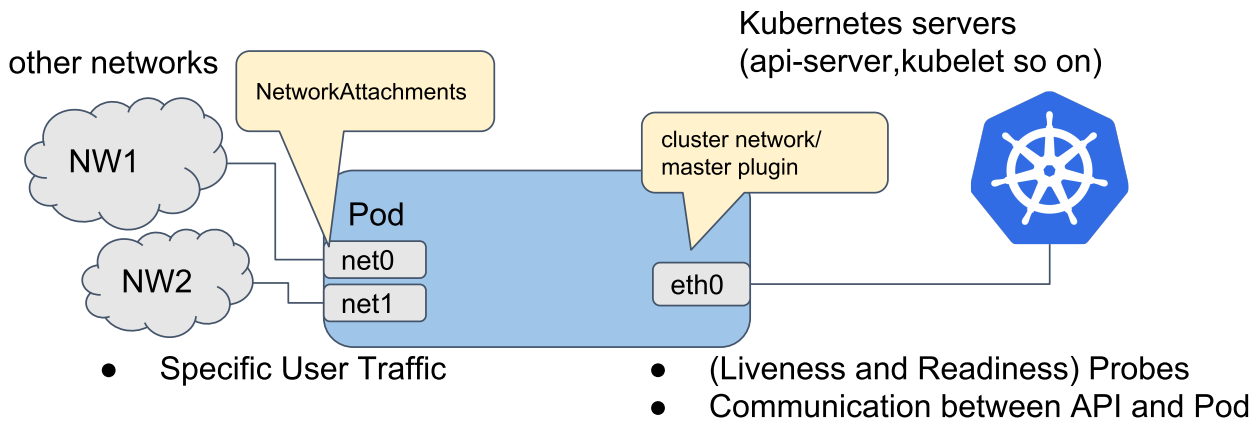
7.1

Kube-OVN CNI macvlan vlan host-device IPAM Kube-OVN IP

Kube-OVN Kube-OVN

7.1.1

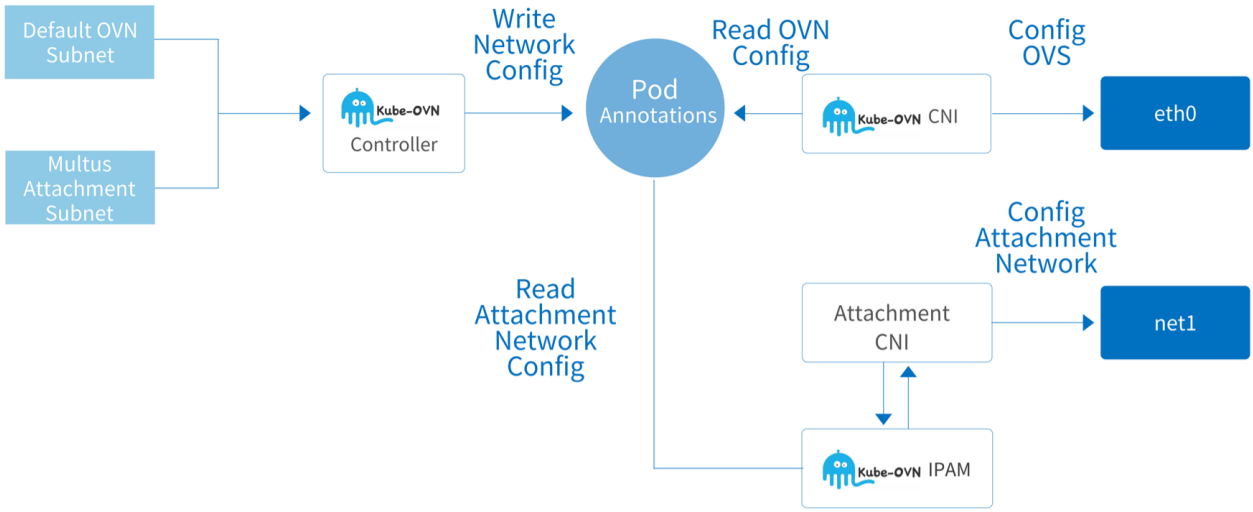
Multus CNI pod pod eth0 net0 net1 eth0 kubernetes kubernetes kubernetes api-server kubelet net0 net1 CNI vlan/vxlan/ptp



IPAM

Multus CNI Pod IP Kube-OVN Subnet IP CRD IP IP Kube-OVN IPAM

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Kube-OVN IP eth0 OVN net1 CNI net1 multus-cni NetworkAttachmentDefinition Pod `kube-ovn-controller` Pod Pod annotation Subnet IP Pod Pod annotation CNI `kube-ovn-cni` ipam `kube-ovn-cni` Pod annotation CNI CNI

7.1.2

`NetworkAttachmentDefinition` `spec` multus `defaultConfDir` CNI

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: macvlan-conf-2
```

`kube-ovn-controller` `NetworkAttachmentDefinition` `provider` `spec` `kube-ovn-controller` `provider` `spec` Kube-OVN IPAM

7.1.3

Kube-OVN Multus

Kube-OVN Multus how to use Kube-OVN Multus-CNI

CNI IPAM

Kube-OVN CNI

NETWORKATTACHMENTDEFINITION

macvlan ipam `kube-ovn`

```
#    macvlan
sudo modprobe macvlan
```


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```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: macvlan
  namespace: default
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "macvlan",
      "master": "eth0",
      "mode": "bridge",
      "ipam": {
        "type": "kube-ovn",
        "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
        "provider": "macvlan.default"
      }
    }'
```

- `spec.config.ipam.type` : `kube-ovn` kube-ovn
- `server_socket` : Kube-OVN socket `/run/openvswitch/kube-ovn-daemon.sock`
- `provider` : NetworkAttachmentDefinition `<name>.<namespace>` Kube-OVN Subnet
- `master` :

Info

`provider` Underlay `ProviderNetwork`

KUBE-OVN SUBNET

Kube-OVN Subnet `cidrBlock` `exclude_ips` `provider` NetworkAttachmentDefinition `<name>.<namespace>` macvlan Subnet

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: macvlan
spec:
  protocol: IPv4
  provider: macvlan.default
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
  - 172.17.0.0..172.17.0.10
```

> `gateway` `private` `nat` `provider` ovn attachment network

Pod

Pod annotation `k8s.v1.cni.cncf.io/networks` NetworkAttachmentDefinition `<namespace>/<name>`

```
apiVersion: v1
kind: Pod
metadata:
  name: samplepod
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/macvlan
spec:
  containers:
  - name: samplepod
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

IP Pod

IP Pod `<networkAttachmentName>.<networkAttachmentNamespace>.kubernetes.io/ip_address` annotation

```
apiVersion: v1
kind: Pod
metadata:
  name: static-ip
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/macvlan
    ovn.kubernetes.io/ip_address: 10.16.0.15
    ovn.kubernetes.io/mac_address: 00:00:00:53:6B:B6
```


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```
    macvlan.default.kubernetes.io/ip_address: 172.17.0.100
    macvlan.default.kubernetes.io/mac_address: 00:00:00:53:6B:BB
spec:
  containers:
  - name: static-ip
    image: docker.io/library/nginx:alpine
```

IP

ippool `<networkAttachmentName>.<networkAttachmentNamespace>.kubernetes.io/ip_pool` annotation

```
apiVersion: apps/v1
kind: Deployment
metadata:
  namespace: default
  name: static-workload
  labels:
    app: static-workload
spec:
  replicas: 2
  selector:
    matchLabels:
      app: static-workload
  template:
    metadata:
      labels:
        app: static-workload
      annotations:
        k8s.v1.cni.cncf.io/networks: default/macvlan
        ovn.kubernetes.io/ip_pool: 10.16.0.15,10.16.0.16,10.16.0.17
        macvlan.default.kubernetes.io/ip_pool: 172.17.0.200,172.17.0.201,172.17.0.202
    spec:
      containers:
      - name: static-workload
        image: docker.io/library/nginx:alpine
```

macvlan Pod

macvlan Pod Pod annotation `default-route`

```
apiVersion: v1
kind: Pod
metadata:
  name: samplepod-route
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: '[{
      "name": "macvlan",
      "namespace": "default",
      "default-route": ["172.17.0.1"]
    }]'
spec:
  containers:
  - name: samplepod-route
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

macvlan Pod

macvlan Pod annotation `v1.multus-cni.io/default-network` NetworkAttachmentDefinition `<namespace>/<name>`

```
apiVersion: v1
kind: Pod
metadata:
  name: samplepod-macvlan
  namespace: default
  annotations:
    v1.multus-cni.io/default-network: default/macvlan
spec:
  containers:
  - name: samplepod-macvlan
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

KUBE-OVN SUBNET PROVIDER OVN

`provider` ovn subnet IP `provider` ovn Kube-OVN Subnet `cidrBlock` `exclude_ips` Subnet

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: macvlan
spec:
  protocol: IPv4
  provider: ovn
```


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```
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
  - 172.17.0.0..172.17.0.10
```

Pod

`provider` ovn subnet IP Pod annotation `k8s.v1.cni.cncf.io/networks` `<networkAttachmentName>.<networkAttachmentNamespace>.kubernetes.io/logical_switch`

```
apiVersion: v1
kind: Pod
metadata:
  name: samplepod
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/macvlan
    macvlan.default.kubernetes.io/logical_switch: macvlan
spec:
  containers:
  - name: samplepod
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

- `k8s.v1.cni.cncf.io/networks`: NetworkAttachmentDefinition `<namespace>/<name>`
- `macvlan.default.kubernetes.io/logical_switch`:

> - `<networkAttachmentName>.<networkAttachmentNamespace>.kubernetes.io/logical_switch` provider
> - ovn subnet ipam IP Pod IP macvlan Pod
> - macvlan Pod

Kube-OVN

Kube-OVN

NETWORKATTACHMENTDEFINITION

`provider` `ovn`

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: attachnet
  namespace: default
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "attachnet.default.ovn"
    }'
```

- `spec.config.type`: `kube-ovn` CNI Kube-OVN
- `server_socket`: Kube-OVN socket `/run/openvswitch/kube-ovn-daemon.sock`
- `provider`: NetworkAttachmentDefinition `<name>.<namespace>.ovn` Kube-OVN Subnet ovn

KUBE-OVN SUBNET

Kube-OVN `provider` NetworkAttachmentDefinition `<name>.<namespace>.ovn` `ovn` Kube-OVN Subnet

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: attachnet
spec:
  protocol: IPv4
  provider: attachnet.default.ovn
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
  - 172.17.0.0..172.17.0.10
```


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Pod

Pod annotation `k8s.v1.cni.cncf.io/networks` NetworkAttachmentDefinition `<namespace>/<name>`

```
apiVersion: v1
kind: Pod
metadata:
  name: samplepod
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/attachnet
spec:
  containers:
  - name: samplepod
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

IP/MAC

NetworkAttachmentDefinition IP/MAC `k8s.v1.cni.cncf.io/networks` `interface` annotation `<nadName>.<nadNamespace>.kubernetes.io/ip_address.<interfaceName>` `<nadName>.<nadNamespace>.kubernetes.io/mac_address.<interfaceName>` - IP/MAC

Kube-OVN Pod `<networkAttachmentName>.<networkAttachmentNamespace>.ovn.kubernetes.io/routes` annotation

```
apiVersion: v1
kind: Pod
metadata:
  name: custom-routes-attach
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/attachnet
    attachnet.default.ovn.kubernetes.io/routes: |
      [{
        "dst": "192.168.0.101/24",
        "gw": "172.17.0.254"
      }, {
        "gw": "172.17.0.254"
      }]
spec:
  containers:
  - name: custom-routes-attach
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

> `dst`

Deployment DaemonSet StatefulSet Annotation `.spec.template.metadata.annotations`

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: custom-routes-attach
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
      annotations:
        k8s.v1.cni.cncf.io/networks: default/attachnet
        attachnet.default.ovn.kubernetes.io/routes: |
          [{
            "dst": "192.168.0.101/24",
            "gw": "172.17.0.254"
          }, {
            "gw": "172.17.0.254"
          }]
    spec:
      containers:
      - name: nginx
        image: docker.io/library/nginx:alpine
```

KUBE-OVN SUBNET PROVIDER OVN

`provider` ovn subnet IP `provider` ovn Kube-OVN Subnet `cidrBlock` `exclude_ips` Subnet

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```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: attachnet
spec:
  protocol: IPv4
  provider: ovn
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
  - 172.17.0.0..172.17.0.10
```

Pod

`provider` ovn subnet IP Pod annotation `k8s.v1.cni.cncf.io/networks` `<networkAttachmentName>.<networkAttachmentNamespace>.ovn.kubernetes.io/logical_switch`

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: samplepod
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/attachnet
    attachnet.default.ovn.kubernetes.io/logical_switch: attachnet
spec:
  containers:
  - name: samplepod
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

- `k8s.v1.cni.cncf.io/networks` : NetworkAttachmentDefinition `<namespace>/<name>`
- `attachnet.default.ovn.kubernetes.io/logical_switch` :

> - `<networkAttachmentName>.<networkAttachmentNamespace>.ovn.kubernetes.io/logical_switch` provider
> - Kube-OVN Pod IP Pod IP macvlan Pod Kube-OVN Pod
> - Kube-OVN Pod `<networkAttachmentName>.<networkAttachmentNamespace>.ovn.kubernetes.io/routes` annotation

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7.1.4

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7.2

Kube-OVN

Kube-OVN

7.2.1

1. CNI CNI
2. 10ns 20ns
3. CPU CPU
4. Macvlan SR-IOV

7.2.2

Overlay

- Kubernetes: 1.22.0
- OS: CentOS 7
- Kube-OVN: 1.8.0 *Overlay*
- CPU: Intel(R) Xeon(R) E-2278G
- Network: 2*10Gbps, xmit_hash_policy=layer3+4

`qperf -t 60 <server ip> -ub -oo msg_size:1 -vu tcp_lat tcp_bw udp_lat udp_bw` 1 tcp/udp

Type	tcp_lat (us)	udp_lat (us)	tcp_bw (Mb/s)	udp_bw(Mb/s)
Kube-OVN Default	25.7	22.9	27.1	1.59
Kube-OVN Optimized	13.9	12.9	27.6	5.57
HOST Network	13.1	12.4	28.2	6.02

Overlay Underlay

Kube-OVN Overlay Underlay

Environment:

- Kubernetes: 1.22.0
- OS: CentOS 7
- Kube-OVN: 1.8.0
- CPU: AMD EPYC 7402P 24-Core Processor
- Network: Intel Corporation Ethernet Controller XXV710 for 25GbE SFP28

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`qperf -t 60 <server ip> -ub -oo msg_size:1 -vu tcp_lat tcp_bw udp_lat udp_bw`

Type	tcp_lat (us)	udp_lat (us)	tcp_bw (Mb/s)	udp_bw(Mb/s)
Kube-OVN Overlay	15.2	14.6	23.6	2.65
Kube-OVN Underlay	14.3	13.8	24.2	3.46
HOST Network	16.6	15.4	24.8	2.64

`qperf -t 60 <server ip> -ub -oo msg_size:1K -vu tcp_lat tcp_bw udp_lat udp_bw`

Type	tcp_lat (us)	udp_lat (us)	tcp_bw (Gb/s)	udp_bw(Gb/s)
Kube-OVN Overlay	16.5	15.8	10.2	2.77
Kube-OVN Underlay	15.9	14.5	9.6	3.22
HOST Network	18.1	16.6	9.32	2.66

`qperf -t 60 <server ip> -ub -oo msg_size:4K -vu tcp_lat tcp_bw udp_lat udp_bw`

Type	tcp_lat (us)	udp_lat (us)	tcp_bw (Gb/s)	udp_bw(Gb/s)
Kube-OVN Overlay	34.7	41.6	16.0	9.23
Kube-OVN Underlay	32.6	44	15.1	6.71
HOST Network	35.9	45.9	14.6	5.59

netfilter `kube-proxy` netfilter

7.2.3

CPU

CPU CPU

```
cpupower frequency-set -g performance
```

```
# ethtool -g eno1
 Ring parameters for eno1:
 Pre-set maximums:
 RX:             4096
 RX Mini:        0
 RX Jumbo:       0
 TX:             4096
 Current hardware settings:
 RX:             255
 RX Mini:        0
 RX Jumbo:       0
 TX:             255
```

```
ethtool -G eno1 rx 4096
ethtool -G eno1 tx 4096
```


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tuned

tuned profile

```
tuned-adm profile network-latency
```

```
tuned-adm profile network-throughput
```

`irqbalance` CPU CPU

OVN LB

OVN L2 LB `conntrack` recirculate CPU 20% CPU Overlay `kube-proxy` Service Pod-to-Pod `kube-ovn-controller`

```
command:
- /kube-ovn/start-controller.sh
args:
...
- --enable-lb=false
...
```

> Underlay `kube-proxy` iptables ipvs LB Service

conntrack

OVN LB Service Service NetworkPolicy Subnet A Pod Subnet B Service kube-ovn-controller `--skip-conntrack-dst-cidrs` conntrack

```
--skip-conntrack-dst-cidrs="10.17.0.0/16,169.254.169.245/32"
```

FastPath

network ns netfilter 20% CPU netfilter `FastPath` netfilter CPU

> netfilter iptables ipvs nftables

FastPath

`insmod kube_ovn_fastpath.ko` `FastPath` `dmesg`

```
# dmesg
...
[619631.323788] init_module,kube_ovn_fastpath_local_out
[619631.323798] init_module,kube_ovn_fastpath_post_routing
[619631.323800] init_module,kube_ovn_fastpath_pre_routing
[619631.323801] init_module,kube_ovn_fastpath_local_in
...
```

OVS

OVS flow 10% CPU x86 CPU `popcnt` `sse4.2` flow CPU 5%

`FastPath`

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CPU

```
cat /proc/cpuinfo  | grep popcnt
cat /proc/cpuinfo  | grep sse4_2
```

CENTOS

```
yum install -y gcc kernel-devel-$(uname -r) python3 autoconf automake libtool rpm-build openssl-devel
```

OVS RPM :

```
git clone -b branch-3.5 --depth=1 https://github.com/openvswitch/ovs.git
cd ovs
curl -s  https://github.com/kubeovn/ovs/commit/2d2c83c26d4217446918f39d5cd5838e9ac27b32.patch |  git apply
./boot.sh
./configure --with-linux=/lib/modules/$(uname -r)/build CFLAGS="-g -O2 -mpopcnt -msse4.2"
make rpm-fedora-kmod
cd rpm/rpmbuild/RPMS/x86_64/
```

RPM

```
rpm -i openvswitch-kmod-3.5.1-1.el7.x86_64.rpm
```

Kube-OVN OVS

UBUNTU

```
apt install -y autoconf automake libtool gcc build-essential libssl-dev
```

OVS

```
apt install -y autoconf automake libtool gcc build-essential libssl-dev

git clone -b branch-3.5 --depth=1 https://github.com/openvswitch/ovs.git
cd ovs
curl -s  https://github.com/kubeovn/ovs/commit/2d2c83c26d4217446918f39d5cd5838e9ac27b32.patch |  git apply
./boot.sh
./configure --prefix=/usr/ --localstatedir=/var --enable-ssl --with-linux=/lib/modules/$(uname -r)/build
make -j `nproc`
make install
make modules_install

cat > /etc/depmod.d/openvswitch.conf << EOF
override openvswitch * extra
override vport-* * extra
EOF

depmod -a
cp debian/openvswitch-switch.init /etc/init.d/openvswitch-switch
/etc/init.d/openvswitch-switch force-reload-kmod
```

Kube-OVN OVS

STT

> **Warning**
>
> OpenVswitch 3.6 STT Tunnel

Geneve Vxlan UDP UDP TCP TCP offload TCP CPU TCP

STT TCP TCP TCP TCP TCP

STT OVS OVS

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STT

```
kubectl set env daemonset/ovs-ovn -n kube-system TUNNEL_TYPE=stt

kubectl delete pod -n kube-system -lapp=ovs
```

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7.2.4

7.3 FastPath

Profile `Netfilter` 20% CPU FastPath `Netfilter` CPU FastPath

7.3.1

```
git clone --depth=1 https://github.com/kubeovn/kube-ovn.git
```

7.3.2

RPM CentOS RHEL

```
yum install -y kernel-devel-$(uname -r) gcc elfutils-libelf-devel
```

DEB Ubuntu Debian

```
apt install -y linux-headers-$(uname -r) build-essential
```

7.3.3

3.x

```
cd kube-ovn/fastpath/3.x
make all
```

4.x ~ 6.x

```
cd kube-ovn/fastpath/4.x-6.x
make all
```

7.3.4

`kube_ovn_fastpath.ko`

```
insmod kube_ovn_fastpath.ko
```

`dmesg`

```
# dmesg
[619631.323788] init_module,kube_ovn_fastpath_local_out
[619631.323798] init_module,kube_ovn_fastpath_post_routing
[619631.323800] init_module,kube_ovn_fastpath_pre_routing
[619631.323801] init_module,kube_ovn_fastpath_local_in
```

```
rmmod kube_ovn_fastpath.ko
```

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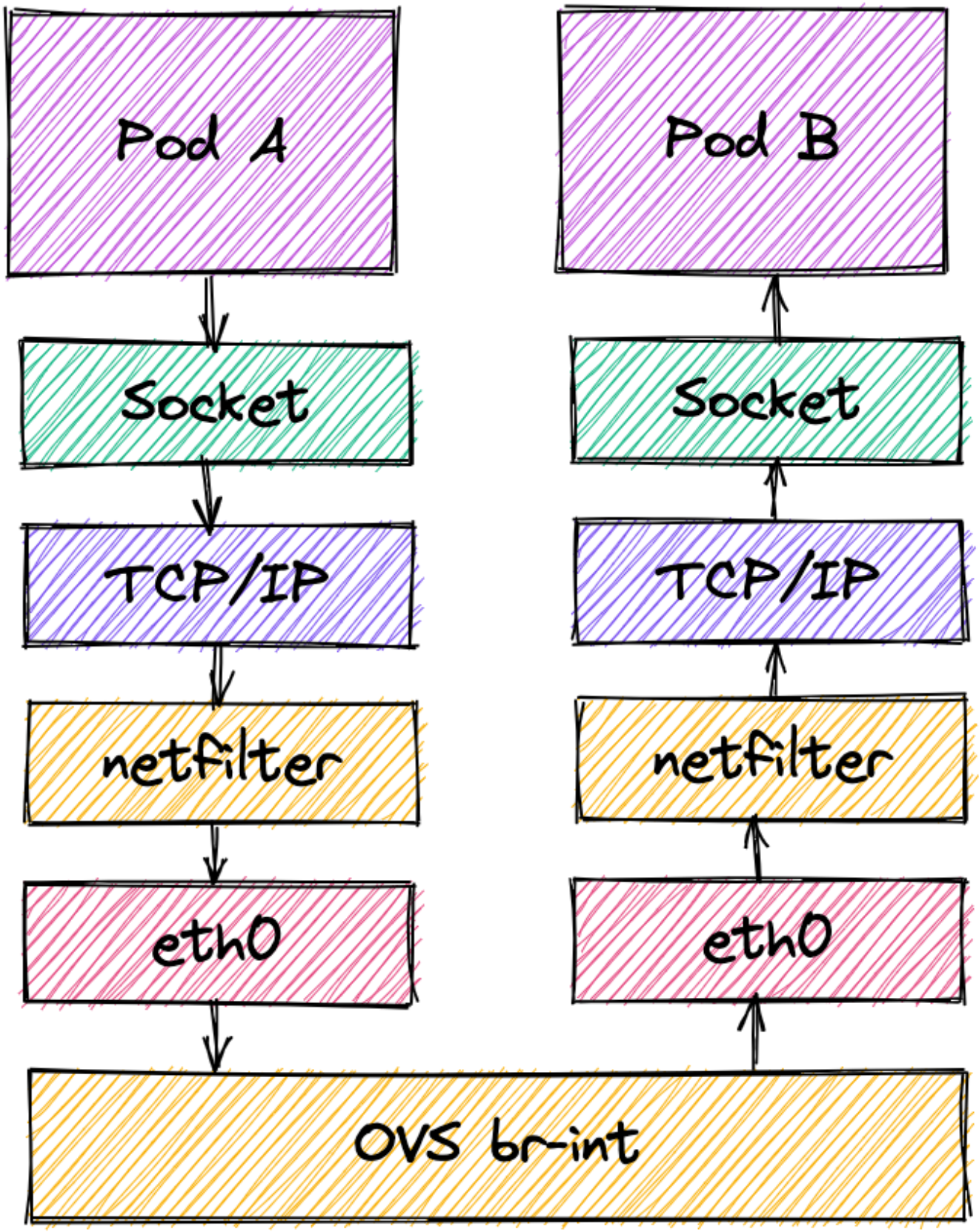
7.3.5

7.4 eBPF TCP

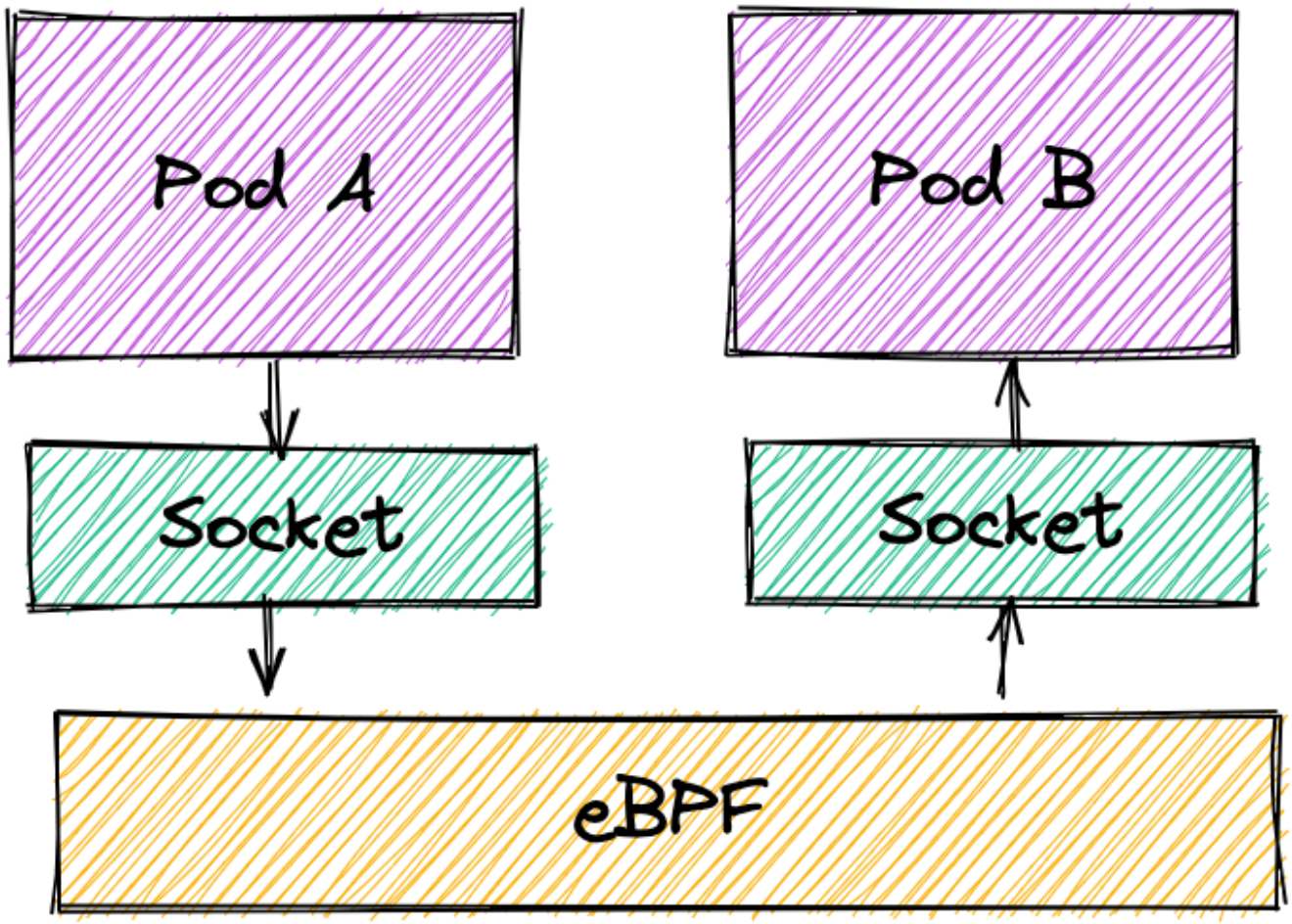
5G Pod TCP Intel istio-tcpip-bypass Pod eBPF TCP/IP socket

7.4.1

Pod TCP TCP/IP, netfilter OVS



istio-tcpip-bypass	TCP	socket



TCP Service Mesh Service Mesh

Tanzu Service Mesh Acceleration using eBPF

7.4.2

eBPF Ubuntu 20.04 Linux 5.4.0-74-generic

7.4.3

Pod `nodeSelector`

```
# kubectl create deployment perf --image=kubeovn/perf:dev --replicas=2
deployment.apps/perf created
# kubectl get pod -o wide
NAME                    READY   STATUS    RESTARTS   AGE   IP           NODE     NOMINATED NODE   READINESS GATES
perf-7697bc6ddf-b2cpv   1/1     Running   0          28s   100.64.0.3   sealos   <none>           <none>
perf-7697bc6ddf-p2xpt   1/1     Running   0          28s   100.64.0.2   sealos   <none>           <none>
```

Pod qperf server Pod qperf client

```
# kubectl exec -it perf-7697bc6ddf-b2cpv sh
/ # qperf

# kubectl exec -it perf-7697bc6ddf-p2xpt sh
/ # qperf -t 60 100.64.0.3 -ub -oo msg_size:1:16K:*4 -vu tcp_lat tcp_bw
```

istio-tcpip-bypass

```
kubectl apply -f https://raw.githubusercontent.com/intel/istio-tcpip-bypass/main/bypass-tcpip-daemonset.yaml
```


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perf client

```
# kubectl exec -it perf-7697bc6ddf-p2xpt sh
/ # qperf -t 60 100.64.0.3 -ub -oo msg_size:1:16K:*4 -vu tcp_lat tcp_bw
```

7.4.4

TCP 40% ~ 60% 1024 40% ~ 80%

Packet Size (byte)	eBPF tcp_lat (us)	Default tcp_lat (us)	eBPF tcp_bw (Mb/s)	Default tcp_bw(Mb/s)
1	20.2	44.5	1.36	4.27
4	20.2	48.7	5.48	16.7
16	19.6	41.6	21.7	63.5
64	18.8	41.3	96.8	201
256	19.2	36	395	539
1024	18.3	42.4	1360	846
4096	16.5	62.6	4460	2430
16384	20.2	58.8	9600	6900

> 512 eBPF TCP eBPF eBPF TCP

7.4.5

1. istio-tcpip-bypass
2. Deep Dive TCP/IP Bypass with eBPF in Service Mesh
3. Tanzu Service Mesh Acceleration using eBPF

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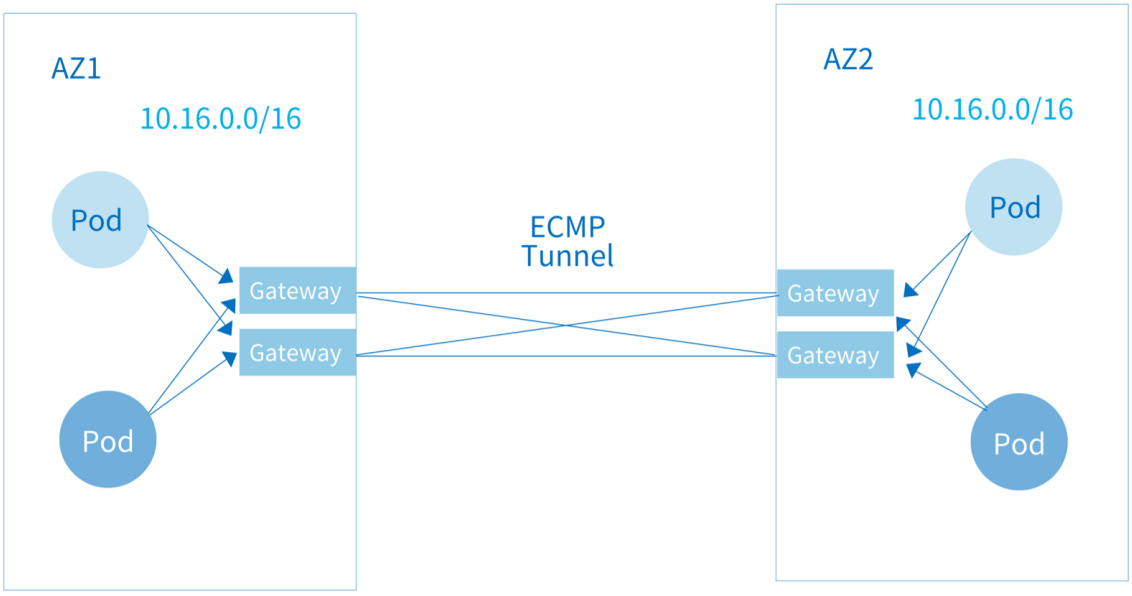
GitHub 

7.4.6

7.5 OVN-IC

Kube-OVN OVN-IC Kubernetes Pod Pod Pod IP Kube-OVN IP

Overlay Underlay



Limitation

OVN-IC Pod IP Service DNS NetworkPolicy Istio

7.5.1

1. 1.11.16 `install.sh`

```
ENABLE_IC=true
```

deployment ovn-ic-controller

2. CIDR
3. `kube-ovn-controller` IP
4. IP
5. VPC VPC

7.5.2 OVN-IC

1

1 Kube-OVN v1.11.16

" " " " Deployment master master 1 master

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`install-ovn-ic.sh`

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/install-ic-server.sh
```

`TS_NUM` ECMP Path

```
sed 's/VERSION=.*/VERSION=v1.16.16/' dist/images/install-ic-server.sh | TS_NUM=3 bash
```

```
deployment.apps/ovn-ic-server created
Waiting for deployment spec update to be observed...
Waiting for deployment "ovn-ic-server" rollout to finish: 0 out of 3 new replicas have been updated...
Waiting for deployment "ovn-ic-server" rollout to finish: 0 of 3 updated replicas are available...
Waiting for deployment "ovn-ic-server" rollout to finish: 1 of 3 updated replicas are available...
Waiting for deployment "ovn-ic-server" rollout to finish: 2 of 3 updated replicas are available...
deployment "ovn-ic-server" successfully rolled out
OVN IC Server installed Successfully
```

`kubectl ko icsbctl show`

```
kubectl ko icsbctl show
availability-zone az0
    gateway 059b5c54-c540-4d77-b009-02d65f181a02
        hostname: kube-ovn-worker
        type: geneve
            ip: 172.18.0.3
        port ts-az0
            transit switch: ts
            address: ["00:00:00:B4:8E:BE 169.254.100.97/24"]
    gateway 74ee4b9a-ba48-4a07-861e-1a8e4b9f905f
        hostname: kube-ovn-worker2
        type: geneve
            ip: 172.18.0.2
        port ts1-az0
            transit switch: ts1
            address: ["00:00:00:19:2E:F7 169.254.101.90/24"]
    gateway 7e2428b6-344c-4dd5-a0d5-972c1ccec581
        hostname: kube-ovn-control-plane
        type: geneve
            ip: 172.18.0.4
        port ts2-az0
            transit switch: ts2
            address: ["00:00:00:EA:32:BA 169.254.102.103/24"]
availability-zone az1
    gateway 034da7cb-3826-4318-81ce-6a877a9bf285
        hostname: kube-ovn1-worker
        type: geneve
            ip: 172.18.0.6
        port ts-az1
            transit switch: ts
            address: ["00:00:00:25:3A:B9 169.254.100.51/24"]
    gateway 2531a683-283e-4fb8-a619-bdbcb33539b8
        hostname: kube-ovn1-worker2
        type: geneve
            ip: 172.18.0.5
        port ts1-az1
            transit switch: ts1
            address: ["00:00:00:52:87:F4 169.254.101.118/24"]
    gateway b0efb0be-e5a7-4323-ad4b-317637a757c4
        hostname: kube-ovn1-control-plane
        type: geneve
            ip: 172.18.0.8
        port ts2-az1
            transit switch: ts2
            address: ["00:00:00:F6:93:1A 169.254.102.17/24"]
```

2

`kube-ovn-controller` IP `OVN-IC`

`docker` `OVN-IC`

```
docker run --name=ovn-ic-db -d --env "ENABLE_OVN_LEADER_CHECK="false"" --network=host --privileged  -v /etc/ovn/:/etc/ovn -v /var/run/ovn:/var/run/ovn -v /
var/log/ovn:/var/log/ovn kubeovn/kube-ovn:v1.16.16 bash start-ic-db.sh
```

`containerd` `docker`

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```
ctr -n k8s.io run -d --env "ENABLE_OVN_LEADER_CHECK="false"" --net-host --privileged --mount="type=bind,src=/etc/ovn/,dst=/etc/ovn,options=rbind:rw" --
mount="type=bind,src=/var/run/ovn,dst=/var/run/ovn,options=rbind:rw" --mount="type=bind,src=/var/log/ovn,dst=/var/log/ovn,options=rbind:rw" docker.io/kubeovn/
kube-ovn:v1.16.16 ovn-ic-db bash start-ic-db.sh
```

7.5.3

VPC Subnet CIDR `OVN-IC` Subnet CIDR

`kube-system` Namespace `ovn-ic-config` ConfigMap

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-ic-config
  namespace: kube-system
data:
  enable-ic: "true"
  az-name: "az1"
  ic-db-host: "192.168.65.3"
  ic-nb-port: "6645"
  ic-sb-port: "6646"
  gw-nodes: "az1-gw"
  auto-route: "true"
```

- `enable-ic` :
- `az-name` :
- `ic-db-host` : `OVN-IC`
- `ic-nb-port` : `OVN-IC` 6645
- `ic-sb-port` : `OVN-IC` 6646
- `gw-nodes` :
- `auto-route` :

`ovn-ic-config` ConfigMap ConfigMap ConfigMap

`ovn-ic` `ts`

```
# ovn-ic-sbctl show
availability-zone az1
    gateway deee03e0-af16-4f45-91e9-b50c3960f809
        hostname: az1-gw
        type: geneve
            ip: 192.168.42.145
        port ts-az1
            transit switch: ts
            address: ["00:00:00:50:AC:8C 169.254.100.45/24"]
availability-zone az2
    gateway e94cc831-8143-40e3-a478-90352773327b
        hostname: az2-gw
        type: geneve
            ip: 192.168.42.149
        port ts-az2
            transit switch: ts
            address: ["00:00:00:07:4A:59 169.254.100.63/24"]
```

```
# kubectl ko nbctl lr-route-list ovn-cluster
IPv4 Routes
                10.42.1.1            169.254.100.45 dst-ip (learned)
                10.42.1.3                100.64.0.2 dst-ip
                10.16.0.2                100.64.0.2 src-ip
                10.16.0.3                100.64.0.2 src-ip
                10.16.0.4                100.64.0.2 src-ip
                10.16.0.6                100.64.0.2 src-ip
             10.17.0.0/16            169.254.100.45 dst-ip (learned)
            100.65.0.0/16            169.254.100.45 dst-ip (learned)
```

1 Pod `ping` 2 Pod IP

Subnet `disableInterConnection`

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
```


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```
  name: no-advertise
spec:
  cidrBlock: 10.199.0.0/16
  disableInterConnection: true
```

7.5.4

CIDR

`kube-system` Namespace `ovn-ic-config` ConfigMap `auto-route` `false`

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-ic-config
  namespace: kube-system
data:
  enable-ic: "true"
  az-name: "az1"
  ic-db-host: "192.168.65.3"
  ic-nb-port: "6645"
  ic-sb-port: "6646"
  gw-nodes: "az1-gw"
  auto-route: "false"
```

```
[root@az1 ~]# kubectl ko nbctl show
switch a391d3a1-14a0-4841-9836-4bd930c447fb (ts)
    port ts-az1
        type: router
        router-port: az1-ts
    port ts-az2
        type: remote
        addresses: ["00:00:00:4B:E2:9F 169.254.100.31/24"]

[root@az2 ~]# kubectl ko nbctl show
switch da6138b8-de81-4908-abf9-b2224ec4edf3 (ts)
    port ts-az2
        type: router
        router-port: az2-ts
    port ts-az1
        type: remote
        addresses: ["00:00:00:FB:2A:F7 169.254.100.79/24"]
```

`az1` `az2` `169.254.100.31` `az2` `az1` `169.254.100.79`

`az1` CIDR `10.16.0.0/24` `az2` CIDR `10.17.0.0/24`

`az1` `az2`

```
kubectl ko nbctl lr-route-add ovn-cluster 10.17.0.0/24 169.254.100.31
```

`az2` `az1`

```
kubectl ko nbctl lr-route-add ovn-cluster 10.16.0.0/24 169.254.100.79
```

7.5.5 OVN-IC

1

1 Kube-OVN v1.11.16

1

2

`OVN-IC` Raft 3

`OVN-IC` leader

`docker`

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```
docker run --name=ovn-ic-db -d --env "ENABLE_OVN_LEADER_CHECK="false"" --network=host --privileged -v /etc/ovn/:/etc/ovn -v /var/run/ovn:/var/run/ovn -v /var/
log/ovn:/var/log/ovn -e LOCAL_IP="192.168.65.3"  -e NODE_IPS="192.168.65.3,192.168.65.2,192.168.65.1"   kubeovn/kube-ovn:v1.16.16 bash start-ic-db.sh
```

`containerd`

```
ctr -n k8s.io run -d --env "ENABLE_OVN_LEADER_CHECK="false"" --net-host --privileged --mount="type=bind,src=/etc/ovn/,dst=/etc/ovn,options=rbind:rw" --
mount="type=bind,src=/var/run/ovn,dst=/var/run/ovn,options=rbind:rw" --mount="type=bind,src=/var/log/ovn,dst=/var/log/ovn,options=rbind:rw"  --
env="NODE_IPS="192.168.65.3,192.168.65.2,192.168.65.1"" --env="LOCAL_IP="192.168.65.3"" docker.io/kubeovn/kube-ovn:v1.16.16 ovn-ic-db bash start-ic-db.sh
```

- `LOCAL_IP` IP
- `NODE_IPS` `OVN-IC` IP

`OVN-IC` follower

`docker`

```
docker run --name=ovn-ic-db -d --network=host --privileged -v /etc/ovn/:/etc/ovn -v /var/run/ovn:/var/run/ovn -v /var/log/ovn:/var/log/ovn -e
LOCAL_IP="192.168.65.2"  -e NODE_IPS="192.168.65.3,192.168.65.2,192.168.65.1" -e LEADER_IP="192.168.65.3"  kubeovn/kube-ovn:v1.16.16 bash start-ic-db.sh
```

`containerd`

```
ctr -n k8s.io run -d --net-host --privileged --mount="type=bind,src=/etc/ovn/,dst=/etc/ovn,options=rbind:rw" --mount="type=bind,src=/var/run/ovn,dst=/var/run/
ovn,options=rbind:rw" --mount="type=bind,src=/var/log/ovn,dst=/var/log/ovn,options=rbind:rw"  --env="NODE_IPS="192.168.65.3,192.168.65.2,192.168.65.1"" --
env="LOCAL_IP="192.168.65.2"" --env="LEADER_IP="192.168.65.3"" docker.io/kubeovn/kube-ovn:v1.16.16 ovn-ic-db bash start-ic-db.sh
```

- `LOCAL_IP` IP
- `NODE_IPS` `OVN-IC` IP
- `LEADER_IP`: `OVN-IC` leader IP

`ovn-ic-config` `OVN-IC`

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-ic-config
  namespace: kube-system
data:
  enable-ic: "true"
  az-name: "az1"
  ic-db-host: "192.168.65.3,192.168.65.2,192.168.65.1"
  ic-nb-port: "6645"
  ic-sb-port: "6646"
  gw-nodes: "az1-gw"
  auto-route: "true"
```

7.5.6 ECMP

1

ECMP ECMP path 3 ECMP path

```
kubectl edit deployment ovn-ic-server -n kube-system
```

'TS_NUM' `TS_NUM` ECMP Path

7.5.7

`ovn-ic-config` Configmap

```
kubectl -n kube-system delete cm ovn-ic-config
```

`ts`

```
kubectl ko nbctl ls-del ts
```

7.5.8 az-name

`kubectl edit` `ovn-ic-config` configmap `az-name` ovn-cni pod 10

```
ovn-appctl -t ovn-controller inc-engine/recompute
```

7.5.9

`ovn-ic-config` Configmap

```
kubectl -n kube-system delete cm ovn-ic-config
```

`ts`

```
kubectl ko nbctl ls-del ts
```

OVN-IC

`docker`

```
docker stop ovn-ic-db
docker rm ovn-ic-db
```

`containerd`

```
ctr -n k8s.io task kill ovn-ic-db
ctr -n k8s.io containers rm ovn-ic-db
```

deployment `ovn-ic-server`

```
kubectl delete deployment ovn-ic-server -n kube-system
```

master DB

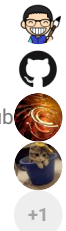
```
rm -f /etc/origin/ovn/ovn_ic_nb_db.db
rm -f /etc/origin/ovn/ovn_ic_sb_db.db
```

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7.5.10

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7.6 Submariner

Submariner Kubernetes Pod Service Kube-OVN

OVN-IC Submariner Kube-OVN Kube-OVN Service Submariner

7.6.1

- Service CIDR CIDR

7.6.2 Submariner

Note

Submariner v0.22.0 nftables Kube-OVN Submariner

```
kubectl create namespace submariner-operator
kubectl create configmap submariner-global --namespace=submariner-operator --from-literal=use-nftables=false
```

`subctl`

```
curl -Ls https://get.submariner.io | bash
export PATH=$PATH:~/.local/bin
echo export PATH=\$PATH:~/.local/bin >> ~/.profile
```

`kubeconfig` `submariner-broker`

```
subctl deploy-broker
```

`cluster0` CIDR `10.16.0.0/16` join CIDR `100.64.0.0/16` `cluster1` CIDR `11.16.0.0/16` join CIDR `100.68.0.0/16`

`kubeconfig` `cluster0` broker :

```
subctl  join broker-info.subm --clusterid  cluster0 --clustercidr 100.64.0.0/16,10.16.0.0/16  --natt=false --cable-driver vxlan --health-check=false
kubectl label nodes cluster0 submariner.io/gateway=true
```

`kubeconfig` `cluster1` broker :

```
subctl  join broker-info.subm --clusterid  cluster1 --clustercidr 100.68.0.0/16,11.16.0.0/16  --natt=false --cable-driver vxlan --health-check=false
kubectl label nodes cluster1 submariner.io/gateway=true
```

`join` `gateway, routeagent` pod , `submariner-operator` `clusterrole` :

```
- apiGroups:
  - "apps"
  resources:
  - daemonsets
  verbs:
  - create
  - get
  - list
  - watch
  - update
```

`subnet` `ovn-default` `centralized` `gatewayNode`

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: ovn-default
spec:
  gatewayType: centralized
  gatewayNode: "node1,node2"   #   submariner    gateway
```

submariner `gateway` `gatewayNode`

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Pod IP

`subctl`

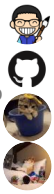
```
subctl show all
subctl diagnose all
```

Submariner Submariner

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7.6.3

7.7 Overlay

Underlay Pod IP

> VPC Overlay Pod IP IP

7.7.1

- `ip_forward`
- iptables forward Drop
- ct `INVALID`

7.7.2

`natOutgoing` `false` nat Pod IP

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: routed
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
  - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: distributed
  natOutgoing: false
```

Pod

Kubernetes

```
ip route add 10.166.0.0/16 via 192.168.2.10 dev eth0
```

`10.166.0.0/16` `192.168.2.10` Kubernetes

IP Keepalived VIP VIP

Subnet `gatewayType` `centralized` `gatewayNode` IP

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: routed
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
  - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: centralized
  gatewayNode: "node1"
  natOutgoing: false
```

nat VPC NAT

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7.7.3

7.8 Overlay

Overlay tunnel

-
-
-

7.8.1

Kube-OVN Kubernetes Node IP

```
IFACE=eth1
```

`ens[a-z0-9]*,eth[a-z0-9]*`

`kube-ovn-cni` DaemonSet

```
args:
- --iface=eth1
```

annotation `ovn.kubernetes.io/tunnel_interface` annotation `iface` annotation

```
kubectl annotate node no1 ovn.kubernetes.io/tunnel_interface=ethx
```

7.8.2

Kube-OVN IP tunnel

- IP
-
- Overlay

1. Node IP
2. Subnet `nodeNetwork`
3. kube-ovn-daemon IP OVS
4. Pod `nodeNetwork` Pod OVS IP

`ovn.kubernetes.io/node_networks` JSON key value IP

```
kubectl annotate node <node-name> ovn.kubernetes.io/node_networks='{"storage": "192.168.100.10", "app": "172.16.0.10"}'
```

- `storage` IP `192.168.100.10` IP
- `app` IP `172.16.0.10` IP

IP

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`spec.nodeNetwork`

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: storage-subnet
spec:
  protocol: IPv4
  cidrBlock: 10.100.0.0/16
  gateway: 10.100.0.1
  nodeNetwork: storage
```

`storage-subnet` Pod `storage` tunnel

`nodeNetwork` IP `IFACE` IP

OVS IP

OVS IP

```
ovs-vsctl get open . external-ids:ovn-encap-ip
```

IP

```
"192.168.1.10,192.168.100.10,172.16.0.10"
```

IP

```
ovs-vsctl get open . external-ids:ovn-encap-ip-default
```

POD IP

Pod Pod OVS IP

```
ovs-vsctl --columns=external_ids find interface external-ids:iface-id="<pod-name>.<namespace>"
```

`nodeNetwork` `encap-ip`

```
external_ids        : {encap-ip="192.168.100.10", iface-id="test-pod.default", ...}
```

1.

- `eth0` IP `192.168.1.10` `192.168.1.11`
- `eth1` IP `10.10.10.10` `10.10.10.11`

```
kubectl annotate node node1 ovn.kubernetes.io/node_networks='{"storage": "10.10.10.10"}'
kubectl annotate node node2 ovn.kubernetes.io/node_networks='{"storage": "10.10.10.11"}'
```

2.

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: storage-net
spec:
  protocol: IPv4
  cidrBlock: 10.200.0.0/16
  gateway: 10.200.0.1
  nodeNetwork: storage
```


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```
  namespaces:
  - storage-namespace
```

3. POD

```
apiVersion: v1
kind: Pod
metadata:
  name: storage-pod
  namespace: storage-namespace
spec:
  containers:
  - name: app
    image: docker.io/library/nginx:alpine
```

Pod `eth1`

1.
2. IP IP
3. Overlay Underlay Underlay Underlay
4. `nodeNetwork` IP
5. Pod `nodeNetwork` Pod
6. `ovn.kubernetes.io/node_networks` Pod

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7.9 BGP

Kube-OVN Pods Subnets Services EIPs IP BGP IP `kube-ovn-speaker` Pod Subnet annotation

EIP BGP VPC NAT Gateway EIPs

7.9.1 kube-ovn-speaker

`kube-ovn-speaker` GoBGP

`kube-ovn-speaker`

```
kubectl label nodes speaker-node-1 ovn.kubernetes.io/bgp=true
kubectl label nodes speaker-node-2 ovn.kubernetes.io/bgp=true
```

kube-ovn-speaker ECMP

yaml:

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/yamls/speaker.yaml
```

yaml

```
- --neighbor-address=10.32.32.254
- --neighbor-ipv6-address=2409:AB00:AB00:2000::AFB:8AFE
- --neighbor-as=65030
- --cluster-as=65000
```

```
- --neighbor-address=10.32.32.252,10.32.32.253
- --neighbor-ipv6-address=2409:AB00:AB00:2000::AFB:8AFC,2409:AB00:AB00:2000::AFB:8AFD
- --neighbor-as=65030
- --cluster-as=65000
```

- `neighbor-address` : BGP Peer
- `neighbor-as` : BGP Peer AS
- `cluster-as` : AS

yaml:

```
kubectl apply -f speaker.yaml
```

7.9.2 Pod/Subnet

BGP Subnet `natOutgoing` `false` Pod IP

annotation

```
kubectl annotate pod sample ovn.kubernetes.io/bgp=true
kubectl annotate subnet ovn-default ovn.kubernetes.io/bgp=true
```

annotation

```
kubectl annotate pod sample ovn.kubernetes.io/bgp-
kubectl annotate subnet ovn-default ovn.kubernetes.io/bgp-
```

BGP

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7.9.3 `ClusterIP` Service

Service ClusterIP `kube-ovn-speaker` `--announce-cluster-ip` `true` BGP

annotation

```
kubectl annotate service sample ovn.kubernetes.io/bgp=true
```

annotation

```
kubectl annotate service sample ovn.kubernetes.io/bgp-
```

7.9.4 EIPs

EIPs VPC NAT Gateway `VpcNatGateway` BGP BGP Sidecar

VPC NAT Gateway BGP BGP Speaker Sidecar `NetworkAttachmentDefinition` NAD VPC Subnet Sidecar Kubernetes API EIPs VPC CoreDNS NAD

`NetworkAttachmentDefinition` `Subnet` `provider` `{nadName}.{nadNamespace}.ovn`

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: api-ovn-nad
  namespace: default
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "api-ovn-nad.default.ovn"
    }'
---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: vpc-apiserver-subnet
spec:
  protocol: IPv4
  cidrBlock: 100.100.100.0/24
  provider: api-ovn-nad.default.ovn
```

`ovn-vpc-nat-config` ConfigMap `apiNadProvider` BGP Speaker :

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-vpc-nat-config
  namespace: kube-system
data:
  apiNadProvider: api-ovn-nad.default.ovn              # What NetworkAttachmentDefinition provider to use so that the sidecar
                                                       # can access the K8S API, as it can't by default due to VPC segmentation
  bgpSpeakerImage: docker.io/kubeovn/kube-ovn:v1.13.0  # Sets the BGP speaker image used
  image: docker.io/kubeovn/vpc-nat-gateway:v1.13.0
```

`ovn-default` `provider`

```
provider: api-ovn-nad.default.ovn
```

VPC NAT Gateway BGP

```
kind: VpcNatGateway
apiVersion: kubeovn.io/v1
metadata:
  name: vpc-natgw
spec:
  vpc: vpc1
  subnet: net1
  lanIp: 10.0.1.10
  bgpSpeaker:
    enabled: true
    asn: 65500
    remoteAsn: 65000
    neighbors:
      - 100.127.4.161
      - fd:01::1
    enableGracefulRestart: true # Optional
```


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```
    routerId: 1.1.1.1           # Optional
    holdTime: 1m                # Optional
    password: "password123"     # Optional
    extraArgs:                  # Optional, passed directly to the BGP speaker
      - -v5                     # Enables verbose debugging of the BGP speaker sidecar
  selector:
    - "kubernetes.io/os: linux"
  externalSubnets:
  - ovn-vpc-external-network # Network on which we'll speak BGP and receive/send traffic to the outside world
                             # BGP neighbors need to be on that network
```

BGP EIP

```
kubectl annotate eip sample ovn.kubernetes.io/bgp=true
```

7.9.5

`kube-ovn-speaker` :

- **Cluster**: speaker Pod IPs/Subnet CIDRs IP CIDR Pod speaker Pod Pod Subnet
- **Local**: Pod IPs Pod Cluster Pod

> : `Local` `kube-ovn-speaker` Pod speaker
>
> `Cluster` Pod/Subnet annotation `ovn.kubernetes.io/bgp`

- `ovn.kubernetes.io/bgp=cluster` `ovn.kubernetes.io/bgp=true` `Cluster`
- `ovn.kubernetes.io/bgp=local` `Local`

> Service `kube-proxy` `ClusterIP` Service `Cluster`

7.9.6 BGP

`kube-ovn-speaker` BGP

- `announce-cluster-ip` : Service `false`
- `auth-password` : BGP peer
- `holdtime` : BGP 90
- `graceful-restart` : BGP Graceful Restart
- `graceful-restart-time` : BGP Graceful restart time RFC4724 3
- `graceful-restart-deferral-time` : BGP Graceful restart deferral time RFC4724 4.1
- `passivemode` : Speaker passive peer
- `ebgp-multihop` : ebgp ttl 1

7.9.7 BGP routes debug

```
# show peer neighbor
gobgp neighbor

# show announced routes to one peer
gobgp neighbor 10.32.32.254 adj-out
```

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7.9.8

7.10 MetalLB Kube-OVN Underlay

MetalLB Kubernetes Kubernetes Service

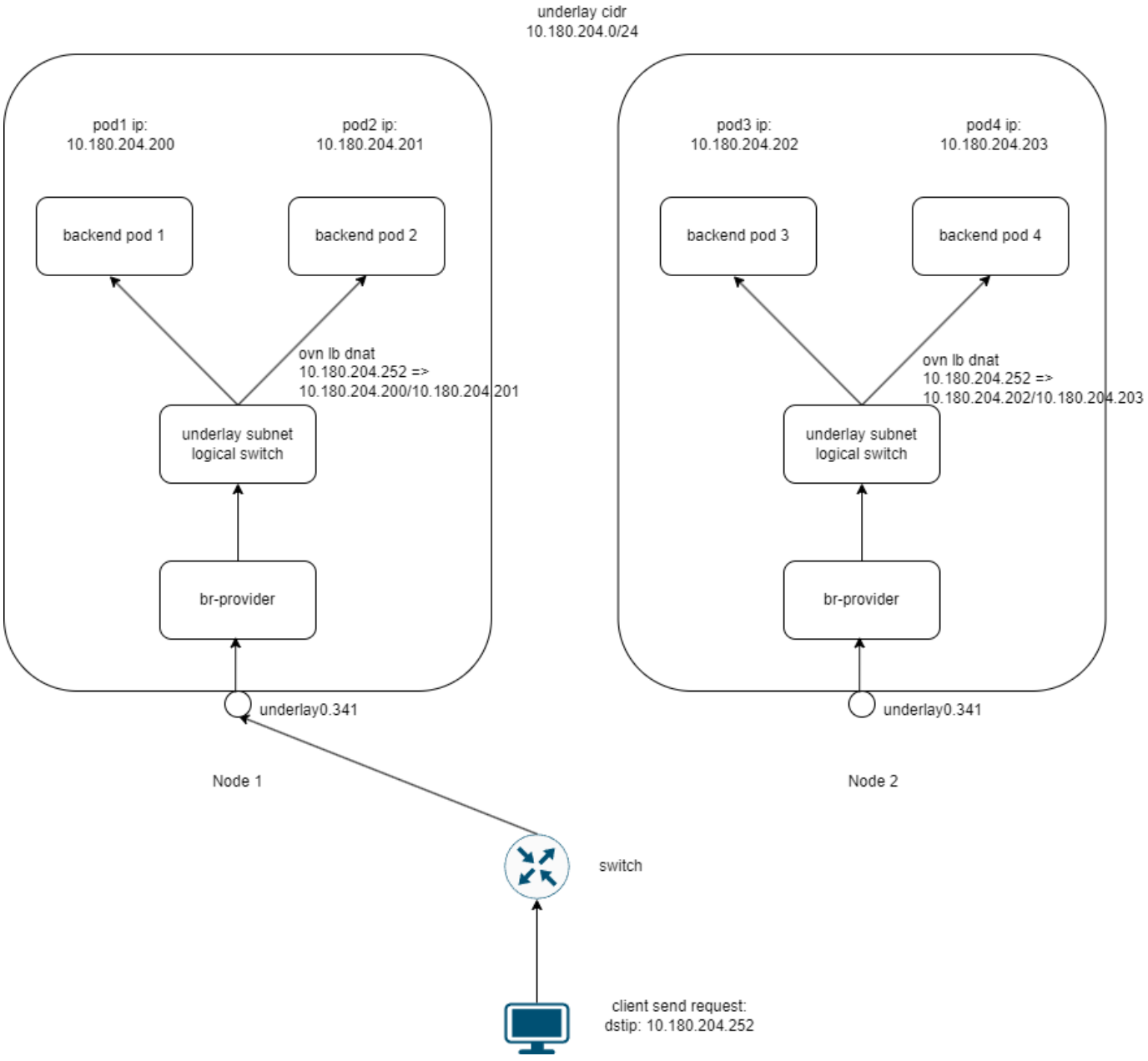
MetalLB Kube-OVN Underlay

7.10.1

Kube-OVN 1.14.0 MetalLB Underlay

- MetalLB IP
- Pod MetalLB VIP Underlay
- IP SNAT

7.10.2



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1 MetalLB VIP Kube-OVN Underlay

MetalLB Kube-OVN Underlay

1. VIP 10.180.204.252 IP MetalLB L2 Node1 metallb VIP
2. VIP `underlay0.341`
3. `br-provider` Underlay
4. `br-provider` OpenFlow OVN
5. `underlay subnet` OVN `ovn lb dnat`
6. OVN Pod

10.180.204.0/24 VIP Pod IP

7.10.3

- Kube-OVN `--enable-ovn-lb-prefer-local=true`
- Underlay `enableExternalLBAddress=true`
- Underlay `excludeIps` MetalLB IP

7.10.4

1. Kube-OVN

Kube-OVN Kube-OVN `--enable-ovn-lb-prefer-local=true` `--ls-ct-skip-dst-lport-ips=false`

```
#   kube-ovn-controller   Deployment
kubectl edit deployment -n kube-system kube-ovn-controller
```

```
--enable-ovn-lb-prefer-local=true
--ls-ct-skip-dst-lport-ips=false
```

2. Underlay

Underlay LoadBalancer excludeIps MetalLB IP

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: underlay-subnet
spec:
  protocol: IPv4
  provider: ovn
  cidrBlock: 10.180.204.0/24  #
  gateway: 10.180.204.1
  excludeIps:
  - 10.180.204.250
  - 10.180.204.251
  - 10.180.204.252  # MetalLB            VIP 10.180.204.252
  natOutgoing: false
  enableExternalLBAddress: true   #     subnet   cidr    ip      metallb vip
```

3. MetalLB

MetalLB MetalLB

```
kubectl apply -f https://raw.githubusercontent.com/metallb/metallb/v0.13.7/config/manifests/metallb-native.yaml
```

MetalLB L2

```
apiVersion: metallb.io/v1beta1
kind: IPAddressPool
metadata:
```


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```
  name: underlay-pool
  namespace: metallb-system
spec:
  addresses:
  - 10.180.204.250-10.180.204.254  #       VIP 10.180.204.252
---
apiVersion: metallb.io/v1beta1
kind: L2Advertisement
metadata:
  name: l2-advert
  namespace: metallb-system
spec:
  ipAddressPools:
  - underlay-pool
```

4. LoadBalancer Service

LoadBalancer Service Underlay Pod

```
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: nginx
  name: deploy-169402624
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      annotations:
        ovn.kubernetes.io/logical_switch: underlay-subnet
      labels:
        app: nginx
    spec:
      containers:
      - args:
        - netexec
        - --http-port
        - "80"
        image: kubeovn/agnhost:2.47
        imagePullPolicy: IfNotPresent
        name: nginx
---
apiVersion: v1
kind: Service
metadata:
  name: nginx-lb
spec:
  externalTrafficPolicy: Local
  ipFamilies:
  - IPv4
  ipFamilyPolicy: PreferDualStack
  ports:
  - port: 80
    protocol: TCP
    targetPort: 80
  selector:
    app: nginx
  type: LoadBalancer
```

7.10.5

1. Service MetalLB IP

```
kubectl get svc nginx-lb
```

EXTERNAL-IP IP 10.180.204.252

1. Service IP

```
curl http://10.180.204.252
```

1. Pod

Service endpoints Pod

```
#    Service   endpoints
kubectl get endpoints nginx-lb
```


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```
#    Pod
kubectl get pods -l app=nginx -o wide
```

1. IP

nginx Pod IP IP SNAT IP

```
kubectl exec -it $(kubectl get pods -l app=nginx -o name | head -n1) -- cat /var/log/nginx/access.log
```

7.10.6

IP

MetalLB Underlay CIDR Underlay `excludeIps` IP

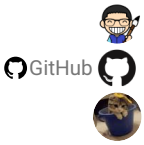
MetalLB Kube-OVN Underlay `underlay0.341` VLAN VLAN ARP MetalLB VIP

- Kube-OVN `--enable-ovn-lb-prefer-local=true` - Service `externalTrafficPolicy: Local`

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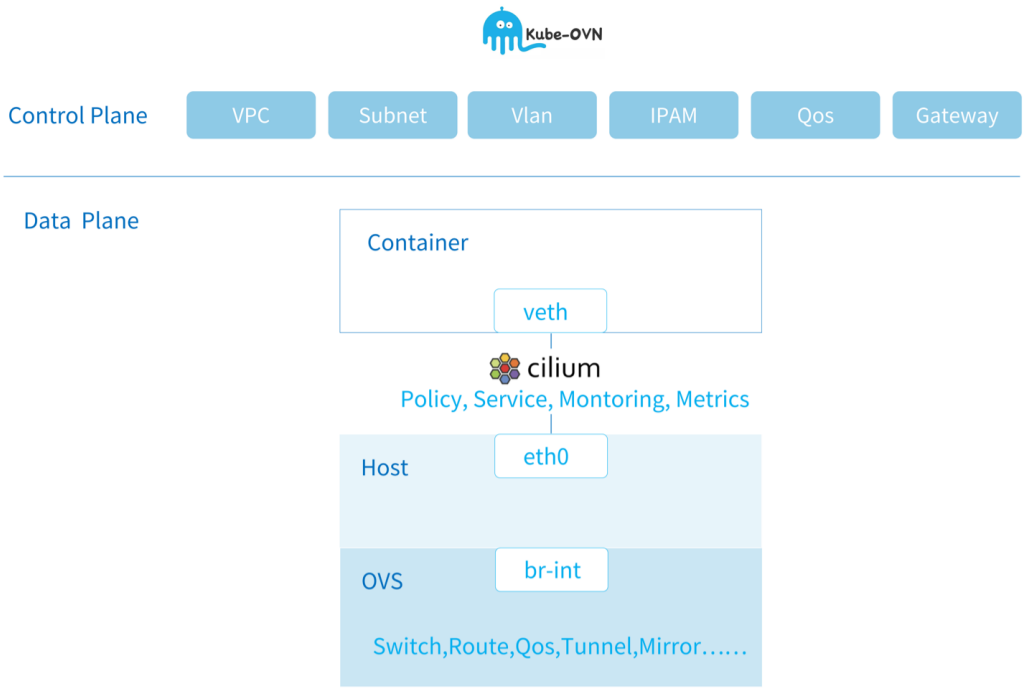
7.10.7

7.11 Cilium

Cilium eBPF Kube-OVN CNI Chaining Kube-OVN eBPF

Cilium Kube-OVN

-
- Hubble



7.11.1

1. Linux 4.19 eBPF
2. Helm Cilium Helm Installing Helm

7.11.2 Kube-OVN

Cilium Kube-OVN `networkpolicy` CNI

`install.sh`

```
ENABLE_NP=false
CNI_CONFIG_PRIORITY=10
```

`kube-ovn-controller` `networkpolicy`

```
args:
- --enable-np=false
```

`kube-ovn-cni` CNI

```
args:
- --cni-conf-name=10-kube-ovn.conflist
```


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Kube-OVN Cilium

```
mv /etc/cni/net.d/01-kube-ovn.conflist /etc/cni/net.d/10-kube-ovn.conflist
```

7.11.3 Cilium

`chaining.yaml` Cilium generic-veth

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: cni-configuration
  namespace: kube-system
data:
  cni-config: |-
    {
      "name": "generic-veth",
      "cniVersion": "0.3.1",
      "plugins": [
        {
          "type": "kube-ovn",
          "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
          "ipam": {
              "type": "kube-ovn",
              "server_socket": "/run/openvswitch/kube-ovn-daemon.sock"
          }
        },
        {
          "type": "portmap",
          "snat": true,
          "capabilities": {"portMappings": true}
        },
        {
          "type": "cilium-cni"
        }
      ]
    }
```

```
kubectl apply -f chaining.yaml
```

Helm Cilium

```
helm repo add cilium https://helm.cilium.io/
helm install cilium cilium/cilium --version 1.11.6 \
    --namespace kube-system \
    --set cni.chainingMode=generic-veth \
    --set cni.customConf=true \
    --set cni.configMap=cni-configuration \
    --set tunnel=disabled \
    --set enableIPv4Masquerade=false \
    --set devices="eth+ ovn0 genev_sys_6081 vxlan_sys_4789" \
    --set enableIdentityMark=false
```

Cilium

```
# cilium  status
    /¯¯\
 /¯¯\__/¯¯\    Cilium:         OK
 \__/¯¯\__/    Operator:       OK
 /¯¯\__/¯¯\    Hubble:         disabled
 \__/¯¯\__/    ClusterMesh:    disabled
    \__/

DaemonSet         cilium             Desired: 2, Ready: 2/2, Available: 2/2
Deployment        cilium-operator    Desired: 2, Ready: 2/2, Available: 2/2
Containers:       cilium             Running: 2
                  cilium-operator    Running: 2
Cluster Pods:     8/11 managed by Cilium
Image versions    cilium             quay.io/cilium/cilium:v1.11.6: 2
                  cilium-operator    quay.io/cilium/operator-generic:v1.11.6: 2
```

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7.11.4

7.12 Cilium NetworkPolicy

Kube-OVN Cilium Cilium

Cilium Cilium Cilium L3 L4

7.12.1

Pod

namespace `test` yaml test namespace label `app=test` Pod Pod

```
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: test
  name: test
  namespace: test
spec:
  replicas: 1
  selector:
    matchLabels:
      app: test
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
    type: RollingUpdate
  template:
    metadata:
      labels:
        app: test
    spec:
      containers:
      - image: docker.io/library/nginx:alpine
        imagePullPolicy: IfNotPresent
        name: nginx
```

yaml default namespace label `app=dynamic` Pod Pod

```
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: dynamic
  name: dynamic
  namespace: default
spec:
  replicas: 2
  selector:
    matchLabels:
      app: dynamic
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
    type: RollingUpdate
  template:
    metadata:
      creationTimestamp: null
      labels:
        app: dynamic
    spec:
      containers:
      - image: docker.io/library/nginx:alpine
        imagePullPolicy: IfNotPresent
        name: nginx
```

Pod Label :

```
# kubectl get pod -o wide --show-labels
NAME                       READY   STATUS    RESTARTS   AGE   IP           NODE                   NOMINATED NODE   READINESS GATES   LABELS
dynamic-7d8d7874f5-9v5c4   1/1     Running   0          28h   10.16.0.35   kube-ovn-worker        <none>           <none>            app=dynamic,pod-
template-hash=7d8d7874f5
dynamic-7d8d7874f5-s8z2n   1/1     Running   0          28h   10.16.0.36   kube-ovn-control-plane   <none>           <none>            app=dynamic,pod-
template-hash=7d8d7874f5
# kubectl get pod -o wide -n test --show-labels
NAME                       READY   STATUS    RESTARTS   AGE     IP           NODE                   NOMINATED NODE   READINESS GATES   LABELS
dynamic-7d8d7874f5-6dsg6   1/1     Running   0          7h20m   10.16.0.2    kube-ovn-control-plane   <none>           <none>            app=dynamic,pod-
template-hash=7d8d7874f5
```


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```
dynamic-7d8d7874f5-tjgtp       1/1     Running   0          7h46m   10.16.0.42   kube-ovn-worker          <none>           <none>            app=dynamic,pod-
template-hash=7d8d7874f5
label-test1-77b6764857-swq4k   1/1     Running   0          3h43m   10.16.0.12   kube-ovn-worker          <none>           <none>            app=test1,pod-
template-hash=77b6764857

#           Pod
test-54c98bc466-mft5s          1/1     Running   0          8h      10.16.0.41   kube-ovn-worker          <none>           <none>            app=test,pod-
template-hash=54c98bc466
```

L3

yaml `CiliumNetworkPolicy` :

```yaml
apiVersion: "cilium.io/v2"
kind: CiliumNetworkPolicy
metadata:
  name: "l3-rule"
  namespace: test
spec:
  endpointSelector:
    matchLabels:
      app: test
  ingress:
  - fromEndpoints:
    - matchLabels:
        app: dynamic
```

default namespace Pod Pod test namespace Pod

default namespace :

```
# kubectl exec -it dynamic-7d8d7874f5-9v5c4 -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 0 packets received, 100% packet loss
```

test namespace Pod :

```
# kubectl exec -it -n test dynamic-7d8d7874f5-6dsg6 -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes
64 bytes from 10.16.0.41: seq=0 ttl=64 time=2.558 ms
64 bytes from 10.16.0.41: seq=1 ttl=64 time=0.223 ms
64 bytes from 10.16.0.41: seq=2 ttl=64 time=0.304 ms

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.223/1.028/2.558 ms
```

Cilium `CiliumNetworkPolicy` `Namespace` Cilium

Namespace Pod **Namespace** Pod

Namespace Namespace

`CiliumNetworkPolicy` namespace :

```yaml
  ingress:
  - fromEndpoints:
    - matchLabels:
        app: dynamic
        k8s:io.kubernetes.pod.namespace: default    #      Namespace    Pod
```

`CiliumNetworkPolicy` :

```
# kubectl get cnp -n test  -o yaml l3-rule
apiVersion: cilium.io/v2
kind: CiliumNetworkPolicy
metadata:
  name: l3-rule
  namespace: test
spec:
  endpointSelector:
    matchLabels:
      app: test
  ingress:
  - fromEndpoints:
    - matchLabels:
```


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```
        app: dynamic
    - matchLabels:
        app: dynamic
        k8s:io.kubernetes.pod.namespace: default
```

default namespace Pod Pod :

```
# kubectl exec -it dynamic-7d8d7874f5-9v5c4 -n test -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes
64 bytes from 10.16.0.41: seq=0 ttl=64 time=2.383 ms
64 bytes from 10.16.0.41: seq=1 ttl=64 time=0.115 ms
64 bytes from 10.16.0.41: seq=2 ttl=64 time=0.142 ms

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.115/0.880/2.383 ms
```

Kubernetes networkpolicy Cilium Namespace Namespace

Kube-OVN Kube-OVN k8s Namespace **Pod** Pod Namespace Namespace Pod Pod

L4

yaml L4 :

```
apiVersion: "cilium.io/v2"
kind: CiliumNetworkPolicy
metadata:
  name: "l4-rule"
  namespace: test
spec:
  endpointSelector:
    matchLabels:
      app: test
  ingress:
  - fromEndpoints:
    - matchLabels:
        app: dynamic
    toPorts:
    - ports:
      - port: "80"
        protocol: TCP
```

Namespace Pod

```
# kubectl exec -it -n test dynamic-7d8d7874f5-6dsg6 -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 0 packets received, 100% packet loss
bash-5.0#
bash-5.0# curl 10.16.0.41:80
<html>
<head>
        <title>Hello World!</title>
        <link href='//fonts.googleapis.com/css?family=Open+Sans:400,700' rel='stylesheet' type='text/css'>
        <style>
        body {
                background-color: white;
                text-align: center;
                padding: 50px;
                font-family: "Open Sans","Helvetica Neue",Helvetica,Arial,sans-serif;
        }
        #logo {
                margin-bottom: 40px;
        }
        </style>
</head>
<body>
                <h1>Hello World!</h1>
                                <h3>Links found</h3>
        <h3>I am on  test-54c98bc466-mft5s</h3>
        <h3>Cookie                  =</h3>
                                    <b>KUBERNETES</b> listening in 443 available at tcp://10.96.0.1:443<br />
                                            <h3>my name is hanhouchao!</h3>
                        <h3> RequestURI='/'</h3>
</body>
</html>
```


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Namespace Pod

```
# kubectl exec -it -n test label-test1-77b6764857-swq4k -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 0 packets received, 100% packet loss
bash-5.0#
bash-5.0# curl -v 10.16.0.41:80 --connect-timeout 10
*   Trying 10.16.0.41:80...
* After 10000ms connect time, move on!
* connect to 10.16.0.41 port 80 failed: Operation timed out
* Connection timeout after 10001 ms
* Closing connection 0
curl: (28) Connection timeout after 10001 ms
```

Namespace L3

L4 ping TCP

ICMP L4

L7

chaining L7 Cilium Generic Veth Chaining

issue 12454

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7.12.2

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7.13 Cilium

Kube-OVN Cilium Cilium

Cilium Hubble Hubble

7.13.1 Hubble

Cilium Hubble Hubble

helm Hubble

```
helm upgrade cilium cilium/cilium --version 1.11.6 \
   --namespace kube-system \
   --reuse-values \
   --set hubble.relay.enabled=true \
   --set hubble.ui.enabled=true
```

Hubble `cilium status` Hubble

```
# cilium status
    /¯¯\
 /¯¯\__/¯¯\    Cilium:         OK
 \__/¯¯\__/    Operator:       OK
 /¯¯\__/¯¯\    Hubble:         OK
 \__/¯¯\__/    ClusterMesh:    disabled
    \__/

Deployment        hubble-relay       Desired: 1, Ready: 1/1, Available: 1/1
Deployment        cilium-operator    Desired: 2, Ready: 2/2, Available: 2/2
DaemonSet         cilium             Desired: 2, Ready: 2/2, Available: 2/2
Deployment        hubble-ui          Desired: 1, Ready: 1/1, Available: 1/1
Containers:       cilium             Running: 2
                  hubble-ui          Running: 1
                  hubble-relay       Running: 1
                  cilium-operator    Running: 2
Cluster Pods:     16/17 managed by Cilium
Image versions    hubble-relay       quay.io/cilium/hubble-relay:v1.11.6@sha256:fd9034a2d04d5b973f1e8ed44f230ea195b89c37955ff32e34e5aa68f3ed675a: 1
                  cilium-operator    quay.io/cilium/operator-generic:v1.11.6@sha256:9f6063c7bcaede801a39315ec7c166309f6a6783e98665f6693939cf1701bc17: 2
                  cilium             quay.io/cilium/cilium:v1.11.6@sha256:f7f93c26739b6641a3fa3d76b1e1605b15989f25d06625260099e01c8243f54c: 2
                  hubble-ui          quay.io/cilium/hubble-ui:v0.9.0@sha256:0ef04e9a29212925da6bdfd0ba5b581765e41a01f1cc30563cef9b30b457fea0: 1
                  hubble-ui          quay.io/cilium/hubble-ui-backend:v0.9.0@sha256:000df6b76719f607a9edefb9af94dfd1811a6f1b6a8a9c537cba90bf12df474b: 1
apple@bogon cilium %
```

Hubble Hubble CLI :

```
curl -L --fail --remote-name-all https://github.com/cilium/hubble/releases/download/v0.10.0/hubble-linux-amd64.tar.gz
sudo tar xzvfC hubble-linux-amd64.tar.gz /usr/local/bin
```

7.13.2

Cilium

`cilium connectivity test` Cilium `cilium-test` Namespace cilium-test

cilium-test namespace

```
# kubectl get all -n cilium-test
NAME                                    READY   STATUS    RESTARTS   AGE
pod/client-7df6cfbf7b-z5t2j             1/1     Running   0          21s
pod/client2-547996d7d8-nvgxg            1/1     Running   0          21s
pod/echo-other-node-d79544ccf-h14gg     2/2     Running   0          21s
pod/echo-same-node-5d466d5444-ml7tc     2/2     Running   0          21s

NAME                      TYPE       CLUSTER-IP      EXTERNAL-IP   PORT(S)          AGE
service/echo-other-node   NodePort   10.109.58.126   <none>        8080:32269/TCP   21s
service/echo-same-node    NodePort   10.108.70.32    <none>        8080:32490/TCP   21s

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/client              1/1     1            1           21s
deployment.apps/client2             1/1     1            1           21s
deployment.apps/echo-other-node     1/1     1            1           21s
deployment.apps/echo-same-node      1/1     1            1           21s

NAME                                           DESIRED   CURRENT   READY   AGE
```


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```
replicaset.apps/client-7df6cfbf7b           1         1         1       21s
replicaset.apps/client2-547996d7d8          1         1         1       21s
replicaset.apps/echo-other-node-d79544ccf   1         1         1       21s
replicaset.apps/echo-same-node-5d466d5444   1         1         1       21s
```

7.13.3

Cilium `kube-system namespace` Cilium pod `hubble observe`

```
# kubectl get pod -n kube-system -o wide
NAME                                      READY   STATUS    RESTARTS   AGE     IP           NODE                    NOMINATED NODE   READINESS GATES
cilium-d6h56                              1/1     Running   0          2d20h   172.18.0.2   kube-ovn-worker         <none>           <none>
cilium-operator-5887f78bbb-c7sb2          1/1     Running   0          2d20h   172.18.0.2   kube-ovn-worker         <none>           <none>
cilium-operator-5887f78bbb-wj8gt          1/1     Running   0          2d20h   172.18.0.3   kube-ovn-control-plane  <none>           <none>
cilium-tq5xb                              1/1     Running   0          2d20h   172.18.0.3   kube-ovn-control-plane  <none>           <none>
kube-ovn-pinger-7lgk8                     1/1     Running   0          21h     10.16.0.19   kube-ovn-control-plane  <none>           <none>
kube-ovn-pinger-msvcn                     1/1     Running   0          21h     10.16.0.18   kube-ovn-worker         <none>           <none>

# kubectl exec -it -n kube-system cilium-d6h56 -- bash
root@kube-ovn-worker:/home/cilium# hubble observe --from-namespace kube-system
Jul 29 03:24:25.551: kube-system/kube-ovn-pinger-msvcn:35576 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 29 03:24:25.561: kube-system/kube-ovn-pinger-msvcn:35576 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: RST)
Jul 29 03:24:25.561: kube-system/kube-ovn-pinger-msvcn:35576 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK, RST)
Jul 29 03:24:25.572: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: SYN)
Jul 29 03:24:25.572: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.651: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 29 03:24:25.661: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: RST)
Jul 29 03:24:25.661: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK, RST)
Jul 29 03:24:25.761: kube-system/kube-ovn-pinger-msvcn:52004 -> 172.18.0.3:6443 to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 29 03:24:25.779: kube-system/kube-ovn-pinger-msvcn -> kube-system/kube-ovn-pinger-7lgk8 to-stack FORWARDED (ICMPv4 EchoRequest)
Jul 29 03:24:25.779: kube-system/kube-ovn-pinger-msvcn <- kube-system/kube-ovn-pinger-7lgk8 to-endpoint FORWARDED (ICMPv4 EchoReply)
Jul 29 03:24:25.866: kube-system/hubble-ui-7596f7ff6f-7j6f2:55836 <- kube-system/hubble-relay-959988db5-zc5vv:4245 to-stack FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.866: kube-system/hubble-ui-7596f7ff6f-7j6f2:55836 <- kube-system/hubble-relay-959988db5-zc5vv:80 to-endpoint FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.866: kube-system/hubble-ui-7596f7ff6f-7j6f2:55836 -> kube-system/hubble-relay-959988db5-zc5vv:4245 to-stack FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.866: kube-system/hubble-ui-7596f7ff6f-7j6f2:55836 -> kube-system/hubble-relay-959988db5-zc5vv:4245 to-endpoint FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.975: kube-system/kube-ovn-pinger-7lgk8 -> kube-system/kube-ovn-pinger-msvcn to-endpoint FORWARDED (ICMPv4 EchoRequest)
Jul 29 03:24:25.975: kube-system/kube-ovn-pinger-7lgk8 <- kube-system/kube-ovn-pinger-msvcn to-stack FORWARDED (ICMPv4 EchoReply)
Jul 29 03:24:25.979: kube-system/kube-ovn-pinger-msvcn -> 172.18.0.3 to-stack FORWARDED (ICMPv4 EchoRequest)
Jul 29 03:24:26.037: kube-system/coredns-6d4b75cb6d-lbgjg:36430 -> 172.18.0.3:6443 to-stack FORWARDED (TCP Flags: ACK)
Jul 29 03:24:26.282: kube-system/kube-ovn-pinger-msvcn -> 172.18.0.2 to-stack FORWARDED (ICMPv4 EchoRequest)
```

Hubble Relay Hubble

Hubble API Hubble Service `kubectl port-forward deployment/hubble-relay -n kube-system 4245:4245`

`kubectl port-forward`

`hubble status`

```
# hubble status
Healthcheck (via localhost:4245): Ok
Current/Max Flows: 8,190/8,190 (100.00%)
Flows/s: 22.86
Connected Nodes: 2/2
```


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`hubble observe` cilium-test

```
apple@bogon ovn-test % hubble observe
Jul 28 08:00:07.033: cilium-test/client-7df6cfbf7b-qn7q6:56906 (ID:15432) -> kube-system/coredns-6d4b75cb6d-b444j:53 (ID:11307) to-endpoint FORWARDED (UDP)
Jul 28 08:00:07.033: cilium-test/client-7df6cfbf7b-qn7q6:56906 (ID:15432) <- kube-system/coredns-6d4b75cb6d-b444j:53 (ID:11307) to-stack FORWARDED (UDP)
Jul 28 08:00:07.095: 100.64.0.3:43037 (world) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: SYN)
Jul 28 08:00:07.095: 100.64.0.3:43037 (world) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.096: 100.64.0.3:43037 (world) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.096: 100.64.0.3:43037 (world) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.096: 100.64.0.3:43037 (world) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.097: 100.64.0.3:43037 (world) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.249: 100.64.0.2:33419 (host) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: SYN)
Jul 28 08:00:07.249: 100.64.0.2:33419 (host) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.250: 100.64.0.2:33419 (host) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.251: 100.64.0.2:33419 (host) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.251: 100.64.0.2:33419 (host) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.251: 100.64.0.2:33419 (host) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.323: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) <- 172.18.0.3:31514 (kube-apiserver) to-endpoint FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.323: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) -> 172.18.0.3:31514 (kube-apiserver) to-stack FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.324: 100.64.0.2:34927 (world) -> cilium-test/echo-same-node-5d466d5444-4vllm:8080 (ID:15976) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.324: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) -> 172.18.0.3:31514 (kube-apiserver) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.324: 100.64.0.2:34927 (world) -> cilium-test/echo-same-node-5d466d5444-4vllm:8080 (ID:15976) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.324: 100.64.0.2:34927 (world) <- cilium-test/echo-same-node-5d466d5444-4vllm:8080 (ID:15976) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.324: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) <- 172.18.0.3:31514 (kube-apiserver) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.324: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) -> 172.18.0.3:31514 (kube-apiserver) to-stack FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.324: 100.64.0.2:34927 (world) -> cilium-test/echo-same-node-5d466d5444-4vllm:8080 (ID:15976) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.340: kube-system/hubble-relay-959988db5-zc5vv:46938 (ID:53347) -> 172.18.0.2:4244 (host) to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.340: kube-system/hubble-relay-959988db5-zc5vv:44656 (ID:53347) -> 172.18.0.3:4244 (kube-apiserver) to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.340: kube-system/hubble-relay-959988db5-zc5vv:46938 (ID:53347) <- 172.18.0.2:4244 (host) to-endpoint FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.341: kube-system/hubble-relay-959988db5-zc5vv:44656 (ID:53347) <- 172.18.0.3:4244 (kube-apiserver) to-endpoint FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.409: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: SYN)
Jul 28 08:00:07.409: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: SYN)
Jul 28 08:00:07.409: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.410: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.410: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.410: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.410: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.411: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.411: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.412: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.412: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.412: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.412: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
apple@bogon ovn-test %
```

`hubble observe` `hubble help observe` Hubble CLI

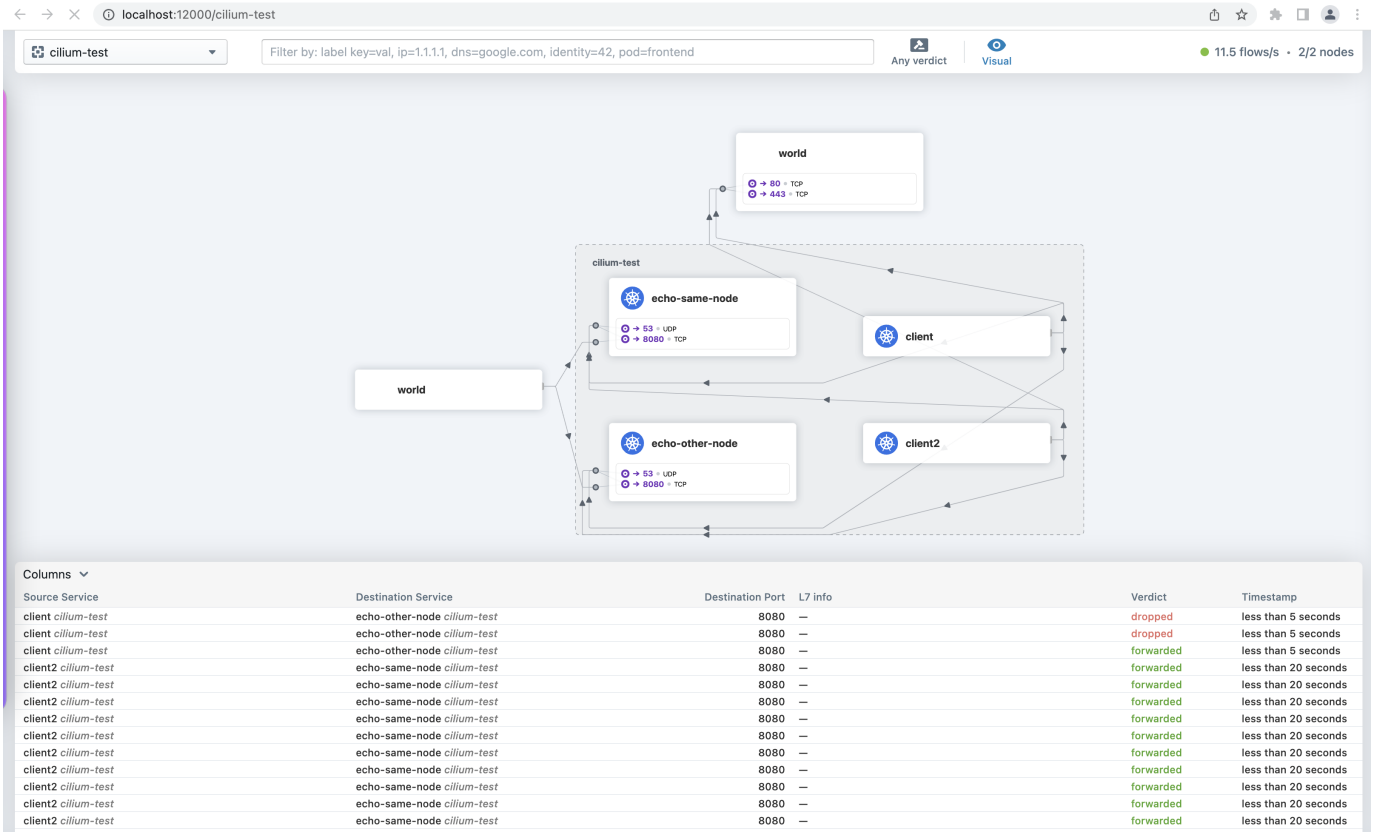
7.13.4 UI

`cilium status` Hubble UI Hubble UI

`cilium hubble ui` `hubble-ui service` Hubble UI `http://localhost:12000` UI

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`cilium-test` namespace Cilium



7.13.5 Hubble

Hubble Pod Hubble

`hubble.metrics.enabled` :

```
helm upgrade cilium cilium/cilium --version 1.11.6 \
   --namespace kube-system \
   --reuse-values \
   --set hubble.relay.enabled=true \
   --set hubble.ui.enabled=true \
   --set hubble.metrics.enabled="{dns,drop,tcp,flow,icmp,http}"
```

kube-system namespace `hubble-metrics` Endpoints Hubble :

```
# curl 172.18.0.2:9091/metrics
# HELP hubble_drop_total Number of drops
# TYPE hubble_drop_total counter
hubble_drop_total{protocol="ICMPv6",reason="Unsupported L3 protocol"} 2
# HELP hubble_flows_processed_total Total number of flows processed
# TYPE hubble_flows_processed_total counter
hubble_flows_processed_total{protocol="ICMPv4",subtype="to-endpoint",type="Trace",verdict="FORWARDED"} 335
hubble_flows_processed_total{protocol="ICMPv4",subtype="to-stack",type="Trace",verdict="FORWARDED"} 335
hubble_flows_processed_total{protocol="ICMPv6",subtype="",type="Drop",verdict="DROPPED"} 2
hubble_flows_processed_total{protocol="TCP",subtype="to-endpoint",type="Trace",verdict="FORWARDED"} 8282
hubble_flows_processed_total{protocol="TCP",subtype="to-stack",type="Trace",verdict="FORWARDED"} 6767
hubble_flows_processed_total{protocol="UDP",subtype="to-endpoint",type="Trace",verdict="FORWARDED"} 1642
hubble_flows_processed_total{protocol="UDP",subtype="to-stack",type="Trace",verdict="FORWARDED"} 1642
# HELP hubble_icmp_total Number of ICMP messages
# TYPE hubble_icmp_total counter
hubble_icmp_total{family="IPv4",type="EchoReply"} 335
hubble_icmp_total{family="IPv4",type="EchoRequest"} 335
hubble_icmp_total{family="IPv4",type="RouterSolicitation"} 2
# HELP hubble_tcp_flags_total TCP flag occurrences
# TYPE hubble_tcp_flags_total counter
hubble_tcp_flags_total{family="IPv4",flag="FIN"} 2043
hubble_tcp_flags_total{family="IPv4",flag="RST"} 301
hubble_tcp_flags_total{family="IPv4",flag="SYN"} 1169
hubble_tcp_flags_total{family="IPv4",flag="SYN-ACK"} 1169
```

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7.13.6

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7.14

Kube-OVN

7.14.1

```yaml
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: external
spec:
  cidrBlock: 172.31.0.0/16
  gatewayType: centralized
  natOutgoing: false
  externalEgressGateway: 192.168.0.1
  policyRoutingTableID: 1000
  policyRoutingPriority: 1500
```

- `natOutgoing`: `false`
- `externalEgressGateway`
- `policyRoutingTableID` TableID
- `policyRoutingPriority`

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7.14.2

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7.15 VIP IP

VIP IP IP VIP kube-ovn IP POD IP IP VIP Openstack neutron Allowed-Address-Pairs AAP Openstack octavia POD IP aliyun terway neutron IP VIP VIP POD IP VIP IP OVN Switch LB IP LB VIP VIP OVN Switch LB Rule VIP

- Allowed-Address-Pairs VIP
- Switch LB rule VIP
- Pod VIP IP

7.15.1 1. Allowed-Address-Pairs VIP

IP Pod

- Kubernetes Kubernetes Kubernetes Underlay Subnet
- LB Subnet IP Pod

VIP Allowed-Address-Pairs IP IP

- Keepalived IP

1.1 VIP

IP IP yaml

```yaml
apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: vip-dynamic-01
spec:
  subnet: ovn-default
  type: ""
```

- `subnet`: Subnet IP
- `type`: ipam ip `switch_lb_vip` vip switch lb vip ip

VIP

```
# kubectl get vip
NAME             V4IP         PV4IP   MAC               PMAC   V6IP   PV6IP   SUBNET       READY
vip-dynamic-01   10.16.0.12           00:00:00:F0:DB:25                       ovn-default   true
```

VIP `10.16.0.12` IP

1.2 VIP

VIP IP yaml

```yaml
apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: static-vip01
spec:
  subnet: ovn-default
  v4ip: "10.16.0.121"
```

- `subnet`: Subnet IP
- `v4ip`: IP `subnet` CIDR

VIP

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```
# kubectl get vip
NAME             V4IP         PV4IP   MAC                PMAC   V6IP   PV6IP   SUBNET        READY
static-vip01   10.16.0.121           00:00:00:F0:DB:26                        ovn-default   true
```

VIP IP

1.3 Pod VIP AAP

Pod annotation VIP AAP Pod labels VIP `selector`

Pod annotation VIP ovn.kubernetes.io/aaps: vip-aap,vip-aap2,vip-aap3

AAP Pod AAP Pod VIP subnet Port

1.3.1 VIP AAP

```yaml
apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: vip-aap
spec:
  subnet: ovn-default
  namespace: default
  selector:
    - "app: aap1"
```

VIP

- `namespace` : AAP VIP VIP AAP
- `selector` : AAP VIP Pod `namespace` Pod `key: value` AND

VIP Port

```
# kubectl ko nbctl show ovn-default
switch e32e1d3b-c539-45f4-ab19-be4e33a061f6 (ovn-default)
    port aap-vip
        type: virtual
```

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: busybox
  annotations:
    ovn.kubernetes.io/aaps: vip-aap
  labels:
    app: aap1
spec:
  containers:
    - name: busybox
      image: busybox
      command: ["sleep", "3600"]
      securityContext:
        capabilities:
          add:
            - NET_ADMIN
```

AAP

```
# kubectl ko nbctl list logical_switch_port aap-vip
_uuid               : cd930750-0533-4f06-a6c0-217ddac73272
addresses           : []
dhcpv4_options      : []
dhcpv6_options      : []
dynamic_addresses   : []
enabled             : []
external_ids        : {ls=ovn-default, vendor=kube-ovn}
ha_chassis_group    : []
mirror_rules        : []
name                : aap-vip
options             : {virtual-ip="10.16.0.100", virtual-parents="busybox.default"}
parent_name         : []
port_security       : []
tag                 : []
tag_request         : []
type                : virtual
up                  : false
```

virtual-ip VIP IP virtual-parents AAP Pod Port

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Pod

```
# kubectl exec -it busybox -- ip addr add 10.16.0.100/16 dev eth0
# kubectl exec -it busybox01 -- ip addr show eth0
35: eth0@if36: <BROADCAST,MULTICAST,UP,LOWER_UP,M-DOWN> mtu 1400 qdisc noqueue
    link/ether 00:00:00:e2:ab:0c brd ff:ff:ff:ff:ff:ff
    inet 10.16.0.7/16 brd 10.16.255.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet 10.16.0.100/16 scope global secondary eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fee2:ab0c/64 scope link
       valid_lft forever preferred_lft forever
```

Pod IP VIP IP subnet Pod IP

7.15.2 2. Switch LB rule vip

```
apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: slr-01
spec:
  subnet: ovn-default
  type: switch_lb_vip
```

- `subnet`: Subnet IP
- `type`: ipam ip `switch_lb_vip` vip switch lb vip ip

7.15.3 3. Pod VIP IP

> v1.12

IP

```
apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: pod-use-vip
spec:
  subnet: ovn-default
  type: ""
```

annotation VIP Pod

```
apiVersion: v1
kind: Pod
metadata:
  name: static-ip
  annotations:
    ovn.kubernetes.io/vip: pod-use-vip #    vip
  namespace: default
spec:
  containers:
    - name: static-ip
      image: docker.io/library/nginx:alpine
```

3.1 StatefulSet Kubevirt VM VIP

`StatefulSet` `VM` Pod VIP

VM VIP `kube-ovn-controller` `keep-vm-ip` `true` Kubevirt VM

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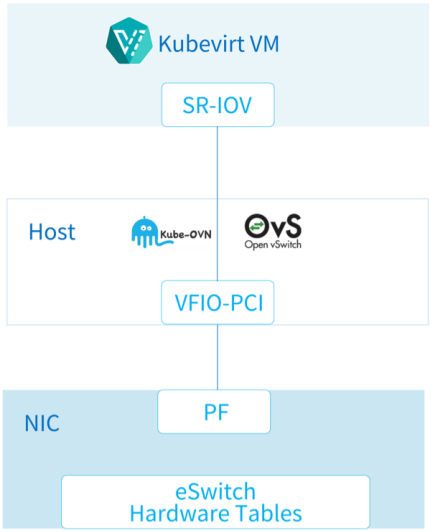
GitHub

+2

7.15.4

7.16 Mellanox Offload

Kube-OVN OVS CPU CPU Mellanox Accelerated Switching And Packet Processing (ASAP²) OVS eSwitch OVS CPU



Note

2022

7.16.1

- Mellanox CX5/CX6/CX7/BlueField ASAP²
- CentOS 8 Stream Linux 5.7
- `dp_hash` `hash` OVN LB
- bond

7.16.2 SR-IOV Device Plugin

Mellanox offload SR-IOV Device Plugin sriov-network-operator

SR-IOV Device Plugin

SR-IOV

ID `84:00.0` `84.00.1`

```
# lspci -nn | grep ConnectX-5
84:00.0 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5] [15b3:1017]
84:00.1 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5] [15b3:1017]
```


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ID

```
# ls -l /sys/class/net/ | grep 84:00.0
lrwxrwxrwx 1 root root 0 Feb 4 16:16 enp132s0f0np0 -> ../../devices/pci0000:80/0000:80:08.0/0000:84:00.0/net/enp132s0f0np0
# ls -l /sys/class/net/ | grep 84:00.1
lrwxrwxrwx 1 root root 0 Feb 4 16:16 enp132s0f1np1 -> ../../devices/pci0000:80/0000:80:08.0/0000:84:00.1/net/enp132s0f1np1
```

bond

enp132s0f0np0 enp132s0f1np1 bond1

```
# ip link show enp132s0f0np0 | grep bond
160: enp132s0f0np0: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 1500 qdisc mq master bond1 state UP mode DEFAULT group default qlen 1000
# ip link show enp132s0f1np1 | grep bond
169: enp132s0f1np1: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 1500 qdisc mq master bond1 state UP mode DEFAULT group default qlen 1000
```

bond VF

```
ifenslave -d bond1 enp132s0f0np0
ifenslave -d bond1 enp132s0f1np1
echo 0 > /sys/class/net/enp132s0f0np0/device/sriov_numvfs
echo 0 > /sys/class/net/enp132s0f1np1/device/sriov_numvfs
ip link set enp132s0f0np0 down
ip link set enp132s0f1np1 down
```

OVS SMFS DMFS

- SMFS (software-managed flow steering)
- DMFS (device-managed flow steering)

sysfs devlink API

```
#    sysfs
echo <smfs|dmfs> > /sys/class/net/enp132s0f0np0/compat/devlink/steering_mode
echo <smfs|dmfs> > /sys/class/net/enp132s0f1np1/compat/devlink/steering_mode
#    devlink
devlink dev param set pci/84:00.0 name flow_steering_mode value smfs cmode runtime
devlink dev param set pci/84:00.1 name flow_steering_mode value smfs cmode runtime
```

VF

```
# cat /sys/class/net/enp132s0f0np0/device/sriov_totalvfs
127
# cat /sys/class/net/enp132s1f0np1/device/sriov_totalvfs
127
```

VF

```
# echo '4' > /sys/class/net/enp132s0f0np0/device/sriov_numvfs
# echo '4' > /sys/class/net/enp132s1f0np1/device/sriov_numvfs
# ip link show enp132s0f0np0
160: enp132s0f0np0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN mode DEFAULT group default qlen 1000
    link/ether 08:c0:eb:74:c3:4a brd ff:ff:ff:ff:ff:ff
    vf 0 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 1 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 2 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 3 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
# ip link show enp132s0f1np1
169: enp132s0f1np1: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN mode DEFAULT group default qlen 1000
    link/ether 08:c0:eb:74:c3:4b brd ff:ff:ff:ff:ff:ff
    vf 0 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 1 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 2 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 3 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
# ip link set enp132s0f0np0 up
# ip link set enp132s0f1np1 up
```

VF ID

```
# lspci -nn | grep ConnectX-5 | grep Virtual
84:00.2 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:00.3 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:00.4 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
```

```
84:00.5 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:00.6 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:00.7 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:01.0 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:01.1 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
```

VF

```
echo 0000:84:00.2 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.3 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.4 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.5 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.6 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.7 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:01.0 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:01.1 > /sys/bus/pci/drivers/mlx5_core/unbind
```

eSwitch

```
devlink dev eswitch set pci/0000:84:00.0 mode switchdev
devlink dev eswitch set pci/0000:84:00.1 mode switchdev
ethtool -K enp132s0f0np0 hw-tc-offload on
ethtool -K enp132s0f1np1 hw-tc-offload on
```

SR-IOV VF

SR-IOV VF LAG PF OVS e-switch bond

- Active-backup
- XOR
- LACP

> SR-IOV VF LAG LAG bond bond PF VF VF bond PF
> VF PF XOR LACP PF VF PF

LACP

```
modprobe bonding mode=802.3ad
ip link set enp132s0f0np0 master bond1
ip link set enp132s0f1np1 master bond1
ip link set enp132s0f0np0 up
ip link set enp132s0f1np1 up
ip link set bond1 up
```

> bond

VF

```
echo 0000:84:00.2 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.3 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.4 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.5 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.6 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.7 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:01.0 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:01.1 > /sys/bus/pci/drivers/mlx5_core/bind
```

`NetworkManager` `NetworkManager`

```
systemctl stop NetworkManager
systemctl disable NetworkManager
```

DEVICE PLUGIN

VF Pod VF SR-IOV Device Plugin

SR-IOV Configmap

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: sriovdp-config
  namespace: kube-system
data:
  config.json: |
```

```
    {
      "resourceList": [{
          "resourcePrefix": "mellanox.com",
          "resourceName": "cx5_sriov_switchdev",
          "selectors": {
                  "vendors": ["15b3"],
                  "devices": ["1018"],
                  "drivers": ["mlx5_core"]
              }
      }
      ]
    }
```

SR-IOV Device Plugin ConfigMap ConfigMap name sriovdp-config

- `selectors`: VF
- `vendors`:
- `devices`:
- `drivers`:

`selectors` `pciAddresses` `acpiIndexes` VF SR-IOV ConfigMap

SR-IOV

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/sriov-network-device-plugin/v3.6.2/deployments/sriovdp-daemonset.yaml
```

SR-IOV Kubernetes Node

```
kubectl describe node kube-ovn-01  | grep mellanox

mellanox.com/cx5_sriov_switchdev:  8
mellanox.com/cx5_sriov_switchdev:  8
mellanox.com/cx5_sriov_switchdev  0           0
```

sriov-network-operator SR-IOV Device Plugin

node-feature-discovery

```
kubectl apply -k https://github.com/kubernetes-sigs/node-feature-discovery/deployment/overlays/default?ref=v0.11.3
```

offload annotation:

```
kubectl label nodes [offloadNicNode] feature.node.kubernetes.io/network-sriov.capable=true
```

Operator

```
git clone --depth=1 https://github.com/kubeovn/sriov-network-operator.git
kubectl apply -k sriov-network-operator/deploy
```

Operator

```
# kubectl get -n kube-system all | grep sriov
NAME                                          READY   STATUS    RESTARTS   AGE
pod/sriov-network-config-daemon-bf9nt         1/1     Running   0          8s
pod/sriov-network-operator-54d7545f65-296gb   1/1     Running   0          10s

NAME                                        DESIRED   CURRENT   READY   UP-TO-DATE   AVAILABLE   NODE
SELECTOR                                              AGE
daemonset.apps/sriov-network-config-daemon   1         1         1       1            1           beta.kubernetes.io/os=linux,feature.node.kubernetes.io/
network-sriov.capable=true   8s

NAME                                    READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/sriov-network-operator   1/1     1            1           10s

NAME                                               DESIRED   CURRENT   READY   AGE
replicaset.apps/sriov-network-operator-54d7545f65   1         1         1       10s
```

`SriovNetworkNodeState` `node1` Mellanox

```
# kubectl get sriovnetworknodestates.sriovnetwork.openshift.io -n kube-system node1 -o yaml
apiVersion: sriovnetwork.openshift.io/v1
kind: SriovNetworkNodeState
spec: ...
```


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```
status:
  interfaces:
  - deviceID: "1017"
    driver: mlx5_core
    mtu: 1500
    pciAddress: "0000:5f:00.0"
    totalvfs: 8
    vendor: "15b3"
    linkSeed: 25000Mb/s
    linkType: ETH
    mac: 08:c0:eb:f4:85:bb
    name: ens41f0np0
  - deviceID: "1017"
    driver: mlx5_core
    mtu: 1500
    pciAddress: "0000:5f:00.1"
    totalvfs: 8
    vendor: "15b3"
    linkSeed: 25000Mb/s
    linkType: ETH
    mac: 08:c0:eb:f4:85:bb
    name: ens41f1np1
```

`SriovNetworkNodePolicy` `nicSelector`

```yaml
apiVersion: sriovnetwork.openshift.io/v1
kind: SriovNetworkNodePolicy
metadata:
  name: policy
  namespace: kube-system
spec:
  nodeSelector:
    feature.node.kubernetes.io/network-sriov.capable: "true"
  eSwitchMode: switchdev
  numVfs: 3
  nicSelector:
    pfNames:
    - ens41f0np0
    - ens41f1np1
  resourceName: cx_sriov_switchdev
```

`SriovNetworkNodeState` `status`

```
# kubectl get sriovnetworknodestates.sriovnetwork.openshift.io -n kube-system node1 -o yaml

...
spec:
  interfaces:
  - eSwitchMode: switchdev
    name: ens41f0np0
    numVfs: 3
    pciAddress: 0000:5f:00.0
    vfGroups:
    - policyName: policy
      vfRange: 0-2
      resourceName: cx_sriov_switchdev
  - eSwitchMode: switchdev
    name: ens41f1np1
    numVfs: 3
    pciAddress: 0000:5f:00.1
    vfGroups:
    - policyName: policy
      vfRange: 0-2
      resourceName: cx_sriov_switchdev
status:
  interfaces
  - Vfs:
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.2
      vendor: "15b3"
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.3
      vendor: "15b3"
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.4
      vendor: "15b3"
    deviceID: "1017"
    driver: mlx5_core
    linkSeed: 25000Mb/s
    linkType: ETH
    mac: 08:c0:eb:f4:85:ab
    mtu: 1500
    name: ens41f0np0
    numVfs: 3
    pciAddress: 0000:5f:00.0
    totalvfs: 3
    vendor: "15b3"
  - Vfs:
```

```
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.5
      vendor: "15b3"
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.6
      vendor: "15b3"
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.7
      vendor: "15b3"
    deviceID: "1017"
    driver: mlx5_core
    linkSeed: 25000Mb/s
    linkType: ETH
    mac: 08:c0:eb:f4:85:bb
    mtu: 1500
    name: ens41f1np1
    numVfs: 3
    pciAddress: 0000:5f:00.1
    totalvfs: 3
    vendor: "15b3"
```

VF

```
# lspci -nn | grep ConnectX
5f:00.0 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5] [15b3:1017]
5f:00.1 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5] [15b3:1017]
5f:00.2 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.3 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.4 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.5 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.6 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.7 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
```

PF

```
# cat /sys/class/net/ens41f0np0/compat/devlink/mode
switchdev
```

7.16.3 Multus-CNI

SR-IOV Device Plugin ID Multus-CNI Kube-OVN Multus-CNI

Multus-CNI

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/v4.0.2/deployments/multus-daemonset-thick.yml
```

> multus Thin Thick SR-IOV Thick

`NetworkAttachmentDefinition`

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: sriov
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/resourceName: mellanox.com/cx5_sriov_switchdev
spec:
  config: '{
    "cniVersion": "0.3.1",
    "name": "kube-ovn",
    "plugins":[
        {
            "type":"kube-ovn",
            "server_socket":"/run/openvswitch/kube-ovn-daemon.sock",
            "provider": "sriov.default.ovn"
        },
        {
            "type":"portmap",
            "capabilities":{
                "portMappings":true
            }
        }
    ]
}'
```

- `provider`: `NetworkAttachmentDefinition` {name}.{namespace}.ovn

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7.16.4 Overlay

Kube-OVN

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/install.sh
```

`IFACE` IP

```
ENABLE_MIRROR=${ENABLE_MIRROR:-false}
HW_OFFLOAD=${HW_OFFLOAD:-true}
ENABLE_LB=${ENABLE_LB:-false}
IFACE="bond1"
#      SR-IOV   Device Plugin           bond    IFACE     bond1      bond     IFACE     enp132s0f0np0   enp132s0f1np1
```

Kube-OVN

```
bash install.sh
```

VF Pod

yaml VF Pod:

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx-overlay
  annotations:
    v1.multus-cni.io/default-network: default/sriov
    sriov.default.ovn.kubernetes.io/logical_switch: ovn-default
spec:
  containers:
  - name: nginx-overlay
    image: docker.io/library/nginx:alpine
    resources:
      requests:
        mellanox.com/cx5_sriov_switchdev: '1'
      limits:
        mellanox.com/cx5_sriov_switchdev: '1'
```

- `v1.multus-cni.io/default-network`: `NetworkAttachmentDefinition` {namespace}/{name}
- `sriov.default.ovn.kubernetes.io/logical_switch`: Pod Pod

7.16.5 Underlay

Kube-OVN

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/install.sh
```

`IFACE` IP

```
ENABLE_MIRROR=${ENABLE_MIRROR:-false}
HW_OFFLOAD=${HW_OFFLOAD:-true}
ENABLE_LB=${ENABLE_LB:-false}
IFACE=""
#     Underlay    IFACE         PF      IFACE          K8s                 PF
```

Kube-OVN

```
bash install.sh
```

VF Pod

yaml VF Pod:

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```
apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: underlay-offload
spec:
  defaultInterface: bond1

---
apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
  name: vlan0
spec:
  id: 0
  provider: underlay-offload

---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: vlan0
spec:
  protocol: IPv4
  provider: ovn
  cidrBlock: 10.10.204.0/24
  gateway: 10.10.204.254
  vlan: vlan0
  excludeIps:
  - 10.10.204.1..10.10.204.100

---
apiVersion: v1
kind: Pod
metadata:
  name: nginx-underlay
  annotations:
    k8s.v1.cni.cncf.io/networks: '[{
      "name": "sriov",
      "namespace": "default",
      "default-route": ["10.10.204.254"]
    }]'
    sriov.default.ovn.kubernetes.io/logical_switch: vlan0
spec:
  containers:
  - name: nginx-underlay
    image: docker.io/library/nginx:alpine
    resources:
      requests:
        mellanox.com/cx5_sriov_switchdev: '1'
      limits:
        mellanox.com/cx5_sriov_switchdev: '1'
```

- `v1.multus-cni.io/default-network`: `NetworkAttachmentDefinition` {namespace}/{name}

multus VF Pod VF Pod VF Pod multus

yaml VF Pod:

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-underlay-noVF
  annotations:
    ovn.kubernetes.io/logical_switch: vlan0
spec:
  containers:
  - name: nginx-underlay-noVF
    image: docker.io/library/nginx:alpine
```

VF Pod ovs-kernel e-switch

7.16.6

Pod `ovs-ovn`

```
# ovs-appctl dpctl/dump-flows -m type=offloaded
ufid:91cc45de-e7e9-4935-8f82-1890430b0f66, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0x1),recirc_id(0),dp_hash(0/
0),in_port(5b45c61b307e_h),packet_type(ns=0/0,id=0/
0),eth(src=00:00:00:c5:6d:4e,dst=00:00:00:e7:16:ce),eth_type(0x0800),ipv4(src=0.0.0.0/0.0.0.0,dst=0.0.0.0/0.0.0.0,proto=0/0,tos=0/0,ttl=0/0,frag=no),
packets:941539, bytes:62142230, used:0.260s, offloaded:yes, dp:tc, actions:54235e5753b8_h
ufid:e00768d7-e652-4d79-8182-3291d852b791, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0x1),recirc_id(0),dp_hash(0/
0),in_port(54235e5753b8_h),packet_type(ns=0/0,id=0/
0),eth(src=00:00:00:e7:16:ce,dst=00:00:00:c5:6d:4e),eth_type(0x0800),ipv4(src=0.0.0.0/0.0.0.0,dst=0.0.0.0/0.0.0.0,proto=0/0,tos=0/0,ttl=0/0,frag=no),
packets:82386659, bytes:115944854173, used:0.260s, offloaded:yes, dp:tc, actions:5b45c61b307e_h
```


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`offloaded:yes, dp:tc`

PDF Slack Support

2025 9 10

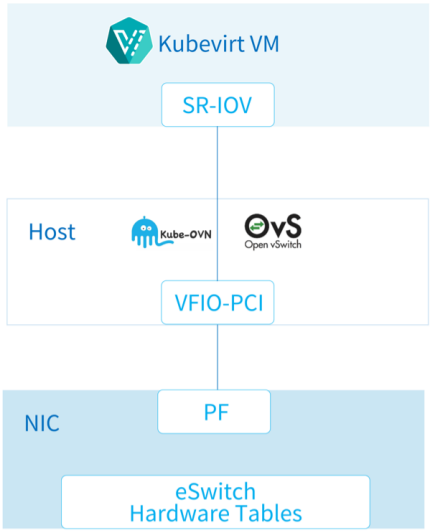
2022 5 24

GitHub 

7.16.7

7.17 Offload

Kube-OVN OVS CPU CPU Agilio CX OVS OVS CPU



Note

2022

7.17.1

- Agilio CX
- CentOS 8 Stream Linux 5.7
- `dp_hash` `hash` OVN LB

7.17.2 SR-IOV

Agilio Open vSwitch TC User Guide

```
#!/bin/bash
DEVICE=${1}
DEFAULT_ASSY=scan
ASSY=${2:-${DEFAULT_ASSY}}
APP=${3:-flower}

if [ "x${DEVICE}" = "x" -o ! -e /sys/class/net/${DEVICE} ]; then
    echo Syntax: ${0} device [ASSY] [APP]
    echo
    echo This script associates the TC Offload firmware
    echo with a Netronome SmartNIC.
    echo
    echo device: is the network device associated with the SmartNIC
```


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```
    echo ASSY: defaults to ${DEFAULT_ASSY}
    echo APP: defaults to flower. flower-next is supported if updated
    echo      firmware has been installed.
    exit 1
fi

# It is recommended that the assembly be determined by inspection
# The following code determines the value via the debug interface
if [ "${ASSY}x" = "scanx" ]; then
    ethtool -W ${DEVICE} 0
    DEBUG=$(ethtool -w ${DEVICE} data /dev/stdout | strings)
    SERIAL=$(echo "${DEBUG}" | grep "^SN:")
    ASSY=$(echo ${SERIAL} | grep -oE AMDA[0-9]{4})
fi

PCIADDR=$(basename $(readlink -e /sys/class/net/${DEVICE}/device))
FWDIR="/lib/firmware/netronome"

# AMDA0081 and AMDA0097 uses the same firmware
if [ "${ASSY}" = "AMDA0081" ]; then
    if [ ! -e ${FWDIR}/${APP}/nic_AMDA0081.nffw ]; then
       ln -sf nic_AMDA0097.nffw ${FWDIR}/${APP}/nic_AMDA0081.nffw
   fi
fi

FW="${FWDIR}/pci-${PCIADDR}.nffw"
ln -sf "${APP}/nic_${ASSY}.nffw" "${FW}"

# insert distro-specific initramfs section here...
```

```
./agilio-tc-fw-select.sh ens47np0 scan
rmmod nfp
modprobe nfp
```

VF VF

```
# cat /sys/class/net/ens3/device/sriov_totalvfs
65

# echo 4 > /sys/class/net/ens47/device/sriov_numvfs
```

7.17.3 SR-IOV Device Plugin

VF Pod VF SR-IOV Device Plugin

SR-IOV Configmap

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: sriovdp-config
  namespace: kube-system
data:
  config.json: |
    {
      "resourceList": [{
          "resourcePrefix": "corigine.com",
          "resourceName": "agilio_sriov",
          "selectors": {
                  "vendors": ["19ee"],
                  "devices": ["6003"],
                  "drivers": ["nfp_netvf"]
              }
      }
      ]
    }
```

SR-IOV Network Device Plugin intel/sriov-network-device-plugin k8snetworkplumbingwg

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/sriov-network-device-plugin/master/deployments/sriovdp-daemonset.yaml
```

SR-IOV Kubernetes Node

```
kubectl describe no containerserver  | grep corigine

corigine.com/agilio_sriov:  4
corigine.com/agilio_sriov:  4
corigine.com/agilio_sriov  0           0
```

7.17.4 Multus-CNI

SR-IOV Device Plugin ID Multus-CNI Kube-OVN Multus-CNI

Multus-CNI

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/master/deployments/multus-daemonset.yml
```

`NetworkAttachmentDefinition`

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: default
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/resourceName: corigine.com/agilio_sriov
spec:
  config: '{
    "cniVersion": "0.3.1",
    "name": "kube-ovn",
    "plugins":[
        {
            "type":"kube-ovn",
            "server_socket":"/run/openvswitch/kube-ovn-daemon.sock",
            "provider": "default.default.ovn"
        },
        {
            "type":"portmap",
            "capabilities":{
                "portMappings":true
            }
        }
    ]
}'
```

- `provider`: `NetworkAttachmentDefinition` {name}.{namespace}.ovn

7.17.5 Kube-OVN

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/install.sh
```

`IFACE` IP

```
ENABLE_MIRROR=${ENABLE_MIRROR:-false}
HW_OFFLOAD=${HW_OFFLOAD:-true}
ENABLE_LB=${ENABLE_LB:-false}
IFACE="ensp01"
```

Kube-OVN

```
bash install.sh
```

7.17.6 VF Pod

yaml VF Pod:

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx
  namespace: default
  annotations:
    v1.multus-cni.io/default-network: default/default
spec:
  containers:
    - name: nginx
      image: docker.io/library/nginx:alpine
      resources:
        requests:
          corigine.com/agilio_sriov: '1'
```


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```
        limits:
          corigine.com/agilio_sriov: '1'
```

- `v1.multus-cni.io/default-network`： `NetworkAttachmentDefinition` {namespace}/{name}

Pod `ovs-ovn`

```
# ovs-appctl dpctl/dump-flows -m type=offloaded
ufid:91cc45de-e7e9-4935-8f82-1890430b0f66, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0x1),recirc_id(0),dp_hash(0/
0),in_port(5b45c61b307e_h),packet_type(ns=0/0,id=0/
0),eth(src=00:00:00:c5:6d:4e,dst=00:00:00:e7:16:ce),eth_type(0x0800),ipv4(src=0.0.0.0/0.0.0.0,dst=0.0.0.0/0.0.0.0,proto=0/0,tos=0/0,ttl=0/0,frag=no),
packets:941539, bytes:62142230, used:0.260s, offloaded:yes, dp:tc, actions:54235e5753b8_h
ufid:e00768d7-e652-4d79-8182-3291d852b791, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0x1),recirc_id(0),dp_hash(0/
0),in_port(54235e5753b8_h),packet_type(ns=0/0,id=0/
0),eth(src=00:00:00:e7:16:ce,dst=00:00:00:c5:6d:4e),eth_type(0x0800),ipv4(src=0.0.0.0/0.0.0.0,dst=0.0.0.0/0.0.0.0,proto=0/0,tos=0/0,ttl=0/0,frag=no),
packets:82386659, bytes:115944854173, used:0.260s, offloaded:yes, dp:tc, actions:5b45c61b307e_h
```

`offloaded:yes, dp:tc`

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7.17.7

7.18 Offload

Kube-OVN OVS CPU CPU metaScale OVS OVS CPU

> **Note**
>
> 1. 2024
> 2. 1.11 Kube-OVN

7.18.1

- metaScale
- MCR
- BIOS SR-IOV VT-d

7.18.2

hw-offload Kube-OVN

1.

```
wget https://raw.githubusercontent.com/yunsilicon/kube-ovn/release-1.11/dist/images/install.sh
```

1.

`/opt/ovs-config/ovs-dpdk-config`

```
# specify log level for ovs dpdk, the value is info or dbg, default is info
VLOG=info
# specify nic offload, the value is true or false, default is true
HW_OFFLOAD=true
# specify cpu mask for ovs dpdk, not specified by default
CPU_MASK=0x02
# specify socket memory, not specified by default
SOCKET_MEM="2048,2048"
# specify encap IP
ENCAP_IP=6.6.6.208/24
# specify pci device
DPDK_DEV=0000:b3:00.0
# specify mtu, default is 1500
PF_MTU=1500
# specify bond name if bond enabled, not specified by default
BR_PHY_BOND_NAME=bond0
```

1. Kube-OVN

```
bash install.sh
```

SR-IOV

1. metaScale ID `b3:00.0`:

```
[root@k8s-master ~]# lspci -d 1f67:
b3:00.0 Ethernet controller: Device 1f67:1111 (rev 02)
b3:00.1 Ethernet controller: Device 1f67:1111 (rev 02)
```

1. ID `p3p1`

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```
ls -l /sys/class/net/ | grep b3:00.0
lrwxrwxrwx 1 root root 0 May  7 16:30 p3p1 -> ../../devices/pci0000:b2/0000:b2:00.0/0000:b3:00.0/net/p3p1
```

1. VF

```
cat /sys/class/net/p3p1/device/sriov_totalvfs
512
```

1. VF

```
echo '10' > /sys/class/net/p3p1/device/sriov_numvfs
```

1. VF

```
lspci -d 1f67:
b3:00.0 Ethernet controller: Device 1f67:1111 (rev 02)
b3:00.1 Ethernet controller: Device 1f67:1111 (rev 02)
b3:00.2 Ethernet controller: Device 1f67:1112
b3:00.3 Ethernet controller: Device 1f67:1112
b3:00.4 Ethernet controller: Device 1f67:1112
b3:00.5 Ethernet controller: Device 1f67:1112
b3:00.6 Ethernet controller: Device 1f67:1112
b3:00.7 Ethernet controller: Device 1f67:1112
b3:01.0 Ethernet controller: Device 1f67:1112
b3:01.1 Ethernet controller: Device 1f67:1112
b3:01.2 Ethernet controller: Device 1f67:1112
b3:01.3 Ethernet controller: Device 1f67:1112
```

1. switchdev

```
devlink dev eswitch set pci/0000:b3:00.0 mode switchdev
```

SR-IOV Device Plugin

1. SR-IOV ConfigMap

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: sriovdp-config
  namespace: kube-system
data:
  config.json: |
    {
        "resourceList": [{
                "resourceName": "xsc_sriov",
                "resourcePrefix": "yunsilicon.com",
                "selectors": {
                    "vendors": ["1f67"],
                    "devices": ["1012", "1112"]
                }}
        ]
    }
```

1. SR-IOV Device Plugin DevicePlugin
2. SR-IOV

```
# kubectl describe node <node name> | grep yunsilicon.com/xsc_sriov
  yunsilicon.com/xsc_sriov:  10
  yunsilicon.com/xsc_sriov:  10
  yunsilicon.com/xsc_sriov  0            0
```

Multus-CNI

1. Multus-CNI Multus-CNI

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/master/deployments/multus-daemonset.yml
```

1. NetworkAttachmentDefinition

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
```


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```
metadata:
  name: sriov-net1
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/resourceName: yunsilicon.com/xsc_sriov
spec:
  config: '{
    "cniVersion": "0.3.1",
    "name": "kube-ovn",
    "plugins":[
        {
            "type":"kube-ovn",
            "server_socket":"/run/openvswitch/kube-ovn-daemon.sock",
            "provider": "sriov-net1.default.ovn"
        },
        {
            "type":"portmap",
            "capabilities":{
                "portMappings":true
            }
        }
    ]
}'
```

SR-IOV Pod

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx
  annotations:
    v1.multus-cni.io/default-network: default/sriov-net1
spec:
  containers:
    - name: nginx
      image: docker.io/library/nginx:alpine
      resources:
        requests:
          yunsilicon.com/xsc_sriov: '1'
        limits:
          yunsilicon.com/xsc_sriov: '1'
```

Offload

Pod `ovs-ovn`

```
ovs-appctl dpctl/dump-flows type=offloaded
flow-dump from pmd on cpu core: 9
ct_state(-new+est-rel+rpl+trk),ct_mark(0/
0x3),recirc_id(0x2d277),in_port(15),packet_type(ns=0,id=0),eth(src=00:00:00:9d:fb:1a,dst=00:00:00:ce:cf:b9),eth_type(0x0800),ipv4(dst=10.16.0.14,frag=no),
packets:6, bytes:588, used:7.276s, actions:ct(zone=4,nat),recirc(0x2d278)
ct_state(-new+est-rel-rpl+trk),ct_mark(0/
0x3),recirc_id(0x2d275),in_port(8),packet_type(ns=0,id=0),eth(src=00:00:00:ce:cf:b9,dst=00:00:00:9d:fb:1a),eth_type(0x0800),ipv4(dst=10.16.0.18,frag=no),
packets:5, bytes:490, used:7.434s, actions:ct(zone=6,nat),recirc(0x2d276)
ct_state(-new+est-rel-rpl+trk),ct_mark(0/0x1),recirc_id(0x2d276),in_port(8),packet_type(ns=0,id=0),eth(src=00:00:00:ce:cf:b9,dst=00:00:00:9d:fb:1a/
01:00:00:00:00:00),eth_type(0x0800),ipv4(frag=no), packets:5, bytes:490, used:7.434s, actions:15
recirc_id(0),in_port(15),packet_type(ns=0,id=0),eth(src=00:00:00:9d:fb:1a/
01:00:00:00:00:00,dst=00:00:00:ce:cf:b9),eth_type(0x0800),ipv4(dst=10.16.0.14/255.192.0.0,frag=no), packets:6, bytes:588, used:7.277s,
actions:ct(zone=6,nat),recirc(0x2d277)
recirc_id(0),in_port(8),packet_type(ns=0,id=0),eth(src=00:00:00:ce:cf:b9/01:00:00:00:00:00,dst=00:00:00:9d:fb:1a),eth_type(0x0800),ipv4(dst=10.16.0.18/255.192.0
packets:6, bytes:588, used:7.434s, actions:ct(zone=4,nat),recirc(0x2d275)
ct_state(-new+est-rel+rpl+trk),ct_mark(0/
0x1),recirc_id(0x2d278),in_port(15),packet_type(ns=0,id=0),eth(dst=00:00:00:ce:cf:b9/01:00:00:00:00:00),eth_type(0x0800),ipv4(frag=no), packets:6, bytes:588,
used:7.277s, actions:8
```

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7.18.3

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7.19 Offload

> **Note**
>
> 2024

7.19.1

- 2200E
- HADOS
- BIOS SR-IOV

7.19.2

SR-IOV

1. 2200E vendor ID 1f47 ID 00:0a.0 00:0b.0 2200E

```
lspci | grep 1f47
00:0a.0 Ethernet controller: Device 1f47:1001 (rev 10)
00:0b.0 Ethernet controller: Device 1f47:1001 (rev 10)
```

1. VF

```
cat /sys/bus/pci/devices/0000\:00\:0a.0/sriov_totalvfs
256
```

1. VF VF

```
echo 7 > /sys/bus/pci/devices/0000\:00\:0a.0/sriov_numvfs
```

1. VF

```
lspci | grep 1f47
00:0a.0 Ethernet controller: Device 1f47:1001 (rev 10)
00:0a.1 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.2 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.3 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.4 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.5 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.6 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.7 Ethernet controller: Device 1f47:110f (rev 10)
00:0b.0 Ethernet controller: Device 1f47:1001 (rev 10)
```

SR-IOV Device Plugin

1. SR-IOV Configmap SR-IOV Device Plugin VF Pod

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: sriovdp-config
  namespace: kube-system
data:
  config.json: |
    {
        "resourceList": [{
                "resourceName": "sriov_dpu",
                "resourcePrefix": "yusur.tech",
                "selectors": {
                    "vendors": ["1f47"],
                    "devices": ["110f"]
                }}
```


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```
        ]
    }
```

1. SR-IOV Device Plugin

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/sriov-network-device-plugin/v3.6.2/deployments/sriovdp-daemonset.yaml
```

1. SR-IOV kubernetes Node

```
kubectl describe node node1 | grep yusur
  yusur.tech/sriov_dpu:  7
  yusur.tech/sriov_dpu:  7
  yusur.tech/sriov_dpu  0           0
```

7.19.3 Multus-CNI

Multus-CNI Kube-OVN SRIOV Device ID

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/v4.0.2/deployments/multus-daemonset-thick.yml
```

`NetworkAttachmentDefinition`

```
apiVersion:
  "k8s.cni.cncf.io/v1"
kind:
  NetworkAttachmentDefinition
metadata:
  name: test
  namespace: kube-system
  annotations:
    k8s.v1.cni.cncf.io/resourceName: yusur.tech/sriov_dpu
spec:
  config: '{
    "cniVersion": "0.3.1",
    "name": "kube-ovn",
    "plugins":[
        {
            "type":"kube-ovn",
            "server_socket":"/run/openvswitch/kube-ovn-daemon.sock",
            "provider": "test.kube-system.ovn"
        },
        {
            "type":"portmap",
            "capabilities":{
                "portMappings":true
            }
        }
    ]
}'
```

- `provider`: NetworkAttachmentDefinition `{name}.{namespace}.ovn`

7.19.4 Kube-OVN

1.

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.12/dist/images/install.sh
```

1. IFACE IP

```
ENABLE_MIRROR=${ENABLE_MIRROR:-false}
HW_OFFLOAD=${HW_OFFLOAD:-true}
ENABLE_LB=${ENABLE_LB:-false}
IFACE="p0"
```

1. kube-ovn

```
bash install.sh
```

VF pod

yaml VF Pod

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```yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx
  namespace: default
  annotations:
    v1.multus-cni.io/default-network: kube-system/test
spec:
  containers:
    - name: nginx
      image: docker.io/library/nginx:alpine
      resources:
        requests:
          yusur.tech/sriov_dpu: '1'
        limits:
          yusur.tech/sriov_dpu: '1'
```

- `v1.multus-cni.io/default-network`： `NetworkAttachmentDefinition` `{namespace}/{name}`

Offload

Pod `ovs-ovn`

```
# ovs-appctl dpctl/dump-flows -m type=offloaded
ufid:67c2e10f-92d4-4574-be70-d072815ff166, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0),recirc_id(0),dp_hash(0/
0),in_port(d85b161b6840_h),packet_type(ns=0/0,id=0/
0),eth(src=0a:c9:1c:70:01:09,dst=8a:18:a4:22:b7:7d),eth_type(0x0800),ipv4(src=10.0.1.10,dst=10.0.1.6,proto=6,tos=0/0x3,ttl=0/
0,frag=no),tcp(src=60774,dst=9001), packets:75021, bytes:109521630, offload_packets:75019, offload_bytes:109521498, used:3.990s,offloaded:yes,dp:tc,
actions:set(tunnel(tun_id=0x5,dst=192.168.201.12,ttl=64,tp_dst=6081,geneve({class=0x102,type=0x80,len=4,0xa0006}),flags(csum|key))),genev_sys_6081
ufid:7940666e-a0bd-42a5-8116-1e84e81bb338, skb_priority(0/0),tunnel(tun_id=0x5,src=192.168.201.12,dst=192.168.201.11,ttl=0/
0,tp_dst=6081,geneve({class=0x102,type=0x80,len=4,0x6000a}),flags(+key)),skb_mark(0/0),ct_state(0/0),ct_zone(0/0),ct_mark(0/0),ct_label(0/
0),recirc_id(0),dp_hash(0/0),in_port(genev_sys_6081),packet_type(ns=0/0,id=0/
0),eth(src=8a:18:a4:22:b7:7d,dst=0a:c9:1c:70:01:09),eth_type(0x0800),ipv4(src=10.0.1.6,dst=10.0.1.10,proto=6,tos=0/0,ttl=0/
0,frag=no),tcp(src=9001,dst=60774), packets:6946, bytes:459664, offload_packets:6944, offload_bytes:459532, used:4.170s,
dp:tc,offloaded:yes,actions:d85b161b6840_h
```

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7.19.5

7.20 DPDK

Kube-OVN OVS-DPDK KubeVirt DPDK

> KubeVirt OVS-DPDK patchVhostuser implementation KubeVirt KVM Device Plugin OVS-DPDK

7.20.1

- DPDK
- Hugepages

7.20.2 DPDK

`driverctl` DPDK

```
driverctl set-override 0000:00:0b.0 uio_pci_generic
```

7.20.3

OVS-DPDK Kube-OVN

```
kubectl label nodes <node> ovn.kubernetes.io/ovs_dp_type="userspace"
```

OVS-DPDK `/opt/ovs-config` `ovs-dpdk-config`

```
ENCAP_IP=192.168.122.193/24
DPDK_DEV=0000:00:0b.0
```

- `ENCAP_IP`:
- `DPDK_DEV`: PCI ID

7.20.4 Kube-OVN

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/install.sh
```

DPDK

```
bash install.sh --with-hybrid-dpdk
```

7.20.5

vhostuser OVS-DPDK

KVM Device Plugin KVM Device Plugin

```
kubectl apply -f https://raw.githubusercontent.com/kubevirt/kubernetes-device-plugins/master/manifests/kvm-ds.yml
```

NetworkAttachmentDefinition

```
apiVersion: k8s.cni.cncf.io/v1
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-dpdk
  namespace: default
spec:
  config: >-
    {
        "cniVersion": "0.3.0",
        "type": "kube-ovn",
```

```
        "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
        "provider": "ovn-dpdk.default.ovn",
        "vhost_user_socket_volume_name": "vhostuser-sockets",
        "vhost_user_socket_name": "sock"
    }
```

Dockerfile VM

```
FROM quay.io/kubevirt/virt-launcher:v0.46.1

# wget http://cloud.centos.org/centos/7/images/CentOS-7-x86_64-GenericCloud.qcow2
COPY CentOS-7-x86_64-GenericCloud.qcow2 /var/lib/libvirt/images/CentOS-7-x86_64-GenericCloud.qcow2
```

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: vm-config
data:
  start.sh: |
    chmod u+w /etc/libvirt/qemu.conf
    echo "hugetlbfs_mount = \"/dev/hugepages\"" >> /etc/libvirt/qemu.conf
    virtlogd &
    libvirtd &

    mkdir /var/lock

    sleep 5

    virsh define /root/vm/vm.xml
    virsh start vm

    tail -f /dev/null
  vm.xml: |
    <domain type='kvm'>
      <name>vm</name>
      <uuid>4a9b3f53-fa2a-47f3-a757-dd87720d9d1d</uuid>
      <memory unit='KiB'>2097152</memory>
      <currentMemory unit='KiB'>2097152</currentMemory>
      <memoryBacking>
        <hugepages>
          <page size='2' unit='M' nodeset='0'/>
        </hugepages>
      </memoryBacking>
      <vcpu placement='static'>2</vcpu>
      <cputune>
        <shares>4096</shares>
        <vcpupin vcpu='0' cpuset='4'/>
        <vcpupin vcpu='1' cpuset='5'/>
        <emulatorpin cpuset='1,3'/>
      </cputune>
      <os>
        <type arch='x86_64' machine='pc'>hvm</type>
        <boot dev='hd'/>
      </os>
      <features>
        <acpi/>
        <apic/>
      </features>
      <cpu mode='host-model'>
        <model fallback='allow'/>
        <topology sockets='1' cores='2' threads='1'/>
        <numa>
          <cell id='0' cpus='0-1' memory='2097152' unit='KiB' memAccess='shared'/>
        </numa>
      </cpu>
      <on_reboot>restart</on_reboot>
      <devices>
        <emulator>/usr/libexec/qemu-kvm</emulator>
        <disk type='file' device='disk'>
          <driver name='qemu' type='qcow2' cache='none'/>
          <source file='/var/lib/libvirt/images/CentOS-7-x86_64-GenericCloud.qcow2'/>
          <target dev='vda' bus='virtio'/>
        </disk>

        <interface type='vhostuser'>
          <mac address='00:00:00:0A:30:89'/>
          <source type='unix' path='/var/run/vm/sock' mode='server'/>
           <model type='virtio'/>
          <driver queues='2'>
            <host mrg_rxbuf='off'/>
          </driver>
        </interface>
        <serial type='pty'>
          <target type='isa-serial' port='0'>
            <model name='isa-serial'/>
          </target>
        </serial>
        <console type='pty'>
          <target type='serial' port='0'/>
```

```
        </console>
        <channel type='unix'>
          <source mode='bind' path='/var/lib/libvirt/qemu/channel/target/domain-1-vm/org.qemu.guest_agent.0'/>
          <target type='virtio' name='org.qemu.guest_agent.0' state='connected'/>
          <alias name='channel0'/>
          <address type='virtio-serial' controller='0' bus='0' port='1'/>
        </channel>

      </devices>
    </domain>
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: vm-deployment
  labels:
    app: vm
spec:
  replicas: 1
  selector:
    matchLabels:
      app: vm
  template:
    metadata:
      labels:
        app: vm
      annotations:
        k8s.v1.cni.cncf.io/networks: default/ovn-dpdk
        ovn-dpdk.default.ovn.kubernetes.io/ip_address: 10.16.0.96
        ovn-dpdk.default.ovn.kubernetes.io/mac_address: 00:00:00:0A:30:89
    spec:
      nodeSelector:
        ovn.kubernetes.io/ovs_dp_type: userspace
      securityContext:
        runAsUser: 0
      volumes:
        - name: vhostuser-sockets
          emptyDir: {}
        - name: xml
          configMap:
            name: vm-config
        - name: hugepage
          emptyDir:
            medium: HugePages-2Mi
        - name: libvirt-runtime
          emptyDir: {}
      containers:
        - name: vm
          image: vm-vhostuser:latest
          command: ["bash", "/root/vm/start.sh"]
          securityContext:
            capabilities:
              add:
                - NET_BIND_SERVICE
                - SYS_NICE
                - NET_RAW
                - NET_ADMIN
            privileged: false
            runAsUser: 0
          resources:
            limits:
              cpu: '2'
              devices.kubevirt.io/kvm: '1'
              memory: '8784969729'
              hugepages-2Mi: 2Gi
            requests:
              cpu: 666m
              devices.kubevirt.io/kvm: '1'
              ephemeral-storage: 50M
              memory: '4490002433'
          volumeMounts:
            - name: vhostuser-sockets
              mountPath: /var/run/vm
            - name: xml
              mountPath: /root/vm/
            - mountPath: /dev/hugepages
              name: hugepage
            - name: libvirt-runtime
              mountPath: /var/run/libvirt
```

Pod

```
# virsh set-user-password vm root 12345
Password set successfully for root in vm

# virsh console vm
Connected to domain 'vm'
Escape character is ^] (Ctrl + ])

CentOS Linux 7 (Core)
Kernel 3.10.0-1127.el7.x86_64 on an x86_64
```

```
localhost login: root
Password:
Last login: Fri Feb 25 09:52:54 on ttyS0
```

```
ip link set eth0 mtu 1400
ip addr add 10.16.0.96/16 dev eth0
ip ro add default via 10.16.0.1
ping 114.114.114.114
```

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7.20.6

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7.21 OpenStack

OpenStack Kubernetes OpenStack Neutron OVN Kube-OVN OVN OpenStack Kubernetes

7.21.1

OVN-IC Kubernetes OpenStack Kubernetes

1. OpenStack Kubernetes CIDR
2.
3. IP
4. Kubernetes OpenStack VPC

OVN-IC

`OVN-IC`

```
docker run --name=ovn-ic-db -d --network=host -v /etc/ovn/:/etc/ovn -v /var/run/ovn:/var/run/ovn -v /var/log/ovn:/var/log/ovn kubeovn/kube-ovn:v1.16.16 bash
start-ic-db.sh
```

Kubernetes

`kube-system` Namespace `ovn-ic-config` ConfigMap

```yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-ic-config
  namespace: kube-system
data:
  enable-ic: "true"
  az-name: "az1"
  ic-db-host: "192.168.65.3"
  ic-nb-port: "6645"
  ic-sb-port: "6646"
  gw-nodes: "az1-gw"
  auto-route: "true"
```

- `enable-ic`:
- `az-name`:
- `ic-db-host`: `OVN-IC`
- `ic-nb-port`: `OVN-IC` 6645
- `ic-sb-port`: `OVN-IC` 6646
- `gw-nodes`:
- `auto-route`:

OpenStack

Kubernetes

```
# openstack router create router0
# openstack router list
+--------------------------------------+---------+--------+-------+---------------------------------+
| ID                                   | Name    | Status | State | Project                         |
+--------------------------------------+---------+--------+-------+---------------------------------+
| d5b38655-249a-4192-8046-71aa4d2b4af1 | router0 | ACTIVE | UP    | 98a29ab7388347e7b5ff8bdd181ba4f9 |
+--------------------------------------+---------+--------+-------+---------------------------------+
```


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OpenStack OVN

```
ovn-nbctl set NB_Global . name=op-az
```

`OVN-IC` `OVN-IC`

```
/usr/share/ovn/scripts/ovn-ctl --ovn-ic-nb-db=tcp:192.168.65.3:6645 \
  --ovn-ic-sb-db=tcp:192.168.65.3:6646 \
  --ovn-northd-nb-db=unix:/run/ovn/ovnnb_db.sock \
  --ovn-northd-sb-db=unix:/run/ovn/ovnsb_db.sock \
  start_ic
```

- `ovn-ic-nb-db` `ovn-ic-sb-db` : OVN-IC
- `ovn-northd-nb-db` `ovn-northd-sb-db` : OVN

```
ovs-vsctl set open_vswitch . external_ids:ovn-is-interconn=true
```

OpenStack OVN

`ts` `router0`

```
ovn-nbctl lrp-add router0 lrp-router0-ts 00:02:ef:11:39:4f 169.254.100.73/24
ovn-nbctl lsp-add ts lsp-ts-router0 -- lsp-set-addresses lsp-ts-router0 router \
  -- lsp-set-type lsp-ts-router0 router \
  -- lsp-set-options lsp-ts-router0  router-port=lrp-router0-ts
ovn-nbctl lrp-set-gateway-chassis lrp-router0-ts {gateway chassis} 1000
ovn-nbctl set NB_Global . options:ic-route-adv=true options:ic-route-learn=true
```

Kubernetes

```
# ovn-nbctl lr-route-list router0
IPv4 Routes
               10.0.0.22            169.254.100.34 dst-ip (learned)
            10.16.0.0/16            169.254.100.34 dst-ip (learned)
```

`router0` Kubernetes Pod

7.21.2 OVN

OpenStack Kubernetes OVN VPC Subnet

Kube-OVN OVN OpenStack Neutron OVN OpenStack networking-ovn Neutron

Neutron

Neutron `/etc/neutron/plugins/ml2/ml2_conf.ini`

```
[ovn]
...
ovn_nb_connection = tcp:[192.168.137.176]:6641,tcp:[192.168.137.177]:6641,tcp:[192.168.137.178]:6641
ovn_sb_connection = tcp:[192.168.137.176]:6642,tcp:[192.168.137.177]:6642,tcp:[192.168.137.178]:6642
ovn_l3_scheduler = OVN_L3_SCHEDULER
```

- `ovn_nb_connection` `ovn_sb_connection` : Kube-OVN `ovn-central`

OVS

```
ovs-vsctl set open . external-ids:ovn-remote=tcp:[192.168.137.176]:6642,tcp:[192.168.137.177]:6642,tcp:[192.168.137.178]:6642
ovs-vsctl set open . external-ids:ovn-encap-type=geneve
ovs-vsctl set open . external-ids:ovn-encap-ip=192.168.137.200
```

- `external-ids:ovn-remote` : Kube-OVN `ovn-central`
- `ovn-encap-ip` : IP

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Kubernetes OpenStack

Kubernetes OpenStack OpenStack Pod

> **Note**
>
> `kube-ovn-controller` args `--enable-external-vpc=true`

OpenStack

```
# openstack router list
+--------------------------------------+---------+--------+-------+----------------------------------+
| ID                                   | Name    | Status | State | Project                          |
+--------------------------------------+---------+--------+-------+----------------------------------+
| 22040ed5-0598-4f77-bffd-e7fd4db47e93 | router0 | ACTIVE | UP    | 62381a21d569404aa236a5dd8712449c |
+--------------------------------------+---------+--------+-------+----------------------------------+
# openstack network list
+--------------------------------------+----------+--------------------------------------+
| ID                                   | Name     | Subnets                              |
+--------------------------------------+----------+--------------------------------------+
| cd59e36a-37db-4c27-b709-d35379a7920f | provider | 01d73d9f-fdaa-426c-9b60-aa34abbfacae |
+--------------------------------------+----------+--------------------------------------+
# openstack subnet list
+--------------------------------------+-------------+--------------------------------------+----------------+
| ID                                   | Name        | Network                              | Subnet         |
+--------------------------------------+-------------+--------------------------------------+----------------+
| 01d73d9f-fdaa-426c-9b60-aa34abbfacae | provider-v4 | cd59e36a-37db-4c27-b709-d35379a7920f | 192.168.1.0/24 |
+--------------------------------------+-------------+--------------------------------------+----------------+
# openstack server list
+--------------------------------------+-------------------+--------+----------------------+--------+--------+
| ID                                   | Name              | Status | Networks             | Image  | Flavor |
+--------------------------------------+-------------------+--------+----------------------+--------+--------+
| 8433d622-a8d6-41a7-8b31-49abfd64f639 | provider-instance | ACTIVE | provider=192.168.1.61 | ubuntu | m1     |
+--------------------------------------+-------------------+--------+----------------------+--------+--------+
```

Kubernetes VPC

```
# kubectl get vpc
NAME                                    STANDBY   SUBNETS
neutron-22040ed5-0598-4f77-bffd-e7fd4db47e93   true      ["neutron-cd59e36a-37db-4c27-b709-d35379a7920f"]
ovn-cluster                             true      ["join","ovn-default"]
```

`neutron-22040ed5-0598-4f77-bffd-e7fd4db47e93` OpenStack VPC

Kube-OVN VPC Subnet Pod

VPC, Subnet Namespace `net2` Pod:

```yaml
apiVersion: v1
kind: Namespace
metadata:
  name: net2
---
apiVersion: kubeovn.io/v1
kind: Vpc
metadata:
  creationTimestamp: "2021-06-20T13:34:11Z"
  generation: 2
  labels:
    ovn.kubernetes.io/vpc_external: "true"
  name: neutron-22040ed5-0598-4f77-bffd-e7fd4db47e93
  resourceVersion: "583728"
  uid: 18d4c654-f511-4def-a3a0-a6434d237c1e
spec:
  namespaces:
  - net2
---
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net2
spec:
  vpc: neutron-22040ed5-0598-4f77-bffd-e7fd4db47e93
  namespaces:
    - net2
  cidrBlock: 12.0.1.0/24
  natOutgoing: false
---
apiVersion: v1
kind: Pod
metadata:
  name: ubuntu
```


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```
  namespace: net2
spec:
  containers:
    - image: docker.io/kubeovn/kube-ovn:v1.8.0
      command:
        - "sleep"
        - "604800"
      imagePullPolicy: IfNotPresent
      name: ubuntu
  restartPolicy: Always
```

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7.21.3

7.22 IPsec

Kube-OVN v1.13.0 OVN/OVS IPsec Geneve/Vxlan/STT

7.22.1

- UDP 500 IKE UDP 4500 NAT-T
- `xfrm` / `af_key` strongSwan/libreswan

7.22.2

`kube-ovn-cni` CertificateSigningRequest `kube-ovn-controller` approve `kube-ovn-cni` IPsec ipsec

7.22.3 IPsec

`kube-ovn-controller` Deployment `kube-ovn-cni` DaemonSet `--enable-ovn-ipsec=false` `--enable-ovn-ipsec=true`

```
ENABLE_OVN_IPSEC=true
```

7.22.4 cert-manager

cert-manager IPsec `--cert-manager-ipsec-cert=true` kube-ovn-cni cert-manager IPsec Kube-OVN CA

```
args:
- --enable-ovn-ipsec=true
- --cert-manager-ipsec-cert=true
```

7.22.5

- `kube-ovn-cni` `ovs-appctl ipsec/show` IPsec SA SPI
- `kubectl get csr` kube-ovn-cni CSR IPsec CSR approve controller `signer.go`
- IPsec `false` IPsec SA `kube-ovn-cni` Pod

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7.22.6

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7.23 OVN

Pod GRE/ERSPAN

> Kube-OVN v1.12

7.23.1 Multus-CNI

Multus-CNI Multus

7.23.2

```yaml
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: attachnet
  namespace: default
spec:
  config: |
    {
      "cniVersion": "0.3.1",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "attachnet.default.ovn"
    }
```

`provider` `<NAME>.<NAMESPACE>.ovn`

7.23.3 Underlay

MTU LSP/Pod Underlay

Underlay

```yaml
apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: net1
spec:
  defaultInterface: eth1
---
apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
  name: vlan1
spec:
  id: 0
  provider: net1
---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  protocol: IPv4
  cidrBlock: 172.19.0.0/16
  excludeIps:
  - 172.19.0.2..172.19.0.20
  gateway: 172.19.0.1
  vlan: vlan1
  provider: attachnet.default.ovn
```

`provider` `provider`

7.23.4 Pod

Pod

```yaml
apiVersion: v1
kind: Pod
```


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```
metadata:
  name: pod1
  annotations:
    k8s.v1.cni.cncf.io/networks: default/attachnet
spec:
  containers:
  - name: bash
    image: docker.io/kubeovn/kube-ovn:v1.16.16
    args:
    - bash
    - -c
    - sleep infinity
    securityContext:
      privileged: true
```

Pod IP

```
$ kubectl get ips | grep pod1
pod1.default                        10.16.0.12   00:00:00:FF:34:24  kube-ovn-worker  ovn-default
pod1.default.attachnet.default.ovn  172.19.0.21  00:00:00:A0:30:68  kube-ovn-worker  subnet1
```

IP `172.19.0.21`

7.23.5 OVN

OVN

```
kubectl ko nbctl mirror-add mirror1 gre 99 from-lport 172.19.0.21
kubectl ko nbctl lsp-attach-mirror coredns-787d4945fb-gpnkb.kube-system mirror1
```

`coredns-787d4945fb-gpnkb.kube-system` OVN LSP `<POD_NAME>.<POD_NAMESPACE>`

OVN

```
ovn-nbctl mirror-add <NAME> <TYPE> <INDEX> <FILTER> <IP>

NAME   - add a mirror with given name
TYPE   - specify TYPE 'gre' or 'erspan'
INDEX  - specify the tunnel INDEX value
         (indicates key if GRE, erpsan_idx if ERSPAN)
FILTER - specify FILTER for mirroring selection
         ('to-lport' / 'from-lport')
IP     - specify Sink / Destination i.e. Remote IP

ovn-nbctl mirror-del [NAME]         remove mirrors
ovn-nbctl mirror-list               print mirrors

ovn-nbctl lsp-attach-mirror PORT MIRROR   attach source PORT to MIRROR
ovn-nbctl lsp-detach-mirror PORT MIRROR   detach source PORT from MIRROR
```

7.23.6 Pod

Pod

```
root@pod1:/kube-ovn# ip link add mirror1 type gretap local 172.19.0.21 key 99 dev net1
root@pod1:/kube-ovn# ip link set mirror1 up
```

Pod

```
root@pod1:/kube-ovn# tcpdump -i mirror1 -nnve
tcpdump: listening on mirror1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
05:13:30.328808 00:00:00:a3:f5:e2 > 00:00:00:97:0f:6e, ethertype ARP (0x0806), length 42: Ethernet (len 6), IPv4 (len 4), Request who-has 10.16.0.7 tell
10.16.0.4, length 28
05:13:30.559167 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 212: (tos 0x0, ttl 64, id 57364, offset 0, flags [DF], proto UDP (17),
length 198)
    10.16.0.4.53 > 10.16.0.6.50472: 34511 NXDomain*- 0/1/1 (170)
05:13:30.559343 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 212: (tos 0x0, ttl 64, id 57365, offset 0, flags [DF], proto UDP (17),
length 198)
    10.16.0.4.53 > 10.16.0.6.45177: 1659 NXDomain*- 0/1/1 (170)
05:13:30.560625 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 200: (tos 0x0, ttl 64, id 57367, offset 0, flags [DF], proto UDP (17),
length 186)
    10.16.0.4.53 > 10.16.0.6.43848: 2636*- 0/1/1 (158)
05:13:30.562774 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 191: (tos 0x0, ttl 64, id 57368, offset 0, flags [DF], proto UDP (17),
length 177)
    10.16.0.4.53 > 10.16.0.6.37755: 48737 NXDomain*- 0/1/1 (149)
05:13:30.563523 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 187: (tos 0x0, ttl 64, id 57369, offset 0, flags [DF], proto UDP (17),
length 173)
    10.16.0.4.53 > 10.16.0.6.53887: 45519 NXDomain*- 0/1/1 (145)
05:13:30.564940 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 201: (tos 0x0, ttl 64, id 57370, offset 0, flags [DF], proto UDP (17),
```


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```
length 187)
    10.16.0.4.53 > 10.16.0.6.40846: 25745 NXDomain*- 0/1/1 (159)
05:13:30.565140 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 201: (tos 0x0, ttl 64, id 57371, offset 0, flags [DF], proto UDP (17),
length 187)
    10.16.0.4.53 > 10.16.0.6.45214: 61875 NXDomain*- 0/1/1 (159)
05:13:30.566023 00:00:00:a3:f5:e2 > 00:00:00:55:e4:4e, ethertype IPv4 (0x0800), length 80: (tos 0x0, ttl 64, id 45937, offset 0, flags [DF], proto UDP (17),
length 66)
    10.16.0.4.44116 > 172.18.0.1.53: 16025+ [1au] AAAA? kube-ovn.io. (38)
```

7.23.7

1. ERSPAN OVN Linux 4.14 ERSPAN IPv6 Linux 4.16
2. OVN

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7.23.8

7.24 DNS Kube-OVN

NodeLocal DNSCache DaemonSet DNS DNS Kube-OVN

7.24.1 DNS

Kubernetes DNS

Kubernetes Nodelocaldnscache

```
#!bin/bash

localdns=169.254.20.10
domain=cluster.local
kubedns=10.96.0.10

wget https://raw.githubusercontent.com/kubernetes/kubernetes/master/cluster/addons/dns/nodelocaldns/nodelocaldns.yaml
sed -i "s/__PILLAR__LOCAL__DNS__/$localdns/g; s/__PILLAR__DNS__DOMAIN__/$domain/g; s/,__PILLAR__DNS__SERVER__//g; s/__PILLAR__CLUSTER__DNS__/$kubedns/g"
nodelocaldns.yaml

kubectl apply -f nodelocaldns.yaml
```

kubelet `/var/lib/kubelet/config.yaml` clusterDNS DNS IP 169.254.20.10 kubelet

Kube-OVN DNS

Kubernetes Nodelocal DNScache Kube-OVN

UNDERLAY SUBNET U2O

Underlay Subnet DNS U2O `kubectl edit subnet {your subnet}` `spec.u2oInterconnection = true` , Overlay Subnet

KUBE-OVN-CONTROLLER DNS IP

```
kubectl edit deployment kube-ovn-controller -n kube-system
```

`spec.template.spec.containers.args` `--node-local-dns-ip=169.254.20.10`

POD

Pod `/etc/resolv.conf` nameserver DNS IP Pod nameserver DNS ClusterIP u2o Pod Pod

7.24.2 DNS

Pod Pod DNS 169.254.20.10

```
# kubectl exec -it pod1 -- nslookup github.com
Server:         169.254.20.10
Address:        169.254.20.10:53


Name:   github.com
Address: 20.205.243.166
```

DNS ovn0 DNS DNS

```
# tcpdump -i any port 53

06:20:00.441889 659246098c56_h P   ifindex 17 00:00:00:73:f1:06 ethertype IPv4 (0x0800), length 75: 10.16.0.2.40230 > 169.254.20.10.53: 1291+ A? baidu.com.
(27)
06:20:00.441889 ovn0  In  ifindex 7 00:00:00:50:32:cd ethertype IPv4 (0x0800), length 75: 10.16.0.2.40230 > 169.254.20.10.53: 1291+ A? baidu.com. (27)
06:20:00.441950 659246098c56_h P   ifindex 17 00:00:00:73:f1:06 ethertype IPv4 (0x0800), length 75: 10.16.0.2.40230 > 169.254.20.10.53: 1611+ AAAA?
baidu.com. (27)
06:20:00.441950 ovn0  In  ifindex 7 00:00:00:50:32:cd ethertype IPv4 (0x0800), length 75: 10.16.0.2.40230 > 169.254.20.10.53: 1611+ AAAA? baidu.com. (27)
06:20:00.442203 ovn0  Out ifindex 7 00:00:00:52:99:d8 ethertype IPv4 (0x0800), length 145: 169.254.20.10.53 > 10.16.0.2.40230: 1611* 0/1/0 (97)
```


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```
06:20:00.442219 659246098c56_h Out ifindex 17 00:00:00:ea:b3:5e ethertype IPv4 (0x0800), length 145: 169.254.20.10.53 > 10.16.0.2.40230: 1611* 0/1/0 (97)
06:20:00.442273 ovn0  Out ifindex 7 00:00:00:52:99:d8 ethertype IPv4 (0x0800), length 125: 169.254.20.10.53 > 10.16.0.2.40230: 1291* 2/0/0 A 39.156.66.10, A
110.242.68.66 (77)
06:20:00.442278 659246098c56_h Out ifindex 17 00:00:00:ea:b3:5e ethertype IPv4 (0x0800), length 125: 169.254.20.10.53 > 10.16.0.2.40230: 1291* 2/0/0 A
39.156.66.10, A 110.242.68.66 (77)
```

7.24.3

⚠️

NetworkPolicy NetworkPolicy DNS IP 169.254.20.10 CIDR NetworkPolicy DNS Pod

NetworkPolicy

Pod DNS NetworkPolicy

```yaml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-local-dns-and-node-cidr
  namespace: default  #
spec:
  podSelector: {}  #       Pod
  policyTypes:
  - Ingress
  - Egress
  egress:
  #       DNS
  - to:
    - ipBlock:
        cidr: 169.254.20.10/32
  #       CIDR          CIDR
  - to:
    - ipBlock:
        cidr: 10.0.0.0/8  #      CIDR
  ingress:
  #       DNS
  - from:
    - ipBlock:
        cidr: 169.254.20.10/32
  #       CIDR              CIDR
  - from:
    - ipBlock:
        cidr: 10.0.0.0/8  #      CIDR
```

- `169.254.20.10/32` DNS IP
- `10.0.0.0/8` CIDR

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7.24.4

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7.25 VPC NAT

7.25.1

VPC Overlay `natOutgoing` Subnet Pod SNAT IP Pod SNAT

NAT CIDR IP SNAT

7.25.2

`subnet.Spec` `natOutgoing` `natOutgoingPolicyRules`

```yaml
spec:
  natOutgoing: true
  natOutgoingPolicyRules:
    - action: forward
      match:
        srcIPs: 10.0.11.0/30,10.0.11.254
    - action: nat
      match:
        srcIPs: 10.0.11.128/26
        dstIPs: 114.114.114.114,8.8.8.8
```

NAT

1. IP 10.0.11.0/30 10.0.11.254 SNAT
2. IP 10.0.11.128/26 IP 114.114.114.114 8.8.8.8 SNAT

`action` `match` action, action `forward` `nat` `forward` SNAT, `nat` SNAT natOutgoingPolicyRules SNAT

`match` `srcIPs` `dstIPs` IP IP `match.srcIPs` `match.dstIPs` CIDR IP

match `natOutgoingPolicyRules` action

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7.25.3

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8.

8.1

Kube-OVN

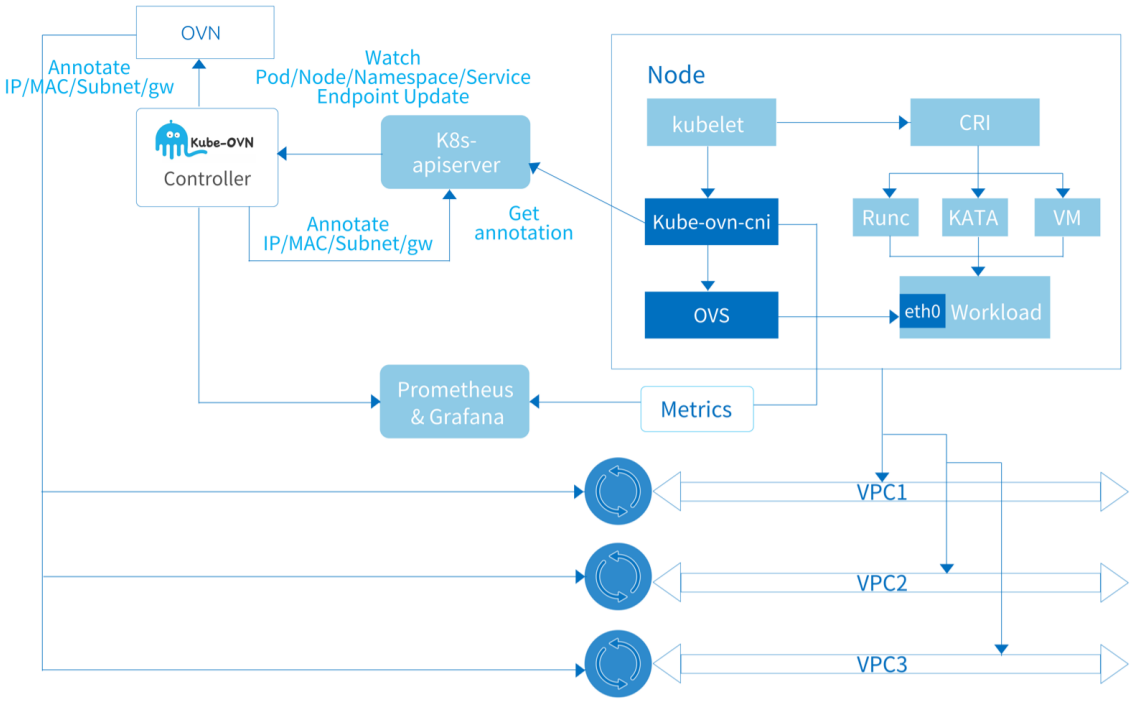
Kube-OVN Kubernetes OVN SDN Kube-OVN OVN Kubernetes CNI Service Networkpolicy SDN VPC QoS ACL

Kube-OVN Cilium Submariner Prometheus KubeVirt

8.1.1

Kube-OVN

- OVN/OVS
- Agent
-



OVN/OVS

OVN/OVS Kube-OVN OVN/OVS SDN Kube-OVN ovn-architecture(7) OVN Kube-OVN OVN Kubernetes

OVN/OVS Kubernetes

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OVN-CENTRAL

`ovn-central` Deployment OVN `ovn-nb`, `ovn-sb`, `ovn-northd`

- `ovn-nb` API `kube-ovn-controller` `ovn-nb`
- `ovn-sb` `ovn-nb`
- `ovn-northd` `ovn-nb` `ovn-sb`

`ovn-central` Raft

OVS-OVN

`ovs-ovn` DaemonSet Pod `openvswitch`, `ovsdb`, `ovn-controller` `ovn-central` Agent

Agent

Kube-OVN OVN Kubernetes

KUBE-OVN-CONTROLLER

Deployment Kubernetes OVN Kube-OVN `kube-ovn-controller` OVN
Pod Service Endpoint Node NetworkPolicy VPC Subnet Vlan ProviderNetwork

Pod `kube-ovn-controller` Pod IPAM `ovn-central` ACL `kube-ovn-controller`
CIDR Pod annotation annotation `kube-ovn-cni`

KUBE-OVN-CNI

DaemonSet CNI OVS

DaemonSet `kube-ovn` `kubelet` `kube-ovn-cni` CNI `kube-ovn-cni` `/opt/cni/bin`

`kube-ovn-cni`

1. `ovn-controller` `vswitchd`
2. CNI add/del
   a. veth OVS
   b. OVS
   c. iptables/ipset/route
3. QoS.
4. `ovn0`
5. Vlan/Underlay/EIP
6.

Kube-OVN

KUBE-OVN-SPEAKER

DaemonSet Pod IP

BGP

KUBE-OVN-PINGER

DaemonSet OVS Kube-OVN

KUBE-OVN-MONITOR

Deployment OVN Kube-OVN

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KUBECTL-KO

kubectl kubectl

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8.1.2

8.2 What's Next

This document lists the features merged into the master branch for the next minor release.

8.2.1 Post-v1.15.0

- NetworkPolicy now supports provider-scoped policies for multi-network pods using the `ovn.kubernetes.io/policy-for` annotation. #6223
- Support static IP/MAC for multiple interfaces on the same logical switch. #6060
- MetalLB underlay integration now supports IPv6 and dual-stack. #6159
- KubeVirt live-migration multi-chassis options now apply to all VM NICs, not just the primary one. #6241
- Add human-readable descriptions to all Kube-OVN CRD fields for better `kubectl explain` output. #6133 #6147
- Support BGP and EVPN (L3VPN) for VPC Egress Gateway, running FRR in the egress gateway Pod. #6224
- Support per-pod NIC granular DHCP control via annotations, overriding subnet-level DHCP settings. #6475
- Extend SecurityGroup API with tiers, larger priority range (1-16384), localAddress and port matches. #6330
- Use BigInt for SubnetStatus IP count fields to support large address ranges. #6403
- Improve hairpin logic to support FIP, SNAT from any CIDR within the VPC and LoadBalancers. #6445
- VPC NAT Gateway
- Support user-defined annotations on NAT gateway Pod template. #6256
- Allow SNAT EIP to FIP EIP traffic. #6329
- Allow any EIP to share external subnet gateway in one native VLAN. #6426
- VPC Egress Gateway
- Support custom resources and bandwidth limits. #6407
- Accept integer Mbps values and `M`, `Mi`, `G`, or `Gi` bit-rate quantities for per-replica bandwidth limits. #7123
- Interconnection
- Add vendor ID to transit switches to avoid conflicts with other OVN controllers. #6186
- Reliability
- OpenFlow synchronization: detect and recover from stale or missing OVS flows automatically. #6117
- OVN DB: back up Raft header before rejoining cluster to improve recovery. #6106
- Performance
- Strip `managedFields` from informer cache to reduce memory usage. #6119
- Add field selectors to informer factory to reduce API server load. #6091
- Security
- Replace wildcard RBAC verbs with explicit verb lists. #6233
- Specify ephemeral storage limits for containers. #6259
- Helm Chart
- Make DaemonSet update strategy configurable via `values.yaml`. #6136
- Introduce `extraEnv` variable for all components. #6142
- Add `affinity` and `nodeSelector` support for ovs-ovn and ovs-ovn-dpdk DaemonSets. #6308
- Add `external-gateway-config-ns` option for controller. #6211
- Introduce ServiceMonitor and labels to services. #6340

8.2.2 Post-v1.14.0

- ACL log supports ratelimiting. #5938

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- Subnet with centralized gateway now supports nodeSelectors. #5956
- Overlay encapsulation NIC selection. #5946
- Performace: skip conntrack for specific dst CIDRs. #5821
- NetworkPolicy supports `lax` mode which only deny traffic type of TCP, UDP and SCTP. That means ARP, ICMP and DHCP traffic are alaways allowed. #5745
- Remove internal-port type interface code. #5794
- IPPool
- Multiple IPPools now can bind to the same Namespace. #5731
- Pods in a bound namespace will only get IPs from the bound pool(s), not other ranges in the subnet. #5731
- IPPool will create an AddressSet that can be work with VPC Policy Route and ACL. #5920
- `AdminNetworkPolicy` now supports specify egress peers using FQDNs. #5703
- Using ARP for IPv4 network ready check: now you don't need ACL allow rules for gateway to make Pod running. #5716
- Non-primary CNI mode: you can run Kube-OVN as the secondary only network, without annoying unused annotations and logical switch port allocations. #5618
- VPC NAT Gateway:
- No default EIP mode: the secondary interface can initialize without a default EIP to avoid the waste. #5605
- Custom routes: you can control the route rules within the vpc-nat-gateway Pods to control traffic paths. #5608
- Gratuitous ARP: VPC NAT Gateway automatically sends gratuitous ARP packets during initialization to accelerate network convergence. #5607
- Healthchecks for static endpoints in `SwitchLBRules`: SLR with both selector or endpoints key can support healthchecks. #5435
- Underlay
- Node Selectors for `ProviderNetwork`: instead of adding/removing nodes to the `ProviderNetwork` one by one, you can use node selectors to simplify the workflow. #5518
- Different `NetworkProvider`s can now share the same VLAN. #5471
- Auto create VLAN sub-interfaces. #5966
- Auto move VLAN sub-interfaces to OVS bridges. #5949
- Adding `pod_name` and `pod_namespace` labels to interface metrics. #5463
- IPSec
- Support `cert-manager` to issue certificates. #5365
- Request new certificate if current certificate is not trusted. #5710
- kubectl-ko
- Collect IPSec and xFRM information. #5472
- Replace `Endpoint` with `EndpointSlice`. #5425
- NetworkAttachment caching: reduce APIServer load in large-scale deployments with Multus. #5386
- Upgrade `OVS` to 3.5 and `OVN` to 25.03. #5537

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8.2.3

8.3

Kube-OVN

8.3.1

Kube-OVN

1. `default` Pod IP CIDR `10.16.0.0/16` `10.16.0.1`
2. `join` Node Pod , CIDR `100.64.0.0/16` `100.64.0.1`

```
POD_CIDR="10.16.0.0/16"
POD_GATEWAY="10.16.0.1"
JOIN_CIDR="100.64.0.0/16"
EXCLUDE_IPS=""
```

`EXCLUDE_IP` `POD_CIDR` `192.168.10.20..192.168.10.30`

Overlay Service CIDR

Join

8.3.2 Service

`kube-proxy` iptables Kube-OVN Kube-OVN Service CIDR

```
SVC_CIDR="10.96.0.0/12"
```

`kube-ovn-controller` Deployment

```
args:
- --service-cluster-ip-range=10.96.0.0/12
```

8.3.3 Overlay

Kube-OVN Kubernetes Node IP

```
IFACE=eth1
```

`ens[a-z0-9]*,eth[a-z0-9]*`

`kube-ovn-cni` DaemonSet

```
args:
- --iface=eth1
```

annotation `ovn.kubernetes.io/tunnel_interface` annotation `iface` annotation

```
kubectl annotate node no1 ovn.kubernetes.io/tunnel_interface=ethx
```

Overlay

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8.3.4 MTU

Overlay Kube-OVN MTU MTU Overlay Pod MTU MTU - 100 Underlay Pod MTU

Overlay MTU `kube-ovn-cni` DaemonSet

```
args:
- --mtu=1333
```

8.3.5

Kube-OVN `mirror0` tcpdump

```
ENABLE_MIRROR=true
```

`kube-ovn-cni` DaemonSet :

```
args:
- --enable-mirror=true
```

8.3.6 LB

Underlay kube-proxy Service OVN L2 LB ClusterIP Service LB ClusterIP kube-proxy kube-proxy

```
ENABLE_LB=false
```

`kube-ovn-controller` Deployment

```
args:
- --enable-lb=false
```

LB

Kube-OVN v1.12.0 subnet crd spec `enableLb` Kube-OVN LB LB `kube-ovn-controller` Deployment `enable-lb` load-balancer `enableLb` load-balancer v1.12.0 `enableLb`

8.3.7 NetworkPolicy

Kube-OVN OVN ACL NetworkPolicy NetworkPolicy Cilium Chain eBPF NetworkPolicy Kube-OVN NetworkPolicy

```
ENABLE_NP=false
```

`kube-ovn-controller` Deployment

```
args:
- --enable-np=false
```

NetworkPolicy

8.3.8 EIP SNAT

EIP SNAT `kube-ovn-controller`

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```
ENABLE_EIP_SNAT=false
```

`kube-ovn-controller` Deployment

```
args:
- --enable-eip-snat=false
```

EIP SNAT EIP SNAT

8.3.9 Load Balancer Service

VPC Load Balancer Service LoadBalancer Service

```
ENABLE_LB_SVC=true
```

`kube-ovn-controller` Deployment

```
args:
- --enable-lb-svc=true
```

8.3.10 ECMP

ECMP ECMP `kube-ovn-controller` Deployment :

```
args:
- --enable-ecmp=true
```

Kube-OVN v1.12.0 subnet crd spec `enableEcmp` ECMP ECMP `kube-ovn-controller` Deployment `enable-ecmp` v1.12.0

8.3.11 Kubevirt VM

Kubevirt VM `kube-ovn-controller` StatefulSet Pod IP VM

1.10.6 `kube-ovn-controller` Deployment

```
args:
- --keep-vm-ip=false
```

8.3.12 CNI

Kube-OVN `/opt/cni/bin` CNI `/etc/cni/net.d` CNI `01-kube-ovn.conflist` CNI

```
CNI_CONF_DIR="/etc/cni/net.d"
CNI_BIN_DIR="/opt/cni/bin"
CNI_CONFIG_PRIORITY="01"
```

`kube-ovn-cni` DaemonSet Volume

```
volumes:
- name: cni-conf
  hostPath:
    path: "/etc/cni/net.d"
- name: cni-bin
  hostPath:
    path:"/opt/cni/bin"
...
args:
- --cni-conf-name=01-kube-ovn.conflist
```


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8.3.13

Kube-OVN Overlay Geneve Vxlan STT

```
TUNNEL_TYPE="vxlan"
```

`ovs-ovn` DaemonSet

```
env:
- name: TUNNEL_TYPE
  value: "vxlan"
```

STT ovs

8.3.14 SSL

OVN DB API SSL :

```
ENABLE_SSL=true
```

SSL

8.3.15 conntrack CIDR

conntrack NAT/ `kube-ovn-controller` CIDR

```
SKIP_CONNTRACK_DST_CIDRS="10.10.0.0/16,fd00:10::/64"
```

```
args:
- --skip-conntrack-dst-cidrs=10.10.0.0/16,fd00:10::/64
```

8.3.16 OVN IPSec

Kube-OVN OVN IPSec Geneve/Vxlan/STT

```
ENABLE_OVN_IPSEC=true
```

daemon

```
args:
- --enable-ovn-ipsec=true
```

OVN IPSec

8.3.17 AdminNetworkPolicy / DNS

AdminNetworkPolicy ANP/CNP DNS `DNSNameResolver` CRD

```
ENABLE_ANP=true
ENABLE_DNS_NAME_RESOLVER=true
```

```
args:
- --enable-anp=true
- --enable-dns-name-resolver=true
```


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8.3.18 VPC NAT Gateway

VPC NAT Gateway controller

```
ENABLE_NAT_GW=true
```

`ovn-vpc-nat-gw-config` ConfigMap `enable-vpc-nat-gw`

8.3.19 ovn-northd

`ovn-northd`

```
OVN_NORTHD_N_THREADS=4
```

1 CPU

8.3.20 NetworkPolicy

Kube-OVN `standard` / `lax` NetworkPolicy

```
NP_ENFORCEMENT=standard
```

```
args:
- --np-enforcement=standard
```

NetworkPolicy `ovn.kubernetes.io/network_policy_enforcement` annotation NetworkPolicy

8.3.21 VXLAN TX checksum offload

VXLAN checksum offload

```
SET_VXLAN_TX_OFF=true
```

```
args:
- --set-vxlan-tx-off=true
```

8.3.22 KubeVirt

KubeVirt

```
ENABLE_LIVE_MIGRATION_OPTIMIZE=true
```

8.3.23 OVN LB endpoint

OVN LoadBalancer endpoint `externalTrafficPolicy=Local`

```
ENABLE_OVN_LB_PREFER_LOCAL=true
```

```
args:
- --enable-ovn-lb-prefer-local=true
```

8.3.24 ConfigMap

ConfigMap `external-gw-config` `kube-system`

```
EXTERNAL_GATEWAY_CONFIG_NS=kube-system
```


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8.3.25 Non-Primary CNI

Kube-OVN attachment network Pod IP CNI controller daemon

```
args:
- --non-primary-cni-mode=true
```

Helm `cni_conf.NON_PRIMARY_CNI=true` chart v1 / `cni.nonPrimaryCNI=true` chart v2 Non-Primary CNI

8.3.26 ip

kube-ovn-controller/kube-ovn-cni/kube-ovn-monitor ip 0.0.0.0 ip

```
ENABLE_BIND_LOCAL_IP=true
```

kube-ovn-monitor pod ip

```
# netstat -tunlp |grep kube-ovn
tcp        0      0 172.18.0.5:10661        0.0.0.0:*               LISTEN      2612/./kube-ovn-mon
```

deployment daemonSet

```
env:
- name: ENABLE_BIND_LOCAL_IP
  value: "false"
```

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8.3.27

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8.4

Kube-OVN Minor Patch Minor OVN/OVS API Patch Bug API

8.4.1

Kube-OVN `master`, `release-1.12` `release-1.11` `release-1.12` Bug backport

`release-1.11` backport Bug

8.4.2

Minor Patch Bug Bug

8.4.3 Patch

Patch hack/release.sh

1. Build
2. tag Docker Hub
3. tag Github
4.
5.
6. Release Note PR
7. Release Note ( )
8. Merge github action Release Note PR
9. Github Release
10. Github Release Release `v1.12.12` Release Note Release

8.4.4 Minor

Minor

1. Github `release-1.13` ( )
2. `VERSION`, `dist/images/install.sh`, `charts/kube-ovn/values.yaml` `charts/kube-ovn/Chart.yaml` Minor `v1.14.0` ( )
3. tag Docker Hub ( )
4. tag Github ( )
5. `v1.13` `mkdocs.yml` `version` `branch` ( )
6. Release Note PR
7. Release Note ( )
8. Merge github action Release Note PR
9. Github Release
10. Github Release Release `v1.13.0` Release Note Release
11. `VERSION` Patch `v1.13.1`

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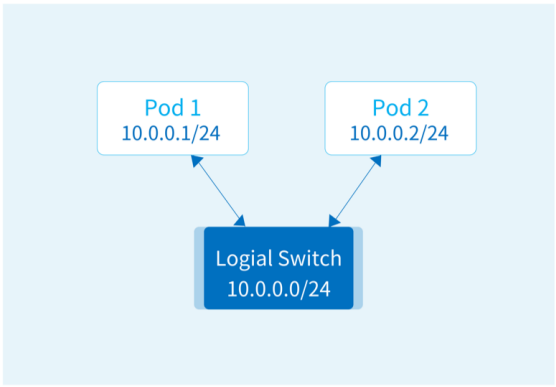


8.4.5

8.5 Underlay

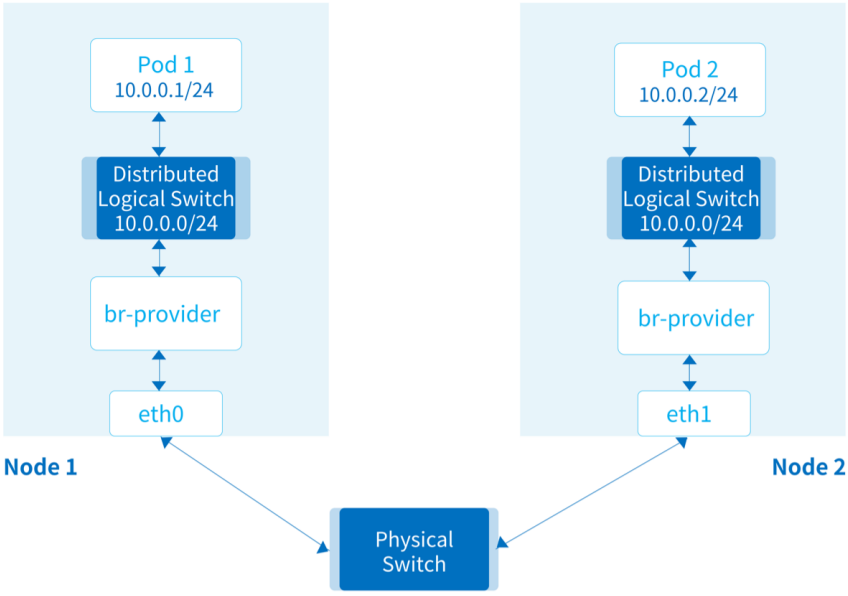
Underlay

8.5.1

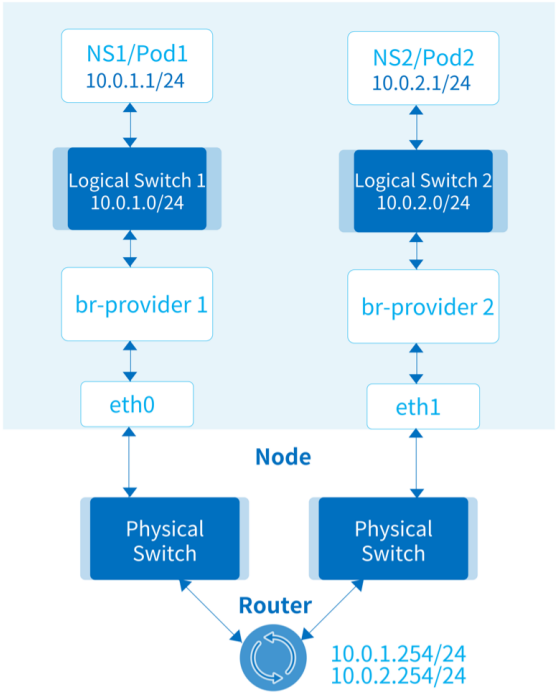


8.5.2

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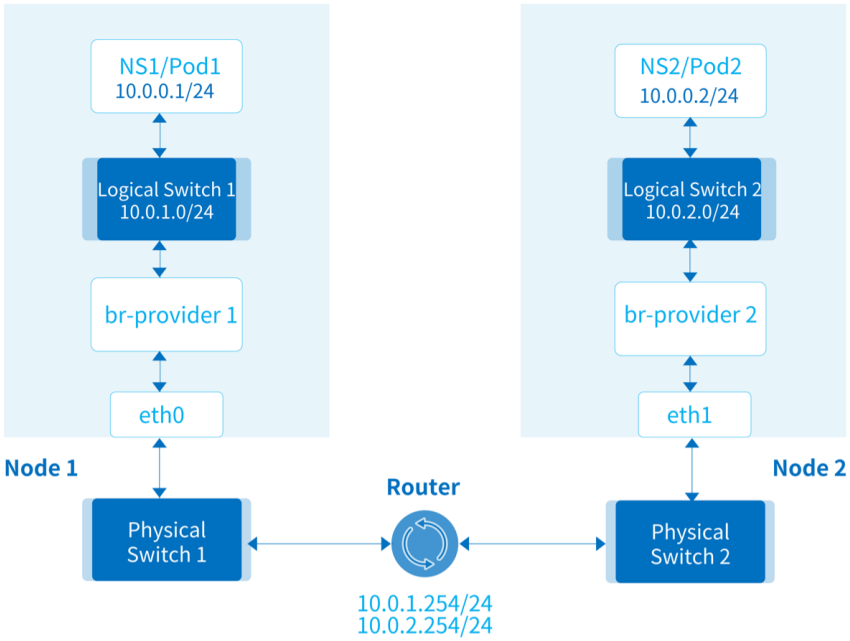
8.5.3



br-provider-1 br-provider-2 OVS Provider Network

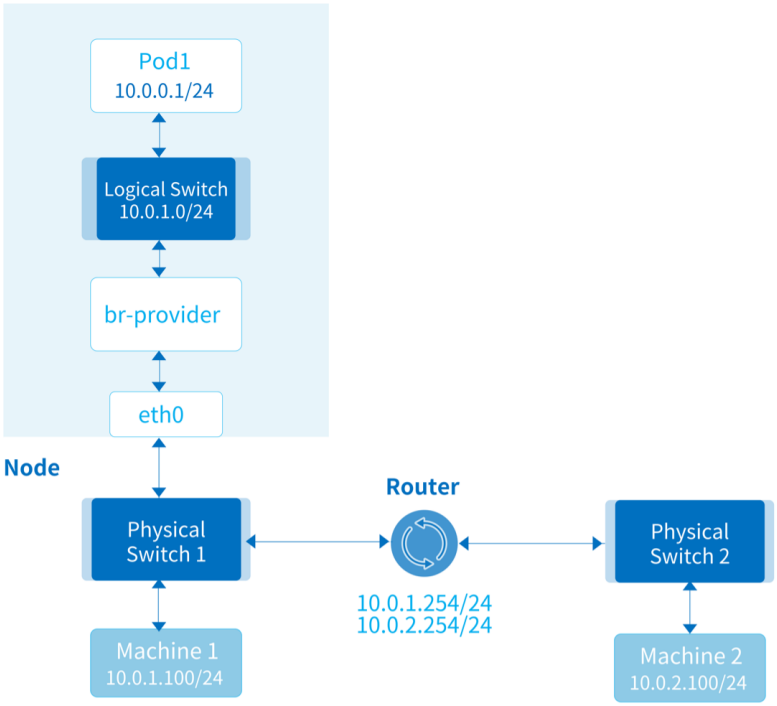
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8.5.4



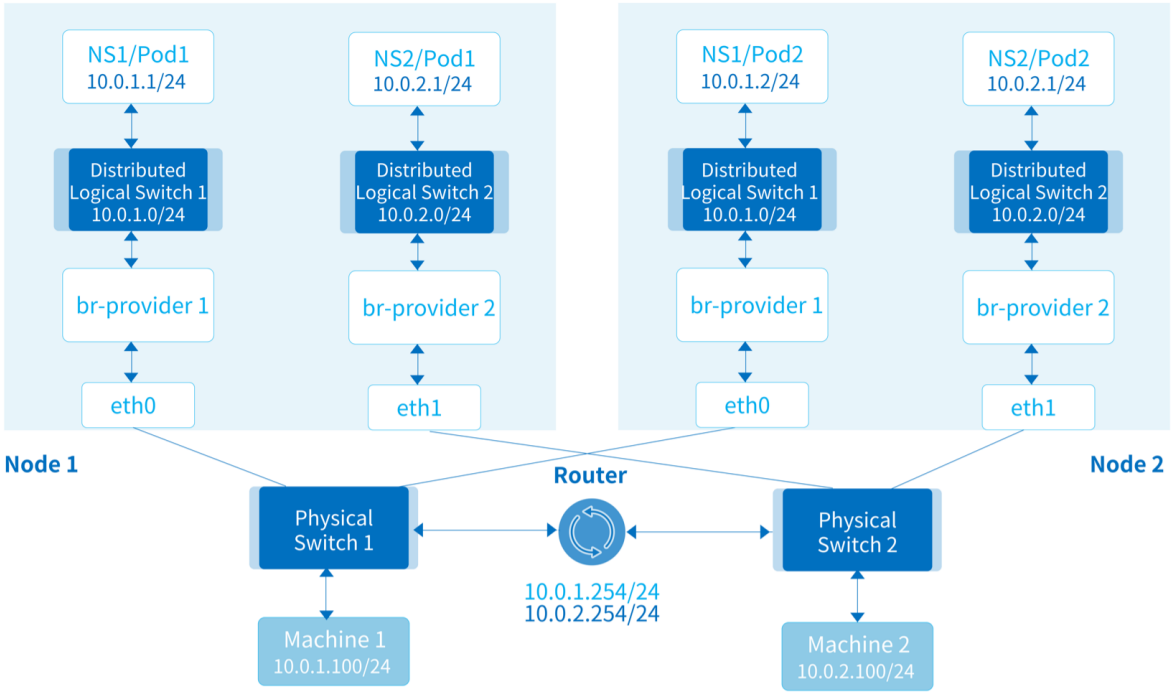
8.5.5

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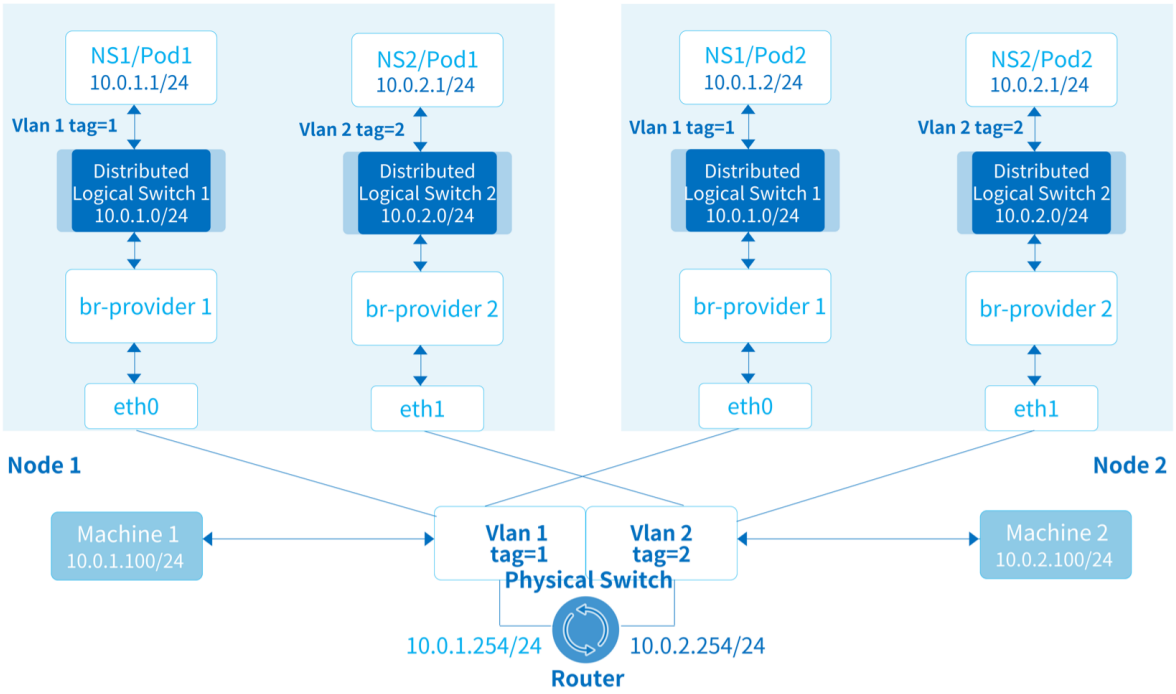


Pod

8.5.6 Vlan Tag



8.5.7 VLAN

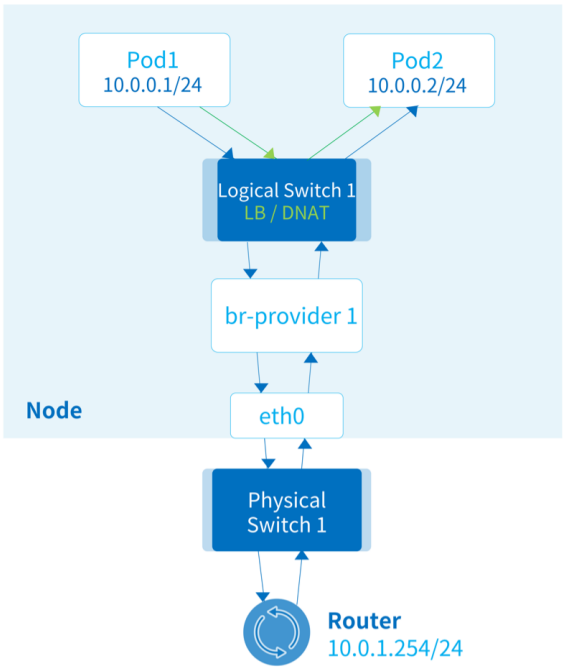


8.5.8 Pod Service IP

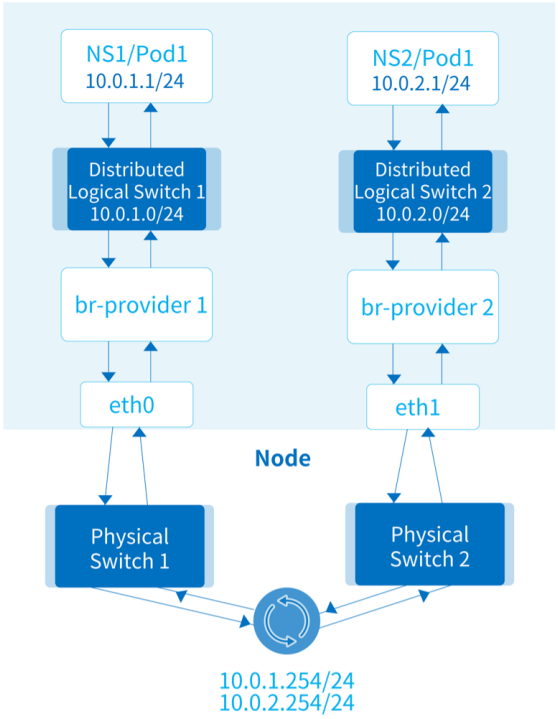
Kube-OVN Kubernetes Service Pod Service IP Pod Service IP MAC MAC
DNAT IP Service Endpoint IP MAC

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Service **Pod**



Service **Pod**



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8.5.9

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8.6 Iptables

Kube-OVN ipset iptables VPC Overlay NAT

ipset

IPv4/IPv6		
ovn40services/ovn60services	hash:net	Service
ovn40subnets/ovn60subnets	hash:net	Overlay NodeLocal DNS IP
ovn40subnets-nat/ovn60subnets-nat	hash:net	NatOutgoing Overlay
ovn40subnets-distributed-gw/ovn60subnets-distributed-gw	hash:net	Overlay
ovn40other-node/ovn60other-node	hash:net	IP
ovn40local-pod-ip-nat/ovn60local-pod-ip-nat	hash:ip	
ovn40subnets-nat-policy	hash:net	natOutgoingPolicyRules
ovn40natpr-418e79269dc5-dst	hash:net	natOutgoingPolicyRules rule dstIPs
ovn40natpr-418e79269dc5-src	hash:net	natOutgoingPolicyRules rule srcIPs

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iptables IPv4

filter	INPUT	-m set --match-set ovn40services src -j ACCEPT	k8s Service Pod	--
filter	INPUT	-m set --match-set ovn40services dst -j ACCEPT		--
filter	INPUT	-m set --match-set ovn40subnets src -j ACCEPT		--
filter	INPUT	-m set --match-set ovn40subnets dst -j ACCEPT		--
filter	FORWARD	-m set --match-set ovn40services src -j ACCEPT		--
filter	FORWARD	-m set --match-set ovn40services dst -j ACCEPT		--
filter	FORWARD	-m set --match-set ovn40subnets src -j ACCEPT		--
filter	FORWARD	-m set --match-set ovn40subnets dst -j ACCEPT		--
filter	FORWARD	-s 10.16.0.0/16 -m comment --comment "ovn-subnet-gateway,ovn-default"	subnet	10.16.0.0/16 subnet cidr comment ovn-subnet-gateway iptables subnet ovn-default subnet
filter	FORWARD	-d 10.16.0.0/16 -m comment --comment "ovn-subnet-gateway,ovn-default"	subnet	
filter	OUTPUT	-p udp -m udp --dport 6081 -j MARK --set-xmark 0x0	SNAT	UDP: bad checksum on VXLAN interface
nat	PREROUTING	-m comment --comment "kube-ovn prerouting rules" -j OVN-PREROUTING	OVN-PREROUTING	--
nat	POSTROUTING	-m comment --comment "kube-ovn postrouting rules" -j OVN-POSTROUTING	OVN-POSTROUTING	--
nat	OVN-PREROUTING	-i ovn0 -m set --match-set ovn40subnets src -m set --match-set ovn40services dst -j MARK --set-xmark 0x4000/0x4000	Pod Service masquerade	LB
nat	OVN-PREROUTING	-p tcp -m addrtype --dst-type LOCAL -m set --match-		kube-proxy ipvs

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		set KUBE-NODE-PORT-LOCAL-TCP dst -j MARK --set-xmark 0x80000/0x80000	ExternalTrafficPolicy Local Service TCP
nat	OVN-PREROUTING	-p udp -m addrtype --dst-type LOCAL -m set --match-set KUBE-NODE-PORT-LOCAL-UDP dst -j MARK --set-xmark 0x80000/0x80000	ExternalTrafficPolicy Local Service UDP
nat	OVN-POSTROUTING	-m set --match-set ovn40services src -m set --match-set ovn40subnets dst -m mark --mark 0x4000/0x4000 -j SNAT --to-source	Service IP kube-proxy ipvs Overlay Pod IP IP
nat	OVN-POSTROUTING	-m mark --mark 0x4000/0x4000 -j MASQUERADE	SNAT --
nat	OVN-POSTROUTING	-m set --match-set ovn40subnets src -m set --match-set ovn40subnets dst -j MASQUERADE	Pod Service -- SNAT
nat	OVN-POSTROUTING	-m mark --mark 0x80000/0x80000 -m set --match-set ovn40subnets-distributed-gw dst -j RETURN	ExternalTrafficPolicy -- Local Service Endpoint SNAT
nat	OVN-POSTROUTING	-m mark --mark 0x80000/0x80000 -j MASQUERADE	ExternalTrafficPolicy -- Local Service Endpoint SNAT
nat	OVN-POSTROUTING	-p tcp -m tcp --tcp-flags SYN NONE -m conntrack --ctstate NEW -j RETURN	Pod IP SNAT --
nat	OVN-POSTROUTING	-s 10.16.0.0/16 -m set ! --match-set ovn40subnets dst -j SNAT --to-source 192.168.0.101	Pod 10.16.0.0/16 192.168.0.101 NatOutgoing IP IP SNAT
nat	OVN-POSTROUTING	-m set --match-set ovn40subnets-nat src -m set ! --match-set ovn40subnets dst -j MASQUERADE	Pod -- NatOutgoing SNAT
nat	OVN-POSTROUTING	-m set --match-set ovn40subnets-nat-policy src -m set ! --match-set ovn40subnets dst -j OVN-NAT-POLICY	Pod natOutgoingPolicyRules OVN NAT-POLICY natOutgoingPolicyRules SNAT
nat			

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	OVN-POSTROUTING	-m mark --mark 0x90001/0x90001 -j MASQUERADE --random-fully		OVN-NAT-POLICY tag 0x90001/0x90001 SNAT
nat	OVN-POSTROUTING	-m mark --mark 0x90002/0x90002 -j RETURN		OVN-NAT-POLICY , tag 0x90002/0x90002 SNAT
nat	OVN-NAT-POLICY	-s 10.0.11.0/24 -m comment --comment natPolicySubnet-net1 -j OVN-NAT-PSUBNET-aa98851157c5		10.0.11.0/24 net1 CIDR OVN-NAT-PSUBNET-aa98851157c5 natOutgoingPolicyRules
nat	OVN-NAT-PSUBNET-xxxxxxxxxxx	-m set --match-set ovn40natpr-418e79269dc5-src src -m set --match-set ovn40natpr-418e79269dc5-dst dst -j MARK --set-xmark 0x90002/0x90002		418e79269dc5 natOutgoingPolicyRules ID status.natOutgoingPolicyRules[index].Rule srcIPs ovn40natpr-418e79269dc src dstIPS ovn40natpr-418e79269dc5-dst tag 0x90002
mangle	OVN-OUTPUT	-d 10.241.39.2/32 -p tcp -m tcp --dport 80 -j MARK --set-xmark 0x90003/0x90003	kubelet tproxy	
mangle	OVN-PREROUTING	-d 10.241.39.2/32 -p tcp -m tcp --dport 80 -j TPROXY --on-port 8102 --on-ip 172.18.0.3 --tproxy-mark 0x90004/0x90004	kubelet tproxy	

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8.6.1

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8.7

8.7.1

Kube-OVN Github / Github Issue PR Maintainer Review Github Action

8.7.2

Kube-OVN Go Go Modules `GO111MODULE="on"`

golangci-lint local-installation

Kube-OVN Docker buildx Docker buildx:

```
docker buildx create --use
```

8.7.3

Kube-OVN

```
git clone https://github.com/kubeovn/kube-ovn.git
cd kube-ovn
make release
```

ARM

```
make release-arm
```

8.7.4 base

OVS/OVN base

base Dockerfile `dist/images/Dockerfile.base`

```
# build x86 base image
make base-amd64

# build arm base image
make base-arm64
```

8.7.5 E2E

Kube-OVN :

- KIND Kubernetes `go install sigs.k8s.io/kind@latest`
- jinjanator : `pip install jinjanator`
- Ginkgo `go install github.com/onsi/ginkgo/v2/ginkgo; go get github.com/onsi/gomega/...`

E2E

```
make kind-init
make kind-install
make e2e
```

Underlay E2E

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```
make kind-init
make kind-install-underlay
make e2e-underlay-single-nic
```

ovn vpc nat gw eip, fip, snat, dnat

```
make kind-init
make kind-install
make ovn-vpc-nat-gw-conformance-e2e
```

iptables vpc nat gw eip, fip, snat, dnat

```
make kind-init
make kind-install-vpc-nat-gw
make iptables-vpc-nat-gw-conformance-e2e
```

loadbalancer service

```
make kind-init
make kind-install-lb-svc
make kube-ovn-lb-svc-conformance-e2e
```

```
make kind-clean
```

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8.7.6

8.8 OVS/OVN

OVN/OVS SDN Kubernetes Kube-OVN Kube-OVN OVN/OVS
OVN/OVS Kube-OVN

- 4228eab1d7 vswitchd ofport_usage
- 54056ea65d timer
- 6b4dcb311f fdb
- f627b7721e hairpin fdb
- 3f3e3a436f ovsdb-tool join-cluster Server ID
- a6cb8215a8 QoS
- d4d76ddb2e ovsdb-tool fix-cluster
- ffd2328d4a netdev CPU
- d088c5d8c2 ovs-router kube-ipvs0
- 1b31f07dc6
- 54b7678229 ovs-sandbox docker run
- 9ee66bd91b
- e889d46924 Underlay resubmit
- f9e97031b5 ovn-controller Kube-OVN localnet GARP
- 78cade0187 conntrack
- 85aa6263ad northd DNS IP conntrack
- 34dc3e3fcf lflow lport conntrack
- a297b840c2 DNAT lsp
- 03e35ed9c5 ovn-controller
- e7d3ba53cd ACL DNS IP conntrack
- 9286e1fd57
- e5916eb53a lr-lb DNAT
- e4e6ea9c5f BFD LRP
- e76880e792 northd nb version_compatibility
- 477695a010 northd localnet lrp arp/nd
- 2209d9a335 ovn-controller LSP
- 52b727b331 ovn-northd
- 53b9837f7f LB ACL
- 737d9f932e GW
- sbrec-delete-null-check ovn-controller sbrec

- 20626ea909 LB ACL
- a2d9ff3ccd Deb

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8.8.1

8.9

Kube-OVN OVN/OVS `Geneve` `Vxlan` `STT` OVN
OVN Architecture Design Decision

8.9.1 Geneve

`Geneve` Kube-OVN OVN Offload `Geneve` 24bit
datapath datapath 32768

Mellanox OVS `Geneve` 5.4 backport

UDP TCP over UDP TCP CPU

8.9.2 Vxlan

`Vxlan` OVN Offload OVN datapath 4096 datapath
datapath 4096 `inport` ACL

Mellanox OVS `Vxlan`

UDP TCP over UDP TCP CPU

8.9.3 STT

> **Warning**
>
> OpenVswitch 3.6 STT Tunnel

`STT` OVN TCP TCP TCP OVN datapath

OVS

OVS

8.9.4

- VXLAN vs GENEVE: Understand The Difference
- OVN FAQ
- What is Geneve

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8.9.5

8.10 Kube-OVN

Kube-OVN

8.10.1 ovn-monitor

OVN

Gauge	kube_ovn_ovn_status	OVN (2) follower (1) leader, (0)
Gauge	kube_ovn_failed_req_count	OVN
Gauge	kube_ovn_log_file_size_bytes	OVN
Gauge	kube_ovn_db_file_size_bytes	OVN
Gauge	kube_ovn_chassis_info	OVN chassis (1) (0)
Gauge	kube_ovn_db_status	OVN , (1) (0)
Gauge	kube_ovn_logical_switch_info	OVN logical switch (1) logical switch
Gauge	kube_ovn_logical_switch_external_id	OVN logical switch external_id (1) external-id
Gauge	kube_ovn_logical_switch_port_binding	OVN logical switch logical switch port (1)
Gauge	kube_ovn_logical_switch_tunnel_key	OVN logical switch tunnel key
Gauge	kube_ovn_logical_switch_ports_num	OVN logical switch logical port
Gauge	kube_ovn_logical_switch_port_info	OVN logical switch port (1)
Gauge	kube_ovn_logical_switch_port_tunnel_key	OVN logical switch port tunnel key
Gauge	kube_ovn_cluster_enabled	(1) OVN (0) OVN
Gauge	kube_ovn_cluster_role	(1)
Gauge	kube_ovn_cluster_status	(1)
Gauge	kube_ovn_cluster_term	RAFT term
Gauge	kube_ovn_cluster_leader_self	leader (1) (0)
Gauge	kube_ovn_cluster_vote_self	leader (1) (0)
Gauge	kube_ovn_cluster_election_timer	election timer
Gauge	kube_ovn_cluster_log_not_committed	commit RAFT
Gauge	kube_ovn_cluster_log_not_applied	apply RAFT
Gauge	kube_ovn_cluster_log_index_start	RAFT
Gauge	kube_ovn_cluster_log_index_next	RAFT
Gauge	kube_ovn_cluster_inbound_connections_total	
Gauge	kube_ovn_cluster_outbound_connections_total	
Gauge	kube_ovn_cluster_inbound_connections_error_total	
Gauge	kube_ovn_cluster_outbound_connections_error_total	

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8.10.2 ovs-monitor

`ovsdb` `vswitchd`

Gauge	ovs_status	OVS (1) (0)
Gauge	ovs_info	OVS (1)
Gauge	failed_req_count	OVS
Gauge	log_file_size_bytes	OVS
Gauge	db_file_size_bytes	OVS
Gauge	datapath	Datapath (1)
Gauge	dp_total	OVS datapath
Gauge	dp_if	Datapath (1)
Gauge	dp_if_total	datapath port
Gauge	dp_flows_total	Datapath flow
Gauge	dp_flows_lookup_hit	Datapath flow
Gauge	dp_flows_lookup_missed	Datapath flow
Gauge	dp_flows_lookup_lost	Datapath userspace
Gauge	dp_masks_hit	Datapath mask
Gauge	dp_masks_total	Datapath mask
Gauge	dp_masks_hit_ratio	Datapath mask
Gauge	interface	OVS (1)
Gauge	interface_admin_state	(0) down, (1) up, (2)
Gauge	interface_link_state	(0) down, (1) up, (2)
Gauge	interface_mac_in_use	OVS Interface MAC
Gauge	interface_mtu	OVS Interface MTU
Gauge	interface_of_port	OVS Interface OpenFlow Port ID
Gauge	interface_if_index	OVS Interface Index
Gauge	interface_tx_packets	OVS Interface
Gauge	interface_tx_bytes	OVS Interface
Gauge	interface_rx_packets	OVS Interface
Gauge	interface_rx_bytes	OVS Interface
Gauge	interface_rx_crc_err	OVS Interface
Gauge	interface_rx_dropped	OVS Interface
Gauge	interface_rx_errors	OVS Interface
Gauge	interface_rx_frame_err	OVS Interface
Gauge	interface_rx_missed_err	OVS Interface miss
Gauge	interface_rx_over_err	OVS Interface overrun
Gauge	interface_tx_dropped	OVS Interface

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Gauge	interface_tx_errors	OVS Interface
Gauge	interface_collisions	OVS interface

8.10.3 kube-ovn-pinger

Gauge	pinger_ovs_up	OVS
Gauge	pinger_ovs_down	OVS
Gauge	pinger_ovn_controller_up	ovn-controller
Gauge	pinger_ovn_controller_down	ovn-controller
Gauge	pinger_inconsistent_port_binding	OVN-SB portbinding OVS interface
Gauge	pinger_apiserver_healthy	kube-ovn-pinger apiserver
Gauge	pinger_apiserver_unhealthy	kube-ovn-pinger apiserver
Histogram	pinger_apiserver_latency_ms	kube-ovn-pinger apiserver
Gauge	pinger_internal_dns_healthy	kube-ovn-pinger
Gauge	pinger_internal_dns_unhealthy	kube-ovn-pinger
Histogram	pinger_internal_dns_latency_ms	kube-ovn-pinger
Gauge	pinger_external_dns_healthy	kube-ovn-pinger
Gauge	pinger_external_dns_unhealthy	kube-ovn-pinger
Histogram	pinger_external_dns_latency_ms	kube-ovn-pinger
Histogram	pinger_pod_ping_latency_ms	kube-ovn-pinger ping Pod
Gauge	pinger_pod_ping_lost_total	kube-ovn-pinger ping Pod
Gauge	pinger_pod_ping_count_total	kube-ovn-pinger ping Pod
Histogram	pinger_node_ping_latency_ms	kube-ovn-pinger ping Node
Gauge	pinger_node_ping_lost_total	kube-ovn-pinger ping Node
Gauge	pinger_node_ping_count_total	kube-ovn-pinger ping Node
Histogram	pinger_external_ping_latency_ms	kube-ovn-pinger ping
Gauge	pinger_external_ping_lost_total	kube-ovn-pinger ping

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8.10.4 kube-ovn-controller

`kube-ovn-controller`

Histogram	rest_client_request_latency_seconds	apiserver
Counter	rest_client_requests_total	apiserver
Counter	lists_total	API list
Summary	list_duration_seconds	API list
Summary	items_per_list	API list
Counter	watches_total	API watch
Counter	short_watches_total	API watch
Summary	watch_duration_seconds	API watch
Summary	items_per_watch	API watch
Gauge	last_resource_version	resource version
Histogram	ovs_client_request_latency_milliseconds	OVN
Gauge	subnet_available_ip_count	IP
Gauge	subnet_used_ip_count	IP

8.10.5 kube-ovn-cni

`kube-ovn-cni`

Histogram	cni_op_latency_seconds	CNI
Counter	cni_wait_address_seconds_total	CNI
Counter	cni_wait_connectivity_seconds_total	CNI
Counter	cni_wait_route_seconds_total	CNI
Histogram	rest_client_request_latency_seconds	apiserver
Counter	rest_client_requests_total	apiserver
Counter	lists_total	API list
Summary	list_duration_seconds	API list
Summary	items_per_list	API list
Counter	watches_total	API watch
Counter	short_watches_total	API watch
Summary	watch_duration_seconds	API watch
Summary	items_per_watch	API watch
Gauge	last_resource_version	resource version
Histogram	ovs_client_request_latency_milliseconds	OVN

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8.10.6

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8.11 Kube-OVN

Kube-OVN Kube-OVN CRD CRD

8.11.1 Condition

type	String	
status	String	`True` `False` `Unknown`
reason	String	
message	String	
observedGeneration	Int64	
lastUpdateTime	Time	
lastTransitionTime	Time	

CRD Status Condition

8.11.2

Subnet

SUBNET

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	`Subnet`
metadata	ObjectMeta	Kubernetes	
spec	SubnetSpec	Subnet	
status	SubnetStatus	Subnet	

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SubnetSpec

default	Bool	
vpc	String	VPC ovn-cluster
protocol	String	IP `IPv4` `IPv6` `Dual`
namespaces	[]String	namespace
cidrBlock	String	10.16.0.0/16
gateway	String	CIDRBlock
excludeIps	[]String	
provider	String	OVN NetworkAttachmentDefinition . Kube-OVN Subnet
gatewayType	String	Overlay `distributed` `centralized`
gatewayNode	String	centralized
natOutgoing	Bool	NAT `externalEgressGateway`
externalEgressGateway	String	`natOutgoing`
policyRoutingPriority	Uint32	
policyRoutingTableID	Uint32	TableID
mtu	Uint32	MTU
private	Bool	
allowSubnets	[]String	
vlan	String	Vlan
vips	[]String	virtual lsp virtual-ip
logicalGateway	Bool	
disableGatewayCheck	Bool	Pod
disableInterConnection	Bool	
enableDHCP	Bool	lsp dhcp
dhcpV4Options	String	lsp dhcpv4_options DHCP_Options
dhcpV6Options	String	lsp dhcpv6_options DHCP_Options
enableIPv6RA	Bool	lrp ipv6_ra_configs
ipv6RAConfigs	String	lrp ipv6_ra_configs
acls	[]Acl	logical-switch acls
allowEWTraffic	Bool	
natOutgoingPolicyRules	[]NatOutgoingPolicyRule	NAT
u2oInterconnectionIP	String	Underlay/Overlay IP
u2oInterconnection	Bool	Overlay/Underlay
enableLb	*Bool	logical-switch load-balancer
enableEcmp	Bool	ECMP

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enableMulticastSnoop	Bool
enableExternalLBAddress	Bool
routeTable	String
namespaceSelectors	[]LabelSelector

Acl

direction	String	Acl	`from-lport` `to-lport`
priority	Int	Acl	0 32767
match	String	Acl	
action	String	Acl	`allow-related` `allow-stateless` `allow` `drop` `reject`

NatOutgoingPolicyRule

match	NatOutGoingPolicyMatch
action	String

NatOutGoingPolicyMatch

srcIPs	String	IP
dstIPs	String	IP

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SubnetStatus

conditions	[]SubnetCondition	Condition	
v4availableIPs	BigInt	IPv4 IP	IPv6 float64
v4availableIPrange	String	IPv4	
v4usingIPs	BigInt	IPv4 IP	
v4usingIPrange	String	IPv4	
v6availableIPs	BigInt	IPv6 IP	
v6availableIPrange	String	IPv6	
v6usingIPs	BigInt	IPv6 IP	
v6usingIPrange	String	IPv6	
activateGateway	String		
dhcpV4OptionsUUID	String	lsp dhcpv4_options	DHCP_Options
dhcpV6OptionsUUID	String	lsp dhcpv6_options	DHCP_Options
u2oInterconnectionIP	String	Overlay/Underlay	IP
u2oInterconnectionMAC	String	Overlay/Underlay	MAC
u2oInterconnectionVPC	String	Overlay/Underlay	VPC
natOutgoingPolicyRules	[]NatOutgoingPolicyRuleStatus	NAT	
mcastQuerierIP	String	IP	
mcastQuerierMAC	String	MAC	

IP

IP

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	`IP`
metadata	ObjectMeta	Kubernetes	
spec	IPSpec	IP	

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IPSpec

podName	String	Pod
namespace	String	Pod Namespace
subnet	String	IP Subnet
attachSubnets	[]String	IP
nodeName	String	Pod
ipAddress	String	IP `v4IP v6IP`
v4IpAddress	String	IPv4 IP
v6IpAddress	String	IPv6 IP
attachIps	[]String	IP IP
macAddress	String	Pod MAC
attachMacs	[]String	IP MAC
containerID	String	Pod Container ID
podType	String	Pod `StatefulSet` `VirtualMachine`

Vpc

VPC

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	`Vpc`
metadata	ObjectMeta	Kubernetes	
spec	VpcSpec	Vpc	
status	VpcStatus	Vpc	

VpcSpec

defaultSubnet	String	
namespaces	[]String	Vpc
staticRoutes	[]StaticRoute	
policyRoutes	[]PolicyRoute	
vpcPeerings	[]VpcPeering	VPC
enableExternal	Bool	
extraExternalSubnets	[]String	
enableBfd	Bool	BFD ( )
bfdPort	BFDPort	BFD

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StaticRoute

policy	String	
cidr	String	
nextHopIP	String	IP
ecmpMode	String	ECMP
bfdId	String	BFD ID
routeTable	String	

PolicyRoute

priority	Int	
match	String	
action	String	`allow` `drop` `reroute`
nextHopIP	String	IP action reroute

VpcPeering

remoteVpc	String	VPC
localConnectIP	String	IP

BFDPort

enabled	Bool	BFD
ip	String	BFD IP
nodeSelector	LabelSelector	BFD LRP

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VpcStatus

conditions	[]VpcCondition	Vpc	Condition
standby	Bool	VPC	
default	Bool	VPC	
defaultLogicalSwitch	String		
router	String		
tcpLoadBalancer	String	TCP	
udpLoadBalancer	String	UDP	
sctpLoadBalancer	String	SCTP	
tcpSessionLoadBalancer	String	TCP	
udpSessionLoadBalancer	String	UDP	
sctpSessionLoadBalancer	String	SCTP	
subnets	[]String	VPC	
vpcPeerings	[]String	VPC	
enableExternal	Bool		
extraExternalSubnets	[]String		
enableBfd	Bool	BFD	

8.11.3 Underlay

Vlan

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	`Vlan`
metadata	ObjectMeta	Kubernetes	
spec	VlanSpec	Vlan	
status	VlanStatus	Vlan	

VLANSPEC

id	Int	Vlan tag	0~4096
provider	String	Vlan	ProviderNetwork

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VLANSTATUS

subnets	[]String	Vlan	
conflict	Bool		
conditions	[]VlanCondition	Vlan	Condition

ProviderNetwork

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	`ProviderNetwork`
metadata	ObjectMeta	Kubernetes	
spec	ProviderNetworkSpec	ProviderNetwork	
status	ProviderNetworkStatus	ProviderNetwork	

PROVIDERNETWORKSPEC

defaultInterface	String						
customInterfaces	[]CustomInterface						
nodeSelector	LabelSelector	OVS	matchLabels	matchExpressions	nodeSelector	excludeNodes	
excludeNodes	[]String						
exchangeLinkName	Bool	OVS					

CustomInterface

interface	String	Underlay
nodes	[]String	

PROVIDERNETWORKSTATUS

ready	Bool		
readyNodes	[]String		
notReadyNodes	[]String		
vlans	[]String	Vlan	
conditions	[]ProviderNetworkCondition	ProviderNetwork	Condition

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8.11.4

SecurityGroup

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	`SecurityGroup`
metadata	ObjectMeta	Kubernetes	
spec	SecurityGroupSpec	SecurityGroup	
status	SecurityGroupStatus	SecurityGroup	

SECURITYGROUPSPEC

ingressRules	[]SecurityGroupRule	
egressRules	[]SecurityGroupRule	
allowSameGroupTraffic	Bool	

SecurityGroupRule

ipVersion	String	IP	`ipv4` `ipv6`
protocol	SgProtocol		`all` `icmp` `tcp` `udp`
priority	Int		1-200
remoteType	SgRemoteType		`address` `securityGroup`
remoteAddress	String		
remoteSecurityGroup	String		
portRangeMin	Int		1
portRangeMax	Int		65535
policy	SgPolicy		`allow` `drop`

SECURITYGROUPSTATUS

portGroup	String	
allowSameGroupTraffic	Bool	
ingressMd5	String	MD5
egressMd5	String	MD5
ingressLastSyncSuccess	Bool	
egressLastSyncSuccess	Bool	

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8.11.5 IP

Vip

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	`Vip`
metadata	ObjectMeta	Kubernetes	
spec	VipSpec	Vip	
status	VipStatus	Vip	

VIPSPEC

namespace	String	VIP	
subnet	String	VIP	
type	String	VIP	
v4ip	String	IPv4	
v6ip	String	IPv6	
macAddress	String	MAC	
selector	[]String		
attachSubnets	[]String		

VIPSTATUS

conditions	[]VipCondition	VIP	Condition
type	String	VIP	
v4ip	String	IPv4	
v6ip	String	IPv6	
mac	String	MAC	

SwitchLBRule

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	`SwitchLBRule`
metadata	ObjectMeta	Kubernetes	
spec	SwitchLBRuleSpec	SwitchLBRule	
status	SwitchLBRuleStatus	SwitchLBRule	

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SWITCHLBRULESPEC

vip	String	IP
namespace	String	
selector	[]String	
endpoints	[]String	
sessionAffinity	String	
ports	[]SwitchLBRulePort	

SwitchLBRulePort

name	String
port	Int32
targetPort	Int32
protocol	String

SWITCHLBRULESTATUS

conditions	[]SwitchLBRuleCondition	SwitchLBRule	Condition
ports	String	SwitchLBRule	
service	String	SwitchLBRule	service

8.11.6 QoS IP

QoSPolicy

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	`QoSPolicy`
metadata	ObjectMeta	Kubernetes	
spec	QoSPolicySpec	QoSPolicy	

QOSPOLICYSPEC

bandwidthLimitRules	QoSPolicyBandwidthLimitRules
shared	Bool
bindingType	QoSPolicyBindingType

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IPPool

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	`IPPool`
metadata	ObjectMeta	Kubernetes	
spec	IPPoolSpec	IPPool	
status	IPPoolStatus	IPPool	

IPPOOLSPEC

subnet	String	
namespaces	[]String	
ips	[]String	IP

IPPOOLSTATUS

v4AvailableIPs	BigInt	IPv4 IP	
v4AvailableIPRange	String	IPv4 IP	
v4UsingIPs	BigInt	IPv4 IP	
v4UsingIPRange	String	IPv4 IP	
v6AvailableIPs	BigInt	IPv6 IP	
v6AvailableIPRange	String	IPv6 IP	
v6UsingIPs	BigInt	IPv6 IP	
v6UsingIPRange	String	IPv6 IP	
conditions	[]IPPoolCondition	IP	Condition

8.11.7 NAT IP

IptablesEIP

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	`IptablesEIP`
metadata	ObjectMeta	Kubernetes	
spec	IptablesEIPSpec	IptablesEIP	
status	IptablesEIPStatus	IptablesEIP	

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IPTABLESEIPSPEC

v4ip	String	IPv4
v6ip	String	IPv6
macAddress	String	MAC
natGwDp	String	NAT
qosPolicy	String	QoS
externalSubnet	String	

IPTABLESEIPSTATUS

ready	Bool	IptablesEIP
ip	String	IptablesEIP IP IPv4
redo	String	IptablesEIP CRD
nat	String	IptablesEIP `fip` `snat` `dnat`
qosPolicy	String	QoS
conditions	[]IptablesEIPCondition	IptablesEIP Condition

OvnEip

apiVersion	String	Kubernetes kubeovn.io/v1
kind	String	Kubernetes `OvnEip`
metadata	ObjectMeta	Kubernetes
spec	OvnEipSpec	OvnEip
status	OvnEipStatus	OvnEip

OVNEIPSPEC

externalSubnet	String	
v4Ip	String	IPv4
v6Ip	String	IPv6
macAddress	String	MAC
type	String	lrp lsp nat

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IptablesFIPRule

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	`IptablesFIPRule`
metadata	ObjectMeta	Kubernetes	
spec	IptablesFIPRuleSpec	IptablesFIPRule	

IPTABLESFIPRULESPEC

eip	String	IP
internalIP	String	IP

OvnFip

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	`OvnFip`
metadata	ObjectMeta	Kubernetes	
spec	OvnFipSpec	OvnFip	
status	OvnFipStatus	OvnFip	

OVNFIPSPEC

ovnEip	String	OVN EIP
ipType	String	IP vip ip
ipName	String	IP
vpc	String	VPC
v4Ip	String	IPv4
v6Ip	String	IPv6
type	String	distributed centralized

IptablesDnatRule

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	`IptablesDnatRule`
metadata	ObjectMeta	Kubernetes	
spec	IptablesDnatRuleSpec	IptablesDnatRule	

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IPTABLESDNATRULESPEC

eip	String	IP
externalPort	String	
protocol	String	
internalIP	String	IP
internalPort	String	

OvnDnatRule

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	`OvnDnatRule`
metadata	ObjectMeta	Kubernetes	
spec	OvnDnatRuleSpec	OvnDnatRule	
status	OvnDnatRuleStatus	OvnDnatRule	

OVNDNATRULESPEC

ovnEip	String	OVN EIP
ipType	String	IP vip ip
ipName	String	IP
internalPort	String	
externalPort	String	
protocol	String	
vpc	String	VPC
v4Ip	String	IPv4
v6Ip	String	IPv6

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OVNDNATRULESTATUS

vpc	String	VPC	
v4Eip	String	IPv4 EIP	
v6Eip	String	IPv6 EIP	
externalPort	String		
v4Ip	String	IPv4	
v6Ip	String	IPv6	
internalPort	String		
protocol	String		
ipName	String	IP	
ready	Bool	DNAT	
conditions	[]OvnDnatRuleCondition	OVN DNAT	Condition

IptablesSnatRule

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	`IptablesSnatRule`
metadata	ObjectMeta	Kubernetes	
spec	IptablesSnatRuleSpec	IptablesSnatRule	

IPTABLESSNATRULESPEC

eip	String	IP
internalCIDR	String	CIDR

OvnSnatRule

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	`OvnSnatRule`
metadata	ObjectMeta	Kubernetes	
spec	OvnSnatRuleSpec	OvnSnatRule	
status	OvnSnatRuleStatus	OvnSnatRule	

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OVNSNATRULESPEC

ovnEip	String	OVN EIP
vpcSubnet	String	VPC
ipName	String	IP
vpc	String	VPC
v4IpCidr	String	IPv4 CIDR
v6IpCidr	String	IPv6 CIDR

8.11.8 VPC

VpcNatGateway

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	`VpcNatGateway`
metadata	ObjectMeta	Kubernetes	
spec	VpcNatGatewaySpec	VpcNatGateway	
status	VpcNatGatewayStatus	VpcNatGateway	

VPCNATGATEWAYSPEC

vpc	String	VPC Pod VPC
subnet	String	VPC Pod
externalSubnets	[]String	
lanIp	String	VPC Pod IP
selector	[]String	Kubernetes Selector
tolerations	[]Toleration	Kubernetes
affinity	Affinity	Kubernetes
qosPolicy	String	QoS
bgpSpeaker	VpcBgpSpeaker	BGP speaker
routes	[]Route	

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VpcBgpSpeaker

enabled	Bool	BGP speaker
asn	Uint32	
remoteAsn	Uint32	
neighbors	[]String	BGP
holdTime	Duration	BGP
routerId	String	BGP ID
password	String	BGP
enableGracefulRestart	Bool	
extraArgs	[]String	

Route

cidr	String	CIDR
nextHopIP	String	IP

VPCNATGATEWAYSTATUS

qosPolicy	String	QoS
externalSubnets	[]String	
selector	[]String	Kubernetes Selector
tolerations	[]Toleration	Kubernetes
affinity	Affinity	Kubernetes

VpcEgressGateway

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	`VpcEgressGateway`
metadata	ObjectMeta	Kubernetes	
spec	VpcEgressGatewaySpec	VpcEgressGateway	
status	VpcEgressGatewayStatus	VpcEgressGateway	

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VPCEGRESSGATEWAYSPEC

vpc	String	VPC VPC
bgpConf	String	`BgpConf`
evpnConf	String	`EvpnConf` EVPN
replicas	Int32	1
prefix	String	`<prefix><veg-name>`
image	String	kube-ovn-controller
internalSubnet	String	VPC
externalSubnet	String	
internalIPs	[]String	IP `v4,v6`
externalIPs	[]String	IP `v4,v6`
internalIPPool	String	`IPPool` `internalSubnet` `internalIPs`
externalIPPool	String	`IPPool` `externalSubnet` `externalIPs`
selectors	[]VpcEgressGatewaySelector	/Pod
trafficPolicy	String	`Cluster` `Local` VPC
bfd	VpcEgressGatewayBFDConfig	BFD
policies	[]VpcEgressGatewayPolicy	
nodeSelector	[]VpcEgressGatewayNodeSelector	
tolerations	[]Toleration	Kubernetes
resources	ResourceRequirements	
bandwidth	BandwidthLimit	/ Mbps bit-rate quantity

VpcEgressGatewaySelector

namespaceSelector	LabelSelector	
podSelector	LabelSelector	Pod

VpcEgressGatewayBFDConfig

enabled	Boolean	BFD	VPC `.spec.bfd.enabled=true`	false
minRX	Int32	BFD	ms	1000
minTX	Int32	BFD	ms	1000
multiplier	Int32	BFD		3

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VpcEgressGatewayPolicy

snat	Boolean	SNAT/MASQUERADE	false
ipBlocks	[]String	CIDR	
subnets	[]String		

VpcEgressGatewayNodeSelector

matchLabels	map[String]String
matchExpressions	[]NodeSelectorRequirement
matchFields	[]NodeSelectorRequirement

BandwidthLimit

ingress	Int64	String	VPC	Mbps quantity	`M` `Mi` `G` `Gi` bit/s
egress	Int64	String	VPC	Mbps quantity	`M` `Mi` `G` `Gi` bit/s

VPCEGRESSGATEWAYSTATUS

replicas	Int32	scale
labelSelector	String	scale
ready	Boolean	
phase	String	`Pending` `Processing` `Completed`
internalIPs	[]String	IP
externalIPs	[]String	IP
conditions	[]Condition	
workload	VpcEgressWorkload	

VpcEgressWorkload

apiVersion	String	API
kind	String	`Deployment`
name	String	
nodes	[]String	

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BgpConf

cluster-scoped `VpcEgressGateway.spec.bgpConf`

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	`BgpConf`
metadata	ObjectMeta	Kubernetes	
spec	BgpConfSpec	BgpConf	

BGPCONFSPEC

localASN	UInt32	BGP AS Number
peerASN	UInt32	BGP AS Number
routerId	String	BGP Router ID
neighbours	[]String	BGP IP
password	String	TCP MD5
holdTime	Duration	BGP Hold time
keepaliveTime	Duration	BGP Keepalive
connectTime	Duration	BGP Connect
ebgpMultiHop	Boolean	eBGP MultiHop
gracefulRestart	Boolean	BGP Graceful Restart

EvpnConf

cluster-scoped `VpcEgressGateway.spec.evpnConf` EVPN VNI Route Target

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	`EvpnConf`
metadata	ObjectMeta	Kubernetes	
spec	EvpnConfSpec	EvpnConf	

EVPNCONFSPEC

vni	UInt32	EVPN VNI
routeTargets	[]String	Route Target

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DNSNameResolver

cluster-scoped `enable-dns-name-resolver` ANP/CNP DNS

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	`DNSNameResolver`
metadata	ObjectMeta	Kubernetes	
spec	DNSNameResolverSpec	DNSNameResolver	
status	DNSNameResolverStatus	DNSNameResolver	

DNSNAMERESOLVERSPEC

name	String	DNS	`*.`

DNSNAMERESOLVERSTATUS

resolvedNames	[]DNSNameResolverResolvedName	DNS	IP

DNSNameResolverResolvedName

dnsName	String	DNS
resolvedAddresses	[]DNSNameResolverResolvedAddress	IP TTL
conditions	[]Condition	`Degraded`
resolutionFailures	Int32	5 TTL

DNSNameResolverResolvedAddress

ip	String	IP	
ttlSeconds	Int32	TTL	`lastLookupTime + ttlSeconds`
lastLookupTime	Time		

VpcDns

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	`VpcDns`
metadata	ObjectMeta	Kubernetes	
spec	VpcDNSSpec	VpcDns	
status	VpcDNSStatus	VpcDns	

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VPCDNSSPEC

replicas	Int32	
vpc	String	VPC
subnet	String	

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8.12 Annotation

Kube-OVN Pod Node Annotation Annotation

> Annotation

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8.12.1 Pod Annotation

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Key	Value	Description
ovn.kubernetes.io/allocated	`true` or `false`	Pod
ovn.kubernetes.io/routed	`true` or `false`	Pod OVN
ovn.kubernetes.io/routes	String	Pod
ovn.kubernetes.io/mac_address	String	Pod Mac Pod Annotation Mac
ovn.kubernetes.io/ip_address	String	Pod IP Pod Annotation IP
\<nadName>.\<nadNamespace>.kubernetes.io/ip_address.\<interfaceName>	String	IP interfaceName Multus interface ovn.kubernetes.io/ip_address
\<nadName>.\<nadNamespace>.kubernetes.io/mac_address.\<interfaceName>	String	MAC interfaceName Multus interface
ovn.kubernetes.io/cidr	String	Pod CIDR
ovn.kubernetes.io/gateway	String	Pod Gateway
ovn.kubernetes.io/ip_pool	IP	Pod Workload IP
ovn.kubernetes.io/bgp	`true`, `cluster`, `local`	BGP Pod
ovn.kubernetes.io/snat	String	Pod SNAT
ovn.kubernetes.io/eip	String	Pod EIP
ovn.kubernetes.io/vip	String	Pod VIP Annotation VIP
ovn.kubernetes.io/aaps	String	Pod AAPs (Additional Allowed Addresses Pairs)
ovn.kubernetes.io/virtualmachine	String	Pod VirtualMachineInstance
ovn.kubernetes.io/activation_strategy	String	Pod
ovn.kubernetes.io/logical_router	String	Pod VPC
ovn.kubernetes.io/layer2_forward	`true` or `false`	Pod OVN LSP `unknown`
ovn.kubernetes.io/port_security	`true` or `false`	Pod Port Security
ovn.kubernetes.io/logical_switch	String	Pod
ovn.kubernetes.io/vlan_id	Int	Pod Vlan ID
ovn.kubernetes.io/ingress_rate	Int	Pod Mbits/s
ovn.kubernetes.io/egress_rate	Int	Pod Mbits/s
ovn.kubernetes.io/security_groups	String	Pod Security Group
ovn.kubernetes.io/default_route	`true` or `false`	
ovn.kubernetes.io/provider_network	String	Pod ProviderNetwork
ovn.kubernetes.io/mirror	`true` or `false`	Pod
ovn.kubernetes.io/north_gateway	String	Pod
ovn.kubernetes.io/latency	Int	Pod ms
ovn.kubernetes.io/limit	Int	Pod qdisc
ovn.kubernetes.io/loss	Float	Pod
ovn.kubernetes.io/jitter	Int	Pod ms

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Key	Value	Description
ovn.kubernetes.io/generate-hash	`true` or `false`	Pod
ovn.kubernetes.io/attachmentprovider	String	Pod
ovn.kubernetes.io/ingress_burst	Int	Pod burst
ovn.kubernetes.io/vpc_nat_gw	String	VPC NAT Gateway Pod
ovn.kubernetes.io/vpc_nat_gw_init	`true` or `false`	VPC NAT Gateway Pod
ovn.kubernetes.io/vpc_nat_gw_container_restarted	`true` or `false`	VPC NAT Gateway
ovn.kubernetes.io/vpc_nat_gw_activated	`true` or `false`	VPC NAT Gateway
\<provider>.kubernetes.io/ip_pool	IP	provider IP
\<provider>.kubernetes.io/dhcp-v4-options	String	provider DHCPv4
\<provider>.kubernetes.io/port_security	`true` or `false`	provider

8.12.2 Node Annotation

Key	Value	Description
ovn.kubernetes.io/allocated	`true` or `false`	`ovn0` `join`
ovn.kubernetes.io/mac_address	String	Node `ovn0` Mac
ovn.kubernetes.io/ip_address	String	Node `ovn0` IP
ovn.kubernetes.io/cidr	String	Node `ovn0` `join` CIDR
ovn.kubernetes.io/gateway	String	Node `ovn0` `join` Gateway
ovn.kubernetes.io/chassis	String	Node OVN-SouthBoundDB Chassis ID
ovn.kubernetes.io/port_name	String	Node `ovn0` OVN-NorthboundDB LSP
ovn.kubernetes.io/logical_switch	String	Node `ovn0`
ovn.kubernetes.io/tunnel_interface	String	
ovn.kubernetes.io/node_networks	String	Pod attachment networks NetworkAttachmentDefinition

8.12.3 Namespace Annotation

Key	Value	Description
ovn.kubernetes.io/cidr	CIDR	Namespace CIDR
ovn.kubernetes.io/exclude_ips	excludeIPs	Namespace excludeIPs

8.12.4 Subnet Annotation

Key	Value	Description
ovn.kubernetes.io/bgp	`true`, `cluster`, `local`	BGP
ovn.kubernetes.io/vpc	String	VPC VPC

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8.12.5 Service Annotation

Key	Value	Description
ovn.kubernetes.io/bgp	`true`, `cluster`, `local`	BGP Service ClusterIP Service Pod/ Subnet
ovn.kubernetes.io/switch_lb_vip	String	Service Kube-OVN VIP
ovn.kubernetes.io/vpc	String	Service VPC
ovn.kubernetes.io/ service_external_ip_from_subnet	`true` or `false`	Service IP
ovn.kubernetes.io/service_health_check	`true` or `false`	Service
ovn.kubernetes.io/lb_svc_img	String	

8.12.6 Networkpolicy Annotation

Key	Value	Description
ovn.kubernetes.io/enable_log	`true` or `false`	NetworkPolicy
ovn.kubernetes.io/log_acl_actions	"allow,drop,pass"	Action ACL
ovn.kubernetes.io/acl_log_meter_rate	Int	NetworkPolicy /
ovn.kubernetes.io/ network_policy_enforcement	`standard` or `lax`	NetworkPolicy
ovn.kubernetes.io/network_policy_for	String	NetworkPolicy provider `<nadName>.<nadNamespace>.ovn`

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8.13

8.13.1

Bad	Good

Bad			Good		
Kube-OVN	1.10	Kube-OVN	Kube-OVN	1.10	Kube-OVN

Bad	Good
`wget 127.0.0.1`	`wget 127.0.0.1`

8.13.2

yaml yaml

Bad	Good
```` ```apiVersion: kubeovn.io/v1 kind: Subnet metadata: name: attach-subnet ``` ````	```` ```yaml apiVersion: kubeovn.io/v1 kind: Subnet metadata: name: attach-subnet ``` ````

bash

Bad	Good
```` ```wget 127.0.0.1 ``` ````	```` ```bash wget 127.0.0.1 ``` ````

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`#`

Bad

```
oilbeater@macdeMac-3 ~ ping 114.114.114.114 -c 3
PING 114.114.114.114 (114.114.114.114): 56 data bytes
64 bytes from 114.114.114.114: icmp_seq=0 ttl=83 time=10.429 ms
64 bytes from 114.114.114.114: icmp_seq=1 ttl=79 time=11.360 ms
64 bytes from 114.114.114.114: icmp_seq=2 ttl=76 time=10.794 ms

--- 114.114.114.114 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 10.429/10.861/11.360/0.383 ms
```

Good

```
# ping 114.114.114.114 -c 3
PING 114.114.114.114 (114.114.114.114): 56 data bytes
64 bytes from 114.114.114.114: icmp_seq=0 ttl=83 time=10.429 ms
64 bytes from 114.114.114.114: icmp_seq=1 ttl=79 time=11.360 ms
64 bytes from 114.114.114.114: icmp_seq=2 ttl=76 time=10.794 ms

--- 114.114.114.114 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 10.429/10.861/11.360/0.383 ms
```

`#`

Bad

```
# mv /etc/origin/ovn/ovnnb_db.db /tmp
# mv /etc/origin/ovn/ovnsb_db.db /tmp
```

Good

```
mv /etc/origin/ovn/ovnnb_db.db /tmp
mv /etc/origin/ovn/ovnsb_db.db /tmp
```

8.13.3

`md`

Bad

```
[    ](http://kubeovn.github.io/prepare)
```

Good

```
[    ](./prepare.md)
```

Bad

```
[Kubernetes    ](http://kubernetes.io)
```

Good

```
[Kubernetes    ](http://kubernetes.io){: target="_blank" }
```

8.13.4

Bad

````
```bash
wget 127.0.0.1
```
````

Good

````
```bash
wget 127.0.0.1
```
````

Bad

````
```bash
wget 127.0.0.1
```
````

Good

````
```bash
wget 127.0.0.1
```
````

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8.13.5

9.

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9.1

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- LinkedIn Kube-OVN
- Issue / GitHub Issues GitHub Discussions

9.2

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9.3

-
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9.4

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